RE – ENTRY#01 PROGRAM

AZADEGAN OIL FIELD

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AZNS – 127 (PAD #54), TYPE: IIF

Drilling Department

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1. General Information

1.1 Field Introduction

Azadegan South Oil Field is situated in West Karoun province on a large long-axis anticline with N-S trending with approximate dimensions about 20 km width and 60 km length near to Iran-Iraq border, to the east of Yaran Oil Field.

The trap area of the south Azadegan is limited to 43.2x17.2 km. The Azadegan Oil field was discovered in 1999, when Well AZN-1 was drilled vertically down to the Cretaceous Fahliyan Formation with total depth of 4,643m. The results of this well, revealed the reservoir characteristics of the field. Following to the result of the well no. AZN-1, another 5 exploration wells was drilled in different parts of this huge field down to Fahliyan Formation. In the later stage the well No.10 assumed as exploration well for evaluation of deeper formations. This well was drilled in 2008, down to the depth of 5340m (Neyriz Fm.), mainly to evaluate the Jurassic formations.

Based on the exploration wells results, carbonate reservoir of Sarvak Formation is broadly developed in Azadegan Field. This formation was penetrated in all of seven exploration wells (AZN-1, AZN-2, AZN-3, AZN-4, AZN-5, AZN-6 and AZN-10). Clastic oil-bearing zones of Kazhdumi Formation developed in main anticline of Azadegan Field, extending from north to center of South Azadegan. The Clastic oil-bearing zones of Gadvan Formation developed only in crest area and south parts of South Azadegan Filed, which was discovered by AZN-1 in crest and AZN-4 in south parts. The carbonates of Fahliyan Formation also developed in Azadegan field, but the oil reservoir of this formation developed in South part of South Azadegan.

The Azadegan Field is an elongated anticline, trending in N-S direction, belonging to "Arabian Trend". The stratigraphy sequence of the field includes AghaJari, Gachsaran, Fahliyan, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan and Fahliyan Formations, from top to bottom. This sequence in well no AZN-10, the deepest up to now drilled well of the field, continues with Garau, Gotnia, Najmah, Sargalu, Alan, Mus and Neyriz formations.

The Dominant lithofacies of Sarvak Formation are Rudist-bearing packstone and partially grainstone, which contains abundant bioclasts. Some other facies like matrix-dominated wackestone and mudstone are also found.

The AZADEGAN Oilfield is Located in Khuzestan Province and is Approximately Located 80 Kilometers West of Ahwaz City, near to Iran-Iraq Border. It is in The Northwest of The YADAVARAN Field and in The West of JUFEYR Field. The Average Monthly Temperature is 45ºc in summer, and Highest Temperature is 52ºc. The Lowest Temperature is -4.4ºc in winter. Maximum Rainfall is 78 mm in February, Winter Season.

This Document covers the work over of well AZNS-122. Main headlines of program are de-complete the well and run new completion string.

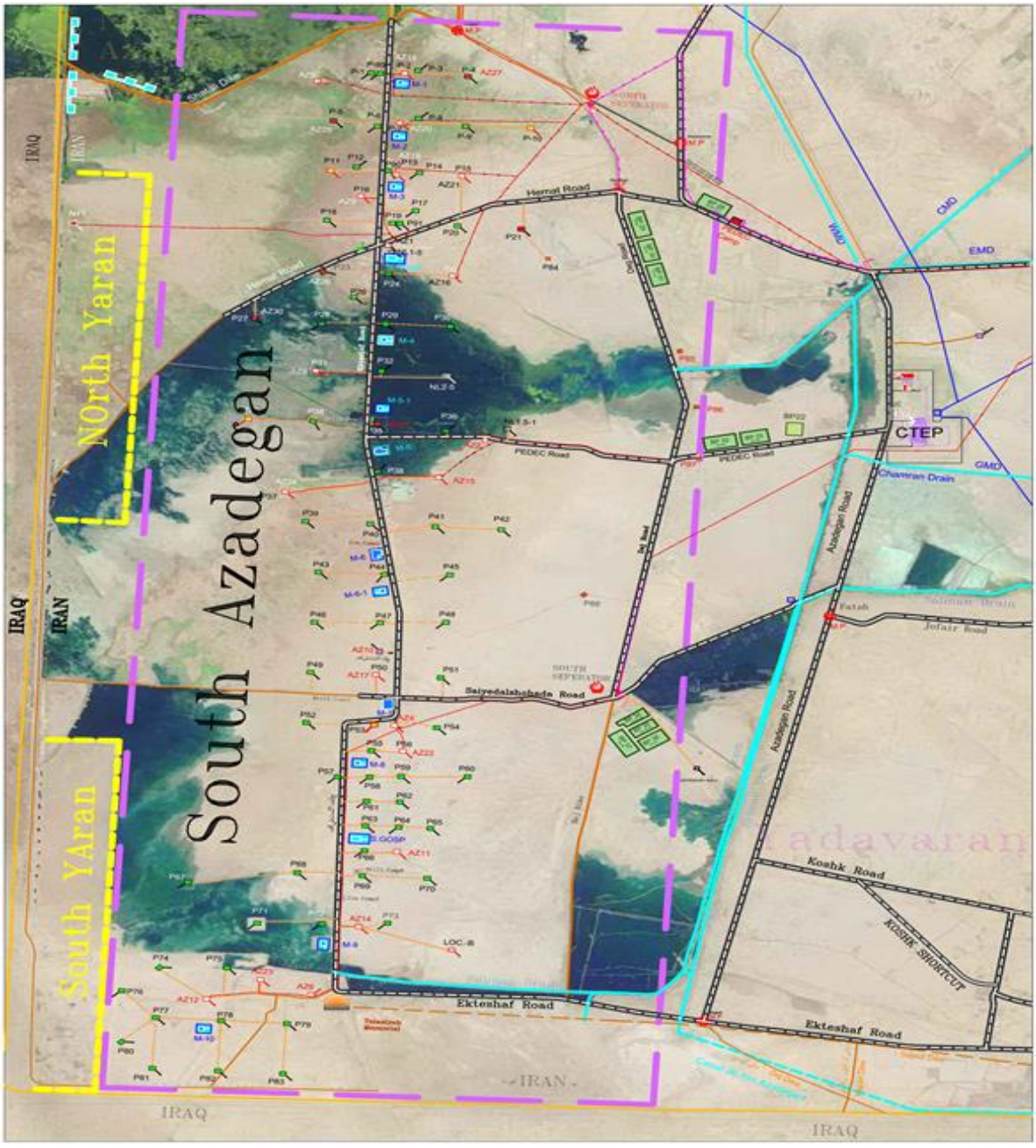


Figure 1-1: AZNS Oilfield Location Map

1.2 Well Information

Country / Area Name:	Iran / Khuzestan
Client	Petro Pars Limited
Field Name:	Azadegan
Type Of Operation:	Re – Entry
Well Name / Type :	AZNS – 127 / Fahliyan well in Block II
Pad Number:	54
Well Type:	Vertical
Wellhead Coordinate:	N: 3,448,561 E: 196,167
Exiting TD Coordinate:	N: 3,448,538.7 E: 196,163.4
New TD Coordinate:	N: 3,448,541.3 E: 196,185.7
Well Objective:	Open The Window & Side Track The Well from 3300 mMD, Drill 8 – 1/2" & 5 – 7/8" Hole & Complete the well.
RTE:	13.45^{m}
GLE:	3.95^{m}
Well Duration :	80 Days (Main Plan)
Total Depth (MD):	4150^{m}
Producer Formation:	Perforation area will be issued in due time in 4 – 1/2" & 7" Liner Through Fahliyan Formation
Formation TD :	Fahliyan
Reservoir :	Fahliyan

Table 1-1 AZNS-127 information

1.3 Area Distribution

According to the drilling history, the CONTRACT AREA is distributed into three blocks as per the geological features with regard to DRILLING OPERATION. This distribution may differ according to knowledge accumulated in the future WORK.
Contract area is as per below:

X (m)	Y (m)
185640	3434275
188020	3477361
205224	3476408
202842	3433327

Table 1-2 Area Distribution

Block I is located on the north part of the CONTRACT AREA in north of well AZN-010. Gachsaran formation is characterized by abnormal high pressure in this block.

Block II is located between well AZN-010 and AZN-011. In this block, Gachsaran formation is characterized by pressure tending to normal

Block III is located on the south of line well AZN-011. In this block, Gachsaran formation is characterized by normal pressure and all drilled wells in this block suffered from complete loss at Pabdeh formation.

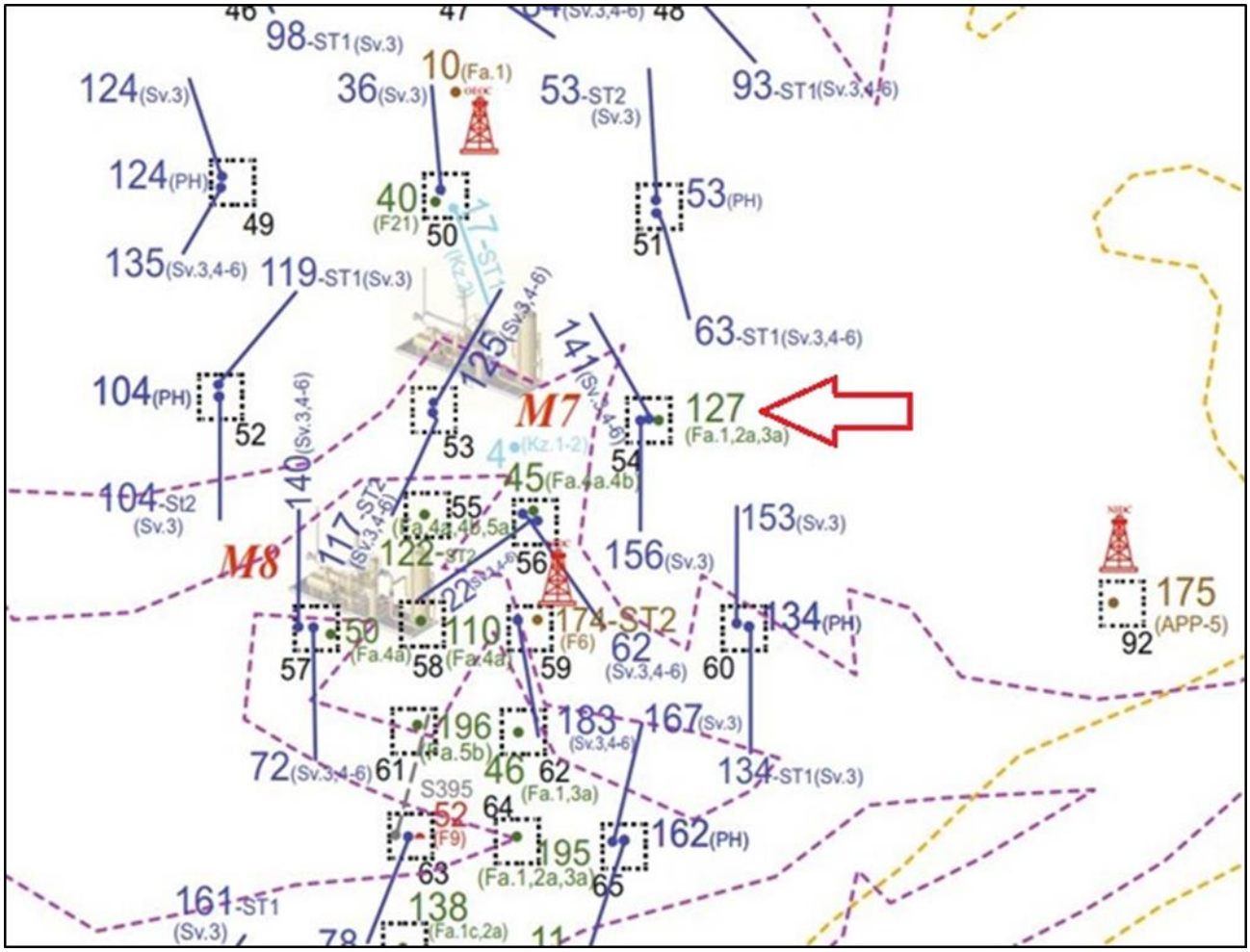


Figure 1-2 AZNS-127 Area Distribution

1.4 Well Pad Specification

Well Pad Type	Well Pad Dimension (m)	Mud Pit Dimension (m)	Cellar Dimension (m)	Conductor Size
Single Well	130 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D	30"
Double Well	145 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D + 4.0L x 3.8W x 3.4D	30"
Triple Well	160 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D + 4.0L x 3.8W x 3.4D + 4.0L x 3.8W x 3.4D	30'

Table 1-3 Well Pad Specification

Compressive Strength for reinforced concrete after 28 days: 280 kg/cm^2

• Fuel tank concrete pond dimension: Length 20^m ×Width 12^m ×Height 0.5^m
• Drainage: Width 30^{cm} × Depth 30^{cm}
• Burning pit: 15^m×15^m square, located on South-East or South-West with at least 350^m distance from well head
• Cellar to cellar distance in multi-well well pad: 15^m
• Conductor pipe depth under cellar bottom is 5.5^m. The conductor pipe will be driven while constructing the well pad.
• Mud material dimension is 25^m*35^m +granular surface
• Mud drying dimension is 25^m*35^m + concrete surface
• Well pad surface is gravel. Access road granular surface. The level of well pads is 3.95 MSL

1. Types of well pads:

• One well in one pad.
• Two wells in one pad. 2 Sarvak horizontal wells or 1 Sarvak horizontal well plus 1 vertical/directional well).
• Three wells in one pad: 2 Sarvak horizontal wells and 1 vertical/directional wells.

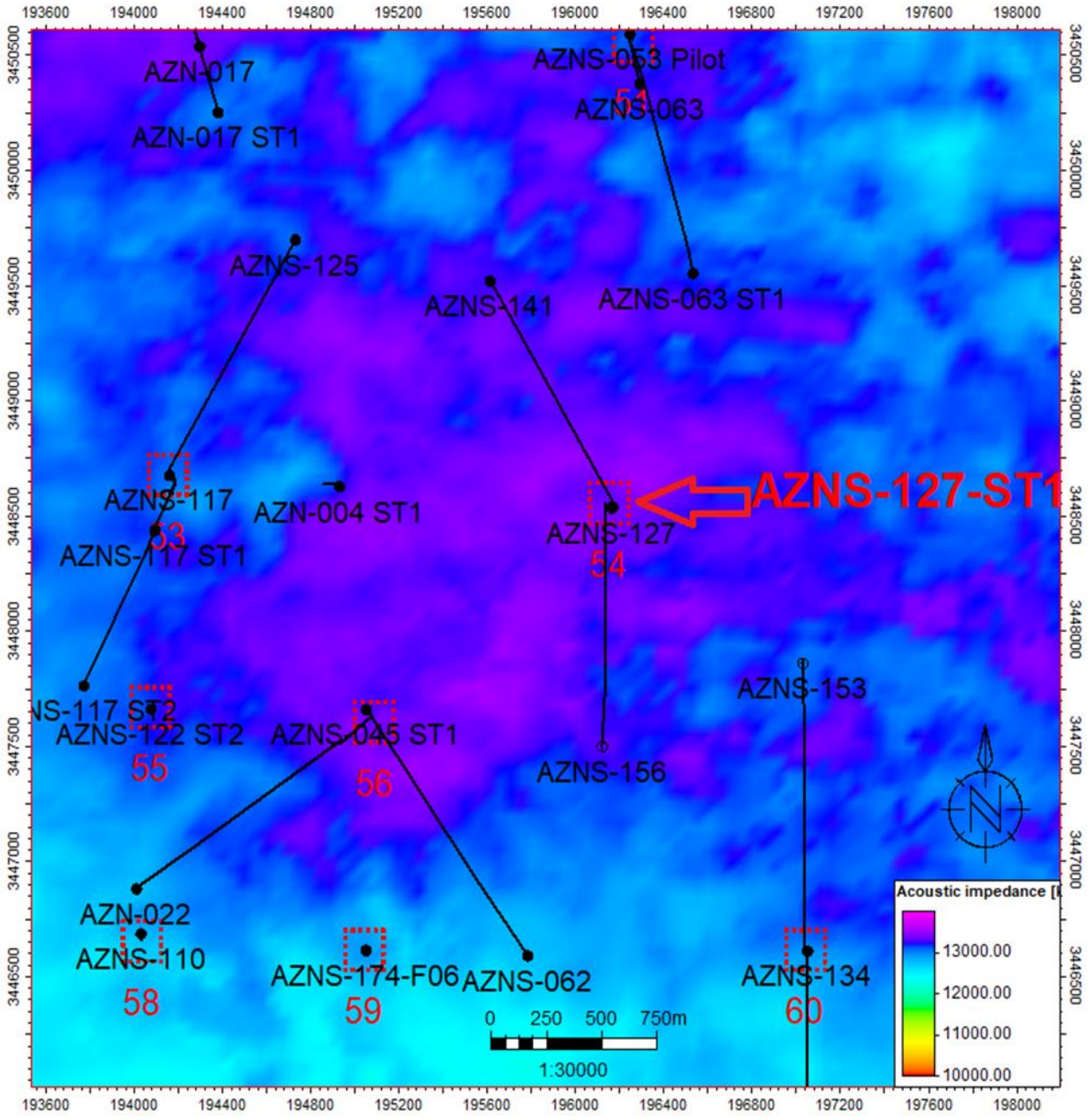


Figure 1-3 The Location of Pad 54 of AZNS - 127

1.5 Reservoir Data

Reservoir Name:	Fahliyan	Unit
Fluids:	Oil	….
Datum Depth:	-4000	M,TVDss
Datum Pressure:	Fah-4a= max 6800 psi Fah-5a= max 7400 psi	Psi
Pressure Gradient:	0.28	Psi/ft
Temperature:	132	C°
Temperature Gradient:	N/A	1.32 F/100ft 0.73 C/100ft 2.4 C/100m
H2S:	H2S in Gas at Atm condition 3000 PPM	PPM
CO2:	Co2 in Gas at Atm Condition 50000 PPM	PPM
Salt Salinity :	143000	ppm
Calcium Content:	8750	(mg/lit)

Table 1-4 Reservoir Data

1.6 Formation Pressure Prediction in Block II

Predicted Formation Pressure in Block II				
Formation	**Interval**	**Pore Pressure**	**Collapse Pressure (Break-Out Pressure)**	**Fracture Pressure (Break-Down Pressure)**
	m	*SG*	*SG*	*SG*
Aghajari	0-1095.9	0.98-1.18	0.06-1.39	1.01-1.89
Gachsaran	1095.9-1461	1.13-1.31	0.78-1.36	1.89-2.01
Asmari	1461-1728	1.14-1.26	0.6-1.28	1.43-1.79
Pabdeh	1728-2196	1.20-1.26	0.82-1.36	1.40-1.78
Gurpi	2196-2536	1.21-1.26	0.66-1.02	1.40-1.76
Ilam	2536-2626	1.18-1.24	0.20-0.9	1.41-1.72
Laffan	2626-2633	1.13-1.18	0.24-1.11	1.41-1.72
Sarvak	2633-3278	1.14-1.26	0.24-1.23	1.44-1.81
Kazhdumi	3278-3481	1.1-1.14	0.46-0.96	1.93-1.94
Dariyan	3481-3647	1.18-1.25	0.55-1.1	1.96-2.06
Gadvan	3647-3937	1.1-1.25	0.55-1.44	1.97-2.11
Fahliyan	3937-T.D.	1.60-1.3	1.20-1.75	2.08-2.12

Table 1-5 Formation Pressure Prediction

Reservoir	Sub-layer	Mid of Sub-Layer Depth in AZNS-127 (mss)	Temp (°F)	Temperature Gradient (°F/100ft)	Current Reservoir Pressure (psia)	Basis for Reservoir Pressure
Fahliyan	Fah-1a	-3966.7	256.3	1.32	9100	MDT 195
	Fah-1b	-3983	257.0		9100	MDT 195
	Fah-1c	-4009	258.0		9200	MDT 195
	Fah-2a	-4035.7	259.3		8912	Dynamic Model
	Fah-2b	-4049.8	259.9		8924	Pressure Gradient
	Fah-3a	-4058	260.3		8440	Dynamic Model
	Fah-3b	-4067	260.6		8447	Pressure Gradient
	Fah-4a	-4077	261.1		7551	Dynamic Model
	Fah-4b	-4091	261.7		8100	MDT 195
	Fah-5a	-4108.6	262.4		8100	MDT 195
	Fah-5b	-4163.7	264.8		8100	MDT 195

Table 1-6 Predicted Pressure and Temperature and TTZ Contact

1.7 Drilling History

26" Hole Section

Rig moving initiated on 95.09.27 and spudded rig 304 on 95.10.08 at 1200hrs.

Commenced drilling 26” hole section from 14m to 150m while experienced low ROP and bit balled up several times. Pulled out to surface, Ran in hole with same rock bit and 26" reamer to 150m and reamed from surface to 150m. Ran 20” casing as following with no major problem:

- 20" stab in shoe 94ppf, k-55, BTC (KTK Co.)
- 2 joints 20" casing 94ppf, k-55, BTC to 25m
- Circulated mud and check float shoe-ok
- Continued run with 10 joints 20" casing 94ppf, K-55, BTC to 146m (Totally run: 12 joints 20" casing 94ppf, K-55, BTC) to 146m. Cemented 20” casing in place as below:
- Sting in, lined up cement line and tested it with 2000psi-ok.
- Pumped 20bbl fresh water as spacer, mixed and pumped 689sxs cement class "G" Kerman (slurry weight: 118pcf, slurry volume: 170bbl), displaced with 7bbl 68pcf mud (1.7bbl short displace).

Note: Got 40bbl cement slurry and 10bbl contaminate as return on surface.

17 1/2" Hole section

Drilling of 17-1/2” hole section started on 12th Dey 1395 by new 17-1/2” rock bit and 17-1/2” string stabilizer placed at 20m, from 150m to 1064m while pumped salt saturated water several times to prevent balled up bit and displaced hole with 74pcf Kcl polymer mud at 502m.

Continued drilling 17-1/2” hole section by 17-1/2” from 1064m to 13-3/8” casing point at 1441m with 3-6bph loss while pumped hi-vis pill several times and pumped salt saturated water several times to prevent balled up bit. Condition hole prior logging, pulled out to surface and ran in hole with Sonic/Caliper/CDR/GR wire line logging tool string to 1438m (3m off bottom), took weight, tried to pass this point-not succeed, logged up from 1438 to 1280m, due to failure of Sonic tools, stopped log up and pulled out of hole to surface, changed sonic tools and ran in hole with same logging tool string

to 1438m (3m off bottom), tried to pass this point-not succeed, logged up from 1438 to 146m, continue logged up with GR to surface.

Rigged up 13-3/8" casing running equipment, held PJSM and ran 13-3/8" casing as following:

- 13-3/8" float shoe (KTK, 68ppf, N-80, BTC)
- 2 joints 13-3/8" casing 68ppf, N-80, BTC as shoe track
- 13-3/8" float collar (KTK, 68ppf, N-80, BTC)
- 2 joints 13-3/8" casing 68ppf, N-80, BTC, circulated mud & tested float shoe & float collar- ok.
- Continued RIH with 13-3/8" casing 68ppf, N-80, BTC (Totally:127 jts) to bottom at 1441m while filled up string every 10 joints& installed centralizer as program, tagged bottom, pulled out to 1440.5m, circulated mud as 1.5 cap of casing with 1-2 bph loss.

Cemented 13-3/8” casing in place as below:

- Pumped 40bbl chemical wash.
- Pumped 80bbl 93pcf spacer.
- Mixed & pumped 2004sxs Cement “G” Kerman as lead (slurry volume: 780bbl, slurry weight: 98pcf).
- Mixed & pumped 582sxs Cement “G” Kerman as tail (slurry volume: 142bbl, slurry weight: 118pcf).
- Dropped top plug (KTK).
- Displaced with 696 bbl 90pcf mud, bumped plug with 1500psi and held pressure for 15 min-ok. Note: Got 150bbl cement slurry and 120bbl contaminate as return on surface and recorded 40bbl loss during cement job.

Ran in hole with 12-1/4” used R. bit and drilled out top plug, float collar, medium to hard cement of shoe track and float shoe to 1441m while worked on junk sub several times.

Note: Confirmed float collar at 1417m and float shoe at 1441m

12 1/4" Hole section

Commenced drilling 12-1/4” hole section from 1441m to 1447m, displaced hole with 80 pcf clay-bond drilling fluid.

Pulled out to surface to change bit and BHA (recovered 4lb metal as junk), ran in hole 12- 1/4" PDC bit and 8" PDM (AKO:0 degree) and continued drilling 12-1/4" hole as performance drilling from 1447m to 1612m while periodically pumped tandem pill with 3-5bph loss. Due to APEK PDM failure, pulled out of hole to surface to change PDM, ran in hole to bottom and resumed performance drilling 12-1/4" hole from 1612m to 1989m with 3-5bph loss while from 1750m periodically pumped FL controlled brine and also pumped hi-vis caustic pill several times.

NOTE: From 1750 m due to change lithology to Arg. limestone, ROP decreased, tried to increase ROP by change in drilling parameters with no success.

Due to pressure dropped from 2100 to 1950psi, stopped drilling, checked surface equipment and pulled out of hole to check string, on surface observed wash out on XOS (6-5/8" Reg. pin x 4-1/2" IF box) between 5" HWDP and 8-1/2" DC.

NOTE: Cherty zone from 1920 m.

Continued drilling 12-1/4" hole from 1989m to 2000m while periodically pumped hi-vis. Pulled out of hole to surface, P/U and M/U 12-1/4" PDC bit to 8" PDM and ran in hole to bottom, resumed performance drilling 12-1/4" hole from 2000m to 2354m, suddenly pressure dropped from 2850psi to 1950psi (900psi), stopped drilling, checked surface equipment, pulled out of hole to surface to check string, on surface observed float sub twist off as (left in hole : 12-1/4" pdc bit, 8" PDM, 0.36m form float sub, length of fish : 9.18m, top of the fish :2344.6m).

P/U & M/U 10-1/8" over shot dressed with 8-5/8" control, 8" spiral grapple and RIH to top of the fish at 2344.6m, engaged and catch fish, pulled out to surface while completely recovered fish.

Ran in hole with 8" PDM to bottom and resumed performance drilling 12-1/4" hole from 2354m to 2371m with 2-3bph loss, due to low ROP stopped performance drilling and pulled out of hole to surface and L/D 8" PDM.

Ran in hole with 12-1/4" B. bit and continued drilling 12-1/4" hole section from 2371m to 2383m, due to low ROP pulled out of hole.

P/U and M/U 12-1/4" PDC bit and 8" PDM and RIH to bottom and resumed performance drilling 12-1/4" hole from 2383m to 3090m while periodically pumped hi-vis pill with 2-6bph loss.

NOTE:

• From 3005m to 3021m, due to change lithology to limestone-shale, decreased in ROP and increase loss to 10bph.

• From 3030m to 3090m due to increase in loss to 7-9bph, spot 70bbl 5 ppb Shell fish LCM pill.

Continued performance drilling 12-1/4" hole from 3090m to 3117m with 9-10bph loss, due to pressure dropped from 2560psi to 2320psi, stopped drilling, checked surface equipment and pulled out of hole to check string, string got at 3040m, worked on stuck with downward & upward by applying maximum 50klbs over pull & pumped 40bbl hi-vis pill several times, pipes got free and continued pulled out of hole to surface. On surface observed wash out on float sub above 8" PDM.

Ran in hole with 12-1/4" PDC bit to bottom and continued drilling 12-1/4" hole from 3117m to 3327m with 6-9bph loss while several times pump 7ppb hi-vis LCM pill (4ppb shell fish+3ppblimestone) and also 16ppb LCM pill (5ppb shell fish med.+7ppb limestone med.+4ppb limestone coarse). At 3327m loss increased from 6-8bph to 15bph, to pump LCM pill pulled out of hole from 3327m to 3050m, spot 100bbl hi-vis LCM pill 20ppb (10ppb shell fish med+10 ppb limestone coarse), ran back from 3050m to bottom at 3327m and hesitated for 1 hr with 1bph loss.

Continued drilling 12-1/4" hole from 3327m to 3705m with 5-9 bph loss while pumped hi-vis 20ppb LCM pill (10ppb shell fish med+ 10 ppb limestone coarse) several times. At 3705m loss increased to 12bph, decreased flow rate from 650gpm to 550gpm & pumped 20ppb hi-vis bentonite LCM pill (10ppb shell fish med+10 ppb limestone coarse).

Continued drilling 12-1/4" hole size from 3786m to planned 9-5/8" casing shoe at 3910m with 5-9 bph loss while periodically pumped 15ppb hi-vis bentonite LCM pill (7.5ppb shell fish med+7.5ppb limestone coarse).

Rigged up logging equipment and ran in hole with first full suite tool string as: MSFL/DLL/SONIC/CALIPER/GR/CDR tools to 3900m, string held up, tried to pass this point with no success, logged up from 3900m to 13-3/8" casing shoe at 1441m, continued POOH to surface and laid down logging tools while decided to repeat DLL log.

Ran second full suite logging tool string as: ZDL/CNC/DSL/WTS/TTRM to 3855m (confirmed by PEDEC witness), logged up from 3855m to 13-3/8" casing shoe at 1441m, continued POOH to surface and laid down logging tools w/1-2 bph loss.

R/U third logging tool string as: DLL/GR, ran in hole to 3630m (confirmed by PEDEC witness), again recorded resistivity log data as client request from 3630m to 1441m and logged up logged up from 3630m to 2150m, and continued pooh to surface and R/D logging equipment.

Ran in hole with 12-1/4" B. bit to bottom at 3910 and pulled out of hole to 13-3/8" casing shoe at 1441m while W & R 3890m to 3910m. Circulated mud at 13-3/8" casing shoe while waited on receive 9-5/8" casing. After 69hrs NPT, received 9-5/8" casing, to condition hole prior run casing RIH to bottom and pulled out of hole to surface with good hole condition.

Rigged up casing running equipment and ran 9-5/8" casing as following:

- PJSM
- 9-5/8" float shoe KTK, 53.5ppf, C-110, TSH Blue
- 2 joints 9-5/8" casing 53.5ppf, T-95, TSH Blue as shoe track.
- 9-5/8" float collar KTK, 53.5ppf, C-110, TSH Blue.
- 2 joints 9-5/8" CSG 53.5#, T-95, TSH Blue.
- Circulated mud & function test float shoe & float collar-ok.
- Continued RIH with 99 joints 9 5/8" casing 53.5ppf, T-95, TSH blue (totally 103 joints) to 1163.6m.
- 9-5/8" XOS TSH blue pin x TPCQ box,
- Continued run with 9-5/8" casing to bottom at 3910m with good hole condition while washed down from 3900m to 3910m while and circulated mud at 1441m & checked casing fill up every 10 jts- ok.
- Ran totally: 103 joints 9-5/8" CSG 53.5ppf, T-95, TSH Blue + 9-5/8" XOS TSH Blue pin x TPCQ box + 226 joints 9-5/8" CSG 53.5ppf, T-95, TPCQ. Circulated mud with 4-5bph loss at bottom while waited on receive cement & additives by NIDC for 65hrs, casing got stuck and unable to move upward and downward.

R/U cement equipment and pressure tested to 4500 psi-ok, cement 9-5/8" casing in place as following:

- Pumped 20bbl 85pcf spacer.

• Lead: mixed & pumped 2282sxs cement “G” delijan with cenosphere, micro silica (slurry volume: 1133bbl, slurry weight: 88pcf).

• Tail: mixed & pumped 363sxs cement “G” delijan with o-silica flour (slurry volume: 116bbl, slurry weight: 118pcf).

• Dropped top plug (KTK), displaced with 902bbl 81pcf mud, bumped plug with 3000psi and held pressure for 15 min-ok.

• After 702bbl displacement, observed cement slurry on surface (got 200bbl slurry cement & 22bbl contaminate on surface), recorded 280bbl loss during job.

• Unable to cut landing joint due to back flow**.**

NOTE:

• After 8hrs opened cement head valve & observed back flow, closed valve, again after 6hrs opened valve & check again, due to back flow closed valve & waited on cement to get hard.

• After 10hrs tail surface sample got hard completely.

• After 41hrs lead surface sample got hard 40%.

P/U and M/U 8-1/2” B. bit and ran in hole from 3700m to 3852m, tag top of cement at 3852 m, drilled out medium to hard cement from 3852m to 2m above 9-5/8" shoe at 3907m. Displaced hole with 104 pcf ultra-carb drill fluid and continued drill out medium to hard cement from 3907m to 3910m while drilled out float shoe at 3909m and worked on junk sub several times.

Note:

• Confirmed 9-5/8" float collar at 3886m and 9-5/8" float shoe at 3909m.

• Recovered 1.5lb metal wool from junk sub

8 1/2" Hole section

Commenced drilling 8-1/2” hole section drilling from 3910m to 3913m, circulate mud and performed FIT as per program by applying 270psi pressure and hold it for 15 min-ok. Pulled out of hole to surface.

Ran in hole with 9-5/8" scraper and scraped from 3600m to 3909m, rigged up logging equipment, ran in hole with logging tool string as: VDL/SBT/CCL/GR, string held up at 3905m, tried to pass several times with no success, logged up from 3905m to 100m above of 13-3/8"casing shoe at 1341m, continued pull out of hole to surface and rigged down logging equipment.

Ran in hole with 8-1/2" PDC bit and commenced drilling 8-1/2" hole section from 3913m to hole TD 4226m with 0-3bph loss with no major problem. Pulled out of hole to surface while condition hole prior run log.

Note: Due to observed tight hole while drilling at 4107m to 4226m, increase mud weight from 104pcf to 105pcf.

Rigged up logging equipment and ran first full suite logging tool string as: DLL/SONIC/MSFL/CDR logging tool string, logged down from 3910m to TD at 4226m, logged up from 4226m to 100 m above shoe at 3810m and pulled out of hole to surface.

Ran second full suite logging tool string as: CNT/LDT/NGS/GR to 3910m, logged down from 3910m to TD at 4226m, logged up to 3900m and continued pull out of hole to surface. Due to low quality of second full suit result, replaced full suite logging tool string modules and again run in hole with same full suite logging tool string to 3910m, logged down from 3910m to TD at 4226m, logged up to 3900m and continued pulled out of hole to surface and rigged down logging equipment.

Held pre job safety meeting and ran 7" liner as following:

- 7" Float shoe 29ppf, C110, TPCQ (PPA Co.)
- 3 joints 7" liner 29ppf, S110, TPCQ
- 7" float collar 29ppf, C110, TPCQ (PPA Co.)
- 1 joint 7" liner 29ppf, S110, TPCQ
- 7" LDC 29ppf, C110, TPCQ (PPA Co)
- 1 joint 7" liner 29ppf, S110, TPCQ to 57m
- Circulated mud & function test float shoe & collar-ok.
- Continued run in hole with 13 joints 7"liner 29ppf, S110, TPCQ to 393m (totally 35 joints)

- Hydraulic liner hanger & acc. as following:
- 7" hydraulic liner hanger 29ppf, C-110, TPCQ (PPA Co)
- POB 29ppf, C-110 (PPA Co)
- Tie back sleeve (ID: 7-3/8", PPA Co) to 399m

Continued ran in hole with 7" liner string followed by 5" DPs from 399m to 4226m w/1 bph loss, tagged bottom at 4226m, pulled out to 0.5m off bottom at 4225.5m, circulated mud and dropped ball, set hydraulic liner hanger w/1800 psi and shear ball seat w/3000 psi, released setting tools.

Cement 7" liner in place as following:

- pumped 9bbl chemical wash, 7 bbl 85pcf mud push
- Mixed & pumped 220 sxs cement class "G" delijan blended with silica flour (SV:72 bbl, SW:118 pcf)
- Dropped pump down plug, displaced with 266bbl 105pcf mud.
- Shearing after 223bbl with 2100psi, plug was not bumped, checked back flow-ok. Note: Surface sample of cement slurry, get 100% hard, after 6 hrs.

Ran in hole with 8-1/2" bit, tagged top of cement at 3680m and drilled out soft to medium cement from 3680m to 3826m.

Note: Confirmed 7" liner lap at 3826m.

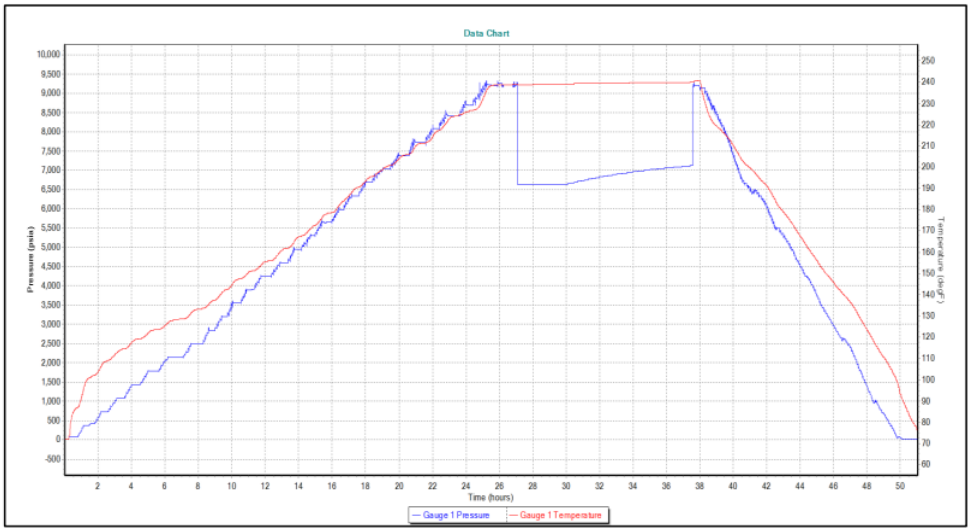
P/U 6-1/8" bit and ran in hole to 3830m, drilled out POB at 3830m, continued RIH to 4167m and resumed drill out hard cement to 6m above LDC at 4173m. Pulled out to surface.

Ran in hole with 6-1/8" used b. bit & 9-5/8" scraper to 7" liner lap 3826.5 m while scraped 9- 5/8" casing from 3650m to 3826.5m several times. Pulled out to surface.

Dry test: 1 st dry test, 7" liner: Rigged up dry test equipment and ran in hole to 3796m while filled up string with 170m 75pcf hi-vis pill and 3400m 85pcf mud as cushion.

L/U dry test line & surface equipment and tested same with 7500psi, set packer, opened hyd. spring valve for 3 hrs while observed no bubble on surface, closed DCIP valve and kept it closed for 6 hrs meanwhile observed annulus-static. Dropped bar, shear reverse sub, reverse circulated and got 208bbl cushion (198bbl 85pcf mud and 10bbl 75pcf hi-vis pill) on surface, continued circulation and balanced

mud unset packer, flow check- well static. Pulled out of hole to surface and laid down dry test equipment.



NOTE: DCIP & hyd. spring sample properties; MW: 105pcf, PH: 10.5, Ca++: 1120 mg/l, Cl: 165000 mg/l.

Ran in hole with 6-1/8" B. bit & 7" scraper and scrap from 7” liner lap at 3826.5m to 4173m several times.

Rigged up logging equipment and ran in hole with VDL-SBT-WTC-GR-CCL logging tool string to top of cement at 4173m, logged up from 4173m to 3826.5m, continued pull out to surface and rigged down logging equipment.

Due to dry test result decided to tie back 7” liner. Tie-back:

P/U and M/U 7-1/4" polish mill & ran in hole to 7” liner lap at 3826.5m, polished liner lap to 3829.5m several times, pulled out of hole to surface and laid down 7-1/4" mill.

Rigged up liner running equipment and ran 7” liner tie back string as below:

- 7-3/8" seal nipple
- 3 joints 7" liner, S-110, TPCQ , 29ppf
- 7-3/4" LDC • 1 joint 7" liner, S-110, TPCQ, 29ppf to 40m, to check return circulated mud-ok.

• Continued ran in hole with 8-1/4" tie back sleeve (PPA) to 7” liner lap 3826.5m Sting in 7-3/8" seal nipple to 3827.5m, observed pressure up from 180psi to 580psi. Line up cement line and tested w/3500 psi-ok and performed cement job as below:

• Pumped 12bbl salt saturated as pre flush

• 7bbl 95pcf as spacer • 62.3sxs cement "G" Delijan (SV: 20bbl, SW: 118 pcf) as program. Dropped pump down plug, displaced cement slurry with 213bbl 105pcf mud, observed shearing with Max 1600psi, continued displacement with 1.8bbl 105pcf mud, bumped plug with 1300psi, sting in 7-3/8" seal nipple to 3829.5m, released setting tools, pulled out of hole and laid down setting tools.

P/U & M/U 8-1/2" bit and ran in hole to 3712m, drilled out medium to hard cement from 3712m to 7” liner lap at 3777m.

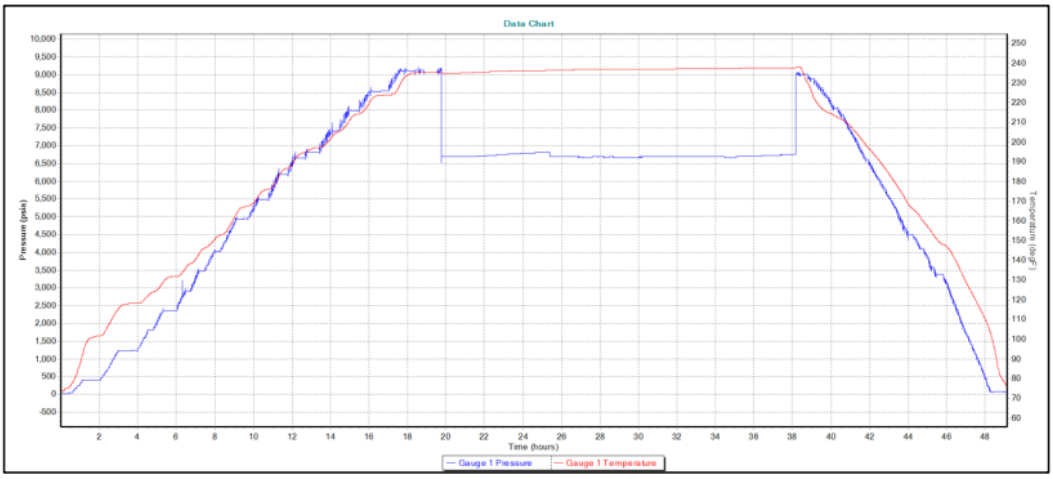
P/U & M/U 6-1/8" used bit to 3780.5 m, drilled out POB at 3780.5m, continued ran in hole to 3792m, drilled out plugs, LDC & hard cement from 3792m to 3826.5m, continued ran in hole to 4137m, pulled out to surface.

Dry test:

2nd dry test, 7” liner tie back:

Ran in hole with 6-1/8" bit and 9-5/8" scraper & ran in hole to 7” liner lap at 3777m while scraped from 3670m to 3777m several times. Pulled out to surface and laid down bit and scraper.

Ran in hole with 9-5/8" dry test tools while filled up string with 170m 75pcf hi-vis pill brine and 3350m 85pcf mud as cushion as per program to 3747m. Set packer at 3747m, 1st stage: opened hyd. spring valve for 2hrs, observed weak bubble. Closed DCIP for 4 hrs. 2nd stage: opened DCIP and kept it open for 3hrs, observed weak bubble, closed DCIP for 6hrs. Opened revers circulating sub, reveres circulated and got 199bbl cushion (189bbl 85pcf mud, 10bbl 75pcf hi-vis pill), continued circulation to balance mud, unset packer, flow check-well static. Pulled out and laid down dry test equipment.



Perforation and Completion

• Perforation

Ran in hole with 6-1/8" bit to top of cement at 4173m, circulated mud, pumped & displaced 33bbl 84pcf CaCl2 hi-vis pill (Length: 273m, from 4173m to 3900m), pulled out of hole to surface with NL-NG.

Loaded guns, performed pre job safety meeting, ran TCP/DST string while drifted DPs and filled up string with 79.5pcf CaCl2 as cushion to 4077m, Rigged up logging equipment and ran in hole with CCL/GR logging tool string to 3925m, logged up to surface and correlated depth, L/D logging equipment.

Set DST packer at 3954m, pressure tested DST packer by applying 400psi pressure in annulus for 15min-ok. Performed pre job safety meeting, N/U flow head, related lines and DST choke manifold, applied 1200psi to annulus, opened tubing tester valve, applied 3200psi pressure through string and step by step increased pressure to 4300psi, fired guns, released string pressure to zero, monitored tubing pressure: zero & annulus pressure: zero psi, applied 2800psi press in annulus, opened AOR valve, reversed circulated & got 217bbl 79.5 CaCl2 meanwhile observed no influx, continued reverse circulation to balance mud, N/D flow head and related lines, flow check-well static, pulled out of hole from 4077m to surface, L/D DST/TCP tools and checked TCP string, observed all bullets were fired.

To check perforated interval depth, R/U running logging equipment and ran CCL/GR logging tools to 4100m, logged up to 3700m and continued pulled out of hole to surface.

According to CCL/GR log result, perforation was done approximately 28m above planned interval, to double check again ran in hole with CCL/GR to 4100m and logged up to surface, the 2nd log showed same result as 1st log.

Planned perforation interval		Actual perforated interval	
Depth (m)	Formation	Depth (m)	Formation
3984m - 4005m	Fahliyan 1a and 1b	3956m – 3977m	Fahliyan top and 1a
4023m - 4026m	Fahliyan 1c	3995m – 3998m	Fahliyan 1b
4071m - 4077m	Fahliyan 3a	4043m – 4049m	Fahliyan 2a

Ran in hole with 6-1/8” E. mill to top of cement at 4173m, circulated mud and pulled out of hole to 4077m, spotted 20bbl 84.5pcf CaCl2 hi-vis pill, continued pulled out pooh from 4077m to surface.

To determine exact depth of perforated interval, rigged up logging equipment and ran in hole with MFC/CCL/GR logging tool string to 4100m, logged up from 4100m to 3930m and confirmed perforated interval as: 3956 to 3977m, 3995 to 3998m, 4043 to 4049m. Pulled out to surface and laid down logging equipment.

Ran in with 6” E. mill to 150m and circulated mud while waited on receive Detonator. After almost 180hrs NPT, received Detonator, rigged up logging equipment and ran in hole with 4-1/2" gun (w/o bullet) & CCL (analog) as dummy run to 4090m, tool string held up, logged up to 7” liner lap at 3777m, pulled out to surface.

1st perforation run (WCP): Ran 1st perforation as: 4-1/2" gun (assembled with 99 No. bullet, phase: 60degree, length:6m) & CCL (Analog) log tool string to 4077m, perforated 6m, from 4071m to 4077m, pulled out to surface and observed all guns fired completely. Changed cable head.

2nd perforation run (WCP): Ran CCL/GR tool string to top of last perforation at 4071m, string held up, tried to pass several times-no success.

Pulled out to surface and changed cable head. Again ran in hole with 4-1/2" gun (assembled with 50 no bullet, phase: 60degree, length: 3m) & CCL log tool string to 4026m while correlated depth.

Perforated 3m from 4026m to 4023m, pulled out to surface and observed all guns fired completely.

3rd perforation run (WCP): Ran in hole with 4-1/2" gun (assembled with 99 no bullet, phase: 60degree, length: 6m) & CCL log tool string to 4030m while correlated & adjusted depth, checked and confirmed last perforated interval-ok, pulled out of hole to 4005m, fired gun & perforated 6m from 4005m to 3999m, pulled out to surface and observed all guns fired completely.

4th perforation run (WCP): Ran in hole with 4-1/2" gun (assembled with 50 no bullet, phase: 60degree, length:3m) & CCL log tool string to 3990.70m while correlated & adjusted depth, perforated 1m from 3999m to 3998m & 2m from 3995m to 3993m, pulled out to surface and observed all guns fired completely.

5th perforation run (WCP): Ran in hole with 4-1/2" gun (assembled with 99 no bullet, phase: 60degree, length: 6m) & CCL log tool string to 3984.7m while correlated & adjusted depth, perforated 6m from 3993m to 3987m, pulled out to surface and observed all guns fired completely.

6th perforation run (WCP): Ran in hole with 4-1/2" gun (assembled with 50 no bullet, phase: 60degree, length: 3m) & CCL log tool string to 3978.69 m while correlated & adjusted depth, perforated 3m from 3987m to 3984m, pulled out of hole to surface and observed all guns fired completely.

Changed cable head, ran CCL/GR logging tools to 3984m, string held up, confirmed top of perforated interval, logged up to 3700m, pulled to surface and rigged down logging equipment.

Ran in hole with 6" E. mill from 1700m to 3984m, took weight, continued ran in hole with rotation & pumping from 3984m to 3994m, continued run in hole with pumping to bottom of perforated interval at 4077m, pumped & displaced 12bbl hi-vis pill and 21bbls 84pcf CaCl2 (277m, from 4077m to 3800m)

After 411hrs wait on receive Wixiqin company completion running crew and equipment, ran in hole with 6” E. mill to 3950m and displaced 105pcf mud in hole with 104pcf CaBr2 as completion fluid w/ NL-NG as following: Pumped 50bbl 105pcf hi-vis pill, 30bbl 85pcf non-solid CaCl2 as spacer, 820bbl 104pcf CabBr2, got 5bbl contaminate, observed well for 5hrs-static.

Run Completion string:

- 1 st Completion running:

Retrieved wear bushing, held pre job safety meeting and ran completion tally as following:

- 3-1/2" mule shoe 10.2ppf, HSM-1, L-80.13Cr
- 1 joint 3-1/2” tubing HSM-1, 10.2ppf, L-80

- 3-1/2" bottom No-Go 10.2ppf, HSM-1, L-80.13Cr (w/2.562" xn plug & prong inside it)
- 3-1/2" pup joint
- 1 joint 3-1/2” tubing
- 7" Permanent production packer model WSDP
- 3-1/2" Anchor seal, 1 joint 3-1/2" pup joint
- 3-1/2" tubing, 1 joint 3-1/2" pup joint, 3-1/2" SSD (in open position w/ 2.750" x plug & prong inside it), 1 joint 3-1/2" pup joint
- 1 joint 3-1/2" tubing, 1 joint 3-1/2"pup joint, 3-1/2" travelling joint (in full open position w/10 ft stroke), 1 joint 3-1/2" pup joint
- 1 joint 3-1/2" tubing, 1 joint 3-1/2" pup joint, 3-1/2" CIV 1, 1 joint 3-1/2" pup joint
- 1 joint 3-1/2" tubing,1 joint 3-1/2" pup joint, 3-1/2" CIV 2, 1 joint 3-1/2" pup joint, 1 joint 3-1/2" tubing. Pressure test completion string w/8000 psi-ok, connected 3/8" CIL to both CIV no1 & CIV no2 and pressure tested same to 4000psi for 15min-ok.
- Continued ran in hole with 17 joint tubing, 3-1/2", 10.2ppf, L-80, HSM-1
- XOS 4 1/2" HSM-1 box X 3 1/2" HSM-1 pin
- 1 joint 4-1/2" tubing, 18.9ppf, L-80, HSM-1 while hold 1000psi pressure inside CIL & filled up string with CaBr2 as completion fluid each 10 joint. Pressure tested completion string at 2530m w/8000 psi for 15 min-ok.
- Rigged up slick line, ran 2.750" gauge ring to above SSD, Laid down slick line equipment
- 133 joint 4 1/2" tubing, 18.9ppf, HSM-1, L-80 while fill up string with CaBr2 as completion fluid each 10 joints with 0-1 bph loss.

Pressure tested completion string at 1480m with 8000psi -no success, rigged up slick line, ran 2" SB-pulling to top of SSD and pushed down prong, pulled out slick line to surface, tested with 8000psi -no success.

Retrieved 2.750" prong & plug from SSD, changed O-ring and packing, again set prong & plug in SSD, tested with 8000psi several times-no success, pulled out completion string to 1200m while pressure tested completion string at 1340m & 1200m with 8000psi- no success, continued pull out to 1050m, pressure tested completion string with 8000psi for 15 min-ok.

• Ran in hole with completion string to 1140m with 0-1 bph loss, pressure tested completion string with 8000psi-no success. Pulled out of hole from 1140m to 1113m, laid down 1 stand 4-1/2" tubing, tested completion string with 8000psi for 15 min-ok.

• Ran in hole to 3856m while tested completion string at 1282m, 1450m, 2135m, 2790m, 3443m and 3865m with 8000psi for 15min-ok.

• Continued ran completion string to 3880m as:

• 1 joint 4-1/2" pup joint, 4-1/2" flow coupling 4ft, 4-1/2" x 3.437" TRSSV, 10k, weiqixin, 4-1/2" flow coupling 6ft, 1 joint 4-1/2" pup joint, 1 joint 4-1/2" tubing, 18.9ppf, l-80, HSM-1. Connected 1/4" control line to TRSSV, tested it with 8000psi-ok, pressure tested completion string with 8000psi for 15min-ok, closed TRSSV and released pressure above TRSSV to 500psi, as inflow test held pressure for 15 min-ok, increased pressure above TRSSV to 6000psi and opened TRSSV by apply 3750psi as equalizing test,.

• Continued ran completion string to 3930m with 5 joints 4-1/2" tubing 18.9ppf, L80 , 4-1/2" fluted swage, double pin Vam top x HSM-1, MSP 11 x 4-1/2" tubing hanger, 10k ACME box up x Vam top box down, tested completion string with 8000psi for 15min-ok. M/U 4-1/2" landing joint, connected 1/4" control line to TRSSV and 2 no 3/8" CIL to hanger, tested 1/4" control line with 8000psi and tested 3/8" CIL with 1000psi- ok, ran completion string to 3933m, landed tubing hanger to THS, to space out pulled out to 3931.5m.

Rigged up slick line equipment, installed bop and slick line lubricator, tested it with 4000psiok and ran with slick line; 1 st run: retrieved prong of 2.750" plug from SSD while circulated cross well head with 0-1bph loss.

As 2nd run, retrieved 2.750" plug "X" type from SSD

As 3rd run, closed SSD sleeve by shifting tools, rigged down slick line equipment, line up test truck, step by step increased completion string pressure to 3500psi, set packer at 3906m, pull test packer with

15klb-ok, landed tubing hanger in THS, closed all tie down screws, pressure tested completion string with 8000psi, observed pressure dropped and also return from annulus, pressure tested annulus with 2000psi observed pressure dropped and also return from completion string.

As 4th run, ran with shifting tools to check SSD is closed, pulled out to surface.

As 5th run, ran with 2.750" selective tester tools to above SSD, to check string. Pressure tested completion string with 8000psi for 15min-ok. Rigged up slick line equipment.

As 6th run, ran with GS pulling, retrieved 2.750" selective tester tools of SSD.

As 7th run, ran with 2" SB-pulling to top of bottom No-go landing nipple, pushed down prong and tested completion string with 2000psi, observed return from annulus.

As 8th run, ran with shifting tools, opened SSD, circulated from string and got return from annulus.

As 9th run, ran with shifting tools and closed SSD, tested completion string with 3000psi, observed return from annulus.

As 10th run, ran with 2.562" SB-pulling, retrieved prong & plug from bottom No-go in two run.

As 11th run, ran with 2.562" selective tester tools to No-go landing nipple while pressurized inside completion string to 7000psi, observed return from annulus.

As 13th run, ran in hole with GS pulling and retrieved 2.562" selective tester tools and changed redress kit. again ran with 2.562" selective tester tools to top of No-go, pressure tested inside completion string with 7000psi, observed return from annulus.

Ran PLT logging equipment, ran with PLT logging tool string as; FCF, GCT, PWH, HSTC to 3918m and line up to 3840m while pressurized string in two stages, first stage applied 7000psi inside completion string and got return from annulus, as second stage applied 2600psi and got return from completion string, PLT result: recorded leakage at 3-1/2" anchor seal assembly and 7" permanent production packer. Pulled out to surface, rigged down PLT logging equipment, r/u slick line equipment.

As 14th run, ran with GS-pulling, retrieved 2.562" selective tester tools from No-go landing nipple, set 2.562" plug & prong in No-go landing nipple in two stage

As 16th run: ran with 2.750" selective tester in SSD. Picked up 1.5m completion string to full stork traveling joint, pulled string with maximum 25klb over pull, slack off weight and applied 2000psi

pressure inside string, ran string to half close traveling joint, applied 50klb weight on anchor seal inside packer, landed tubing hanger and closed tie down screws, released string pressure to zero.

Rigged up slick line equipment and as 17th run, ran with SB-pulling tools, retrieved 2.750" selective tester tools from SSD, tested completion string with 2000psi, observed return from annulus, tested completion annulus with 2000psi-ok.

As 18th run, set protection sleeve in TRSSV, tested completion string, observed return from annulus, tested completion annulus with 2000psi- ok.

As 19th run, ran with 2.750" selective tester tools in SSD. Picked up 1.5m comp string to full open travelling joint, applied 5klb over pull, rotated string 11 round to right, unlatch anchor seal from packer, half close travelling joint and landed tubing hanger in THS while applied 50klb weight on anchor seal & 2000psi pressure in string to latch it in 7" packer again, observed increased pressure to 2300psi, tight tie down screw, released string pressure to zero, pull test completion string with 20klb-ok. Pressure tested from annulus with 2000psi-ok, rigged down slick line equipment, ran with 2.570" SB-puling, retrieved 2.570" selective tester tools from SSD, pressure tested completion string with 2000psi, observed return from annulus, retrieved prong & plug from No-go landing nipple in two, rigged down slick line equipment, opened t.ie down screw, pulled out 1.5m, rotated string 11 round to right, unlatch 3-1/2" anchor seal from packer. Pull out completion string to surface and laid down same.

Ran in hole with 6-1/8" b. bit to 6m above packer at 3900m and circulated cross wellhead, while wait on receive completion accessories with 1bph loss, to check packer depth ran in hole to 3906m, confirmed packer depth at 3906m, circulated bottoms up with 0-1bph loss, continued wait on receive completion equipment while circulated cross well head and pumped through string each 2hrs, pulled out from 3906m to surface.

- 2 nd Completion running:

Rigged up completion running equipment, ran comp string 3-1/2" x 4-1/2" as below:

• 4-1/2" anchor seal, 3-1/2" pup joint, 3-1/2" bottom no-go landing nipple XN 2.56" without plug, 1 joint 3-1/2" pup joint, 1 joint 3-1/2" tubing, 10.2ppf, HSM-1, 1 joint 3-1/2" pup joint, XOS 4-1/2" HSM-1 box x 3-1/2" HSM-1 pin

• 4-1/2" mill out extension, 7" permanent production packer (hs-6 model, weiqixin)

- 4-1/2" anchor seal , 1 joint 3-1/2" pup joint,1 joint tubing, 10.2ppf, HSM-1
- 1 joint 3-1/2" pup joint, 3-1/2" SSD (open position) & 2.750" with check valve
- 1 joint 3-1/2" pup joint, 1 joint tubing, 10.2ppf, HSM-1
- 1 joint 3-1/2" pup joint, expansion joint 3-1/2", HSM-1, L-80, 10 stroke (full open, weiqixin)
- 1 joint 3-1/2" pup joint, 1 joint tubing, 10.2ppf, HSM-1
- 1 joint 3-1/2" pup joint, 3 1/2" CIV 1 tubing retrievable, HSM-1, 10.2ppf with checked valve, 1 joint 3-1/2" pup joint, 1 joint tubing, 10.2ppf, HSM-1, 1 joint 3- 1/2" pup joint, 3 1/2" CIV 2 tubing retrievable, HSM-1, 10.2ppf with check valve
- 1 joint 3-1/2" pup joint, 1 joint tubing, 10.2ppf, HSM-1 to 86m Line up test truck, pressure tested line with 8000psi-ok, pressure tested completion string with 8000psi for 15min-ok, connected 2 no 3/8" CIL to CIV and tested them with 4000psi for 15min, ok, continued run completion string with 17 jts 3-1/2" tubing, 10.2ppf, HSM-1
- XOS 3-1/2" HSM-1 pin x 4-1/2" HSM-1 box, 1 joint 4-1/2" tubing, L-80, 18.9ppf, HSM-1 to 247m. Pressure tested completion string with 8000psi for 15 min-ok, rigged up slick line equipment, ran 2.750" gauge ring to top of SSD-ok, rigged up slick line equipment, continued run completion string from 247m to 950m.
- Continued run completion string from 950m to 3837m with 4-1/2" tubing, L-80, 18.9ppf, HSM-1 meanwhile passed 3.5" drift through all 4-1/2" tubing, pressure tested completion string with 8000psi at 1592m, 2575m and 3837m-ok.

Rigged up slick line, run 2.8" gauge ring to top of SSD, rigged up slick line equipment, ran with 3" SB pulling to top of SSD, tried to catch 2.750" check valve several times, catch and retrieved check valve from SSD after several attempts, reverse circulated with completion fluid, installed new 2.750" check valve inside SSD, rigged down slick line equipment, pressure tested completion string with 8000psi-not ok, pulled out completion string to 3640m, pressure tested completion string with 8000psi- not ok, rigged up slick line equipment, ran 3" SB pulling to top of SSD, tried to engage with 2.750" check valve several times-no success, pulled out to surface, checked tools and ran again with 3" SB pulling to top SSD, catch 2.750" check valve & tried to remove it several times-no success, pulled out to surface.

As 3rd run, retrieved check valve from SSD. Pulled out with completion string from 3640m to 2240m.

1st and 2nd run, tried to install 2.750" plug in SSD several times-no success, pulled out with slick line to surface and changed plug.

3rd run, installed 2.750" plug in SSD,

4th run, installed prong inside plug, rigged down slick line, pressure tested completion string with 8000psi, observed pressure dropped, rigged up slick line, 1st run, tried to engage prong by SB pulling-no success, 2nd run, retrieved prong from SSD plug, 3rd run, retrieved 2.750" plug from SSD, rigged down slick line equipment.

Pulled out with completion string from 2240m to 946m, rigged up slick line equipment, 1st run, installed 2.750" check valve inside SSD, tested completion string with 8000psi-not ok, 2nd run, retrieved 2.750" check valve from SSD, rigged down slick line equipment, due to lack of enough completion fluid on surface, stopped pull out and circulated mud cross well head while wait on receive completion fluid with 0-1bph loss.

Received 68bbl CaBr2, pulled out with completion string from 946m to 76m, rigged up slick line equipment, installed 2.750" check valve inside SSD, pressure tested completion string at 76m with 8000psi, observed 250psi dropped in pressure in 20min, continued pull out to surface while pressure test and visually check completion string on travelling joint and SSD and CIV with 8000psi-ok, tested CIL, observed rupture disks were broken.

- 3 rd completion running:

Connected 2no 3/8" CIL to 1" CIV and pressure tested same with 4000psi for 15min-ok, ran completion string as following:

- 5 stand 3-1/2" tubing, 2 joints 3-1/2" tubing, 10.2ppf, L-80, HSM-1, XOS 4-1/2" HSM-1 box * 3-1/2" HSM-1 pin, 1 joint 4-1/2" tubing, 7 stand 4-1/2" tubing, 18.9ppf, HSM-1 with 0-1bph loss.

- 4-1/2" tubing 18.9ppf, L-80, HSM-1, total: 71 stand, meanwhile ran 2.8" gauge ring by slick line at 640m to top of SSD & pressure tested completion string at 640m & 2240m with 8000psi for 15min-ok.

- continued ran completion string from 2240m to 3837m with 57 stand 4-1/2" tubing 18.9ppf, l-80, HSM-1, total:128 stand, pressure test completion string with 8000psi for 15min-ok. Rigged up slick

line equipment, ran 2.8" gauge ring to top of SSD, rigged down slick line equipment, continued ran in completion string to 3874m as below:

- 1 joint 4-1/2" pup joint, 4 ft flow coupling, 4-1/2" * 3.437" TRSSV-10k 6ft flow coupling, 1 joint 4-1/2" pup joint, 1 stand 4-1/2" tubing, connected 1/4" control line to TRSSV & tested it with 8000psi for 15min-ok with 0-1bph loss. Pressure tested comp string at 3874m with 8000psi for 15min-ok, as inflow test closed TRSSV, released pressure above TRSSV from 8000 to 300psi, monitored pressure for any charge for 15min-ok.

- Continued ran completion string to 3894m with 5 joints 4-1/2" tubing, 4-1/2" fluted swage adapter, MSP tubing hanger 11" * 4 1/2" 10k, pressure tested completion string with 8000psi for 15min-ok.

- Connected 1/4" control line to hanger, tested same with 8000psi-ok, connected 2 no 3/8" CIL to hanger, continued ran completion string to 3906.3m. Sting anchor seal sting in old packer with 0-3klbs, placed tubing hanger 1.5m above THS, set protect sleeve in TRSSV landing nipple, removed check valve from SSD, tried to close SSD in two attempts-no success. Changed shifting tools and tried to close SSD-no success, after 4 attempts, pined shifting tools with steel shear pin, continued ran in hole with same and closed SSD, ran 2" gauge ring as dummy run to No-go landing nipple 3902m, string held up, tried to circulate, pressure increased to 300psi, tried to revers circulate, pressure increased to 300psi- no success, pulled out to surface and rigged down slick line.

Applied 500psi pressure inside string, tried to circulation-no success, applied 700psi pressure in annulus and tried to reverse circulation- no success, released pressure, tried to pull out, observed 30klb overpull, to land tubing hanger in THS, applied 50klbs weight, rigged up slick line and ran 2.3" impression block to 3900m, worked on same and pulled out to surface, no mark on impression block, again ran 2.3" impression block to 3900m, worked on same and pulled out to surface, no mark on impression block.

Worked on completion string upwards with maximum 30klb over pull several-no success, rigged up slick line equipment and ran 1.8" sand bailer to 3905 , worked on bailer and pulled out to surface, recovered no sample, ran with shifting tools to SSD and closed same, pulled out to surface, pressure tested completion string with 500psi-ok.

While waited on Weiqixin Company due to stopped operation by weiqixin crew, unlatched anchor seal from packer by applying 10klb over pull and rotated 12 round to right while held 300psi pressure inside

completion string, pulled out to 0.5m above packer while circulated with NL-NG, latched anchor seal to packer by applying 10klb weight, worked on completion string upwards with maximum 30klb over pull several times- no success, unlatched again anchor seal from packer and pulled out completion string while circulated cross well head with NL-NG.

Ran in hole with 3-1/2" anchor seal to 3890m, latched anchor seal inside packer and tried to recover packer by applying maximum 90klb over pull several times-no success, anchor seal got unlatch, tried to latch several times-no success Pulled out from 3890m to surface, laid down all tubing**.**

- **Milling:**

Ran in hole with 6-1/8" metal muncher junk mill to 5m above top of packer at 3885m, circulated and displaced CaBr2 with 104pcf kcl-polymer mud, continued ran in hole from 3885m to 3890m, tagged top of packer at 3890m. Commenced milling packer from 3890m to 3890.8m, recovered 44lb junk and element rubber of packer from shaker, pulled out from 3890.8m to surface, ran in hole with 2-1/4"*4" taper tap and to top of packer 3891.8m, tap inside packer with 2-4klb weight, catch fish and pulled out with 20-40klb drag to surface, completely recovered tail of packer as : 3-1/2" anchor seal, 3-1/2" pup joint, 3-1/2" bottom no-go, 3-1/2" pup joint, 3-1/2" tubing 10.2ppf, L-80, HSM-1, XOS 3-1/2" * 4-1/2" HSM-1, 4-1/2" mill out extension.

Ran in hole with new 6-1/8" metal muncher junk mill to 3908m, washed down from 3908m to 3910m, tagged top of packer at 3910 m, commenced milling packer from 3910m to 3910.65m, recovered 30lb junk & rubber element of packer from shakers, pulled out to surface.

Ran in hole with 2-1/4" x 4" taper tap to top of packer at 3910.65m, tap inside packer with 2-4klb weight, worked on same and pulled to surface, not recovered fish, ran in hole with same 2-1/4" x 4" taper tap to top of packer at 3910.65m, circulated mud with high flow rate, worked on catch fish with no success. Pulled out to surface and laid down taper tap. Ran in hole with 6-1/8"metal muncher junk mill to top of packer at 3910.65m, commenced milling from 3910.65m to 3910.7m, fish slipped in hole, pushed fish down to 3950.50m and resumed milling to 3951.4m, pulled out to surface.

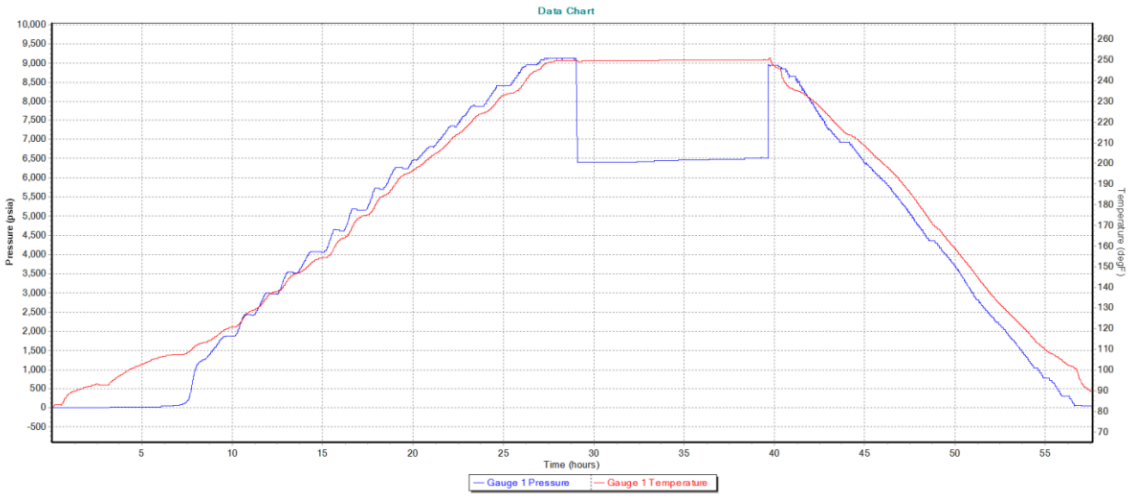
Ran in hole with 2-1/4" x 4" taper tap to top of fish at 3951.4m, break circulation, tap inside packer and worked on catch fish, pulled out from 3951.4m to surface, completely recovered tail of packer as: 3-1/2" mule shoe, 3-1/2" tubing 10.2ppf, L-80, HSM-1, 3-1/2" bottom nogo, 3-1/2" pup joint, 3-1/2" tubing 10.2ppf, L-80, HSM-1, 3-1/2" pup joint, XOS 3-1/2" * 4- 1/2" HSM-1, 4-1/2" mill out extension.

Plug and Suspension:

Ran 6-1/8" b. bit and 7" scraper to 3900m while scraped from 3780m to 3900m, circulated mud while added 0.5 % vol. biocide and 1% vol. brine to mud system and pulled out to surface.

Ran in hole with 7" hydro mechanical bridge plug to 3875m, dropped ball, circulated mud, tried to set 7" bridge plug by applying 3000psi-no success, line up test truck, tried to set hydro mechanical bridge plug by applying pressure to 4000psi-no success, reverse circulated for 15min in 2 stage and several times jerked string, applied 4000psi inside string, set 7" bridge plug at 3875m successfully, released press, checked string upward with 35klbs over pull and slack of 10klb weight-ok, released running tools, pulled out to 3760m, closed top rams & tested bridge plug with 2000psi-ok, continued pull out to surface.

Ran in hole with 7" dry test tools to 3625m while filled up string with 210bbl w/75 pcf non solid brine as water cushion, set packer at 3845m, opened hydro spring and kept open for 3hrs, observed no bubble, closed DCIP valve for 6hrs, dropped bar, shear revers sub and reveres circulated, got water cushion on surface and observed no influx effect, continued circulation to balance mud at 104 pcf, unset packer, pulled out to surface and recovered chart as below:



1st cement plug:

- Ran in hole with 6-1/8" center cut to 3700m, set first cement plug as below:
- Pumped 15bbl spacer.

- Pumped 300 sxs blended silica flour cement "G" Kerman, slurry weight: 118pcf and slurry volume: 35bbl.
- Pumped 3.7bbl behind spacer and displaced with 200bbl mud as balance plug.

Pulled out from 3700m to 3520m, flush DPs, continued pull out to 3350m, circulated bottoms up, and waited on cement to get hard, tagged hard cement with 470gpm and 15klbs-ok at 3513m, pulled out from 3513m to 3500m, pressure tested against cement plug with 2000psiok, pulled out from 3500m to 1480m.

2nd cement plug:

At 1480m, break circulation, line up cement line and tested same with 4000psi-ok, set 2nd cement plug as below:

- Pumped 15bbl 97pcf ahead spacer.
- Pumped 300sxs blended silica flour cement class "G", slurry volume: 118pcf, slurry weight: 35bbl.
- Pumped 3.7bbl behind spacer and displaced with 72bbl mud as balance plug. Pulled out from 1480m to 1300m, flush DPs, continued pulled out to 1000m, circulated bottoms up, waited on cement to get hard, continued ran in hole from 1200m to 1317m, tagged cement with 400gpm &15klb, circulated bottoms up, pulled out to surface, laid down drill pipes and string.

Landed tubing hanger in THS and tight tie down screws, back off 5" landing joint, installed 4" two way in tubing hanger and tested it with 1000 psi-ok.

N/D 13- 5/8" BOP stack (10k) & disconnect outlet valves, installed 11" * 4-1/16" bottom master valve (10k, msp), tested tubing hanger & o-ring & bottom master valve connection & metal seal with 4000psi-ok, removed 4" two way from tubing hanger, pressure tested bottom master valve with 2000 psi-ok, installed 4" one way in tubing hanger, tested it with 2000psi-ok, installed blind flang on bottom master valve and tested it with 2000psi-ok.

Secured well and released rig 304 on 96/06/12 at 24:00hrs.

Re-entry section:

Spudded the rig on 1396.11.03 at 06:00, retrieved 11" * 4-1/16" 10k MSP tubing hanger and checked same-ok, observed the well-static, installed 11" wear bushing, ran in hole with 8- 1/2" bit to 1315m, tagged 1st cement plug at 1315m, circulated mud and balanced mud weight in/out at 104pcf, observed the well- static, drilled out medium cement from 1315m to 1350m and medium to hard cement plug from 1350m to 1480m with NL-NG, circulated bottom's up to clean the well, continued run in hole from 1480m to 3513m, tagged 2nd cement plug at 3513m, circulated to balance mud weight at 104 pcf, observed the wellstatic, drilled out soft to medium cement from 3513m to 3665m and medium cement from 3665m to bottom at 3700m, observed the well-static, ran in hole from 3700m to 7" liner lap at 3777m, circulated mud to balance mud weight in/out at 105pcf, observed the well- static, pulled out from 3777m to surface.

Ran in hole with 6-1/8" new junk mill and tagged 7" bridge plug at 3863.7m, pumped 30bbl hi-vis pill, circulated mud and milled 7" bridge plug from 3863.7 to 3864.08m, due to no progress pulled out from 3864.08m to surface, laid down 80% used 6-1/8" junk mill, ran in hole with 6-1/8" new junk mill to 3864.08m and tagged 7" bridge plug, resumed milling 7" bridge plug from 3864.08 to 3864.38m, 7" bridge plug dropped in hole, observed the wellstatic, ran in hole to top of cement plug at 4173m while pushed down 7" bridge plug with pumping and rotation, circulated mud to balance in/out at 105pcf, spotted 30bbl CaCl2 with NL-NG, pulled out from 4173m to surface and laid down 70% used 6-1/8" junk mill.

Ran in hole with 4-1/8" used junk mill to top of cement plug at 4100m while scraped 7" liner from 3777m to 3930m and 9-5/8" casing from surface to 3777m, circulated bottom's up with NL-NG, pulled out from 4100m to surface and laid down 4-1/8" used junk mill.

Tested BOP and ran in hole with 4-1/8" used junk mill to 3750m and waited on receive CaBr2 while slow circulated through string with NL-NG, pulled out from 3750m to 1400m, rigged up running tubing equipment and stand back 168 joints 4-1/2" tubing while waited on receive CaBr2, pulled out to surface.

Ran in hole with 4-1/8" junk mill to 4173m, circulated bottoms up and spotted 30bbl CaCl2, pulled out from 4173m to 3750m, waited on receive CaBr2 while slow circulated with NL-NG, ran in hole from 3900m to 4100m, line up pump truck and displaced 105pcf mud with 106pcf CaBr2, pumped 30bbls 85pcf hi-vis pill CaCl2as spacer, pumped 840bbls 106pcf CaBr2 and got 10bbls contaminated with NL-NG, observed the well for 4hrs-static, pulled out from 4100m to Surface while laid down DPs.

1st Attempt:

Pressure tested BOP, rigged up OIDS tubing running equipment, held PJSM and ran 3-1/2" x 4-1/2" completion string as below:

- 3-1/2" mule shoe 10.2ppf, L-80, HSM-1, 13Cr
- 1 joint tubing 3-1/2" 10.2ppf, L-80, HSM-1
- 3-1/2" bottom No-go 10.2ppf, L-80, HSM-1, 13Cr with 2.562" seal bore type XN
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- 1 joint tubing 3-1/2" 10.2ppf, L-80, HSM-1
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- XOS 3-1/2" HSM-1 pin X 4-1/2" HSM-1 box
- 4-1/2" mill out extension 18.9ppf, HSM-1
- 7" permanent production packer model WSDP, 3-1/2" Anchor seal, 3-1/2" pup joint 10.2ppf, L80, HSM-1
- 1 joint 3-1/2" tubing 10.2ppf, L-80, HSM-1
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1, 3-1/2" SSD, in open position with 2.750" check valve inside it
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- 1 joint 3-1/2" tubing 10.2ppf, L-80, HSM-1
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1, 3-1/2" expansion joint (10 ft stroke full open) 10.2ppf, L-80, HSM-1, 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- 1 joint 3-1/2" tubing 10.2ppf, L-80, HSM-1
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- 3-1/2" CIV 1 10.2ppf, L-80, HSM-1

- 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- 1 joint 3-1/2" tubing 10.2ppf, L-80, HSM-1
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- 3-1/2" CIV 2, 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- 1 joint 3-1/2" tubing 10.2ppf, L-80, HSM-1 to 93m

Pressure tested completion string with 8000psi for 15min-ok, rigged up slick line, ran 2.8" gauge ring to above check valve inside SSD, rigged down slick line equipment, connected 3/8" CIL to both CIV 1 and CIV 2, press tested CIL with 4000psi for 15min- ok, continued run completion string from 93m to 250m as:

- 13 joints 3-1/2" tubing 10.2ppf, L-80, HSM-1 • XOS 4-1/2" HSM-1 box * 3-1/2" HSM-1 pin
- XOS 4-1/2" TPG-2 box * 4-1/2" HSM-1 pin
- 1 stand 4-1/2" tubing 18.9ppf, L-80, TPG-2

while kept 1500psi pressure inside of CIL Pressure tested completion string at 250m with 8000psi for 15min-ok, rigged up slick line, ran 2.8" gauge ring to above check valve inside SSD, rigged down slick line equipment, continued run completion string to 3800m with 4-1/2" tubing 18.9ppf, L-80, TPG-2 while used cross coupling each joint with NL-NG and while pressure tested completion string at 2422m, 3000m and 3800m with 8000psi for 15min-ok.

Continued run in hole as:

- XOS 4-1/2" TPG-2 pin * 4-1/2" HSM-1 box 18.9ppf, L-80
- 4-1/2" pup joint 18.9ppf, L-80. hsm-1
- 4-1/2" flow coupling 18.9ppf L-80, super 13Cr HSM-1
- 4-1/2" TRSSV 18.9ppf, L-80, super 13 Cr, HSM-1 (profile: 3.437")
- 4-1/2" flow coupling 18.9ppf, L-80, super 13Cr, HSM-1
- 4-1/2" pup joint 18.9ppf, L-80, HSM-1

• XOS 4-1/2" HSM-1 pin * 4-1/2" TPG-2 box 18.9ppf, L-80 to 3820m Tested completion string with 8000psi for 15min- ok meanwhile inflow tested TRSSV-ok, continued run completion string as: • 3 joints 4-1/2" tubing 18.9ppf, L-80, TPG-2

• XOS 4-1/2" TPG-2 pin * 4-1/2" HSM-1 box 18.9ppf, L-80

• 4-1/2" fluted swage adapter 18.9ppf, L-80, Super 13Cr V. top * HSM-1

• 11* 4-1/2" tubing hanger 10k MSP Drilex to 3860m.

Tested completion string with 8000psi -not ok, tested again with 8000psi- not ok, laid down 11* 4-1/2" tubing hanger and 4-1/2" fluted swage adapter 18.9ppf, L-80, Super 13Cr V. top * HSM-1 to 3857m, tested completion string with 8000psi –not ok.

Pulled out from 3857m to 3820m and stand back 1 stand 4-1/2" tubing 18.9ppf, L-80, TPG-2, tested completion string with 8000psi- not ok, rigged up slick line, set 3.437" protection sleeve in TRSSV, ran with SB pulling and tried to retrieve 2.750" check valve inside SSD at several time - not succeed and tried to reverse circulated by applying max 500psi -not succeed. Ran 2.750" impression block to SSD depth at 3787m and worked on same, pulled out to surface no mark observed on impression block, rigged down slick line.

Ran completion string as:

• 3 joints 4-1/2" tubing 18.9ppf, L-80, TPG-2

• XOS 4-1/2" TPG-2 pin * 4-1/2" HSM-1 box 18.9ppf, L-80

• 4-1/2" fluted swage adapter 18.9ppf, L-80, Super 13Cr V. top * HSM-1

• 11* 4-1/2" 10k MSP Drilex tubing hanger

• XOS 4-1/2" Acme pin * 4-1/2" N. Vam box 18.9ppf, L-80

• 4-1/2" tubing 18.9pf, L-80, N. Vam landing joint to 3870m Landed tubing hanger inside 13-5/8" * 11" tubing head spool, pulled out to 3868.5m (1.5 m space out packer position). Nipple up 4-1/16" G. flange, rigged up coil tubing, pressure tested BOP and stuffing box with 5000psi for 20min- ok, tensile tested with 20klb for 20min- ok, flush coil tubing, ran with 1- 1/2" coil tubing to SSD depth 3831m while circulated with max rate 0.7bbl/min, washed and cleaned above check valve several times, pulled from 3831m to surface, rigged down coil tubing with NL-NG, rigged down slick line, ran with SB

pulling and retrieved 2.750" check valve inside SSD, pulled out to surface, rigged down slick line, set 4-1/16" G. flange, rigged up coil tubing, pressure tested BOP and stuffing box with 5000psi for 20min- ok, tensile tested with 20klb for 20 min- ok, flush coil tubing. Ran in hole with 1-1/2" coil tubing to mule shoe at 3870m while pumping, washed & cleaned through string, circulated bottoms up meanwhile added o2 scavenger as per program to completion fluid system with NL-NG. Pulled out from 3870m to surface, flush coil tubing, rigged down coil tubing.

Rigged up slick line, ran with 2.750" check valve to SSD depth at 3831m, set 2.750" check valve in SSD, pulled out to surface, tested completion string with 8000psi- not succeed, ran in hole with SB pulling and retrieved 2.750" check valve, pulled out to surface, ran with 2.750" plug to SSD depth 3831m and set plug in SSD, pulled out to surface, ran with 2.750" prong to 3831m, set prong in plug, pulled out to surface, rigged down slick line, tested completion string with 8000psi- not ok. To check string pulled out comp string from 3870m to 3805m while tested completion string at 3820m and 3805m with 8000psi- not ok, rigged up slick line, ran in hole with SB pulling to SSD depth, tried to retrieved prong and prong in two run, rigged down slick line, pulled out completion string from 3805m to 2478m, rigged up slick line and ran in hole with 2.750" check valve to SSD depth, set check valve, pulled out to surface and rigged down slick line, pressure tested completion string with 8000psi- not ok, rigged up slickline, ran in hole with SB pulling to SSD depth, retrieved check valve, pulled out to surface, rigged down slick line, pulled out of hole from 2475m to 84m, tested completion string with 8000psi- not ok, continued pull out to 58m, pressure tested completion string with 8000psi- not ok, observed leakage from SSD, laid down 3-1/2" expansion joint and 3-1/2" SSD, continued pull out to surface, observed Packer rubber element deformed and damaged, laid down the same, checked bottom no-go profile- ok.

2nd Attempt:

Ran 3-1/2" * 4-1/2" Completion string as following:

- 3-1/2" mule shoe 10.2ppf, L-80, HSM-1, 13Cr
- 1 joint tubing 3-1/2" 10.2ppf, L-80, HSM-1
- 3-1/2" bottom no-go 10.2ppf, L-80, HSM-1, 13Cr with seal bore 2.562" type XN
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- 1 joint tubing 3-1/2" 10.2ppf, L-80, HSM-1

- 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- XOS 3-1/2" HSM-1 * 4-1/2" TPG-2
- 4-1/2" mill out extension 18.9ppf, TPG-2
- 7" permanent production packer model WSDP, 3-1/2" Anchor seal, 3-1/2" pup joint 10.2ppf, L80, HSM-1
- 1 joint 3-1/2" tubing 10.2ppf, L-80, HSM-1 to 36 m

Rigged up slick line, ran run 2.800" gauge ring to top of bottom no-go, continued run completion string as:

- 3-1/2" pup joint 10.2ppf, L-80, HSM-1, 3-1/2" SSD (in close position with 2.750" check valve inside it, 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- 1 joint 3-1/2" tubing 10.2ppf, L-80, HSM-1
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1, 3-1/2" expansion joint (10 ft stroke, full open) 10.2ppf, L-80, HSM-1, 3-1/2"pup joint 10.2ppf, L-80, HSM-1
- 1 joint 3-1/2" tubing 10.2ppf, L-80, HSM-1 to 68m Line up test truck, tested completion string at 40m and 68m with 8000psi for 15min-ok, continued run completion string as:
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1, 3-1/2" CIV1 10.2ppf, L-80, HSM-1, 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- 1 joint 3-1/2" tubing 10.2ppf, L-80, HSM-1
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1, 3-1/2" CIV2, 3-1/2" pup joint 10.2ppf, L-80, HSM-1 to 84m Tested completion string with 8000psi for 15min- ok, connected control lines to both CIV, tested them with 4000psi- ok, observed CIV disks ruptured, worked on change rapture disks, tested again control line and CIVs with 4000psi for 15min- ok, continued run completion string as:
- 14 joints 3-1/2" tubing 10.2ppf, L-80, HSM-1
- XOS 4 1/2" HSM-1 x 3 1/2" HSM-1

• XOS 4-1/2" HSM-1 x TPG-2 p*b to 225m Pressure tested completion string with 8000psi for 15min-ok, rigged up slickline, ran 2.800" gauge ring to above check valve inside SSD, rigged down slick line equipment, continued run comp string with 4-1/2" tubing, 18.9ppf, L-80, TPG2 to 2146m, rigged up slick line, tried to set 2.750" check valve in 3-1/2" SSD- not succeed, pulled out to surface, set plug & prong in SSD, rigged up slickline, pressure tested completion string with 8000psi for 15min-ok, rigged up slick line, retrieved prong and plug from SSD, rigged slick line, continued run completion string with 4-1/2" tubing and:

• XOS TPG-2 pin * 4-1/2" HSM-1 box

• 4-1/2" pup joint 18.9ppf, L-80, HSM-1

• 4-1/2" flow coupling 18.9ppf, L-80, super 13Cr, HSM-1

• 4-1/2" TRSSV 18.9ppf, L-80 super 13Cr, HSM-1 (profile : 3.437")

• 4-1/2" flow coupling 18.9ppf, L-80, super 13Cr, HSM-1

• 4-1/2" pup joint 18.9ppf, L-80, HSM-1

• 4-1/2" fluted swage adapter 18.9ppf, L-80, super 13Cr V. top * HSM-1

• 11* 4-1/2" 10k, MSP tubing hanger

• 4-1/2" Acme pin * 4-1/2" HSM-1 box

• 4-1/2" pup joint 18.9ppf, L-80 HSM0-1 to 3860m Rigged up slickline, set 2.750" check valve in 3-1/2" SSD, tested completion string with 8000psi-not ok, rigged up slick line, retrieved 2.750” check valve from SSD, set plug and prong in 3-1/2" SSD, tested completion string at 3860m with 8000psi-ok, ran slick line to retrieved prong from SSD- not succeed, pulled out to surface, observed 2" SB pulling and slick line accessories left in hole, to recover fish ran 2" SB pulling, retrieved prong from SSD plug, ran 2.5" impression block and checked top of fish, tried to recover 2" SB pulling several times with no success, modified 2" SB pulling and ran in hole to top of fish, worked on same, recovered fish.

Ran 2.750" GS pulling and retrieved plug from 3-1/2" SSD, rigged down slick line, ran in hole with 4-1/2" landing joint N. Vam, connected 1/4" control line to tubing hanger and tested it with 8000psi- ok, connected 2no 3/8" CIL to tubing hanger and tested with 500psi- ok, flush and cleaned THS with water,

ran completion string to 3870.5m landed 4-1/2" x 11" x 4-1/2" tubing hanger in THS, string up 1.5m as space out to 3869m, rigged up slick line, ran 2.750" shifting tools to open and close 3-1/2" SSD, after 2 times ran 2.562" plug to set in No-go landing nipple, ran and set prong in 3-1/2" No-go landing nipple, pressurized inside completion string to 300psi to ensure 3-1/2" expansion joint is full open, increased pressure gradually to 3500psi and set 7" hydro permanent production packer, to test completion string increased pressure to 8000psi and held for 15min- ok and pull tested with 15klbs-ok.

Rigged up r/u slick line, tried to retrieve prong- No success, rigged down slick line, rigged up coiled tubing, flushed coil tubing and ran in hole with 1-1/2" coiled tubing to No-Go at 3860m, circulated and cleaned the completion string, pulled out to surface and rigged down coiled tubing equipment. Rigged up slick line, tried to retrieve prong- no success, rigged down slick line, rigged up coiled tubing, flushed coiled tubing and ran in hole with 1-1/2" coiled tubing to top of No-go at 3860m, circulated and cleaned completion string, pulled out to surface and laid down coiled tubing equipment, rigged up slick line, ran with SB pulling and retrieved plug & prong in 2 stage, ran with GS pulling and retrieved protection sleeve from TRSSV, closed 4-1/2" TRSSV, rigged down slick line, back off 4-1/2" landing joint, installed 4" 2-way valve in tubing hanger then closed blind rams and tested two way with 8000psi- ok, nipple down 13-5/8" 10k BOP stack, checked 11"x4-1/2" 10k hanger neck, control line and CIL- ok, tested body and valves of 10k x-mass tree before installation with 8000psi-ok, nipple up 11"x4-1/16" 10k adaptor flange with MSP BMV and tested s-seals with 7500psi- ok, tested control line with 5000psi- ok, nipple up MSP 4-1/16"*4-1/16"*4-1/16"*4-1/16" 10k x-mass tree, body tested xmass tree and MSP 11"*4-1/16" 10k BMV with 1000psi for 15min- ok, continued nipple up 11"x4-1/16" 10k adaptor flange with BMV, tested s-seals with 7500psi- ok, tested control line with 5000psi- ok, nipple up and body tested x-mass tree and 11"x4-1/16" 10k MSP BMV with 1000psi for 15min- ok, tested metal seal of tubing hanger with 7500psi for 15min- ok, bleed off pressure to 500psi and closed 4-1/16" adaptor flanged, inflow tested- ok, pressure tested p-seals from section B with 1000psi- ok, tested p-seal from section C upper test port with 2500psi- ok, pressure tested lower test port -not ok, observed oil return from 9-5/8"x13-5/8" annulus, retrieved 4" 2-way valve from tubing hanger, set 4"one way valve in tubing hanger, closed 4-1/16" 5k TMV, 4-1/16" BMV, SSV, production and kill valves, released the rig at 2400 on 96/12/23.

Rig less operation was performed on date 17.05.1398.

Tried to RIH with coiled tubing to unload the well. There was an obstruction in the well at depth 3918m. Unable to pass the depth. Several runs were performed by coiled tubing and mill to remove the obstruction.

Rig less operation for unloading and well flowing was performed on date 30.01.1400.

Pressure monitoring was performed on tubing head spool vent port, it was 4,000 psi. Metal seal of tubing hanger is damaged.

Try to run coiled tubing to TD and wash the wellbore with xylene. Run CCL/GR/MFC logging BHA in the well. It was observed 7" liner ID has been decreased to about 3inches in interval 3889m-3911m.

Flow the well, an abnormal annulus pressure was observed.

The other Rig less monitoring well annulus pressure on date 18.09.1400.

It was annulus pressure is 5,300 psi. Sampling was performed from annulus fluid. It was contained with gas and hydrocarbon.

Workover Operation History:

The AZNS – 127 workover well was De-Complete by rig OEOC – 208 on Pad 54, from "08 – May – 2024" as described in the following:

Installed 8" poor boy degasser flare. Pressure test SP & function test TDS, MPs; Ok. Checked section B: P-Seals & ring gasket; Ok. Checked section C: BTM P-Seal; Ok. Top P-Seal & ring gasket; Not Ok. Disconnected adjustable choke of production line. L/U pump truck to XMT kill line & annulus. L/U choke manifold to production line & annulus, meanwhile preparing 200bbl 106pcf mud. Pressure test all line after rectify several leakages. Open Ann & monitor Press; 4700 psi. Bleed off Press to 3000 psi, Build up to3800 psi in 1 Min. Increase Press f/ 3800 to 3900 psi in 5 Min.

P/U 60 Jts 5" DP Jt by Jt, made & R/B 20 Stds to derrick. Monitored annulus pressure build up to 4800psi after 6 hrs. Opened the annulus thru choke manifold to oil storage; after 6min. decreased pressure to 980psi. Decreased pressure gradually to 500psi in 20min. Closed the annulus; Shut in pressure 500psi. Built up to 1000psi in 15min. Opened the annulus, suddenly dropped to zero. After 6min closed the annulus. Opened the 2SV, top master valve & production valve. Tightened XMT cap. Connected hand pump to exit block, tried to open TR3SV. Monitored tubing pressure 3400psi. Opened the well f/ TBG to oil storage tank; dropped pressure to zero in 2min. Closed the well. Built up to 3700psi in 2min. L/U annulus to choke manifold. Obs. annulus pressure built up to 4300psi in 3 hrs. Opened the well & pressure decreased to 600psi. Closed the annulus. Pressure rapidly built up to 3300psi. Opened the well again. Pressure suddenly dropped to zero. Flew the well for 10min. Closed the well, built up to 3300psi. Monitored the well for1 hrs. Pressure built up to 4300psi.

Due to jammed stem of inside SOV of THS in close position, installed pressure gauge on annulus to choke manifold line & pressure test same to 7000psi, Ok. R/U slickline equipment. Installed slickline BOP & lubricator on XMT cap. Tried to pressure test, Obs. leakage between lubricator & BOP. Changed lubricator O-ring & tried to pressure test; No-Success. L/D previous BOP. Installed another slickline BOP & Lubricator on XMT cap. Pressure test slickline BOP & lubricator w/ 7000psi to 15min; Ok. RIH w/ 3.4" gauge cutter to TBG HGR, tried to pass this depth; No-Success. Pulled out to surface & R/D slickline BOP & lubricator.

Installed 4 – 1/16" 10k lubricator on XMT. Pressure test to 7000psi for 15min; Ok. RIH w/ BPV retrieving tools to TBG HGR & retrieved 4" BPV f/ TBG HGR. L/D 4 – 1/16" 10k lubricator. Due to unable to open TR3SV, decided M/U 5" DPs. R/D slickline equipment.

R/U slickline equipment. Pressure test; Ok. Fixed CL port exit block. Ran 3.4" gauge cutter to 206ft, RIH w/ 3.437" protection sleeve to 190ft, opened TR3SV while holding pressure to 1200psi. Applied 7500psi pressure in CL & opened TR3SV. Cont. running tools to 200ft, took weight, tried to set protection sleeve by jarring several times; No-Success. Flew the well with 1/4" choke size & Cont. work on set protection sleeve; No-Success. Pulled out tools to 191ft. Obs. stuck tools. Worked on stuck to max. 60lb over-pull; No-Success. Finally flew the well by annulus & tubing. Tools got free. Cont. pulled out tools to 50ft & stuck tools again. R/D lubricator. Obs. cut the slickline wire from 50ft & left

in hole about 3m tools & 13m wire. RIH w/ 2.5" LIB to 157ft, POOH to surface; Obs. wire mark on same.

While wait on OEOC slickline service, RIH w/ 2.8" spear to TOF. Worked on fish & retrieved 2.45m of wire after 3 attempts. RIH w/ 2.8" LIB to 157ft, POOH to surface. Obs. rope socket marked on same. RIH w/ 2.8" blind box, POOH to surface. Rope socket marked on same. Welded 3.2" skirt on 2.2" JDC, RIH w/ JDC to TOF. Engaged w/ same. Tried to pull out, Obs. indication of shear JDC Shear pin. POOH to surface. Obs. sheared JDC Shear pin. Redressed w/ steel shear pin. RIH w/ JDC to TOF again. Engaged & worked on fish, Pulled to surface. Obs. retrieved 10ft stem of fish. RIH w/ 2.8" LIB to TOF, POOH to surface. Obs. slickline mechanical jar on the same. RIH w/ JDC to engage w/ jar. Worked on fish, jarring up. Pulling out fish 20ft. Cont. worked on stuck. RIH w/ blind box & knock on TOF, without progress. POOH to surface. RIH w/ JDC to engage w/ Jar. Worked on fish, jarring up. POOH to surface. R/D slickline equipment.

In order to work on fish w/ coiled tubing, wait on receive coiled tubing unit. While waiting on OEOC slickline service, R/U CT & M/U 2 – 1/8" dimple connector on 1 – 1/2" CT pipe & performed pull test to max 20klb. Obs. failure in dimple connector, disconnected same. M/U 1 – 11/16" roll on connector on 1 – 1/2" CT pipe & performed pull test to max 20klb, Ok. Installed CT riser on XMT & CT BOP. Close CT blind rams & tried to perform pressure test; No-Success. Tightened between two risers flange & tried to perform pressure test; No-Success. P/U riser & BOP; Tried to perform pressure test; No-Success. Changed 1 Jt CT riser & performed pressure test CT BOP & risers, Ok. RIH CT BHA to 48m & engaged w/ the fish; Obs. stuck the fish. Worked on stuck to max 18klb over-pull; No-Success. While shearing JDC Shear pin @ 2 times & POOH to surface & redressed w/ brass shear pin.

In the 3rd attempt, caught the fish & worked on same upward to 18klb over-pull while bleeding off annulus pressure f/ 4800 to 2000psi. Obs. fish got free, jumped same & sheared the JDC Shear Pin. POOH to surface. Cut 12m 1 – 1/2" CT pipe & changed roll on connector. Performed pull test to 20klb; Ok. RIH w/ 1 – 1/2" CT & 3.2" JDC to 48m. Worked on fish & POOH to surface. In the last attempt caught, engaged the fish & bleed off annulus pressure f/ 4800 to 2000psi. Pulled out f/ 48 to 10m w/ 2 – 3klb drag. Obs. took over-pull. Worked on stuck upward to max 18klb; No-Success. Then Obs. CT got free & POOH to surface. Closed swab valve & unable close TMV. Obs. sheared JDC Shear pin & TOF located in TMV. Installed 1 No 4 – 1/16" 10k gate valve on XMT. Redressed JDC shear pin to steel & M/U on CT BHA. Tightened all bolts. Performed pressure test to 6000psi for 10min; Ok.

Due to refused work on fish & stopped operation from OEOC CT service, wait on CT Service to make a decision. Changed JDC shear pin to brass. RIH w/ 1 – 1/2" CT & 3.2" JDC to 11m & engaged w/ the fish. Worked on stuck upward to max 18klb over-pull; No-Success. Slack off 6klb weight on JDC & sheared JDC Shear Pin. POOH to surface. R/D CT injector head. R/U slickline equipment & worked on fish w/ JDC jarring up several times; No-Success.

Performed injectivity test & monitored the well & checked communication between string & annulus; Ok. Pumped & direct Circ. 30bbl fresh water by applying back pressure in order to check communication between string & annulus; Ok. Commenced killing the well by W&W method,

"pumping pressure f/ 5500 to 3600psi, annulus pressure f/5500 to 2400psi". Meanwhile R/D slickline equipment & CT raiser.

In order to work on fish closed 3SV & released above pressure, M/U & RIH with 3 – 1/2" handmade JDC & tried to engage with fish several times, in the last attempt successfully engaged with fish & recovered w/ max 10klb over-pull. Operate 3SV; Ok. Monitored the well pressure, observed wellhead pressure decreased f/ 500 to 300psi. Bleed off to zero & remained constant. Annulus pressure decreased f/ 2800 to 2000psi, bleed off to 1800psi & remained constant. Performed cavity pressure test to 4000psi for 15min; Ok. Pumped & direct Circ. 125bbl 137pcf mud, "pumping pressure f/ 1300 – 2800psi to 1700psi & annulus pressure 2400psi". Got 78 to 85pcf gas cut mud in return. Stopped pumping.

Bleed off pressure f/ 2200psi to zero step by step & monitored the pressure for any build up in each step; Negative. Wellhead pressure decreased to zero & remained constant. Pumped & direct Circ. 125bbl 145pcf mud. Stopped pumping. Bleed off pressure f/ 700psi to zero step by step & remained constant. Pumped 20bbl 135pcf mud to annulus. Obs. wellhead & annulus pressure decreased to zero. Opened cap of 3/8" CIL & monitored for 1 hr. Ok. Cut the CIL. Closed 3SV & removed XMT cap. Installed TWCV in TBG HGR & test same to 5000psi for 15min; Ok. N/D XMT while passing CIL thru BTM section.

N/D XMT & N/U 13 – 5/8" 10k BOP stack. Connected hydraulic, choke & kill lines. L/U cement line to BOP & performed BOP pressure test as NIOC policy; Ok. Passed CL f/ BOP stack to rig floor. Installed bell nipple. Changed annulus to C/M line, tested same; Ok. P/U & M/U 5" DP, XOS (4 – 1/2" IF * 4 – 1/2" TPCQ), 4 – 1/2" TBG HGR running tools. Retrieved TWCV. Screw TBG HGR running tools + X.O.S (4 – 1/2" TPCQ * 4 – 1/2" IF) + 1 Jt 5" DP + 1 No. Pup Jt on TBG HGR. Performed pressure test of running tool O-ring to 1000psi, Ok. F/U string, opened 3SV & monitored the well, Ok. Back off running tools & R/D same. Close 3SV. R/U Golpa flange, BOP & lubricator of slickline (OEOC Co.), performed pressure test same w/ 1000psi. Obs. leakage between Golpa flange & 4 – 1/16"gate valve. R/D 4 – 1/16" gate valve, changed ring gasket, pressure tested same again to 1000psi, Ok. Ran 3.4" gauge ring to 61m (3SV depth), Obs. drag @59m, pulled out to surface. Ran spear to 61m (3SV depth) to catch possible cable in the well, pulled out to surface, Obs. No cable. Ran w/ protection sleeve to 61m (3SVdepth), held 7500psi pressure inside CL, set protection sleeve inside 3SV, retrieved protection sleeve & pulled out to surface. B/O slickline BOP. R/U 4 – 1/16" gate valve on Golpa flange, M/U slickline BOP & lubricator & performed pressure test same w/1000psi, Ok.

Ran w/ protection sleeve to 61m, held 7500psi pressure inside CL, set protection sleeve inside 3SV & pulled out to surface. Ran w/ 2.230" gauge cutter to 61m (3SV depth), took weight, tried to pull out, Obs. slickline string stuck, tried to pass this point for several times, No Success. Pumped 5bbl 138pcf mud wile working on stuck, No success. Increased CL pressure to 10,000psi step by step, Obs. slickline string got free. Pulled out gauge cutter to surface & L/D same. Ran with GS pulling tools of protection sleeve & retrieved same. Pulled out protection sleeve to surface. Ran back in hole w/ protection sleeve to 61m (3SV depth) & tried to set protection sleeve for several times, No Success. POOH separation sleeve to surface. Decided to extend protection sleeve, RIH w/ new protection sleeve to 61m, set

protection sleeve inside 3SV. RIH w/ 1 – 7/8" stem to 80m, checked opening of 3SV flapper, Ok, POOH stem to surface. RIH w/ 2.230" gauge ring to 3788m (first CIV depth), due to limitation of slickline cable tension decided to POOH to surface. B/O & L/D same.

R/D slickline BOP, lubricator, upper & lower sheaves. Installed X.O.S (4 – 1/16" flange * 4 – 1/2" IF box) on gate valve, pressure tested same w/ 1000psi, Ok. R/D mouse hole, R/U wireline equipment, P/U & M/U CCL & sinker bar. Pumped 100bbl 137pcf mud in string while applying 600 – 700psi pressure, back pressure in annulus. Obs. the well, Ok. B/O X.O.S (4 – 1/16" flange * 4 – 1/2" IF box) & cut box connection, welded 7" new Vam box connection on top of 4 – 1/16" flange & installed same on 4 – 1/16" gate valve. Installed wireline BOP & pressure tested same w/ 1000psi, Ok. Closed Hydrill, unscrew T.H.S tie down, P/U string, recorded string weight, applied allowable over pull to 298klb, got back on 211klb, according to limitation, applied 4klb.ft torque & tried to unlatch, No Success. RIH with CCL & sinker bar to 3850m (5m below PKR depth), POOH CCL & Sinker bar to surface & L/D same.

Held PJSM Assembled & armed colliding tools, RIH to 3850m, spaced out to 3845m, performed colliding, POOH colliding tools to surface, Obs. running tools of colliding tools & CCL left in hole. Pumped 115bbl 137pcf mud, Cont. pumping 685bbl 106pcf mud while dumping return mud, after pumping totally 800bbl mud, get 88pcf mud in return, divert return to poor boy degasser, Cont. pumping 106pcf mud, after pumped totally 850bbl mud, got 97pcf mud in return, all the time applied 700 – 1000psi back pressure in annulus. Opened T.H.S tie down screw, closed Hydrill, pressure tested tie down in open position w/ 1000psi, Ok. B/O & R/D 4 – 1/16" gate valve, Gol. Flang, landing Jt & TBG HGR running tools. P/U & M/U TBG HGR running tools, X.O.S (4 – 1/2" TPCQ * 4 – 1/2" IF) & 2 Jts DP as landing Jt, screw running tools on TBGHGR, tried to release anchor seal with maximum 120klb overpull & applied 5klb.ft torque as forward for several times, No Success. B/O & L/D 2 Jts DP, X.O.S (4 – 1/2" TPCQ * 4 – 1/2" IF) & TBG HGR running tools.

P/U & M/U TBG HGR running tools, landing Jt, Gol. Flange, 4 – 1/16" G.Flange & X.O.S (4 – 1/16" flange * 7" N. Vam), performed break Circ., NL-NG. Obs. the well, Static. R/U wireline equipment (Sealand Co.) & performed pressure test of wireline BOP w/ 1000psi, Ok. RIH w/ CCL & sinker bar to 3786m, took weight, tried to pass this point for several times, No Success. POOH to surface, changed CCL, RIH w/ 1 – 7/8" CCL & sinker bar to 3786m, took weight, tried to pass this point for several times, No Success. POOH to surface. L/D wireline equipment. L/D 4 – 1/16" X.O.S (4 – 1/16" flange * 7" N. Vam), 4 – 1/16" gate valve, Gol. Flange, landing Jt & TBG HGR running tools, closed shear rams & performed break circulation, NL-NG. M/U TBG HGR running tools, X.O.S (4 – 1/12" TPCQ * 4 – 1/2" IF), 2 Jts 5" DP as landing Jt & screw on TBG HGR, at the beginning of P/U string, Obs. 270klb on weight indicator, while pulling the string, indication of EXP.Jt movement not observed (expansion joint is in full open position). Due to condition of EXP.Jt unable to follow releasing mechanism of Anchor seal due to TBG HGR is inside of THS unable to rotate.

P/U string 30cm to be sure TBG HGR is not inside THS (weight: 290klb), set torque limit on 7klb.ft, rotated & reciprocated string between 290 – 300klb weight on weight indicator to transfer TQ for

several times, No Success. Released the TQ, P/U string to 400klb on weight indicator, to release Anchor seal, No success. Established circulation, Obs. gas cut mud in return, dumped 200bbl 50 – 55pcf cut mud, after 200bbl, Obs. 90pcf mud in return, divert return to poor boy degasser. Cont. circulation while getting 90 – 80pcf cut mud in return & applied 400 – 1400psi back pressure. Stopped operation due to poor performance of transfer pump of middle, degasser, desander, desilter tank & also hopper #1 & #2 causing unable to transfer mud to suction tank. Worked on same to rectify problem while holding 1200psi back pressure. Resumed Circ. while applying 1200 – 1400psi back pressure, got 106pcf mud in return. Stopped pumping, Obs. the well, Static. Worked on collided point to get string free with Max. 120klb overpull & applying 6klb.ft TQ as forward, string got free, string weight on weight indicator: 195klb. R/D wireline, R/U CSG running equipment, POOH completion string while L/D landing Jt, TBG HGR, fluted swage, 3SV w/ long & short flow coupling (Obs. protection sleeve was not set in profile) & totally 35 Jts 4 – 1/2" TBG, meanwhile Circ. across wellhead, NL-NG.

Cont. POOH completion string while L/D 94 Jts 4 – 1/2" TBG (totally 294 Jts), while using safety clamps & Circ. across wellhead; NL-NG. Obs. last Jt of string w/ underneath coupling washout, worn & torn in coupling area. Also Obs. 5 No. of cross coupling protectors left in hole & 9 No. of cross coupling protectors parted because of eroded condition. After Jt No. 35 some of cross couplings had been lost, eroded & parted. Obs. some of asphaltene on cross coupling protectors of injection line & slips area of 4 – 1/2" TBGs so need to be washed with steam, which lead delay in POOH completion string.

P/U & M/U 7 – 7/8" Over shot (w/ 4 – 3/4" Basket Grapple Mill Control, 4 – 1/2" Basket Grapple) & Fishing BHA & RIH to 1880m, NL-NG. Estimated TOF: 2706m. Took weight @ 1880m, tried to pass several times; No Success. Established Circ. & pass gently. Cont. wash down to 2660m while trying to pass several intervals w/o wash down; No Success (Asphaltene precipitation indication); NL-NG. Meanwhile got 150bbl 50pcf gas cut mud & dumped same @ 2020m. Established Circ. to uniform Mw in/out on 106pcf, got 300bbl 75 – 95pcf gas cut mud w/ NL-NG. Obs. huge amount of asphaltene precipitation on shakers. Resumed wash down to 2731m while took weight several depth & pass same. Confirmed top of fish @2731m. Tried to catch the fish several times; No Success. Commenced POOH w/ 7 – 7/8" over shot & fishing BHA f/2731m to surface. Circ. CWH; NL-NG. Obs. some scratches on outer body & edge of over shot guide.

R/U slickline equipment & ran 3.8" magnet to 2731m, held up. POOH to surface. 1lb junk recovered (mixed with asphaltene). R/D slickline equipment while waiting on receive 8 – 3/8" Mill. P/U & M/U 12 Jts 6 – 1/2" DC, 3 Jts 5" HWDP & RIH OEDPs to 1250m Jt by Jt. Circ. CWH; NL-NG. POOH string to surface & L/D same. RIH OEDPs by 12 Jts 4 – 3/4" DC & 12 Jts 3 – 1/2" DP. POOH OEDPs to surface & R/B. While Circ. CWH; NL-NG. Difficulty B/O & L/D 1 Jt 5" Pup Jt, 5" FOSV & Circ. swage due to over torque. Tried to B/O 11" plug tester from 1 Jt 5" DP w/ 48klb.ft TQ several times; No Success. Set 11" wear bushing in THS.

P/U & M/U 8 – 3/8" M.M J.Mill, 1 No. 6 – 5/8" Junk Sub, 1 No. 7" Junk Sub, 6 – 1/2" D.Pin, 6 – 1/2" Bit Sub & RIH w/ 4 Stds to 2729.5m & took weight on this depth. Recorded off BTM parameters.

Commenced milling obstruction f/ 2729.5 to 2736m; NL-NG. While got 100bbl 80 – 90pcf gas cut mud @ BTM's up. Cont. milling f/ 2736 to 2739m; NL-NG. Circ. BTM's up, over Circ. till clean shakers w/ Hi-Rate; NL-NG. Obs. metal wool on ditch magnet & also CIL cover on shakers. Worked on junk sub several times. Commenced POOH w/ 8 – 3/8" M.M J.Mill & milling BHA f/ 2739 (TOF) to surface. Circ. CWH; NL-NG. Mill Condition: 80% used. Obs. mark of TBG as top of fish on mill. Both junk subs full of junk including pieces of CIL, cross coupling & metal wool. Recovered 19lbs junk with 2 junk subs.

P/U & M/U 7 – 7/8" overshot, dressed with 4 – 1/2" Basket Grapple & 4 – 3/4" mill control & RIH to 2739m. Worked on Fish & tried to catch & engage with fish several times. Obs. 20 – 50klbs excess weight on M/D & suddenly drop excess weight. Commence POOH w/ 7 – 7/8" overshot & fishing BHA f/ 2739m (TOF) to surface, Circ. CWH; NL-NG. Obs. mark of TOF on 3 wickers of basket grapple & also mark of junk on outer body of guide. P/U & M/U 7 – 5/8" overshot (dressed w/ 4 – 1/2" Basket Grapple, 4 – 1/2" Mill Control & Extension sub) & RIH to 2739m. While W/D last 2 Stds, tried to catch & engage with fish. P/U string Obs. caught & engaged w/ fish. Worked on fish w/ 280 to 450klbs on M/D several times; No Success.

After several more attempts without succeed, it was decided to set a cement plug @3300^{m}, open window & continue drilling directionally to TD @4226^{m}.

1.8 Well Montage

General Information: Well Name: AZNS-127; Pad No.: S4; Well Type: IIF; Well Progile: Side Track; Block: II; Field: South Azadegan; Country / Area: Iran / Khuzestan; Reservoir: Fahlian

Wellhead Coordinate: E: 196,167; N: 3,448,561

TD Coordinate: E: 196,185.7; N: 3,448,541.3

Parties/Refrence: Client: PPL/PEDEC; Contractor: MSA/OEOC; Rig Cont: OEOC; Rig: OEOC 208

Well Montage

AZNS-127 Re-Entry #01

(PAD-54), IIF

Drilling Depth:
TD (MD): 4200m
TD (TVD): 4150m
GLE: 3.95m
RTE: 13.45m

Timing:
Moving Days: 10 Days
Spud Date: TBA
Total Rig Days: 80 Days (With out rigless job)
Rig Released Date: TBA

Field Location/Objective:
*Field Location: The AZNS oilfield is located in Khuzestan province, approximately 80 Km to the west of Ahwaz city.
*Well Objective: Open the window & side track the well from 3337 mMD, drill 8-1/2" & 5-7/8" hole & complete the well

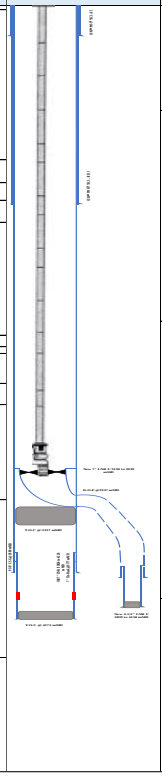
Formation	Formation TOP MD (M)	Formation TOP TVD (M)	Well Schematic	Hole Size Bit	Mud Type - Mw (pcf)	CSG/Liner Design	Cementing	Logging Testing Coring	BOP and Well Head	Remarks (Drilling Issues/Lesson Learned)
Aghajari	Surface	surface		12-1/4" 215 / 517 / 437 PDC	Glycol - Drill fluid 80 - 81	103 Jts 53.5 ppf T-95, TSH Blue + 226 Jts 53.5 ppf, T-95, TPCQ	20 bbl 85 pcf spacer Lead: 2282 sxs Cement G Delijan 88 pcf, 1133 bbl Tail: 363 sxs Cement G Delijan 118 pcf, 116 bbl	Open Hole Logs: Full suite log/ DLL-GR Cased Hole Logs: SBT/CCL/VDL/GR	13-5/8" 5k * 11" 10k THS (MSP Drillex), 13-5/8" * 9-5/8" casing slips	1. PDM failure @1612m. 2. Low ROP f/1750 to 1797m. 3. Low ROP f/1920 to 1989, cherty zone 4. Wash out on XOS @1989 5. PDM twist off @1989m. 6. Low ROP f/ 2371 to 2383 7. stuck pipe @6040. 8. wash out float sub @3371m. 9. loss increased f/ 6 to 15 bph @ 3327m. 10. loss increase @ 3705m to 12 bph.
Gachsaran	1105	1105								
Asmari	1448.8	1449		8-1/2" 3 No. TCI 537 1 No. PDC M323 Directional BHA	Salt Saturated Kcl/Glycol/Polymer/CaCO3 WBM 80 - 85	7" 29ppf, 110S, TPG2 Connection	TBA	Open Hole Logs: MWD Cased Hole Logs: SBT/CBL/VDL	13-5/8", 10K BOPs 4-1/16", 10K X-Tree	1. formation loss 2. tight hole & hole stability 3. Hole instability issues 4. Wellbore Losses 5. Shale
Pabdeh	1740.68	1740.7								
Gurpi	2182.49	2182								
Ilam	2517	2517								
Lafan	2618	2618								
Sarvak	2625	2625		5-7/8" 2 No. MT 537 1 No. PDC M323	Salt Saturated Kcl/Polymer/CaCO3 WBM 100 - 106	4-1/2" 13.5ppf, 110S, TPG2	TBA	Cased Hole Logs: SBT/CBL/VDL CCl-Grafter perforation	13-5/8", 10K BOPs 4-1/16", 10K X-Tree	1. Wellbore Losses 2.Oil/Water influx
Kashkumi	3274	3274								
Dariyan	3479	3479								
Gadvan	3667	3667								
Fahliyan	3939	3939								
TD 4150	TD 4150									

Figure 1-4 AZNS-127 Well Montage

1.9 Casing & Liner Depth

Casing Spec.	20"	K-55, 94#, BTC	0-146 m
	13-3/8"	N-80, 68#, BTC	0-1441 m
	9-5/8"	103 joint 53.5ppf, T-95, TSH Blue, 226 joints 53.5ppf, T-95, TPCQ.	0-3909 m
	7"	110 S, 29#, TPCQ	3777 – 4226 m

Table 1-7 Casing & Liner Data

1.10 Well Schematic before Side Track

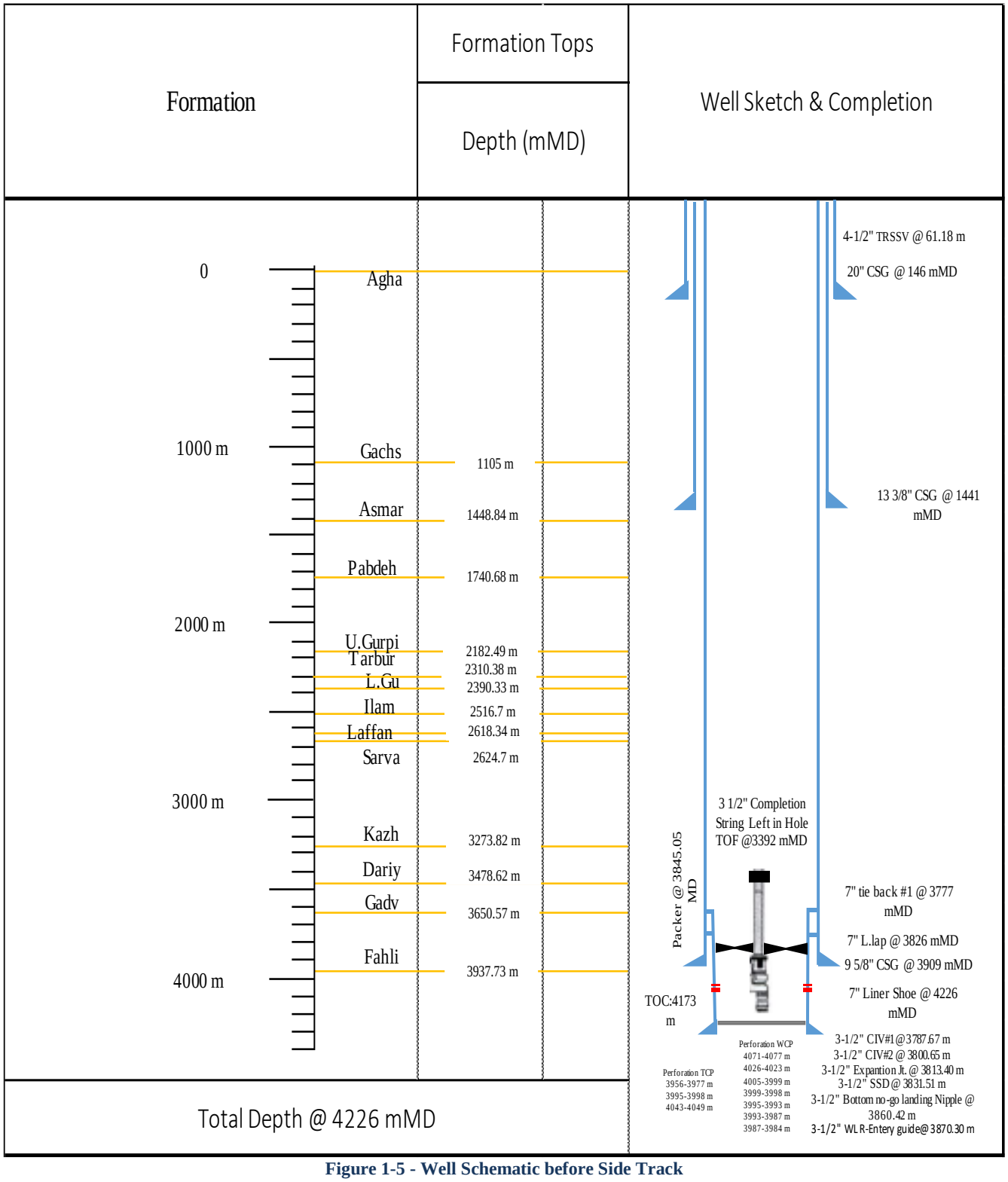


Figure 1-5 - Well Schematic before Side Track

1.11 Well Schematic after Side Track

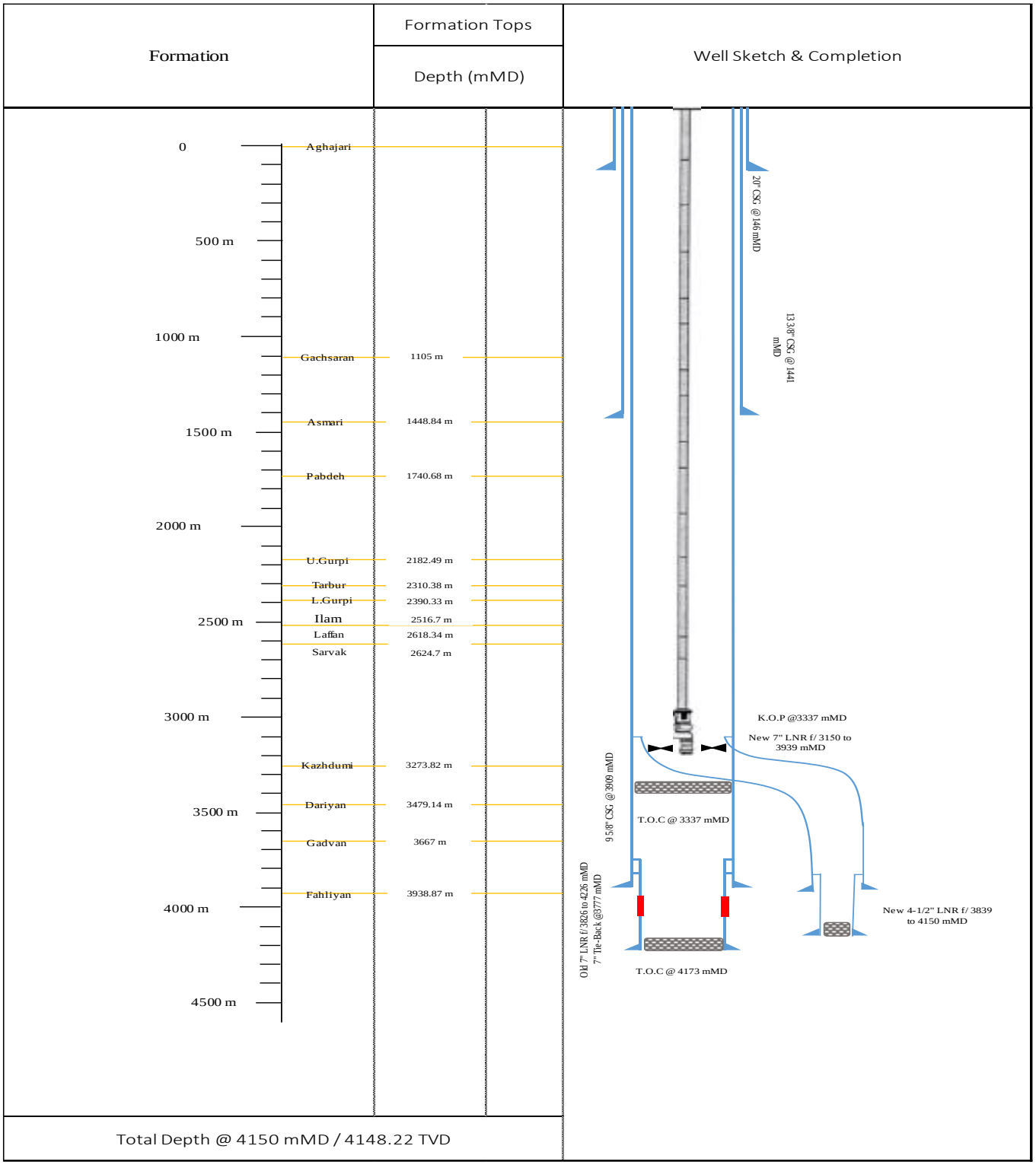


Figure 1-6 - Well Schematic after Side Track

2. Geological Information

2.1 Formation Stratigraphy and Lithology

Stage	Group	Formation	Lithology	Lithology Description	Block I Top (mTVD)	Block II Top (mTVD)	Block III Top (mTVD)
Pliocene Miocene	Fars	Agha Jari		Marl, Sandstone, Anhydrite	Surface	Surface	Surface
Miocene	Fars	Gachsaran		Marl, Anhydrite, Claystone	1380	1150	961
E. Miocene Oligocene		Asmari		Limestone, Sandstone, Claystone	1776	1503	1318
E. Oligocene L. Paleocene		Pabdeh		Limestone, Argillaceous Limestone, Claystone, Dolomite	2079	1786	1525
Maastrichtian Campanian		U. Gurpi / Tarbur / L. Gurpi		Limestone, Argillaceous Limestone	2352	2303	2153
Santonian Coniacian	Bangestan	Ilam		Limestone, Claystone	2606	2534	2549
	Bangestan	Laffan		Claystone	2691	2642	2625
Turonian Cenomanian	Bangestan	Sarvak		Limestone, Claystone	2697	2648	2632
Cenomanian Albian	Bangestan	Kazhdumi		Sandstone, Argillaceous Limestone, Claystone, Shale	3335	3300	3269
Aptian	Khami	Dariyan		Limestone	3538	3499	3471
Aptian Barremian	Khami	Gadvan		Argillaceous Limestone, Claystone, Sandstone	3718	3696	3635
Neocomian	Khami	Fahliyan		Limestone, Marl	4092	3989	3974

Figure 2-1 - Azadegan Stratigraphic Columne

Lithostratigraphic units	Formation Top(m)				Vertical Thickness	Difference (m)	Casing Data	
	Forecast (m)	MD (Sample) (m)	MD (log) (m)	Subsea (msl)	(m)	(Forecast-MD)	Size (Inch)	Depth (m)
Aghajari	Surface	Surface	0	14.45	1059.17		20"	140m
Gachsaran	1107	1105	1084	-1069.67	364.53	-23		
Asmari	1470	1451	1448.84	-1434.2	291.8	-21.16	13 3/8"	1441m
Pabdeh	1747	1724	1740.68	-1726	441.8	-6.32		
U-Gurpi	2203	2189	2182.49	-2167.8	127.9	-20.51		
Tarbur	2335	2314	2310.38	-2295.7	79.9	-24.62		
L-Gurpi	2407	2390	2390.33	-2375.6	126.4	-16.67		
Ilam	2537	2519	2516.7	-2502	101.6	-20.3		
Laffan	2635	2617	2618.34	-2603.6	0.4	-16.66		
Sarvak-1	2602	2595	2624.7	-2610	19.3	22.7		
Sarvak-2			2644.06	-2629.3	8.8			
Sarvak-3			2652.83	-2638.1	31.5			
Sarvak-4			2684.29	-2669.6	94.3			
Sarvak-7			2778.6	-2763.9	21.9			
Sarvak-8			2800.52	-2785.8	59.5			
Sarvak-9			2859.98	-2845.3	37.5			
Sarvak-10			2897.55	-2882.8	22.4			
Sarvak-11			2919.95	-2905.2	32			
Sarvak-12			2951.9	-2937.2	93.6			
Intra Sarvak			3045.5	-3030.8	228.3			
Kazhdumi	3242	3241	3273.82	-3259.1	150.9	31.82		
Kazhdumi-1			3424.78	-3410	23.1			
Kazhdumi-2			3447.89	-3433.1	10.4			
Kazhdumi-3			3458.28	-3443.5	7.7			
Kazhdumi-4			3465.98	-3451.2	7.9			
Kazhdumi-5			3473.83	-3459.1	4.8			
Dariyan	3441	3448	3478.62	-3463.9	171.9	37.62		
Gadvan-1	3596	3605	3650.57	-3635.8	8.8	54.57		
Gadvan-2u			3659.36	-3644.6	54.7			
Khalij Mbr.			3714.04	-3699.3	10.4			
Gadvan-2l			3724.44	-3709.7	4			
Gadvan-3			3728.49	-3713.7	8.2			
Gadvan-4			3736.69	-3721.9	16.3			
Gadvan-5a			3753	-3738.2	5.3			
Gadvan-5b			3758.31	-3743.5	5.4			
Gadvan-5c			3763.62	-3748.9	7.4			
Gadvan-6			3771.11	-3756.3	20.9			
Gadvan-7			3792	-3777.2	24.7			
Gadvan-8			3816.67	-3801.9	121.1		9 5/8"	3909m
Fahliyan	3950	3934	3937.73	-3923	35	-12.27		
Fahliyan-1a			3972.82	-3958	17.5			
Fahliyan-1b			3990.29	-3975.5	15.4			
Fahliyan-1c			4005.63	-3990.9	36.6			
Fahliyan-2a			4042.32	-4027.5	16.4			
Fahliyan-2b			4058.69	-4043.9	11.8			
Fahliyan-3a			4070.51	-4055.7	6.2			
Fahliyan-3b			4076.69	-4061.9	10.4			
Fahliyan-4a			4087.09	-4072.3	10.4			
Fahliyan-4b			4097.52	-4082.7	17.1			
Fahliyan-5a			4114.61	-4099.8	17.8			
Fahliyan-5b			4132.35	-4117.6	93.9			
TD	4226.5	4226.5	4226.5	-4211.5	-	0	7"L	4225.5m

Figure 2-2 Actual Formation Tops of AZNS-127 (F-01)

2.2 Planned Trajectory Data of Side Track

RTE: 14/45m					GLE: 3/95m					
Vertical section was calculted toward azimuth 90deg.										
MD (m)	Inc (°)	Azi (°)	TVD (m)	N/S (m)	E/W (m)	VSEC (m)	DLS (°/30m)	Map N (m)	Map E (m)	Remarks
3273.82	0.45	285.33	3273.57	-19.78	-6.59	-6.59	0.316	3448541.22	196160.41	Kzh. Fm.
3277	0.48	283.6	3276.75	-19.77	-6.61	-6.61	0.316	3448541.23	196160.39	
3307	0.78	274.18	3306.75	-19.73	-6.94	-6.94	0.316	3448541.27	196160.06	
3337	0.64	268.74	3336.74	-19.71	-7.36	-7.36	0	3448541.29	196159.64	K.O.P. inside of Kzh. Fm.
3367	1.36	90	3366.74	-19.72	-7.17	-7.17	2	3448541.28	196159.83	
3397	3.36	90	3396.71	-19.72	-5.94	-5.94	2	3448541.28	196161.06	
3427	5.36	90	3426.63	-19.72	-3.66	-3.66	2	3448541.28	196163.34	
3457	7.36	90	3456.44	-19.72	-0.33	-0.33	2	3448541.28	196166.67	
3479.14	8.83	90	3478.36	-19.72	2.79	2.79	2	3448541.28	196169.79	Dr. Fm.
3487	9.36	90	3486.12	-19.72	4.03	4.03	2	3448541.28	196171.03	
3496.62	10	90	3495.6	-19.72	5.64	5.64	2	3448541.28	196172.64	M.B.U.A.P.
3517	8.64	90	3515.71	-19.72	8.95	8.95	2	3448541.28	196175.95	
3547	6.64	90	3545.45	-19.72	12.93	12.93	2	3448541.28	196179.93	
3577	4.64	90	3575.3	-19.72	15.88	15.88	2	3448541.28	196182.88	
3607	2.64	90	3605.24	-19.72	17.79	17.79	2	3448541.28	196184.79	
3637	0.64	90	3635.22	-19.72	18.65	18.65	2	3448541.28	196185.65	
3646.62	0	0	3644.84	-19.72	18.7	18.7	2	3448541.28	196185.7	
3652.09	0	0	3650.31	-19.72	18.7	18.7	0	3448541.28	196185.7	Drop Point
3667	0	0	3665.22	-19.72	18.7	18.7	0	3448541.28	196185.7	Gd. Fm.
3697	0	0	3695.22	-19.72	18.7	18.7	0	3448541.28	196185.7	
3727	0	0	3725.22	-19.72	18.7	18.7	0	3448541.28	196185.7	
3757	0	0	3755.22	-19.72	18.7	18.7	0	3448541.28	196185.7	
3787	0	0	3785.22	-19.72	18.7	18.7	0	3448541.28	196185.7	
3817	0	0	3815.22	-19.72	18.7	18.7	0	3448541.28	196185.7	
3847	0	0	3845.22	-19.72	18.7	18.7	0	3448541.28	196185.7	
3877	0	0	3875.22	-19.72	18.7	18.7	0	3448541.28	196185.7	
3907	0	0	3905.22	-19.72	18.7	18.7	0	3448541.28	196185.7	
3937	0	0	3935.22	-19.72	18.7	18.7	0	3448541.28	196185.7	
3938.87	0	0	3937.09	-19.72	18.7	18.7	0	3448541.28	196185.7	Fa. Fm.
3939	0	0	3937.22	-19.72	18.7	18.7	0	3448541.28	196185.7	7" Liner Shoe
3967	0	0	3965.22	-19.72	18.7	18.7	0	3448541.28	196185.7	
3997	0	0	3995.22	-19.72	18.7	18.7	0	3448541.28	196185.7	
4027	0	0	4025.22	-19.72	18.7	18.7	0	3448541.28	196185.7	
4057	0	0	4055.22	-19.72	18.7	18.7	0	3448541.28	196185.7	
4087	0	0	4085.22	-19.72	18.7	18.7	0	3448541.28	196185.7	
4117	0	0	4115.22	-19.72	18.7	18.7	0	3448541.28	196185.7	
4147	0	0	4145.22	-19.72	18.7	18.7	0	3448541.28	196185.7	
4150	0	0	4148.22	-19.72	18.7	18.7	0	3448541.28	196185.7	T.D. & 4 1/2" Liner Shoe

3. Logging Program

Hole Size	Casing / Liner	Formation	Open Hole Wire Line Logging	Cased Hole Wire Line Logging
-	9-5/8”	-	-	Corrosion log,
8-1/2”	7”Liner	Sarvak, Kazhdumi, Dariyan, Gadvan	MWD	SBT /CBL/VDL,
6-1/8”	4.5 Liner	Fahliyan		SBT /CBL/VDL, Perforation, After perforation GR-CCL (perforated interval)

Table 3-1 AZNS-127 Logging Program

4. Program Outline

4.1 Contractors Involved

Table 4-1 AZNS-127 Contractor's Involve

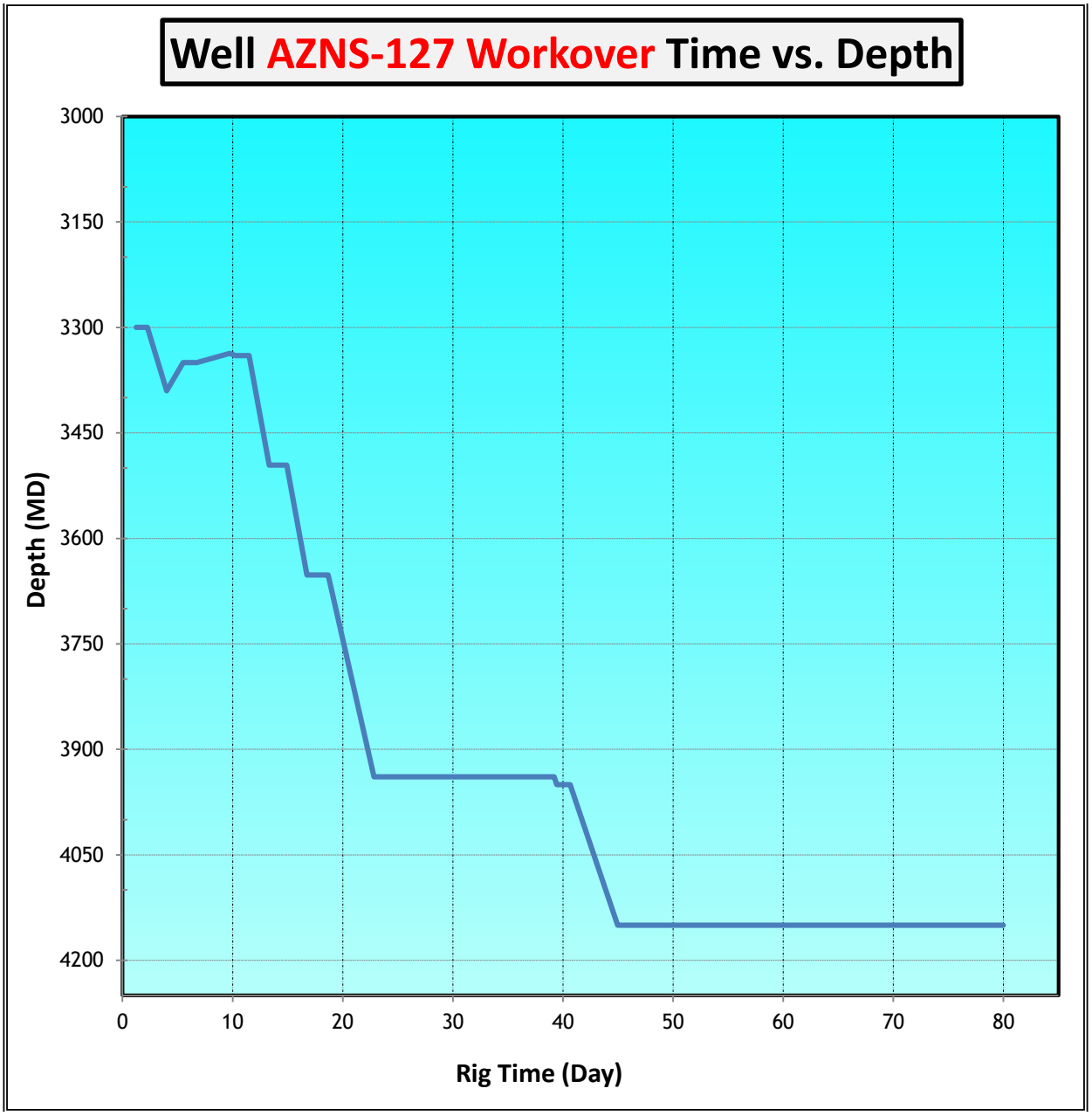
Item	Service Item	Contractor
1	Drilling Rig, Mobilization and Demobilization the Rig, Rig Move Between Well Pads	OEOC 208
2	Drilling Fluids and Completion Fluids	VIRA
3	Waste Management	Pars Hafaran Arvand
4	Electrical Logging and Perforation	SEALAND
5	Mud Logging	MAPSA KISH
6	H_2S Monitoring and Control	BONYAN IMEN
7	Casing and Tubing Running	OEOC
8	Cementing	SPAD ENERGY
9	Dry Test, DST and RTTS	OEOC
10	Slick Line	OEOC
11	Stimulation (Coiled Tubing and Acidizing and Nitrogen Lifting)	OEOC
12	Surface Well Testing	OEOC
13	Liner Hanger	PARS POOLAD
14	Well Head	OEOC
15	Directional Drilling	Paniz Energy

4.2 Time Breakdown

De-Complete, Re-Complete Operation TBD of Well: AZNS-127 Work-over (Side Tracking)	IADC Code	Interval (meters)	Formation	Depth (meters)	ROP	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)
Work-over								
RIH with 8 1/2" bit and 9 5/8" scraper, scrap 9-5/8" casing prior to Corrosion Log, POOH to surface.	Trip		SV	3300		30	1.25	1.3
Change Ram, Install shooting nipple, R/U WL Eq. RIH WL with Corrosion Log to cheack 9-5/8" Csg. , POOH and R/D Logging Eq, change Ram.	Log		SV	3300		24	1.00	2.3
R/U & run 5" & 3 1/2" open end drill pipes to 3390m, MD, set 25 Bbls cement as balance plug. POOH	CMT		SV	3390		42	1.75	4.0
RIH 8-1/2" bit & dress cement to 3350m, MD, Perform Positive Pressure Test CMT Plug,POOH.	D/O		SV	3350		36	1.50	5.5
RIH with 8 1/2" bit and 9 5/8" scraper, scrap whipstock setting depth,& Displace mud in hole by 85 pcf mud, observe the well, POOH.	Side Track		SV	3350		30	1.25	6.8
M/U & R.I.H. with 9 5/8" permanent hydraulic set Whipstock (Suitable for 9-5/8" casing, 53.5#),Set same & open window @ 3337m.	Side Track		SV	3337		72	3.00	9.8
Drill 3m in formation to 3340m,POOH to Surface	Side Track	3	SV	3340		12	0.50	10.3
BOP Test.	BOP Test		GD	3340		6	0.25	10.5
P/U & M/U New 8-1/2" Rock Bit and Directional BHA, RIH to Btm.	Trip		SV	3340		24	1.00	11.5
Cont. Drilling 8-1/2" hole f/ 3340m to 3496m.	Drilling	156	SV KZ DR	3496	4	44	1.83	13.3
Perform Hole condition POOH to surface.	Hole Cond/Circ		DR	3496		20	0.83	14.2
P/U & M/U New 8-1/2" Rock Bit and Directional BHA, RIH to Btm.	Trip		SV	3496		18	0.75	14.9
Cont. Drilling 8-1/2" hole f/ 3496m to 3652m(DOP).	Drilling	156	DR GD	3652	4	44.0	1.83	16.8
Perform Hole condition POOH to Surface to Change BHA	Drilling		DR GD	3652		20.0	0.83	17.6
BOP Test.	BOP Test		GD	3652		6.0	0.25	17.8
P/U & M/U New 8-1/2" PDC Bit and BHA, RIH to Btm.	Trip		GD	3652		20.0	0.83	18.7
Cont.D.Drilling 8-1/2" hole f/ 3652m to section TD @ 3939m.	Drilling	287	GD	3939	3	100.0	4.17	22.8
CBU, Perform condition trip to last CSG Shoe prior to Run Liner, RIH to Btm, CBU, Spot pill POOH to surface, L/D PDC Bit.	Hole Cond/Circ		GD	3939		36.0	1.50	24.3
BOP Test.	BOP Test		GD	3939		10.0	0.42	24.8
R/U CSG running gear, Run 7" Liner to Btm, R/D CSG running gear.	CSG		GD	3939		42	1.75	26.5
Perform 7" Liner cementing operation.	CMT		GD	3939		12	0.50	27.0
POOH w/ L.Hanger setting tool to 2800m, Flush string & CBU.	CSG		GD	3939		6	0.25	27.3
POOH to surface & L/D L.Hanger setting tool.	CSG		GD	3939		16	0.67	27.9
P/U & M/U 8-1/2" MT Bit BHA, RIH to L.Lap @ +/- 3150m while D/Out cement (If any cement: Considered 7STD wet for CMT Job), Circ & clean the well, POOH.	D/Out		GD	3939		36	1.50	29.4
L/D excess 5" DPs/HWDP.	Trip		GD	3939		12	0.50	29.9
P/U & M/U 4-3/4" DCs & 3-1/2" DPs/HWDPS, R/Back same in derrick.	Trip		GD	3939		12	0.50	30.4
P/U & M/U 5-7/8" MT Bit BHA, RIH to tag POB, D/Out POB, RIH to LDC depth @ +/- 3915m while D/Out any cement, work on junk sub and CBU. Perform 7" Liner CIT. (80% of Burst press) with 2000 psi,for 15 min, (If not obs bumping).CBU & POOH to Surface.	D/Out		GD	3939		48	2.00	32.4
RIH with 5-7/8" MT Bit and tandem scraper (7" & 9-5/8") to LDC depth, Scrap inside 7" Liner and 9-5/8" casing before Dry test, CBU, POOH to Surface.	Testing		GD	3939		45	1.88	34.3
BOP Test.	BOP Test		GD	3939		6.0	0.25	34.5
RIH with 9-5/8" Dry test BHA, RIH to L.Lap @ +/- 3150m, Perform 7" Liner lap Dry Test (based on Gadvan Reservoir Datum Pressure), POOH.	Testing		GD	3939		60	2.50	37.0
Change Ram, Install shooting nipple, R/U WL Logging Eq. Perform CMT quality Logging of 7" Liner from 7" liner LDC @ 3915 m to 7" Liner Lap @ 3150 m (SBT or /CBL/VDL), R/D Logging Eq, change Ram.	Log		GD	3939		24	1.00	38.0
P/U & M/U 6-1/8" MT Bit and C/Out BHA, RIH to LDC depth. Displace 7" Liner volume with 100 pcf drilling fluid.	Trip		GD	3939		20	0.83	38.9
Cont' D/Out CMT, Shoe and Rat Hole.	Hole Cond/Circ		GD	3939		8	0.33	39.2
Total Duration of 8-1/2" Hole Section:							*39.21*	*39.2*

Drilling 6-1/8" hole to 3950m.	Drilling	11	FH	3950	2	5.5	0.23	39.44
CBU and balance Mw,,perform FIT with Required Pressure 720 psi (EMW: 108pcf, should be equal to max. planned Mw for this hole section plus 2 pcf),POOH to Surface	Trip		FH	3950		23	0.96	40.40
BOP Test.	BOP Test		FH	3950		6	0.25	40.65
RIH w/ 6-1/8 PDC bit & Drilling 6-1/8" hole f/ 3950m to well TD @ 4150m.	Drilling	200	FH	4150	3	104	4.33	44.98
CBU, Perform condition trip to last CSG Shoe prior to Liner Running, RIH to Btm, CBU, Spot pill.	Hole Cond/Circ		FH	4150		12	0.50	45.48
POOH to surface.	Trip		FH	4150		18	0.75	46.23
BOP Test.	BOP Test		FH	4150		6	0.25	46.48
R/U 4-1/2" Liner running Eq., Run 4-1/2" liner (+/- 311m LNR) to Btm @ 4150m, R/U CMT lines, Set L.Hanger +/- 3793m (100m overlap), release L.Hanger setting tool.	CSG		FH	4150		36	1.50	47.98
Cement 4-1/2" Liner as per program.	CMT		FH	4150		8	0.33	48.31
POOH w/ L.Hanger setting tool to 3500m, Flush string & CBU.	CSG		FH	4150		6	0.25	48.56
POOH to surface & L/D L.Hanger setting tool.	CSG		FH	4150		12	0.50	49.06
P/U & M/U 6-1/8" MT Bit BHA, RIH to L.Lap @ +/- 3793m while D/Out cement (If any cement: Considered 7STD wet for CMT Job), Circ & clean the well, POOH.	D/O		FH	4150		40	1.67	50.73
L/D excess 3-1/2" DPs/HWDP.	Trip		FH	4150		8	0.33	51.06
P/U & M/U 3-1/2" DCs & 2-7/8" DPs/HWDPS, R/Back same in derrick.	Trip		FH	4150		12	0.50	51.56
P/U & M/U 3-1/8" Mill BHA, RIH to tag POB, D/Out POB, RIH to LDC depth @ +/- 4126m while D/Out any cement, work on junk sub and CBU. Perform4-1/2" Liner CIT. (80% of Burst press) with 2000 psi,for 15 min, (If not obs bumping).CBU, POOH to Surface.	D/O		FH	4150		60	2.50	54.06
BOP Test.	BOP Test		FH	4150		6	0.25	54.31
RIH with 3-1/8" Mill and 4-1/2" scraper to LDC depth, Scrap inside 4-1/2" Liner , CBU, POOH to Surface.	Trip		FH	4150		40	1.67	55.98
Change Ram, Install shooting nipple, R/U WL Logging Eq. Perform CMT quality Logging of 4-1/2" Liner from LDC depth @ 4126 m to 4-1/2" Liner Lap @ 3839 m (SBT or CBL/VDL), R/D Logging Eq, change Ram.	Log		FH	4150		24	1.00	56.98
P/U & M/U 3-1/8" Mil and C/Out BHA, RIH to LDC depth. Displace 4-1/2" Liner volume with Non Solid fluid, POOH	Trip		FH	4150		36	1.50	58.48
P/U & M/U W/L Perforation Assy, RIH to perforation depth, correlate depth with WL and perforate Guns as program.POOH to Surface.	Log		FH	4150		48	2.00	60.48
BOP Test.	BOP Test		FH	4150		6	0.25	60.73
P/U & M/U 3-1/8" Mill BHA, RIH to top of LDC depth while smoothing Perforated intervals.	Trip		FH	4150		60	2.50	63.23
Change Ram, Install shooting nipple, R/U WL Eq. RIH WL with CCL/GR Log to confirm Perforated intervals, POOH and R/D Logging Eq, change Ram.	Log		FH	4150		18	0.75	63.98
RIH with 3-1/8" mill & clean out BHA (7" scraper included) to bottom , displace the well with completion fluid (non-solid) CaBr2, POOH & L/D DP's.	Trip		FH	4150		72	3.00	66.98
BOP Test.	BOP Test		FH	4150		6	0.25	67.23
Run 10K completion string as per program. NOTE: Annulus pressure test during completion operation is related to corrosion log result.	Completion		FH	4150		168	7.00	74.23
Set BPV inside TBH, N/D BOP's & install new 10K XMT.	Completion		FH	4150		48	2.00	76.23
Retrive BPV, install TWCV and Pressure test the XMT.	Completion		FH	4150		12	0.50	76.73
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, and Release the Rig.	Completion		FH	4150		8	0.33	77.06
Retrieve TWCV,R/U Slick Line, Set protection sleeve, retrieve plug & prong from NO GO landing nipple, POOH.	Completion		FH	4150		Rig Less	0.00	77.06
RIH with CT to bottom, displace the well with diesel or nitrogen & observ well, POOH.	Completion		FH	4150		Rig Less	0.00	77.06
R/U SL, RIH & retrieve protection sleeve, POOH, R/D SL.	Completion		FH	4150		Rig Less	0.00	77.06
Flow the well as pre-acid clean up till acceptable BSW & reading stable flow rate on 2 different choke size for 72 hrs.	Completion		FH	4150		Rig Less	0.00	77.06
R/U & RIH S.L, Set protection sleeve, POOH & R/D S.L RIH with C/T & perform acid wash as per program, POOH to surface. RIH w/SL and retrieve protection sleeve	Completion		FH	4150		Rig Less	0.00	77.06
Flow the well as post-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for min 96 hrs. Note 1: The time is only for flowing duration (R/U, R/D & preparation time is excluded). Note 2: Acidizing & post-acid clean-up maybe perform as rig less job upon contractor request & client approval but in case of any failure in well integrity the contractor should provide drilling rig & required services in own responsibility & fix the problem.	Completion		FH	4150		Rig Less	0.00	77.06
RIH with S/L, set protection Sleeve, POOH & R/D SL.	Completion		FH	4150		Rig Less	0.00	77.06
RIH with C/T to bottom & displace the well with diesel, POOH.	Completion		FH	4150		Rig Less	0.00	77.06
RIH with S/L, install plug & prong into NO GO landing nipple, POOH.	Completion		FH	4150		Rig Less	0.00	77.06
RIH with S/L, retrieve protection sleeve.	Completion		FH	4150		Rig Less	0.00	77.06
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV.	Completion		FH	4150		Rig Less	0.00	77.06
Total Duration of 6-1/8" Hole Section:							***37.9***	***77.1***
NOTE:								
Total Duration for Cut & Slip DL in all hole section.	Cut & Slip			4150		12	0.50	77.6
Total Duration for Rig Service (Consider 0.5hr per day) in all hole section.	Rig Service			4150		58	2.42	80.0
Total Duration of AZNS-127 Work-over Well:								80.0

4.3 Time vs. Depth Chart



5. Sequence of Operation

5.1 Open Window & Drilling 8 – 1/2" Hole Section

Casing/Liner Point	7" Liner shoe @3939^{m} (7" L.Lap @ ±3150^{m})	
Formation	Asmari, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan	
Risks & Hole Problems	Losses, Tight Hole & Pipe Stuck, Oil/Gas Flow, Shale Instability,Hole Deviation , Complete Loss	
Deviation Control	MWD	
Bit Type & Nozzle	3 No. TCI, 537 & 1 No. PDC M323	
Logging	**Cased Hole: Corroision Log from 9-5/8" Casing** **Open Hole:NA** **Cased Hole:** SBT or CBL//VDL/ from 7" Liner	
Testing	Dry Test	
Drilling Fluid Parameters	**Salt Saturated Kcl/Glycol/Polymer/$CaCO_3$ WBM**	
	MW (PCF)	80 – 85
	FV (Seq/qt)	32 – 52
	PV (CP)	ALAP
	YP (Lbs./100ft^2)	12 – 20
	HPHT FL (CC/30min)	<3
	Gel 10 sec/10 min (Lbs./100ft^2)	2 - 5 / 3 - 7
	PH	9 – 10
Drilling Parameters	GPM	400-500
	WOB (Klbs)	15-20
	RPM	150-170

Table 5-1 AZNS-127 8 – ½" Hole Section General Data

Formation Sampling

Sample would be taken from Gachsaran to Gadvan every 2-5 meters

NOTE: In addition to this general rule, geologist may decide to take more samples.

- Ensure that drilling tools & equipment are on wellsite & are in serviceable condition.
- Ensure that adequate fishing tools for this section of hole are available.
- Ensure that enough mud material, LCM & CMT are available.

1. RIH w/ 8 – 1/2" bit & 9 – 5/8" scraper, scrap 9 – 5/8" casing prior to Corrosion log, POOH to surface.
2. Change ram, install shooting nipple, R/U wireline logging equipment.
3. RIH w/ Corrosion log to check 9 – 5/8" casing, POOH to surface & R/D logging equipment, change ram.
4. R/U & run 5" & 3 – 1/2" open end DPs to 3390^{mMD}, set 25^{bbl} cement as balance plug, POOH.
5. RIH 8 – 1/2" bit & dress cement to 3350^{mMD}. Perform positive pressure test CMT POOH.
6. RIH w/ 8 – 1/2" bit & 9 – 5/8" scraper, scrap whipstock setting depth & displace mud in hole wby 85^{pcf} mud, observe the well, POOH.
7. M/U & RIH w/ 9 – 5/8" permanent hydraulic set whipstock (suitable for 9 – 5/8" casing 53.5^{ppf}), set same & open window @ 3337^{m}.
8. Drill 3^{m} in formation to 3340^{m}, POOH to surface.
9. P/U & M/U new 8 – 1/2" rock bit & directional BHA, RIH to bottom.
10. Continue drilling 8 – 1/2" f/ 3340 to 3496^{m}.
11. Perform hole condition, POOH to surface.
12. P/U & M/U new 8 – 1/2" rock bit & directional BHA, RIH to bottom.
13. Continue drilling 8 – 1/2" hole to 3652^{m} (DOP).
14. Perform hole condition, POOH to surface to change BHA.
15. P/U & M/U new 8 – 1/2" PDC bit & BHA, RIH to bottom.
16. Continue directional drilling 8 – 1/2" hole f/ 3652 to section TD @3939^{m}.

NOTE: Choke manifold & choke line must be lined up for **SOFT SHUT-IN**

17. CBU, perform condition trip to last casing shoe prior to logging, RIH to bottom, CBU, spot pill.
18. POOH to surface & L/D PDC bit.
19. R/U casing running gear, run 7" liner to bottom, R/D casing running gear.
20. Perform 7" liner cementing operation.
21. POOH w/ liner hanger setting tool to 2800^{m}, flush string & CBU.
22. POOH to surface & L/D liner hanger setting tool.
23. P/U & M/U 8 – 1/2" MT bit BHA, RIH to liner lap @ ±3150^{m} while D/O cement, Circulate & clean the well, POOH.

NOTE:

- If any cement: considered 7 STDs wet for cement job.

24. L/D excess 5" DPs/HWDP.

25. P/U & M/U 4 – 3/4" DCs & 3 – 1/2" DPs/HWDP, R/back same in derrick.

26. P/U & M/U 6 – 1/8" MT bit BHA, RIH to tag POB, D/O POB, RIH to LCD depth @ ±3915^{m} while D/O any cement, work on junk sub & CBU. Perform 7" liner CIT (80% of burst pressure) w/ 2000^{psi} for 15min. (if not observed bumping). CBU & POOH to surface.

27. RIH w/ 6 – 1/8" MT bit & tandem scraper (7" & 9 – 5/8") to LDC depth, scrap inside 7" liner & 9 – 5/8" casing before dry test, CBU, POOH to surface.

28. Perform BOP test as NIOC policy.

29. RIH with 9 – 5/8" dry test BHA, RIH to liner lap @ ±3150^{m}, perform 7" liner lap drey test (based on Fahliyan reservoir datum pressure), POOH.

30. Change ram, install shooting nipple, R/U wireline logging equipment. Perform cement quality logging 7" liner from 7" liner LDC @3915^{m} to 7" liner lap @3150^{m} (SBT or CBL/VDL), R/D logging equipment, change ram.

31. P/U & M/U 6 – 1/8" MT bit & C/O BHA, RIH to LDC depth, displace 7" liner volume w/ 100 pcf drilling fluid .

32. Continue D/O cement, shoe & rat hole.

33. Fill up the following forms & & send them to the drilling office:

- 7” Liner Tally
- 7” Liner Report
- 7” Cement Report
- Casing integrity Report
- DAS Report
- Dry test Report
- TMS Report
- Logging Results
- BOP test Report
- Service Company Report
- Kill Sheet (during drilling operaton)

6 – 1/8" Hole Section

Casing/Liner Point	4 – 1/2" Liner shoe @ 4150^{m} (4 – 1/2" L.Lap @ 3839^{m})	
Formation	Fahliyan	
Risks & Hole Problem	Oil Flow, Tight Hole and Differential Sticking, H2S,Complete Loss, Hole Deviation	
Deviation Control	Single/Multi shot Survey	
Bit Type & Nozzle	2 No. MT, 537 & 1 No. PDC, M323	
Logging	**Open Hole:** NA **Cased Hole:** SBT CBL/VDL/ & CCL/GR (after Perforation) from 4-1/2" Liner	
Testing		
Drilling Fluid Parameters	**Salt Saturated Kcl/Polymer/$CaCO_3$ WBM**	
	MW (PCF)	100 – 106
	FV (Seq/qt)	40 – 56
	PV (CP)	ALAP
	YP (Lbs./100ft^2)	14 – 24
	API FL(CC/30min)	< 3
	Gel 10 sec/10 min (Lbs./100ft^2)	3-7 /5-10
Drilling Parameters	GPM	350-450
	WOB	15 – 20
	RPM	150 – 170

Table 5-2 AZNS-127 6 – 1/8" Hole Section General Data

- **Formation Sampling**

Sample would be taken from Fahliyan every 2 meters.

NOTE: In addition to this general rule, geologist may decide to take more samples.

- Ensure that drilling tools & equipment are on wellsite & are in serviceable condition.
- Ensure that adequate fishing tools for this section of hole are available.
- Ensure that enough mud material, LCM & CMT are available.

1. Drilling 6 – 1/8" hole to 3950^{m}.
2. CBU and balance Mw,,perform FIT with Required Pressure 720 psi (EMW: 108pcf, should be equal to max. planned Mw for this hole section plus 2 pcf),POOH to SurfaceRIH w/ 6 – 1/8" PDC bit & drilling 6 – 1/8" hole f/ 3950 to TD @4150m.

NOTE: Choke manifold & choke line must be lined up for **HARD SHUT-IN**

3. CBU, perform condition trip to last casing shoe prior to liner running, RIH to bottom, CBU, spot pill.
4. POOH to surface.
5. Run 4 – ½" liner running equipment. Run 4 – ½" liner (+/- 311^{m} liner) to bottom @4150^{m}, R/U cement lines, set liner hanger +/- 3839^{m} (100^{m} overlap), release liner hange setting tool.
6. Cement 4 – ½" liner as per program.
7. POOH w/ liner hanger setting tool to 3500^{m}, flush string & CBU.
8. POOH to surface & L/D liner hanger setting tool.
9. P/U & M/U 6 – 1/8" MT bit BHA & RIH to liner lap @ ±3839^{m} while D/O cement, Circ. & clean the well, POOH.

NOTE:

↳ If any cement: considered 7 STDs wet from cement job.

10. L/D excess 3 – 1/2" DPs/HWDP.
11. P/U & M/U 3 – 1/2" DC & 2 – 7/8" DPs/HWDP, R/back same in derrick.
12. P/U & M/U 3-1/8" Mill BHA, RIH to tag POB, D/Out POB, RIH to LDC depth @ +/- 4126m while D/Out any cement, work on junk sub and CBU. Perform4-1/2" Liner CIT. (80% of Burst press) with 2000 psi,for 15min, (If not obs bumping).CBU, POOH to Surface.
13. RIH with 3-1/8" Mill and 4-1/2" scraper to LDC depth, Scrap inside 4-1/2" Liner , CBU, POOH to Surface.
14. Change ram, install shotting nipple & R/U wireline logging equipment.
15. Perform cement quality logging of 4 -1/2" liner from LDC depth @4126^{m} to liner lap @3839^{m} (SBT or CBL/VDL). R/D logging equipment & change ram.
16. P/U & M/U 3 – 1/8" mill & C/O BHA, RIH to LDC depth. Displace 4 – 1/2" liner volume w/ non-solid fluid, POOH.
17. P/U & M/U wireline perforation assy & RIH to perforation depth. Correlate depth w/ wirteline & perforate guns as per program. POOH to Surface.
18. P/U & M/U 3 – 1/8" milling BHA, RIH to top of LDC depth while smoothing perforated intervals.
19. Change ram, install shotting nipple & R/U wireline logging equipment.
20. RIH wireline w/ CCL/GR log to confirm perforated intervals, POOH & R/D logging equipment, change ram.

21. RIH with 3-1/8" mill & clean out BHA (7" scraper included) to bottom, displace the well with completion fluid (non-solid), POOH & L/D DP's.
22. Fill up the following forms & & send them to the drilling office:

- 4 – 1/2" Liner Tally
- 4 – 1/2" Liner Report
- 4 – 1/2" Cement Report
- Casing integrity Report
- TMS Report
- Logging Results
- BOP test Report
- Service Company Report
- Kill Sheet (during drilling operaton)

Completion Section

23. Run 10k completion string as per program.
24. Fill up the following forms & & send them to the drilling office:

- Completion Tally
- Completion Running Report
- TMS Data (Completion)
- Test Charts, Xmass tree Installation & Test Report

NOTE:

↳ Annulus pressure test during completion operation is related to corrosion log results.

25. Set BPV inside TBH, N/D BOP's & install new 10k XMT.
26. Retrieve BPV, install TWCV & pressure test the XMT.
27. Secure the well by set BPV inside TBG hanger, close all XMT valves except BMV. Release the rig.

6. Directional Drilling Operation

6.1 Hydraulic Set Whipstock System Operation Procedure

SCOPE

The following procedures are intended to provide concise but definitive steps to follow in operations involving the milling of lateral hole sections from a parent well using the Hydraulic Set Whipstock System.

The Hydraulic Set Whipstock System is a hydraulically set whipstock that requires a hydraulic force to activate the packer assembly, setting itself in the casing.

Milling a window in casing using a whipstock usually consists of one run for setting the Whipstock in place and milling and completing the window through the casing wall and into the formation, while dressing the window to eliminate drag.

EQUIPMENT REQUIRED

Hole Preparation Run Equipment

- A Full Gauge Drift Mill
- Casing Scrapper
- Cross Over Sub
- String Stabilizer
- Pony Drill Collar or 1 Joint of Drill collar
- String Stabilizer
- Drill Collars
- Drilling Jar
- Drill Collars
- HWDP's
- Balance Of Drill Pipes To Surface

The stiffness of this assembly will serve to identify any problem doglegs or restrictions in the hole that may hamper running of the whipstock assembly.

Whipstock Run Equipment

- Hydraulic Set Whipstock
- Tri-Mill (Optional Bi-Mill or a combination of Rugby/Back/Window Mill)
- Running Tools
- Nuzzle plug Assembly

- Flex Joint (1 joint of HWDP)
- Universal Bottom Hole Orientation Sub (UBHO sub) or MWD
- Pup joints (for space-out if necessary)
- Tool kit

PRE-OPERATIONAL PROCEDURES

Pre-Run Checklist

As part of the Pre-operational Procedures, all whipstock jobs should include a completed Pre-Run Preparation Checklist.

Pre-Run Preparation Checklist

† The Packer slip is in place, is in good condition, and is the correct size for the casing size and weight and the Allen screws are tight.

† The shear pins are properly installed and in good condition and the Number of Shear pins are enough.

† The shear screw is the proper size, and it will make up into the Whipface.

† The shear screw will fit through the Rugby/Lead Mill hole.

† Running Tools, Nuzzle plug Assembly and Hydraulic Lines and Fittings are in good condition and properly installed.

† All mills are on location and the OD's are correct for the casing's size/weight. (Check mill OD's with a gauge ring that is OD + 1/32").

† All downhole equipment has been properly measured and OD's, ID's, and lengths have been properly recorded.

† The slots on the gyro match and are compatible with the lug assembly on the orientation sub.

† The survey tools will pass through all drill string tool joints and drill collar ID's.

† All rotary connections are compatible.

† All rotary shouldered connections are made up to the recommended torque.

† Enough Pump Pressure is available on Site for setting the Packer.

Making the Hole Preparation Run

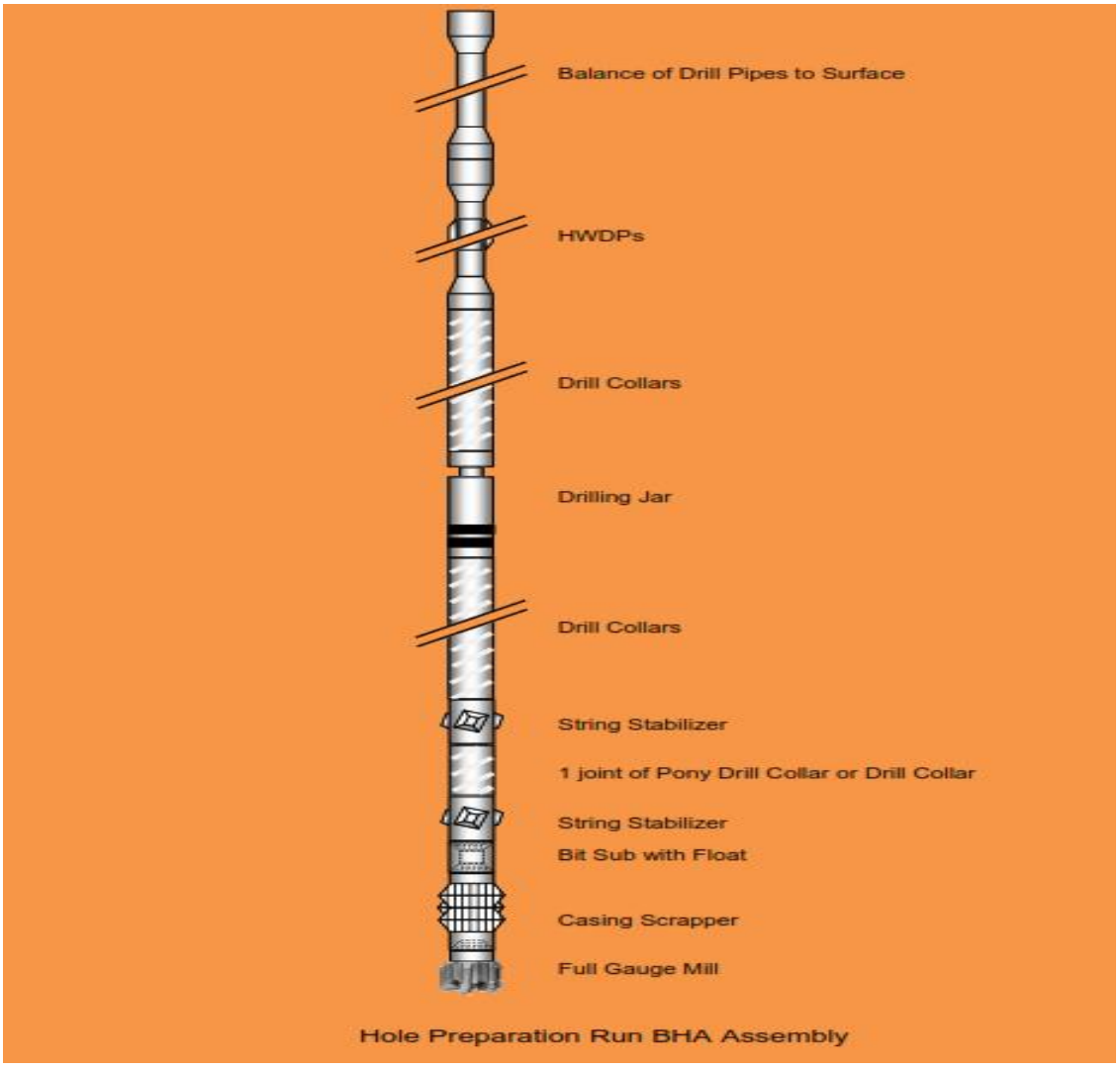
1. Make a run to ensure that the casing is clean and there is smooth and clean place for setting the packer. The run should include the following (from the bottom):

- A Full Gauge Drift Mill
- Casing Scrapper (Do not run casing scrapers across open perforations.)
- Cross Over Sub (Bit Sub with Float)
- String Stabilizer
- Pony Drill Collar or 1 Joint of Drill collar
- String Stabilizer
- Drill Collars
- Drilling Jar
- Drill Collars
- HWDP's
- Balance of Drill Pipes To Surface

2. Run this assembly in the well to the Packer setting depth.
3. Tally the pipe while running in or out of the well to get an accurate measurement to the Packer setting depth.
4. Record the measurements to ensure that:
   - the whipstock can be run and set at the desired depth
   - the depth measurement to the Packer setting depth is accurate
   - the casing wall is clean where the whipstock is to be set

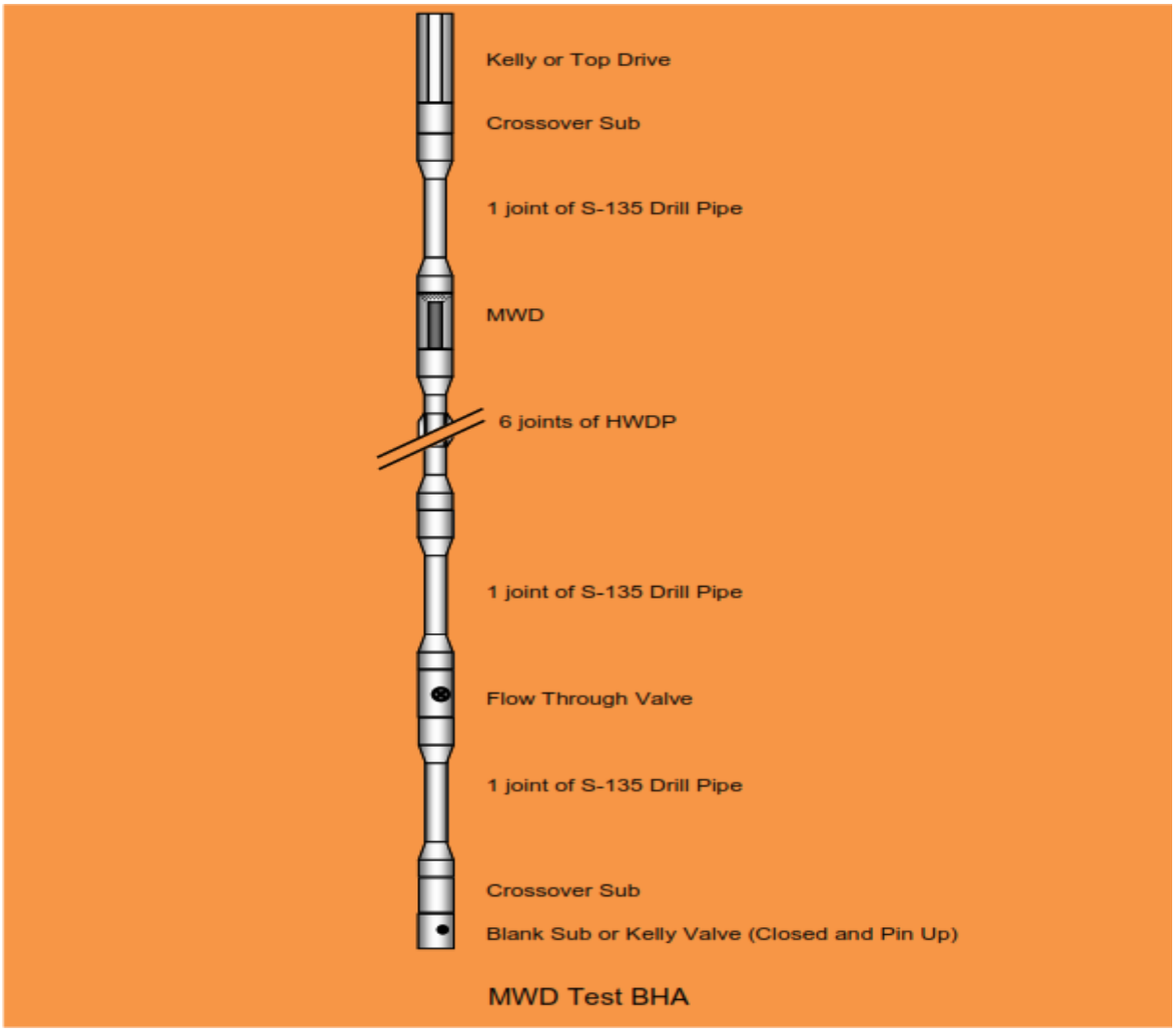
NOTE: Scrape Packer setting depth twice.

Note: It is very important to get an accurate measurement for the packer setting depth so the Whipstock will be oriented as to setting depth as possible. Using this measurement facilitates proper placement of the Whipstock.



Test the MWD tool
prior, to running the assembly in hole.

1. Pick up the following assembly:
   - o Blank Sub or Kelly Valve (valve to be closed and run pin up)
   - o Crossover Sub
   - o One joint of S-135 Drill Pipe
   - o Flow Through Valve
   - o One joint of S-135 Drill Pipe
   - o 6 joints of HWDP
   - o MWD
   - o One joint of S-135 Drill Pipe
   - o Crossover Sub
   - o Kelly or Top Drive
2. RIH to top of Drill Pipe. With top drive or Kelly connected, record the flow rate required to obtain a survey. The Flow Through Valve will remain open.



Preparing the Milling Equipment

1. Lay the whipstock on the catwalk, face up, with the top of the whipstock pointing toward the rig.

WARNING
Never use a nylon sling to lift or move a Whipstock because nylon may be cut. Always us a wire sling or wire rope to lift or move a Whipstock.

2. Complete the whipstock assembly, checking all OD's, ID's, and descriptions.

3. Pick up a joint of S-135 Drill Pipe (FLEXJOINT) and make up the **Flow Through Valve** to the top of the drill pipe and make up the Mill Assembly covered with Running Tool to the bottom of Drill Pipe.

Note: If the window is exiting through two or more strings of casing or into a known hard formation, exclude the Watermelon Mills (in the case of using combination mills), using only the Rugby Mill below the single joint of DP. Depending on the casing size and grade, more than one Rugby Mill may be required to make multiple exits.

Once the casing string(s) has been exited into the formation, pull out of the hole. Make up the Back Mill (and Window Mill) in the assembly as described above and run in the hole, reaming through the window with the Watermelon Mills.

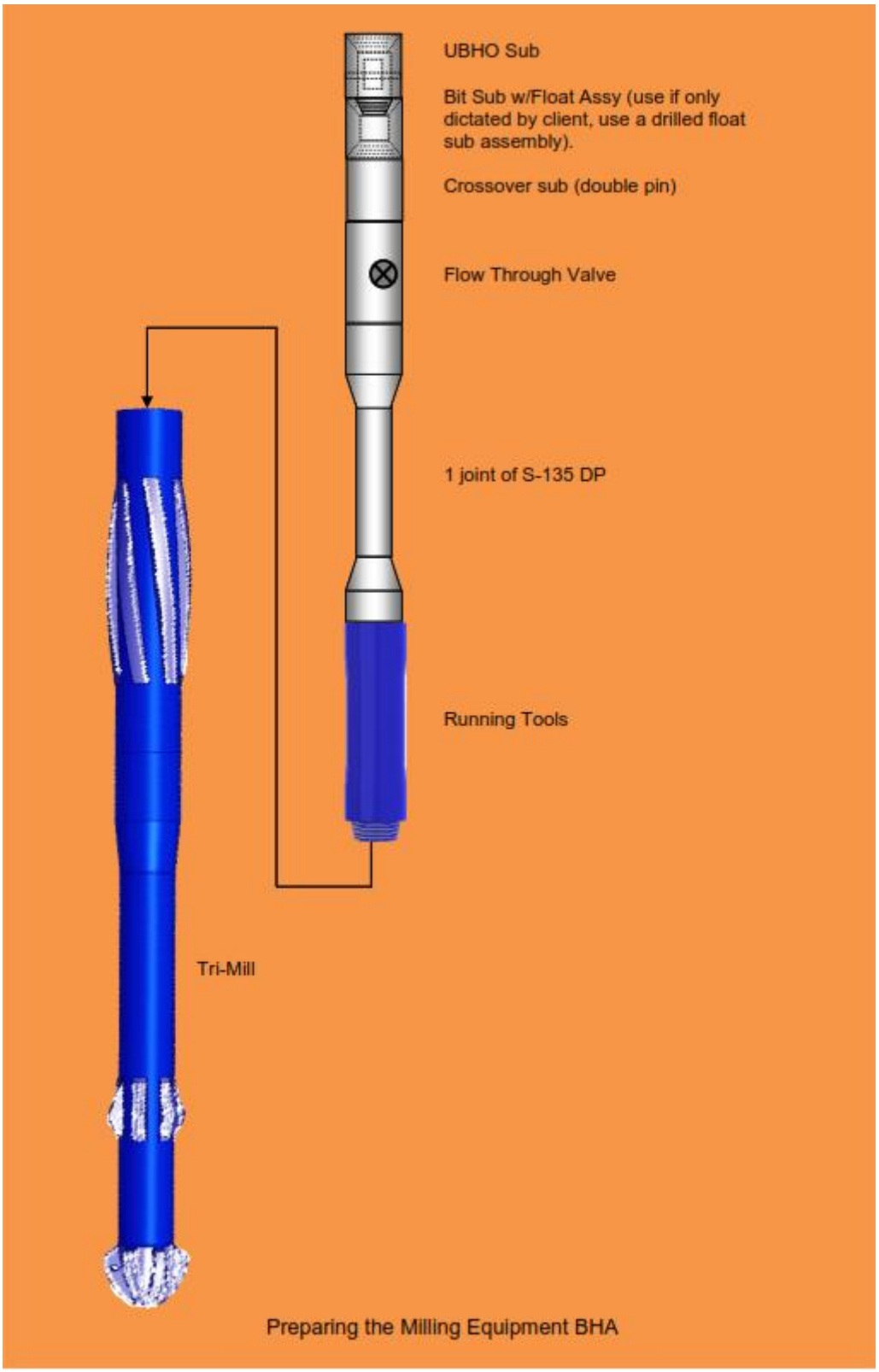
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4. Make up one joint of S-135 Drill Pipe (with cone type DP strainer installed) to the top of Flow Through Valve and make Up MWD to the top of Drill Pipe. (or Make up UBHO Sub to the top of Flow Through Valve).

5. If a bit float have to be used, place the bit float sub above the Flow Through Valve and beneath the UBHO sub.

6. Make up all connections to the recommended torque.

7. Ensure that everything is properly made up before continuing (see Figure).



Preparing the Milling Equipment BHA

There is a hole in a one of Rugby/Lead Mill blades for screwing the shear screw and attaching Rugby Mill to the Whipface. Mark the opposite blade of the blade with shear

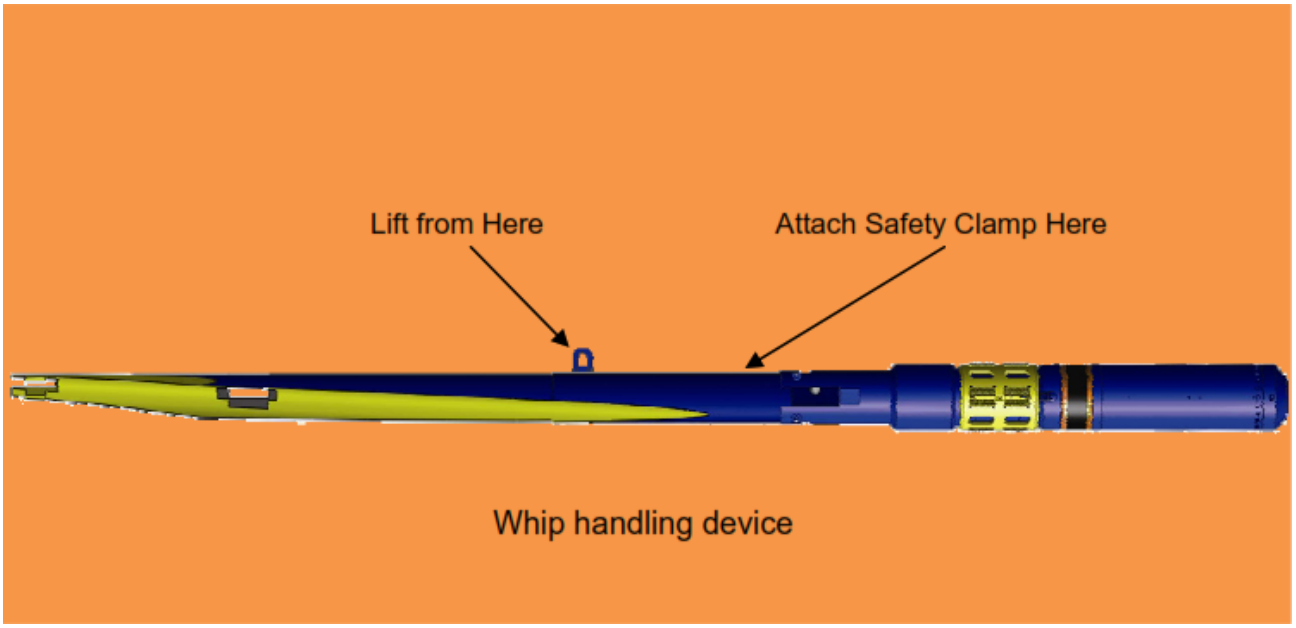
screw hole on the Rugby Mill and scribe a line from this blade, up the Back Mill (and Window Mill), Bit Sub, and the single joint of HWDP to the top of the UBHO Sub.

9. Have the survey company representative back out the set screws that hold the internal key in the UBHO Sub, and rotate the key until it is aligned with the scribed line. After it is in position, tighten the setscrews to lock the key in alignment with the scribed line.

10. Run the Mill assembly through the BOP stack and into the casing liner/hanger (being careful not to damage the wear bushings or strike any closed BOP rams) to ensure that the whipstock's OD is compatible with the drift diameter of the casing. Remove wear bushing if necessary.

11. Pull the assembly out of hole.

12. Secure a safety clamp around the whipstock body about halfway down the face of the Whipface while the whipstock is lying on the pipe deck/catwalk.



13. Pick up the Whipstock with tugger from the catwalk. A whip handling device is provided which is bolted onto the back of whipstock. This is to be used as a false rotary to support the assembly in the Rotary Table when making up the mill to whip via the shear screw. Assist with a crane as necessary.

14. Set Whipstock in the rotary table (rams shut).

15. Lower the whipstock into the mouse hole (or other suitable work hole on the drill floor), allowing both the safety clamp and the winch/tugger line to support the whipstock's weight.

Note: *Larger Whipstocks (9⅝ to 20 inches) are usually too big for the mouse hole and must be set in the rotary to attach to the Mill assembly. Properly sized rotary slips must be used. The 13⅜" whipstock will not fit through the standard rotary (10⅝" ID) and the rotary bushing, in this event must be pulled. Use of an API bowl and slips to hold the whipstock in the rotary is required.*

CAUTION

- **Close the BOP Shear/Blind ram when working on Whipstock.**
- **Do not drop anything in the wellbore when working over the wellbore.**
- **Use extreme caution when handling the whipstock, setting drill collar slips, safety clamps, etc. to prevent dropping the whipstock in the hole.**
- **Run the draw works in low. Pick up and set down the whipstock easily in the slips to prevent damage or premature setting.**
- **Ensure the lead tongs have most of the slack out of the snub line to prevent jerking against the back-up tongs.**

16. Put a light amount of thread compound on the shear screw, adjust Rugby Mill hole and Whipface and connect the whipstock assembly to the mill assembly with shear screw.

Shear screws with a **higher** value should be considered when:

- running the whipstock in a highly deviated hole
- heavy mud is in the well
- other wellbore restrictions such as a liner hanger may be encountered

Shear screws with a **lower** shear value should be considered when:

- whipstock is being set shallow (less than 1500 m)
- running and setting Whipstock with coiled tubing

17. Attach the Hydraulic Line to the connection on the bottom of the Milling Assembly and back of whipstock.

18. Fill Whipstock and Mill Assembly / Running Tool with Hydraulic Oil, allow air to bleed out, fit piston ensuring the top of piston is below the top running tool box connection.

19. Run the Whipstock & Mill Assembly below the rotary table to the top of UBHO sub, Break the connection and top-up with Hydraulic Oil.

20. Pick up the Milling / whipstock assembly.

21. Install the rotary table cover.

22. Remove the safety clamp and whip handling device which is bolted onto the back of whipstock.

23. Remove the rotary table cover.

24. Lower the whipstock slowly into the well, setting the rotary slips on the DP.

25. Check the orientation of the key in the UBHO sub, make any adjustments necessary, and tighten the setscrews again.

Note: For safety purposes, install a drill pipe wiper to prevent dropping the Allen wrench in the hole.

CAUTION

Run the draw works in low. ***Pick up and set down the whipstock gently in the slips to prevent damage or premature setting. Make sure the lead tongs have most of the slack out of the snub line to prevent jerking against the back-up tongs.***

RUNNING PROCEDURE

Running and Orienting the Whipstock

1. Lower the whipstock assembly slowly into the wellbore and **do not pump** anything in the well to prevent premature setting of the whipstock.

Do not alter the orientation of the UBHO sub to the Whipface's face when making the first connection. Do not let the UBHO sub turn or move, or the direction of the milled window could be affected.

2. Refer to dogleg checklist for doglegs greater than 4 degrees. Watch the weight indicator carefully and record up-weights regularly.
3. Monitor trip-tank for proper fill up.
4. Run the whipstock slowly, like a packer assembly, at a rate of 2 to 3 minutes per stand.
5. Do not turn string.
6. Do not Pump in the Well (unless an open Flow Throgh Valve is Run above the Running Tools).
7. Fill the drill string every 5 strands.

Proceed slowly and with caution while running the whipstock assembly through the BOPs, wellhead, liner top, and any other restrictive sections to prevent damage or premature setting. The difference between the whipstock OD and the casing ID is minimal, making it similar to a piston, so lower the assembly very slowly in these small bypass areas. DO NOT PUMP IN THE WELL.

8. Run slowly into the hole until approximately 30 feet off the bottom.

9. Slowly move the whipstock assembly up and down from 5 to 20 feet to remove all trapped torque from the drill string.
10. Record up and down free weights.

11. Run survey tools on wireline to the UBHO sub if using a survey.
12. Take survey readings
13. To orient the whipstock, rotate the drill pipe to the right in ¼ turn increments (no more that ½ turn at a time) until the whipstock assembly is oriented in the correct position.
14. Take another survey, making additional corrections as needed and lowering 4 to 5 feet after each increment.

CAUTION
A crooked hole may require several readings and corrections to remove all of the trapped torque from the drill string. Work slowly and carefully in order to achieve proper orientation.
CAUTION
This Model of Whipstock, because it is hinged, must be set within 30° on either side of the low side to prevent the whipstock's whipface from falling across the wellbore.

15. Continue taking readings until 3 sequential reading are within a 5° range before assuming the direction is correct.
16. After reaching and verifying the proper direction, call a meeting with the customer, driller, and survey/MWD company representative to review the orientation of the whipstock's Whipface once more.
17. Continue the procedure if everyone is in agreement.
18. POOH the survey tools and make a mark on the rotary table and the Drill Pipe, make up the Kelly (or Top Drive), DO NOT ALTER THE ORIENTATION WHILE MAKING UP THE KELLY.
19. Lower the whipstock assembly slowly to the setting depth.
20. Start pump and increase pressure until the packer shear pins are sheared and the packer sets in place (at 1200-1400 psi), hold pressure for 10 minutes to confirm the setting).

Note: In case of Using MWD and Flow Through Valve, increase pump rate until the Flow Through Valve shear pins are sheared and the valve closes, then continue pumping until the packer shear pins are sheared and the packer sets in place.

21. Slack off 10-15klbs and over pull 10-15k lbs to prove packer is set.
22. Pick-up 35,000-55,000 lbs (Depend on whipstock size) over pull, continue over pull on the screw as required to initiate the shear. A sharp drop in the pipe occurs when the shear screw shears.
23. Raise the drill string approximately 1 to 2 feet.
24. Set the drill string back down to make sure the shear screw has sheared. Repeat if necessary.
25. Mark the depth on the pipe each time. It should read the same each time.

26. Set the pipe back down with approximately 20,000 lb of pressure to ensure that the Whipface is over against the casing wall.

27. Raise the drill string so that the Rugby Mill is 5 to 10 feet above the top of the whipstock.

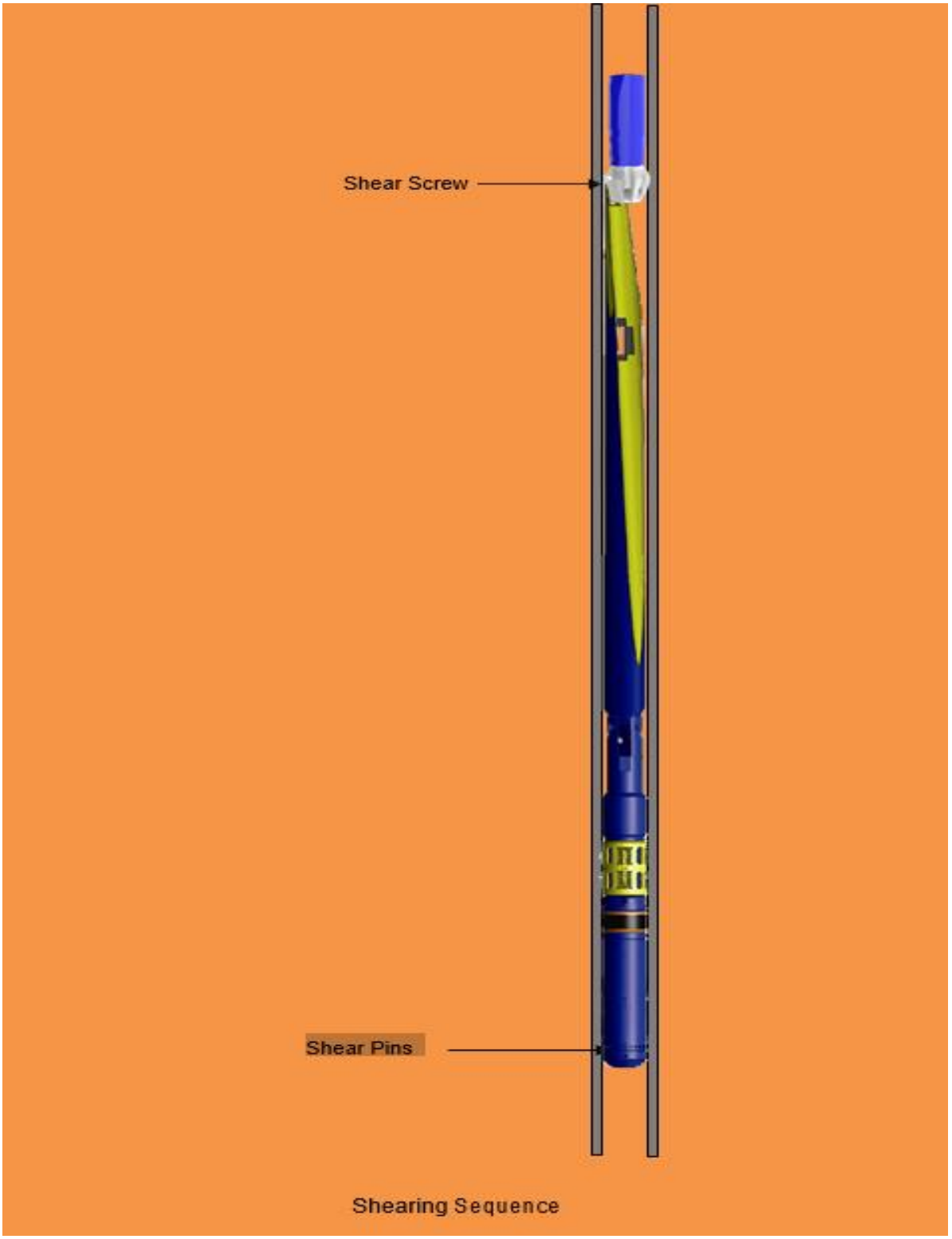
28. Lay down the necessary drill pipe to pick up the kelly.

29. Start pump with maximum allowable GPM (According to the standpipe pressure). Engage with Whipface with Reduced RPM (20-30). Monitor pressure drop due to breaking Nozzle plugs.

Note: Do not Mill and work on Nozzle plugs more than One min with low GPM which leads to Mill burning.

30. Displace the Mud in the well (if needed) and check mud properties for proper rheology for steel removal.

31. The milling operation can now begin.



Shearing Sequence

Running Milling Assembly and Opening the Window

This section assumes that the Rugby Mill has been sheared from the whipstock and raised 5 to 10 feet above the whipstock.

NOTE: Install Ditch Magnets

1. Make up the kelly to the drill string.

2. Record string weight up and down.

3. Engage the pumps, slowly increasing to the planned circulation rate, and record pump pressure and rate.

4. Engage the rotary and slowly increase the speed between 20 and 40 RPM.

5. After the rotary has stabilized, slowly increase between 50 and 75 RPM (depending on the mill size).

6. Record the following information:
   - Free rotary torque
   - Pump pressure
   - Pump rate
   - Free weight – up and down (string weight with zero weight on mill)

7. While rotating, slowly lower the drill string until the Rugby Mill contacts the whipstock and the casing wall, causing an immediate and substantial increase in rotary torque.

8. Mark the kelly. This is the starting point for the Rugby Mill run.

9. After making the initial mark on the kelly, raise the Mill 3 to 5 feet.

10. While rotating, slowly lower the drill string again to see if the increase in rotary torque is indicated at the same place. Work slowly and carefully; repeat if necessary.

11. After marking the kelly with the correct starting point for the Mill, begin milling, using light weights on the Mill (1,000 to 3,000 lb depending on the size of the mill and grade of the casing). As the window milling process begins, an increase in torque and decrease in weight will be indicated.

12. As the torque increases, increase the rotary speed to the recommended RPM (based on the mill size).

13. Continue milling with a constant rotary torque and weight (1,000 to 3,000 lb) on the mill. Additional weight may be required if the rate of penetration (ROP) begins to slow.

Use recommended parameters as follows.

Casing Size (in)	Weight on Mill (klb)	Milling RPM
9 ⅝”	2-15	60-120
7”	2-12	60-120
4-1/2”	1-7	60-120

14. Monitor returns for cuttings during milling operations and **pump high viscosity sweeps** as needed to remove cuttings from the well. This practice is especially important if the window is being milled in a lateral section.

15. The Window Mill run is completed after milling down the Whipface’s face into the formation AND when the Dress Mill is centered at the bottom of the window.

16. Ream back and forth through the window until no signs of additional torque or drag are indicated.

17. Stop the rotary and slide the Mill assembly through the window several times. If any weight or drag is indicated, start the rotary and ream back and forth through the window until weight or drag is eliminated.

18. Stop the rotary and check for weight or drag again. Repeat this process as necessary.

19. Once the window is complete, stop the rotary, **clean the hole of cuttings with high viscosity sweeps**, and pull out of the hole without rotating.

20. Closely inspect and gauge all mills for wear using a gauge ring of OD + 1/32"

21. Determine the estimated size of the window.

22. Review the window requirements with the customer to determine if the Elongation Run is required.

Note: The Elongation run is recommended if:

- *You have run a bi-Mill Assembly for opening the Window and:*
- *The customer requests it.*
- *PDC bit will be used through the window for additional drilling*
- *Drill collars and stabilizers will be used through the window for additional drilling*
- *Liner will be installed through the window*
- *Double bent sub and PDM assembly will be used through the window for additional drilling*
- *Unusual wear is indicated on the Rugby Mill, resulting in an under-gauge hole*

CAUTION

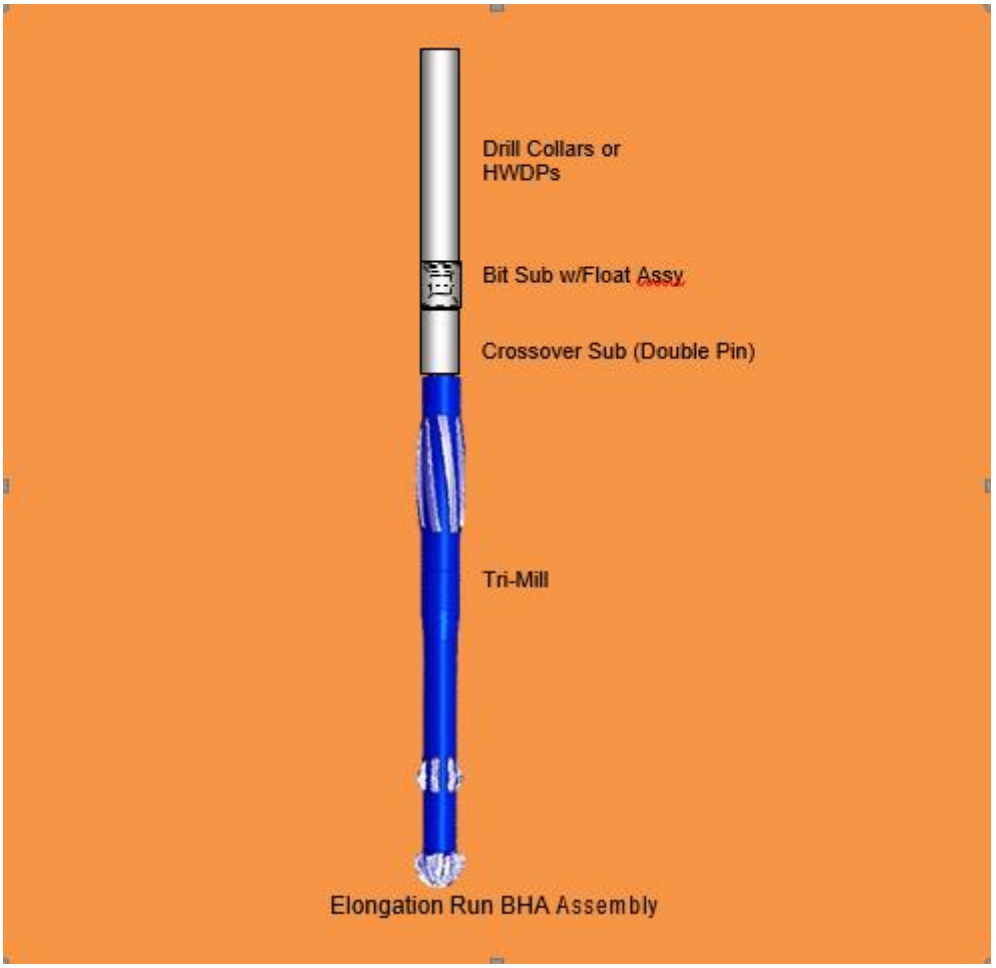
When using a 3° whipface whipstock, do not run the Elongation assembly until AFTER the casing is fully exited. Otherwise, excessive torque will be encountered as the two Watermelon Mills begin elongating the top of the window.



Elongation run

1. Assemble the Elongation run components
   - Balance of drill string
   - One or more drill collars or heavy weight
   - Bit Sub (minimum length, bored for float assembly)
   - Crossover sub (double pin, if Bit sub is used)
   - Tri-Mill

Note: Be sure to eliminate the one joint of drill pipe above the Bit Sub and replace it with one or more drill collars/heavy weight to provide "stiffness" to the milling assembly.



2. Lower the Elongation run assembly into the hole, stopping 5 to 10 feet above the whipstock.

3. Make up the kelly to the drill string.

4. Engage the pumps and slowly increase to the desired circulation rate.

5. Engage the rotary and slowly increase to between 50 and 75 RPM (depending on the mill size).

6. Record the following information:
   - Free rotary torque
   - Free weight – up and down (string weight with zero weight on mill)
   - Pump pressure
   - Pump rate

7. While rotating, slowly lower the drill pipe string and ream back into the window.

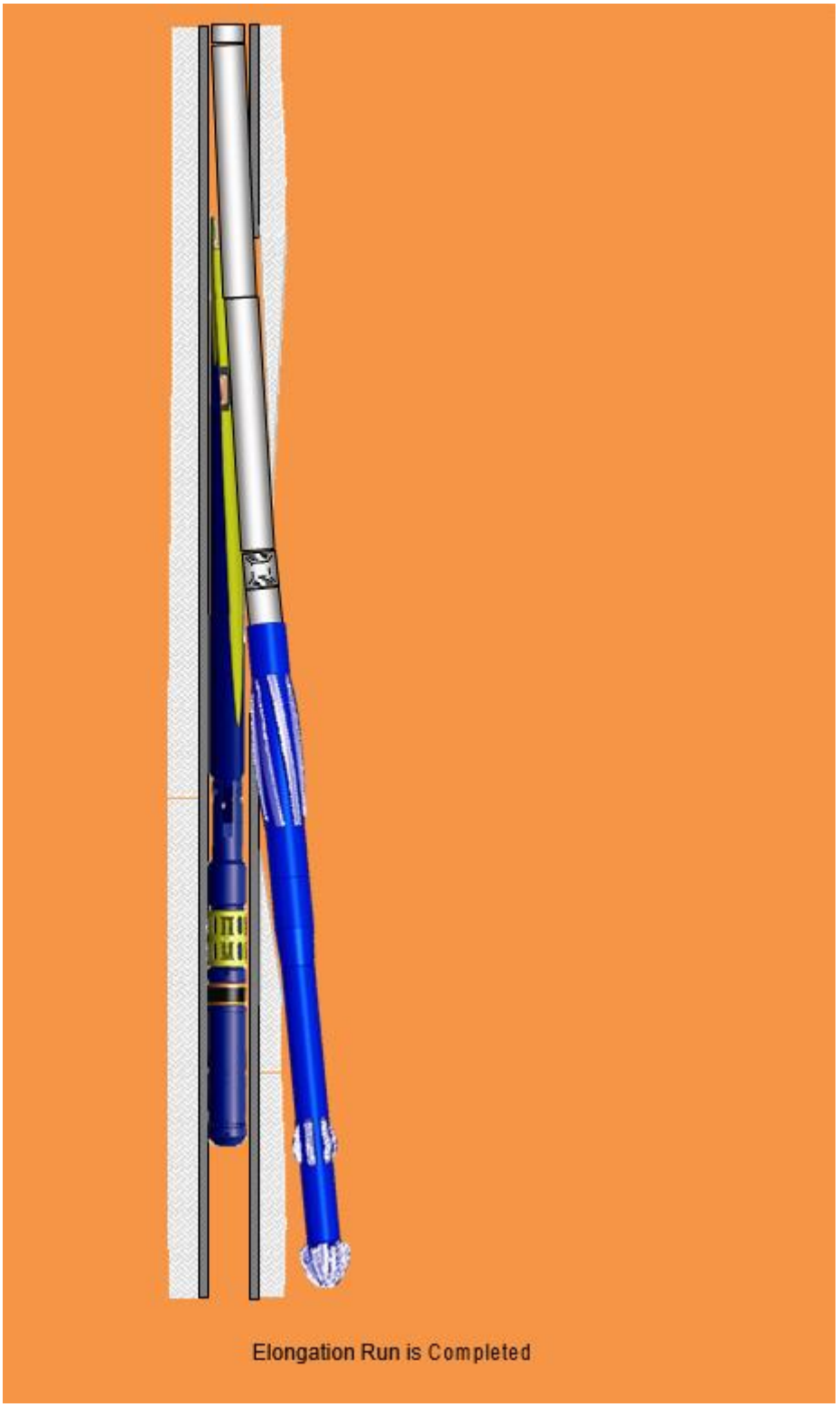
8. As the torque increases, increase the rotary speed to the recommended RPM (based on the mill size).

9. Continue milling with a constant rotary torque and weight on the mill.

10. Monitor returns for cuttings during milling operations. Pump high viscosity sweeps as needed to remove cuttings from the well.

11. Continue milling down the Whipface, elongating the milled window at the top with the Dress Mill.

12. The Elongation run is complete (unless directed otherwise by the customer) when the Dress Mill is centered at the bottom of the window.



13. While rotating, ream back and forth through the window until no additional weight or drag is indicated.

Note: If using stabilizers for additional drilling through the window, it is recommended to drill enough hole into the formation to accommodate the stabilizers. The additional rathole should allow the stabilizers to be in the formation BEFORE drilling commences. Rotating the stabilizers or drill bits while on the Whipface can damage the Whipface and the window.

14. Stop the rotary and slide the Mill assembly through the window several times. If any weight or drag is indicated, start the rotary and ream back and forth through the window until weight or drag is eliminated.

15. Circulate the hole clean with **one or more high viscosity sweeps** to remove all cuttings from the well.

16. When circulation is complete, stop the rotary and pull out of the hole without rotating.

17. Inspect and gauge all mills using a gauge ring of OD + 1/32", to determine wear to the mill and to estimate window/hole size. Review signs of wear with the customer. Repeat Run #3 if the hole appears to be undersized.

18. Lay down the Elongation Run assembly.

Abnormal Conditions

Whipstock stuck

- Try to release the whipstock with 20000klb overpull in 7" hole and 30000klb over pull in 9-5/8" hole.
- Close an annular preventor hydrill and apply 500-1000 psi pressure in annulus to remove casing collapse.
- Apply weight on bit.
- If no success, shear screw and pull out the whipstock out of hole with fishing tool later.

No success in passing the whipstock through liner lab

- Come back to neutral point and turn string 1/4 round in clockwise direction and try to pass through liner lab again.

The packer does not set with designed setting pressure

- Release the pressure and apply the pressure again.
- Pull out of hole to surface and check the tools.

Pressure drop during the milling operation

- Twisting off the string or mill:
- In this case the torque will be constant and progress will be stop.
- Pressure drop due to string washing out:
- In this case the standpipe pressure will decrease gradually.
- Pressure drop due to breaking the nozzle plugs:
- In this case the pressure decreases suddenly and no change in other parameters will happen.
- Pressure drop due to washing out the nozzles:
- In this case the standpipe pressure will decrease gradually.

Note: In all the above items pulling the string out of hole to surface and checking the tools will be necessary.

No progress during milling

- Change the drilling parameters, for example increase or decrease RPM, increase WOB, decrease GPM.

- If you observe no progress in 2 hours, pulling the string out of hole to surface will be necessary.

Observing complete loss during milling

- Don’t continue milling till curing the complete loss.
- If the client insists in continuing the operation, due to working with low GPM, stop the milling operation for 30 minutes after each hour of milling to cool the mill and prevent from thermal cracking.

POH string due to failure of whipstock:

- Do not shock the string while put and pick up slips
- Do not shock the string while wrenching
- Ensure that the rotary table is locked for prevent of rotating string
- Pick up each stand in 5 to 7 minutes
- Disconnect the connections with wrench and air wrench.
- Be careful for 5” and 7” whipstocks , because it will has more swing in larger casing size

Caution: do not use rotary table for disconnect the connections

7. BHA Program

7.1 8-1/2" Hole Section

BIT SIZE (inch)	8-1/2"	Directional BHA Configuration to Drill 8-1/2" Hole Section (PDC Bit)				DC	OD
Depth In/Out (m)	3340-3939					DC-1	6-1/2"
HOLE INCLINATION (deg)	0						
MUD WEIGHT (pcf)	85						
B.FACTOR	0.827						
Weight in Air (lb)	21,393						
Bouyed Weight (lb)	17,682						
Bouyed Weight Below Jar (lb)	15,815						

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	to surface	5	4 1/2	4-1/2" IF Pin x Box	-	
	5" Heavy Weight Drill Pipe	3	5	3	4-1/2" IF Pin x Box	27.94	271.89
	6 -1/4" D. Jar	1	6-1/4	2-1/4	4-1/2" IF Pin x Box	5.83	243.95
	5" Heavy Weight Drill Pipe	24	5	3	4-1/2" IF Pin x Box	219.36	238.12
	6-3/4" NMDC	1	6-3/4	3	4-1/2" IF Pin x Box	9.14	18.76
	6-3/4" MWD Sub	1	6-3/4	2-1/2	4-1/2" IF Pin x Box	0.93	9.62
	6-3/4" Float Sub	1	6-3/4	2-1/4	4-1/2" IF Box x Box	0.42	8.69
	6-3/4" PDM	1	6-3/4	-	4-1/2" Reg B x 4-1/2" IF B	8	8.27
	8-1/2" PDC Bit	1	8-1/2	-	4-1/2" Reg. P	0.27	0.27

Figure 7-1: 8-1/2" Directional Drilling BHA

7.2 6-1/8” Hole Section

BIT SIZE (inch)	6-1/8"	BHA Configuration to C/Out Inside 7" Liner				Collar	OD
Depth In/Out (m)	3915-3950					DC-1	4-3/4"
HOLE INCLINATION	0						
MUD WEIGHT (pcf)	103						
B.FACTOR	0.790						
Weight in Air (lb)	29,642						
Bouyed Weight (lb)	23,410						
Bouyed Weight Below Jar (lb)	16,591						

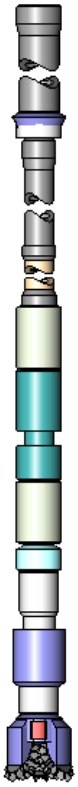
SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	XOS	1	4-3/4	2.25	3-1/2" IF P x 4-1/2" IF B	1	345.61
	3-1/2" Drill Pipe	12	3-1/2	2-7/16	3-1/2" IF P x B	115.2	344.61
	3-1/2" Heavy Weight Drill Pipe	6	3-1/2	2-1/4	3-1/2" IF P x B	54.84	229.41
	4-3/4" Drill Collar	3	4-3/4	2.25	3-1/2" IF P x B	27.42	174.57
	4-3/4" Jar	1	4-3/4	2.25	3-1/2" IF P x B	8	147.15
	4-3/4" Drill Collar	15	4-3/4	2.25	3-1/2" IF P x B	137.1	139.15
	4-3/4" Bit Sub W/ Float Valve	1	4-3/4	2.25	3-1/2" Reg. P x 3-1/2" IF B	0.75	2.05
	Junk Sub	1	4-1/4	1-1/2	3-1/2" Reg. B x P	1	1.3
	6-1/8" MT Bit	1	5-7/8	-	3-1/2" Reg. P x B	0.3	0.3

Figure 7-2: 6 – 1/8” C/Out BHA

BIT SIZE (inch)	6-1/8"	BHA Configuration to Drill 6-1/8" Hole Section (PDC Bit)	Collar	OD
Depth In/Out (m)	3891-4150		DC-1	4-3/4"
HOLE INCLINATION	0			
MUD WEIGHT (pcf)	103			
B.FACTOR	0.790			
Weight in Air (lb)	29,642			
Bouyed Weight (lb)	23,410			
Bouyed Weight Below Jar (lb)	16,591			

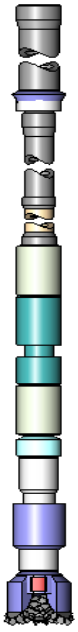
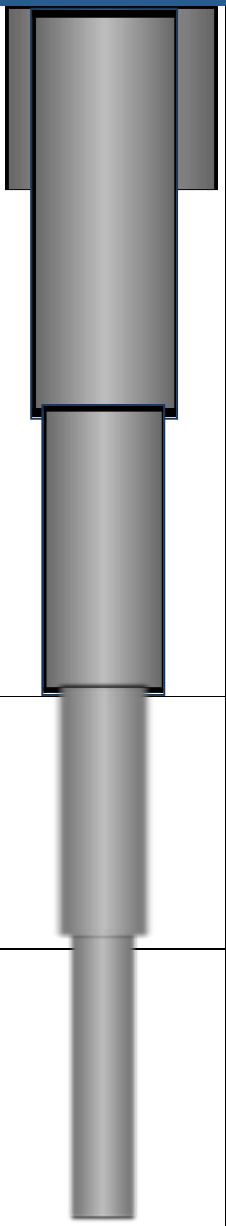
SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	XOS	1	4-3/4	2.25	3-1/2" IF P x 4-1/2" IF B	1	344.61
	3-1/2" Drill Pipe	12	3-1/2	2-7/16	3-1/2" IF P x B	115.2	343.61
	3-1/2" Heavy Weight Drill Pipe	6	3-1/2	2-1/4	3-1/2" IF P x B	54.84	228.41
	4-3/4" Drill Collar	3	4-3/4	2.25	3-1/2" IF P x B	27.42	173.57
	4-3/4" Jar	1	4-3/4	2.25	3-1/2" IF P x B	8	146.15
	4-3/4" Drill Collar	15	4-3/4	2.25	3-1/2" IF P x B	137.1	138.15
	4-3/4" Bit Sub W/ Float Valve	1	4-3/4	2.25	3-1/2" Reg. P x 3-1/2" IF B	0.75	1.05
	6-1/8" PDC Bit	1	6-1/8	-	3-1/2" Reg. P x B	0.3	0.3

Figure 7-3: 6-1/8" Drilling BHA

8. Mud & Cement Program

8.1 Drilling Fluid Summary

Hole Size (in)	Casing Size (in)	Casing Program	Depth (m MD)	Drilling Fluid system	Mud Properties
26”	20”		146 m	**Already Has been Drilled Top of CMT plug @ 55m to 100m**	
17 ½”	13 3/8”		1441 m	**Already Has been Drilled**	
12 ¼”	9 5/8”		3337 m Side-Track Point	**Already Has been Drilled**	
8 1/2”	7” Liner		3939	**KCL Poly Mud**	MW: 80 - 85 pcf PV: ALAP YP: 12 – 20 lb/100 ft² F.L: < 3 cc/30 min PH: 9 - 10 Vis: 40 - 50 sec/qt
6 1/8”	4 1/2” Liner		4150 m	**Caco3/Polymer Mud**	Mw:100 – 106 pcf PV: ALAP YP: 14 – 24 lb/100 ft² F.L: < 3 cc/30 min PH: 9 - 10 Vis: 45 - 55 sec/qt

8.2 8 – ½" Hole Section (3337 – 3939m)

Challenges

- Hole cleaning
- Possible formation losses
- Tight Hole and Hole stability
- Hole instability issues
- Wellbore losses.
- Shale

Recommendation

- It is mandatory to monitor closely the rheological properties in order to ensure adequate hole cleaning, run Hydraulic calculation to ensure that mud carrying capacity is enough to meet this critical goal.

- To combat drilled solids, the shakers should be run with the finest screens possible in accordance with the flow rate and minimize fluid losses from the shakers by checking deck angle and running appropriate screens at all times. And the mud cleaner should be run with the finest mesh screens, making sure that the cones are always clean and discharging heavy liquid into the screen, check the working pressure in order to have the mud cleaner working at the maximum efficiency. The centrifuge should be run as required to maintain the mud weight.

- Use High Pressure Washer Equipment to wash and clean the shale shaker.

- Pre-treat mud received from previous section with Sodium Bicarbonate and Citric Acid to be prevented from cement contamination. Over treating with Sodium bicarbonate could be as detrimental to drilling fluid properties as the cement contamination. Therefore, it is not advisable to pretreat with more than 0.5 lb/bbl of Sodium bicarbonate. One approach to avoid overtreatment is to treat only soluble calcium and wait to treat lime when it goes into solution on subsequent circulations. Treatments should be discontinued when excess lime approaches 0.3 to 0.5 lb/bbl.

- Like in the previous sections, also in this section all the precautions have to be considered in order to take immediate reactions for any risk of circulation losses: Optimize the procedure to fight the losses (see the guide at the end of this mud program) and be ready with LCM material to mix LCM pill as quick as possible.

- Good drilling operation is another important parameter to have proper hole condition particularly whilst Tripping. Gel strength should be kept at minimum range to minimize Surge & Swab pressure.

- The mud weight in this interval is critical for successful drilling, hole condition will dictate the density. Mud weight will be increased by adding Barite.
- If tight hole or packing off problems is encountered increase the frequency and the volume of sweep.
- PH in active system should be maintained between 9.5 – 10
- Good drilling operation is another important parameter to have proper hole condition particularly whilst Tripping. Gel strength should be kept at minimum range to minimize Surge & Swab pressure.
- It is mandatory to monitor closely the rheological properties in order to ensure adequate hole cleaning, run Hydraulic calculation to ensure that mud carrying capacity is enough to meet this critical goal.
- Mud can be treated with additional shale inhibitor if deemed necessary to control shale reactivity and washouts.
- Optimize the procedure to fight the losses (see the guide at the end of this mud program) and be ready with adequate LCM material to mix LCM pill as quick as possible.
- The previouse mud in hole must be dump , due to have enoght surface volume , keep 300 bbl heavy mud in mud pits as kill mud.
- The concentration of KCl in the system is insufficient; shale cuttings will be soft and mushy. If the concentration of KCl is sufficient, they will retain their integrity. As drilling commences the potassium ion becomes depleted and must be replenished. The rate of depletion provides an indication of how reactive the shales are. While drilling gumbo, the appearance of soft mushy cuttings at surface is a good indication that the potassium concentration should be increased. Other indicators include a drilled solids increase, a CEC increase and the occurrence of tight hole or mud rings.
- Since this system is very solids-sensitive, it is best to mix the system from scratch instead of converting an existing drilling fluid (containing drill solids) to a potassium chloride system.
- This system is very sensitive to solids and hardness contamination so the first step in mixing this system is to treat the hardness.
- Due to necessary of directional drilling, it is recommended to pump (at the first 20 bbl Low Vis pill follow by 20 bbl Hi Vis Pill should be pumped every stand) for clean the hole. For better understanding of hole cleaning procedure based on hole angels please refer to technical appendix.

NOTE: In case of need for better hole cleaning PHPA Pill can be used.

- Due to face with torque and drag , it is recommended to mix water base mud lubricant 1 – 3% vol reduce torque and drag while drilling horizontal or vertical
- In the directional drilling section it is recommended to use Glass beads 2 – 3 ppb as due to have torque and drag.
- The mud weight in this interval is critical for successful drilling, hole condition will dictate the density. Mud weight will be increased by adding $CaCO_3$ F.

Interval Description

The 8 1/2" "section will be drilled from 3337 m to an estimated section TD of 3939 m with Sal KCL Polymer Mud system necessitating a minimum requirement of 80 pcf to 85 pcf based on nearby well data.

It is necessary to adjust mud weight and mud properties as hole condition dictate. Some tight hole, well flow and low ROP maybe experienced in this hole section. The drilling fluid engineer will advise the drilling supervisor of any extended circulation or sweep requirement. It uses clarified viscosifier to provide viscosity and carrying capacity and starch to reduce mud filtrate.

8 1/2" HOLE SIZE		**7" Liner**
FROM : 3337 MT		**TO : 3939 MT**
602 meter to drill in kazhdomi, Darian and Gadvanby Caco3 Polymer Mud mud		
DRILLING FLUID SYSTEM		**Salt Saturated/Polymer Mud**
Key products		Limestone , Salt (NaCl), Xanthan gum, Starch LV , Caustic soda , KCL , Soda ash, Biocide
Water type		Drill Water
Solid Control		Shale shakers , Desilters , Desander, Centrifuge

Kazhdomi	:	Shale interbedded with argillaceous limestone, sandy claystone, claystone, marl, sandy limestone, limestone beds
Daryan	:	shale interbedded with argillaceous limestone
Gadvan	:	limestone, claystone, sandy claystone and sandstone

8.2.1.1 Interval Mud Discussion

Sal KCL Polymer Mud are used to prevent excessive hole enlargement while drilling massive salt beds. They can also be used to reduce dispersion and hydration of shales and clays. Solids content (percent by volume) should be maintained within a desired range by mechanical removal or dilution with saturated salt water to prevent unacceptable increases in gel strengths and viscosity.

The chloride content of saturated salt muds is 170,000- 188,000 mg/l at saturation. As the temperature of the mud increases, more salt is able to go into solution. This means that a fluid which is saturated under surface conditions may not be saturated in down hole temperatures and can cause substantial washout in a salt zone due to increased salt solubility.

Control of the pH of saturated salt muds varies widely. Saturated salt muds require larger additions of caustic soda to maintain a pH in Salt saturate muds. Maintaining an±9.5 pH offers several advantages:

- Deflocculants are more effective
- Corrosion is reduced
- Lower concentrations of filtration control additives are required when Ca++ and Mg++ solubility is reduced
- Foaming tendency is lessened
- Mud is generally more stable

Caco3 Polymer Mud will normally contain soluble calcium due to the formations penetrated and the type of makeup water used. Also, the sodium ions from the salt will undergo a Base Exchange and release calcium ions from the clay platelets, thereby contributing to the free calcium. Generally, the presence of calcium does not produce detrimental effects on the mud; except when the pH is increased beyond 12.0 and then the fluid loss will be difficult to control.

Foaming is generally restricted to surface foam and is not troublesome unless aggravated by mechanical agitation. The degree of foaming may sometimes be decreased by increasing the alkalinity of the mud (Pm). A Defoaming agent may be necessary. Unless Mg++ sensitive additives are used, a Saturated Salt Mud is less prone to foam and air cut with a 9.5 - 10 pH. The temperature limitation of Saturated Salt Muds is around 250°F and is normally dictated by the filtration control additive used. Ca++ and Mg++ hardness do not adversely affect filtration control in Saturated Salt Muds when using starch; however.

Usually there are no problems associated with saturated salt mud conversions. Conversion can be accomplished in two circulations. When a freshwater mud is to be converted to a saturated salt mud, it is desirable to use as much of the old mud as possible.

The amount of salt required to saturate the system should be calculated and then added through the hopper. A saturated salt mud will generally weight 75 pcf without weight material addition. Next, a fluid loss additive is added through the hopper while the mud is circulated. Starch is generally used for fluid loss control and as a supplementary viscosifier. Normally, 12 lb/bbl starch will be required to obtain a fluid loss of less than 3 cc/30 minutes. If a higher density is required, Barite which will beuse. When weighting up with Barite, a Lignosulphonate Deflocculant is recommended to avoid excessive viscosity and high gel strengths as contingency.

Potassium is used in areas where inhibition is required to limit chemical alteration of shales. Potassium performance is based on cationic exchange of potassium for sodium or calcium ions on smectites and interlayered clays. The potassium ion compared to calcium ion or other inhibitive ions, fits more closely into the clay lattice structure, thereby greatly reducing hydration of clays. Potassium-based muds perform best on shales containing large quantities of smectites or interlayered clays in the total clay fraction. Potassium interactions with clay surfaces can be traced to two effects: ionic size and hydrational energy. Potassium ions are of the proper size to fit snugly into the spaces between the two silica tetrahedral layers which contact each other in the formation of a three-layer clay packet. The ionic diameter of potassium is 2.66 Å, very close to the available distance of 2.8 Å in the lattice space of the clay structure. A cation slightly smaller than 2.8 Å is desirable to allow for crystalline compaction.

Product Concentration & Function

Salt Saturated/ Polymer mud		
Product N.	**Ppb**	**Product Function**
Caustic Soda	0.5 - 1	Alkalinity Control
Soda Ash	1	Hardness Remover
Salt	Up to 95	non - solid weighting agent
KCL	10.5 – 17	Shale Inhibitor
Limestone Powder	Up to 114	weighting agent
Starch LV	10-12	Fluid loss controller
Xanthan gum	0.5 - 1	Viscosifier
Biocide	As Requirement	Anti-bacteria
Glycol	Optional	Shale Inhibitor
MIXING PROCEDURE		
1. PH , chloride , calcium and total hardness of drill water (that water which will be used for mud making) must be check by Mud engineer		
2. Fill the mud tank with Drill water		
3. add Salt & KCL to reach mentioned mud weight		
4. Remove Hardness by add Soda ash		
5. adjust PH & alkalinity by Lime and Caustic Soda		
6. add Starch LV to control fluid loss		
7. add Xanthan gum as Need to reach good rheology		
8. Add limestone powder to reach Mentioned Weight		
9.Check the pH and keep in range by add Lime		
10. Mix Biocide as need		

Required Properties

Fluid Properties Summary		
MUD PARAMETERS		Interval
		I
Bit diameter	inch	8 1/2”
Density	pcf	80 – 85
Marsh Viscosity	sec	40 – 50
PV	cps	ALAP
Yield Point	lb/100ft²	12 – 20
Gel 10 Sec	lb/100ft²	2 – 5
Gel 10 min	lb/100ft²	3 – 7
API Filtrate	cc/30'	< 3
MBT	ppb	<5
pH	-	9 – 10
Total Hardness	mg/l	< 1000
Ca++	mg/l	< 600
Solid	% vol	< 20
CL^-	Mg/l	170,000 – 180000

****Please note that Rate and controlling of PV and solid percentage are depends on the solid control EQ efficiency****

Mud Volume

DRILLING FLUID TYPE			
Mud Density (pcf)	80 - 85	LGS Below (%vol)	7
Bit Size (inch)	8.375	Solid Control Equipment Eff. %	80
Section TD (m)	3900	Hole Wash Out %	15
Casing OD/ID (inch)	9.625 / 8.835	Dilution Factor (bbl/m)	3.2
Water Depth (m)		Meters To Drilled (m)	600

Volume information (KCL Poly Mud)		
Surface System	800	bbl
Cased Hole Volume	821	bbl
New Hole Volume	134	bbl
Wash Out Volume	20	bbl
Sub Total	1,775	bbl
Dilution Volume	429	bbl
Sub Total	2,204	bbl
Carried from last section	0	bbl
Volume Received	0	bbl
Sub Total	2,204	bbl
New Vol Required	2,204	bbl
Vol transfer out	500	bbl

Product	Unit Size lb or bbl	Concentration	Expected Consumption
Soda Ash	110	1	20
Caustic Soda	55	0.5-1	40
Salt (NaCl)	2,200	up to 95	95
KCL	2,200	3.5 - 10.7	11
Starch LV	55	10-12	481
Limestone Powder	2,200	up to 114	114
Xanthan Gum	55	0.5-1	40
Biocide	55	As Requirement	
Glycol	55	Optional	

Contingency Materials			
Product Name	Function	Unit Size	QTY
Pipe Lax	Pipe freeing agent	55 Gal	4
De foamer	De foamer	55 Gal	4
Lime	Alkalinity control/H2S anti	55 lb	80
Shell fish F,m	Loss control material	55 lb	each size 120 sxs
$CACO_3$ F,M	Loss control material	55 lb	each size 80 sxs
Mica F.M	Loss control material	55 lb	each size 120 sxs
Walnut shell F,M	Loss control material	55 lb	each size 80 sxs
Sodium Bicarbonate	CMT Contaminiation Remover	55 lb	20

1-Please note all calculation are based on normal hole condition and maximum SCE effieciency. Concentration and usage of material may be change as hole condition dictate.

2-All the calculation based on max material Concentration which mentioned.

8.3 6 – 1/8" Hole Section (3939 – 4150m)

Challenges

- Wellbore losses
- Possible HT environment
- Oil/Water Influx

Recommendation

- Hole cleaning is highly related to low shear rate parameters, so an adequate low shear rate yield point should have maintained to ensure hole cleaning. Run Hydraulic calculation to ensure that mud carrying capacity is enough to meet this critical goal.
- The previous mud will be transfer to this section.
- A suitable LCM material for reservoir should be available on location to combat down hole losses if occurred (see the guide at the end of this mud program).
- Since this system is very solids-sensitive, it is best to mix the system from scratch instead of converting an existing drilling fluid (containing drill solids) to a potassium chloride system.
- This system is very sensitive to solids and hardness contamination so the first step in mixing this system is to treat the hardness.
- Due to necessity of directional drilling, it is recommended to pump (at the first 20 bbl Low Vis pill follow by 20 bbl Hi Vis Pill should be pumped every stand) for clean the hole. For better understanding of hole cleaning procedure based on hole angels please refer to technical appendix.
- Due to face with torque and drag , it is recommended to mix WATER BASE MUD Lubricant 1 – 3% vol reduce torque and drag while drilling horizontal or vertical
- In the directional drilling section it is recommended to use Glass beads 2 – 3 ppb as due to have torque and drag.
- The mud weight in this interval is critical for successful drilling, hole condition will dictate the density. Mud weight will be increased by adding CaCO3 F.

Interval Description

After cementing the 7" inch Liner and confirmation of its proper cementing, the 5 7/8"section will be drilled from casing shoe to an estimated section TD $CaCO_3$ Polymer mud system necessitating a minimum requirement of 100^{pcf} to 106^{pcf} based on nearby well data.

It is necessary to adjust mud weight and mud properties as hole condition dictate. Some tight hole, well flow and low ROP maybe experienced in this hole section. The drilling fluid engineer will advise the drilling supervisor of any extended circulation or sweep requirement. It uses clarified viscosifier to provide viscosity and carrying capacity and starch to reduce mud filtrate.

5 7/8" HOLE SIZE		4 1/2" Liner Size
FROM : 3939 MT		TO : 4150 MT
211 meter to drill in Fahliyan, by $CaCo_3$/Polymer mud		
DRILLING FLUID SYSTEM		$CaCo_3$/Polymer Mud
Key products		Limestone Powder , Salt (NaCl), Xanthan gum, Starch LV , Caustic Soda, Soda ash, Biocide
Water type		Drill Water
Solid Control		Shale shakers , Desilters , Desander, Centrifuge

Fahlyan:	Limestone interbedded with green gray, Argillaceous Limestone, sticky Clay stone beds.

Product Concentration & Function

Salt /$CaCo_3$/ Polymer mud		
Product N.	**Ppb**	**Product Function**
Caustic Soda	0.5-1	Alkalinity Control
Soda Ash	1	Hardness Remover
Salt	Up to 95	non - solid weighting agent
KCL (Optional)	3.5 – 10.7	Shale Inhibitor
Limestone Powder	Up to 258	weighting agent
Starch LV	9-10	Fluid loss controller
Starch HT	2-3	Fluid loss controller
Hematite	Up to 70	weighting agent
Xanthan gum	0.3-0.6	Viscosifier
Biocide	As Requirement	Anti-bacteria
MIXING PROCEDURE		
1. PH , chloride , calcium and total hardness of drill water (that water which will be used for mud making) must be check by Mud engineer		
2. Fill the mud tank with Drill water		
3. add Salt to reach mentioned mud weight		
4. Remove Hardness by add Soda ash		
5. adjust PH & alkalinty by Casutic soda		
6. add Starch LV to control fluid loss		
7. add Xanthan gum to reach good rheology		
8. Add Limestone powder to reach Mentioned Weight		
9.Check the pH and keep in range by add caustic soda		
10. Mix Biocide as need		

Required Properties

Fluid Properties Summary		
MUD PARAMETERS		Interval
		II
Bit diameter	inch	5 7/8”
Density	pcf	100-106
Marsh Viscosity	sec	45 – 55
PV	cps	ALAP
Yield Point	lb/100ft²	14 – 24
Gel 10 Sec	lb/100ft²	3-7
Gel 10 min	lb/100ft²	5 - 10
API Filtrate	cc/30'	< 3
pH	-	9 – 10
Ca++	mg/l	< 600
Total Hardness	Mg/l	<1000
MBT	ppb	< 5
LGS (drilled Solids)	% vol	< 10
CL^-	Mg/l	180000-188000

****Please note that Rate and controlling of PV and solid percentage are depends on the solid control EQ efficiency****

Mud Volume

DRILLING FLUID TYPE			
Mud Density (pcf)	100 - 106	LGS Below (%vol)	6
Bit Size (inch)	8.5	Solid Control Equipment Eff. %	80
Section TD (m)	4200	Hole Wash Out %	15
Casing OD/ID (inch)	9.625 / 8.835	Dilution Factor (bbl/m)	2
Water Depth (m)		Meters To Drilled (m)	300

Volume information (Salt Pol Mud)		
Surface System	600	bbl
Cased Hole Volume	871	bbl
New Hole Volume	69	bbl
Wash Out Volume	10	bbl
Sub Total	1,550	bbl
Dilution Volume	138	bbl
Sub Total	1,689	bbl
Carried from last section	500	bbl
Volume Received	0	bbl
Sub Total	1,189	bbl
New Vol Required	1,189	bbl
Vol transfer out	0	bbl

Product	Unit Size lb or bbl	Concentration	Expected Consumption
Soda Ash	110	1	11
Caustic Soda	55	0.5-1	11
Salt (NaCl)	2,200	95	51
KCL (Optional)	2,200	3.5 - 10.7	6
Starch LV	55	9-10	216
Starch HT	55	2-3	65
Limestone Powder	2,200	up to 258	139
Xanthan Gum	55	0.3-.6	13
Ferobar	3,300	Up to 70	36
Biocide	55	As Requirement	

Contingency Materials			
Product Name	Function	Unit Size	QTY
Pipe Lax	Pipe freeing agent	55 Gal	2
Barite	weighting agent	2200 lb	40
De foamer	De foamer	55 Gal	2
Shell fish F,M	Loss control material	55 lb	each size 60 sxs
$CACO_3$ F,M	Loss control material	55 lb	each size 80 sxs
Water Base Mud lubricant	Torque & Drag Reducer/ROP Enhancer	55 Gal	40
Sodium Bicarbonate	CMT Contaminiation Remover	55 lb	20

1-Please note all calculation are based on normal hole condition and maximum SCE effiecieny. Concentration and usage of material may be change as hole condition dictate.

2-All the calculation based on max material Concentration which mentioned.

8.4 Technical Appendix

Loss Control Procedure

Loss of fluid returns will lower the hydrostatic head of the drilling fluid in the wellbore, thereby inducing a kick. The influx fluid will then flow to the surface or into the zone of lower pressure. Lost circulation can occur in naturally occurring fractured, vuggy, cavernous, sub-normally pressured or pressure depleted formations. Induced losses can occur from mechanical fracturing due to pressure surges while breaking circulation. In all cases of lost circulation, attempts should be made to keep the hole full. The hole can be filled with either light drilling fluid or water. A record of the amount of fluid pumped should be made.

The concentration of LCM will vary according to losses. In the event of seepage losses, treat system with a finer grade of LCM at 20 – 30 Kg/m³. In the event of moderate to high rate of well bore loss, prepare pills at different graded concentrations and spot over thief zone. The concentration may vary between 18 to 70 lb/bbl. In order to avoid LCM plugging MWD tools a circulation sub should be installed in the drill string. The MWD engineer is to advise on the size and concentration of LCM that can be pumped through MWD Tools. Ensure that the drilling displace the LCM pill out of the string prior to shutting down the mud pumps.

Important Note: Please note below concentration and recommended materials may be change with same materials on rig site.

In case of having circulation losses:

Losses less than 10 bbls/hr:

- Active system to be loaded with 10 lb/bbl of Fiber Fine.

Losses 10- 40 bbls/hr:

- Active mud to be loaded with 15- 20 lb/bbl of Fiber Medium.
- Sweep the hole frequently with 50 bbls high viscous pill contains;
- 15 lb/bbl Mica Medium - Corse.
- 10 lb/bbl Walnut Shells Medium -Fine.
- 10 lb/bbl Oyster Shell Fine.

Losses 40 – 100 bbl/hr:

- Stop Drilling.
- Pull out string above the loss zone

- Spot 100 bbls pill contains;
- 30 lb/bbl Fiber coarse.
- 30 lb/bbl Walnut Fine.
- 20 lb/bbl Mica Med.
- Repeat the previous pill minimum of two times.

Losses more than 100 bbls:

- Up to company procedure, may be squeeze 50 bbls of Cement squeeze.

8.4.1.1 Mixing LCM Pill

The amount of LCM that can be added to a mud depends very much on the mud properties. High gels and YP will help keep the material in suspension. Since LCM is generally of low gradient, excess material tends to float on top if too much is added. The following procedure may prove useful for mixing LCM pills and avoiding the problem of flotation.

1. ENSURE the pill has enough viscosity to keep the low gradient material in suspension before addition of LCM. ADD appropriate viscosifier if required.
2. ADD the material in stages e g. 5, 10, 15, 20 etc. ppb.
3. STOP after desired concentration has been reached, or when it is observed that the material cannot be held in suspension
4. If a higher LCM content is desired, LEAVE the pill to soak for a period of time and then TRY ADDING more material.

8.4.1.2 LCM Pill Spotting Procedure

1. After spotting LCM pill, pull the bit above the calculated top of the pill. Wait 30minutes and monitor the annulus.
2. After 30 minutes, attempt to fill the annulus, if not already full. If the hole can be filled, close the BOP and squeeze 10 bbls of LCM pill into the formation without exceeding 200 psi.
3. If the hole remains stable, continue to drill ahead.
4. Should the LCM pill fail to reduce the losses or the hole cannot be filled, then mix and spot another 50 bbls of LCM pill.
5. Attempt to fill the annulus. If successful, close the BOP and squeeze 20 bbls of LCM pill into the formation without exceeding 200 psi. If a squeeze pressure develops, hold the pressure for 1 hour before staging the drill string back to bottom.

Notes:

- All time hole must be full, transfer mud to trip tank and asked driller to fill hole from trip tank.
- All LCM pills must be thoroughly mixed and homogenised before being pumped.
- When pumping an LCM pill through a bit with nozzle smaller than 18/64, DO NOT USED LCM Medium and Coarse sizes.
- Remove filter screens from the drill pipe and if necessary from the mud pump suction lines.
- This is particularly true if the mud density is high, 14 lb/gal or greater. Fine loss circulation material which does not adsorb a lot of water should be used in weighted mud.

Guide Line for Fiel Engineers for Shale Inhibition

Shale Hydration takes two forms in water-sensitive shales: surface and osmotic adsorption. Surface hydration occurs when a small volume of water is strongly adsorbed onto the planar surfaces of the clays which causes little softening or swelling, but can lead to excessive stresses if the swelling is confined.

Osmotic swelling occurs when a large volume of weakly held water is attracted to the clay surfaces by electrostatic forces. Osmotic swelling causes softening and significant swelling as the adjacent clay layers hydrate water and expand. Osmostic swelling does not generate excessive stresses, even when confined, and can be substantially reduced if Extremely difficult drilling conditions have led to more systematic approaches to analysing wellbore instabilities.

One such approach that can be used as a guideline for the mud engineer involves three basic steps:

1. Monitor and analyse the fundamental failure mechanism.
2. Determine and understand the factors governing this failure mechanism.
3. Model the problem considering impact of proposed changes.

8.4.1.3 Monitor & Analyze the Failure Mechanism

Monitoring of data and events on the rig is the first step of the process and involves:

• **Collection, organization and processing of drilling data.** Drilling data such as Rate of Penetration (ROP), Revolutions per Minute (RPM), Weight-on-Bit (WOB), Bottom-Hole Assembly (BHA), torque, over pull Trip In Hole (TIH) and Pull Out Of Hole (POOH), swab and surge, flow rate, mud weight, mud Rehology , etc. should be collected, organized and analyzed on a continual basis and compared to the symptoms of wellbore stability being observed. This will help evaluate performance and establish trends or any deviations and measure the effectiveness of treatments and changes.

• **Formation characterization.** The standard procedure in drilling operations is to apply the knowledge

gained from previous wells to subsequent wells drilled in similar geologic regions. Wellbore-instability

investigation requires evaluation of shales during the drilling process. Since rig site shale testing is limited

due to equipment and time limitations, most evaluations are performed in a properly equipped laboratory.

Laboratory tests are conducted to evaluate shale swelling, dispersibility, cation exchange capacity, hydration capacity, colloidal content, capillary suction time, shale hardness tests and mud compatibility properties.

A number of methods have been used to perform rig site tests. These include tests as simple as the methylene blue test (CEC) measurement of shale; shale hardness measurements; hot-rolled dispersion testing with various mud formulations and additives; and sophisticated analytical measurements of various clay properties. It is particularly important to try and determine the failure mechanism: swelling and softening or brittle spalling. Information from these tests, current shaker observations and mud logger

data are used to evaluate drilling fluid performance, adjust mechanical drilling parameters and making casing design decisions. The primary options to counter shale problems at the rig site are:

- Change the drilling fluid density.
- Change to an oil- or synthetic-base system.
- Change to a more inhibited water-base mud.
- Reduce fluid loss to very low levels and increase filtrate viscosity.
- Utilize pore-plugging additives, emulsified insoluble materials or plastering agents.
- Adjust mechanical drilling parameters.
- **Evaluation and analysis of instability symptoms.**

Instability warning signs such as the presence of excessive cavings on the shakers for a given ROP, fill on connections, tripping difficulties, torque, stuck pipe, etc. indicate that something is wrong down hole. The magnitude or volume of failed rock should be estimated and recorded with

the other drilling parameters. The number and location of specific incidents should be recorded, such as pipe

sticking, fill or packing-off, reaming and re-drilling. All of this information can then be organized and correlated with time and depth to detect the location and quantify the severity of critical sections in the well.

8.4.1.4 Determine & Understand Failure Mechanism

All of the monitored and plotted parameters are then evaluated to identify the most probable mechanism causing the instability. The most useful symptoms used to indicate the mechanism responsible for wellbore instability are:

- Cavings analysis.
- Tripping difficulties.
- Drilling conditions.
- Mud system analysis.
- In-situ conditions.

These symptoms should be evaluated based on an interdisciplinary knowledge of the causes of wellbore instability to determine the most likely failure mechanism.

8.4.1.5 Model & Considering Proposed Changes

Once the monitoring and determination steps have produced parameters, organize and analyze this data to obtain qualitative solutions and evaluate proposed changes. One good way to do this is through the use of a decision tree or flowchart approach to evaluate the possible failure mechanism and to identify the most promising remedies. After arriving at a proposed solution, a corrective measure should be applied and

the results used as feedback into the method and decision tree method.

A decision tree or flowchart, such as shown in Figure 9 for evaluating the cause of excessive cavings, provides a logical path for identification of the mechanism and a qualitative solution. Maintaining hole stability while drilling shale sections can be particularly troublesome. Simple solutions do not always exist, but good drilling practices combined with good mud practices are most often successful. Analysis of the mechanical stress/strain relationships has helped to develop the proper density requirements for hole stability.

8.5 Directional Drilling

Drilling Fluid Properties & Selection

It is common knowledge among drilling engineers and drillers that borehole problems can be considerably more severe in a directional hole than in a vertical well. The problems are magnified as the hole angle and horizontal displacement increases. Although most of these reactions are mechanical in nature, much can be done to alleviate them by proper selection and maintenance of the drilling fluid. For example, if a water-base fluid is being used, lubricants can be added to lower the coefficient of friction between the drill string and borehole or casing. Inhibiting salts can be added to reduce the interaction of the drilling fluid with the exposed formations, thereby improving borehole stability.

There are a number of drilling factors influenced by the drilling fluid, but the following are of primary concern in drilling a horizontal well.

- Borehole instability
- Hole cleaning
- Torque and drag
- Formation damage
- Barite sag
- Hydraulics

Borehole Instability

Borehole instability is one of the more costly, but relatively common problem associated with drilling oil and gas wells. Not only can it impede drilling, it can result in loss of the hole. The term normally conjures thoughts of borehole collapse with accompanying stuck pipe. This is one of the more critical reactions of borehole instability, but it can manifest itself in more subtle ways such as hole-fill, bridges, hole enlargement, or hole size reduction.

These reactions lead to other associated problems including increased torque and drag, logging difficulties, poor cement jobs, and increased drilling days, all of which add significantly to the drilling cost. It should be pointed out that the discussions to follow do not cover hole enlargement problems from weakly consolidated sands, or rubble zones that erode from the hydraulic action of the drilling fluid.

Stabilization

The swelling activity is controlled by adjusting the electrolyte content of the fluid to minimize expansion of the clays. Lime and gypsum-treated fluids have long been used for this purpose, but basically do not satisfy the inhibition needs of the more troublesome shales. The potassium ion is an effective swelling inhibitor and is used extensively for this purpose through the addition of potassium salts. The concentration of potassium utilized ranges from several hundred mg/L to over 60,000 mg/L. The most cost-effective potassium concentration, however, appears to be in the 3000 to 7000 mg/L range. In marine environments, seawater or seawater with additional sodium chloride, are good alternates to potassium. The tendency for the shale to disperse or subdivide into discrete particles appears to be best controlled by the addition of an adsorptive, high molecular weight water-soluble polymer.

One of the more effective polymers for this purpose is the partially hydrolyzed poly acrylamide with molecular weight in excess of 15 million. These anionic polymers adsorb on to the surface of the cuttings and borehole wall to act as a binder or elastic coating to retard dispersion. Another approach to stabilizing the hard brittle shales that contain natural or induced micro fractures is the use of water-dispersible asphalt or asphaltines. The addition of these type materials appear to plug the micro-fractures, or perhaps even surface pore throats to retard further fluid invasion. This action has proven very effective in certain shales and can be applied in most water base fluids.

Hole Cleaning

Cleaning cuttings from a vertical hole has been covered extensively by numerous investigators. Hole cleaning in a vertical hole is essentially a matter of overcoming the slip velocity of the cuttings with sufficient annular velocity and carrying capacity of the drilling fluid. Although any velocity greater than the settling velocity of the largest cutting will theoretically lift all the cuttings to surface eventually, too low an annular velocity will lead to an undesirable high concentration of cuttings in the annulus. Experience has shown that cuttings concentrations more than 5% by volume cause tight hole or stuck pipe when circulation is stopped.

Cuttings Bed Formation

Radial slip velocity increases with increasing angle of inclination, forcing more cuttings to the low side of the annulus, which eventually results in the formation of a cuttings bed. reported that bed formation was dramatic in the 40° to 50° angle with the bed sliding downward against the flow, resulting in a very high cuttings concentration.

The fact that a cutting bed forms is a serious problem, but the instability of the bed in this critical angle of the well makes the problem even more severe. High torque and drag resulting from the high cutting concentration at the lower part of the angle can obviously result in stuck pipe or loss of circulation.

Hole Cleaning Sweeps

In static drill string concluded that high-density sweeps were more effective than high-density high-viscosity sweeps based on field measurements. The higher viscosity of the latter caused flow diversion to the high side of the wellbore. Not surprisingly, high-viscosity sweeps also performed poorly. Also concluded that the best hole cleaning sweeps were high-density low viscosity sweeps, with a density 3 – 4 ppg higher than that of the drilling fluid. the following procedure is recommended:

- Calculate sweep height in the vertical part of the annulus, and added hydrostatic head in the annulus.
- Rotate at 80+ rpm while pumping the sweep, 180+ rpm is preferred. It is not recommended to pump a weighted sweep while sliding.
- Keep pumping until the sweep is out of the hole. If pumping stops while the sweep is in the deviated hole, it falls out of the "jet stream" and re-deposits captured fines.
- Avoid increasing the viscosity of the weighted sweep.
- Plan to run 3 – 4 sweeps to achieve desired results when hole cleaning problems are indicated. One sweep will probably not be sufficient to remedy an existing situation.
- Avoid checking for flow or closing the annulus with the weighted sweep in the drill pipe.
- If desired for preventative maintenance: 1) schedule sweeps every 6 – 8 hours or 500 – 1000 feet (whichever comes first, unless the hole dictates greater frequency); barite/silt build- ups are dependent on time and footage: and 2) pump a sweep while circulating bottom up before tripping, and another once back on bottom.

concluded that high-density sweeps were more effective than high-viscosity sweeps.

They strongly advocated the use of sweep reports, which should be standard practice in any drilling fluids operation.

Short Tripping

Another technique that has proven helpful in controlling problems created by the cuttings bed in the high-angle portion of the well is to make periodic short trips. Tripping the bit through the bed not only assures that the bottom hole assembly and bit can be pulled through it, but the action should agitate the bed to the point that a portion will slide down the hole and subsequently be picked up when circulation is commenced. The time intervals for these short trips will be determined by the torque and drag conditions that exist in any particular well. Lacking specific well history, a general rule of thumb is to short trip at least once for every 500 feet of new hole.

Drill String Rotation

The industry is in total agreement that drill string rotation improves hole cleaning, especially in hole sizes less than 17 ½". The general opinion in the industry is that the optimum drill string rotation rate is approximately 120 rpm. It is believed that the drill string rotation improves hole cleaning by mechanically disturbing the cuttings bed on the low side of the hole. There could also be effects due to skewing the velocity profile in the wellbore and the creation of Taylor Vortices. Encouraged the use of rotary steerable systems.

Turbulance

Bottom hole assembly design becomes important because the flow restrictions of some of the components, such as the measurement-while-drilling tools or positive displacement motors, may not allow sufficient flow to reach turbulence in the annulus. Turbulence can, of course, be induced at some constant flow rate by decreasing the annular area. This may be more difficult to accomplish in a horizontal well because of limitations on the design of drill strings and tool assemblies. An increase in the effective viscosity of the drilling fluid will decrease the Reynolds Number, and thus increase the flow rate required to obtain turbulent flow. Thus, a lower effective viscosity fluid may be desirable in the horizontal well, but attention must also be given to cleaning the larger diameter portion of the upper hole or possible washed-out sections where it is impractical to maintain turbulent flow.

It is interesting to point out a basic fact between static fluid pressure and dynamic fluid pressure. Static fluid pressure is a function of true vertical depth, while dynamic fluid pressure, or ECD, is a function of measured depth. Thus, the ECD will continue to increase, albeit slowly, even after the hole becomes essentially horizontal. The fracture pressure could be reached if the hole was extended to sufficient depth and the fluid density is close to the fracture gradient.

Hole Cleaning Guideline

The single most important factor related to hole cleaning in deviated wells is flow rate. During directional drilling operations drilled cuttings will settle on the low side of the hole and form a stationary bed if insufficient annular fluid velocity is used. The critical flow rate required to prevent cuttings bed formation must be determined .

Typically few hole cleaning problems exist in vertical or horizontal sections. Most problems

associated with hole cleaning are seen on deviated wells and occur in the 40 – 60 degree section where gravity effects can cause cuttings beds to slump down the hole.Typical flow rates to aim for in ERD wells:

Hole Size	Typical Flow Rates
17 ½"	800 gpm minimum. Some rigs achieve 850 – 1000 gpm.
12 ¼"	Aim for 800 gpm (although 800 – 1000 gpm is typically achieved). If 800 gpm is not achievable, ensure tripping procedures are in place for poorly cleaned hole.
8 ½"	Aim for 400 gpm.

When correctly designed, both laminar and turbulent flow regimes will effectively clean a deviated well. In general increasing the viscosity of a fluid in laminar flow will improve hole cleaning as will a reduction of the viscosity of a fluid in turbulent flow. It is important that one or other regime is selected and that the transition zone between the two is avoided.

Generally viscous fluids in laminar flow are preferred because:

- It is possible to achieve higher cleaning capacity.
- Viscous fluids give better transport in the near-vertical sections.
- Viscous fluids have better suspension characteristics when circulation is stopped.
- It is difficult to achieve "turbulent flow" except in small hole sizes.

Experience has shown that good drilling fluid rheological properties are extremely important to hole cleaning when drilling a high angle well. Studies show that the effects of drilling fluid

rheological properties and annular flow regime are mutually dependent. In the laminar flow regime, increasing fluid viscosity will improve hole cleaning and this is particularly effective if the Yield Stress is high. In the turbulent regime, however, reducing fluid viscosity will help removing cuttings. Therefore the drilling fluid rheological properties should be designed to avoid the transitional flow regime. For hole sizes above 8 ½", the annular flow is laminar under most circumstances. Therefore it is necessary to specify a minimum Yield Stress. The optimum level required is best established based on field data and experience.

A higher ROP requires a higher flow rate to clean the hole. It is good practice to drill the hole with a steady ROP and select the required flow rate for hole cleaning accordingly. Before making a connection, the hole should be circulated at the normal flow rate to clear the cuttings from around the BHA. Depending upon the hole angle and length of BHA, a circulation time of 5 to 10 minutes is often required.

Before tripping out, the hole should be circulated at the normal flow rate until the shakers are clean, and the drill pipe be rotated at maximum speed/reciprocated in the meantime. This may require up to 3 bottoms-ups, depending on the hole angle and hole size. Below are listed the recommended number of bottoms-up prior to tripping.

Hole Angle, deg	8 ½"	12 ¼"	17 ½"
0 – 10	1.3	1.3	1.5
10 – 30	1.4	1.4	1.7
30 – 60	1.6	1.8	2.5
60+	1.7	2.0	3.0

Overall friction can become so great in an extended-reach well or horizontal section that the drill string can only be rotated with great difficulty. Needless to say, this condition is conducive to stuck pipe. Obviously, this coefficient of friction will be significantly affected by the lubricating properties of the drilling fluid.

any of the causes of torque and drag are mechanical, while others are related to hole geometry. Several, however, can be directly attributed to the drilling fluid. Since wall contact with the drill string will be extremely high in a horizontal hole, it is imperative that the lubricating characteristics of the fluid be considered.

Formation Damage

The obvious reason for drilling a horizontal well is to expose more of the producing zone to the wellbore for increased production. Thus, the importance of preventing formation damage during drilling and completion of the horizontal well is just as important as it is in its vertical counterpart.

In fact, it may be more important since the increased cost of drilling the horizontal well must show additional production for its justification. A 50% damage or permeability loss in a potential 500 bbl/day horizontal well represents a considerably greater loss in production than a 50% damage in a 100 bbl/day vertical hole through the same reservoir.

Sandstone production is by far the most susceptible reservoir rock to damage or permeability loss during drilling and completion operations. Though formation damage may be caused by poor drilling and completion practices, one of the primary causes is due to exposure to the drilling and completion fluid.

This loss, attributable to the fluid, can be caused by one or more of the following occurrences:

- Particle transport from the drilling fluid into the production zone, subsequently plugging pore throats.
- Fluid filtrate reacting with expandable clays in the rock to decrease pore throat diameters.
- Particle movement within the permeable rock due to dispersion of clays and other minerals from quartz surfaces.
- Wettability changes of the formation from exposure to the drilling fluid filtrate.
- Interaction of fluid filtrate with formation fluids to form water - insoluble precipitants.

The effect of these actions on the permeability of sandstones has been the subject of a considerable investigation. In almost all cases, some damage is caused by the drilling and completion fluid. The primary objective is to limit this damage as much as possible. Consequently, selection of the drilling and completion fluid becomes an extremely important function in drilling the horizontal well.

Recommended Fluid Parameters

- Velocity – One of the major obstacles to overcome in horizontal and ERD drilling is hole cleaning. In vertical holes, cuttings have room to settle without causing problems. In horizontal wells, cuttings only have to settle 4+ inches to form a cuttings bed. Research and practical experience has shown that cuttings bed formation is analogous to sand dunes on an ocean beach. The bed moves in a wave fashion up the hole. If sufficient annular velocity is not present, the bed thickens until it reaches equilibrium, which usually occurs when turbulent flow conditions exist. With water, turbulent flow is easily obtained since viscosity is in the denominator of the turbulent flow equation, and water has a viscosity of 1. Fluids other than water require more pump output to approach turbulent flow conditions. Practical experience in other industries has proven that materials can be transported in water slurries if the fluid velocity is between 3 and 7 ft/sec.

- Rheological Properties – Ideally one would want a true thixotropic fluid to drill horizontally. In effect, two different conditions exist simultaneously. One occurs in the vertical portion of the hole where larger diameter casing may be set, or wash-outs in the open hole occur. In that case, a fluid having a high Yield Stress, or if the fluid is visco-elastic, the initial gel strength needs to be elevated. The opposite is needed in the horizontal section from a hole-cleaning point of view. Here one wants water for the reasons pointed out earlier. Compromises are the result. If casing is set into the zone of interest, or if formations up-hole are not a problem, then water may be possibly combined with occasional sweeps.

- Surface Apparatus – Special attention must be placed on the solids removal equipment at the surface. Solids removal efficiency must be continually monitored so that unwanted particulate material is not retained in the fluid system that may negatively affect fluid weight or filter cake quality, resulting in lost circulation or increased torque and drag. The cost of problems that may occur due to not doing this correctly may far outweigh the cost of equipment needed.

8.6 Hydrogen Sulphide Gas in Drilling Fluid

Description of Hydrogen Sulphide Gas

Hydrogen sulphide is a colourless gas, with an offensive odour and a sweetish taste. It is soluble in water, alcohol, oils, and many other solvents. It has a specific gravity of 1.1895 with reference to air. It is considered a weak acid, it is toxic to humans and corrosive to metals. Hydrogen sulphide can be dangerous to personnel on the surface as it is extremely toxic to human and even animal life, and is extremely corrosive to most metals as it can cause cracking of drill pipe and tubular goods, and destruction of testing tools and wire lines. When hydrogen sulphide enters the borehole during drilling, completion, or testing for hydrocarbons, it creates several very detrimental problems. These problems are encountered regardless of the source of the hydrogen sulphide. Hydrogen sulphide has created unsolved problems in the oil field.

Origin of Hydroge Sulphide

During oil and gas operations, hydrogen sulphide can be found as a component of formation gases, dissolved in water, hydrocarbons, or even liquid sulphur. Thermal degradation of organic materials and sulphate reducing bacteria (SRB) can create hydrogen sulphide along with other gases.

Problem Cused by Hydroge Sulphide Gas

8.6.1.1 Health & Safety of the Personnel

One of the main problems hydrogen sulphide causes is in respect to the health and safety and safety of the personnel. A maximum of eight hours exposure to concentrations greater than 100 ppm H2S will cause hemorrhage and death. Concentrations above 600 ppm can be fatal in three to five minutes. Highly stressed, high strength steel can fail in a matter of minutes in the presence of 50 ppm H2S. At high pressure, as little as 0.1 ppm H2S can greatly reduce the time to failure of highly stressed, high strength steel. Relying solely on its odour is not a good idea because at concentrations above 100 ppm it deadens a person's sense of smell within a few minutes. The pure gas is heavier than air and can collect in low areas such as pit rooms and accommodation. below Table shows the potential hazards hydrogen sulphide can cause to the personnel from short-term exposures. There are many historical cases to document the ill effects of hydrogen sulphide exposure.

The documented cases reported in Table 2 indicate that hydrogen sulphide is very toxic and it takes only a matter of minutes to affect unprotected workers.

Health effects from short-term exposure to hydrogen sulphide	
Concentration (ppm)	**Health Effect**
0.01 – 0.3	Odour threshold
1 – 20	Offensive odour, possible nausea, tearing of the eyes or headaches with prolonged exposure
20 – 50	Nose, throat and long irritation; digestive upset and loss of appetite; sense of smell starts to become fatigued; acute conjunctivitis may occur (pain, tearing and light sensitivity)
100 – 200	Severe nose, throat and long irritation; ability to smell odour completely disappears.
500	Severe lung irritation, excitement, headache, dizziness, staggering, sudden collapse knockdown), unconsciousness and death within a few hours, loss of memory for the period of exposure (results in permanent brain damage if not rescued immediately).
500 – 1000	Respiratory paralysis, irregular heart beat, collapse and death without rescue.
> 1000	Rapid collapse and death

8.6.1.2 Effects of Hydrogen Sulphide on Metal

Hydrogen sulphide mainly cause corrosion problems to drill strings, transport pipes, storage tanks etc. Hydrogen sulphide causes sulphide stress cracking, hydrogen embrittlement and pitting corrosion in oil and gas operations. The corrosion of iron in the presence of hydrogen sulphide and water is dependent upon the dissociation of the hydrogen-sulphide molecule. Iron is oxidized to the ferrous form at the anode and hydrogen sulphide under- goes a two-step dissociation at the cathode. Tung et al22propose the following equilibrium reactions below.

At the anode:

$$Fe \longrightarrow Fe^{2+} + 2e^-$$

And at the cathode hydrogen is produced and either enters the steel or forms hydrogen gas and bubbles off:

If H2S is present:

$2H^+ + 2e^- \rightarrow H^\circ + H^\circ$

If H2S is not present:

$H^\circ + H^\circ \rightarrow H_2$

Hydrogen enters the steel first by adsorption onto the water to steel interface and then by being absorbed into the steel as hydrogen. The amount of hydrogen entry depends on the corrosion rate of the steel surface and the concentration of anions such as HS- that reduce the tendency to produce hydrogen gas (as shown in second equation above) and promotes the hydrogen (Ho) to enter the steel. pH also has an influence on amount of corrosion damage. Basically, the term "sulphides" used in oil and gas operations includes all three water-soluble species, H2S, HS- and S2-, which coexist in a sulphide-water system. Shown in the ionization chart for hydrogen sulphide in figure 1, it can be observed that molecular H2S predominates in the acidic range, where the pH is below 7. In the range of 7 to 12, monovalent HS- predominates and above pH 12, divalent S2- predominates.

The illustration in figure 1 indicates that the most important practice is that when H2S enters in alkaline mud, it reacts to form an alkaline sulphide, most often sodium sulphide.17 While elevated pH is a good way to keep H2S from causing problems under certain conditions, it does not remove it from the fluid and any drop in pH can create a significant hazard. This then brought about the advent of sulphide scavengers in the history of oil and gas operations. The only safe method for the total removal of hydrogen sulphide or soluble sulphides is with a sulphide scavenger.

8.6.1.1 Effects of Hydrogen Sulphide on Drilling Fluids

The first positive identification of soluble sulphide in the drilling fluids must be taken as a possible indication of H2S entering the fluid from some source. In the simplest case an increase in mud weight may be sufficient to control minor influxes. If levels persist chemical treatment will be required. This can take two forms; neutralization and scavenging. The two terms must not be confused. Neutralization of H2S in drilling fluid by caustic soda or lime is a common field approach. This alkalinity maintenance method has proved successful when small amounts of H2S enter the drilling fluid, for example, from drilling up pore spaces containing the gas.

This is not a suitable approach to the control on influxes of Hydrogen Sulphide. The alkalinity neutralization of H2S is instantly reversible by reduction in pH. Should the pH of the fluid decrease due to further influxes of H2S, Co2 or brine flows the sulphids may come out of solution as hydrogen sulphide gas. To ensure that such influxes are handled safely, it is essential to employ a sulphide scavenger. This term refers to any additive which is capable of reacting with one or more of the sulphide species and as a result, converting them to a more inert form. To be truly effective an irreversible and complete reaction must occur. As a mutual equilibrium exist between the three sulphide species in solution (H2S, HS- and S2-) irreversible and complete removal of one serves to removal all three.

To remove the problem it is necessary to precipitate the sulphide as an insoluble inorganic sulphide. Zinc or iron compounds are usually chosen because of the relative inertness of the sulphids formed. Zinc source (Zinc oxide or carbonate) has in the past been the preferred treatment. However, continuous treatment with Zinc carbonate will produce significant undesirable carbonate concentration (Carbonate contamination) which can adversely effect on drilling fluid rehology and fluid loss. See the below table:

Contaminant	FV	PV	YP	Gels	FL	pH	Pm	Pf	Mf	Cl^-	Ca^{2+}
Carbonate	↑		↑	↑	↗	↓	↓	↗→	↓		↓
H_2S	↑		↑	↑	↑	↓	↓	↓	↓		↗

Table 8-1 Quick reference for recognizing carbonate and H2S contamination

As a precaution against Hydrogen Sulphide influxes into the drilling fluid system should be pre-treated with Zinc Source. Zinc oxide can be used as an alternative, provided that the possible environmental payloads associated with its use are appreciated. pH and alkalinity tests (Pf/Mf) are a good first indicator of Hydrogen Sulphide influx. However, the use of scavenger is used up. It can be seen from this that while a scavenger increase safety levels, it also makes detection of small Hydrogen Sulphide influxes difficult by conventional methods. Measurement of the concentration of scavenger in the mud system is possible but the analysis method is not safe for use on the rig.

The most effective environmentally acceptable scavengers for Hydrogen Sulphide in water base muds are Zinc sources like Zinc Carbonate or Oxide, primarily because of their solubility which leads to very fast reaction rates compared to other scavengers.

Zinc Sources clearly reacts very much faster than the other scavengers. This is due to the fact that the iron in solution ensures a fast reaction with the sulphides and the chemistry in this case produces elemental sulphur as the reaction product. Elemental sulphur is completely insoluble in water. The iron is reduced from Fe+3 to Fe+2 in the process and some of this will be regenerated to produce Fe+3 when the mud comes in to contact with the air. Any regeneration of Fe+3 will aid in reducing product consumption without affecting is scavenging properties.

Recommended treatment levels of Zinc sources are as follows:

pre- emptive treatment is recommended at 1.0 lb/bbl treatment of a mud system in the event of gas influx should be made on the basis that 2.0 lb/bb; will treat approximately 100 ppm Hydrogen Sulphide.

8.6.1.1 Testing, Reporting, Treatment of Hydrogen Sulphide

The active mud system must be regularly analyzed and monitored for the presence of active soluble Sulphide with the Garrett Gas train. This test must be carried out at the same time as each complete mud check, i.e.: three times per day while drilling.

Alkalinity checks should be carried out intermittently in between these full checks and any significant reduction in the pH and Mf/Pf should be followed up by testing for the presence of active soluble sulphides with the Garrett Gas Train.

For routine checks lead acetate paper can be used to give a qualitative check. Should this test be positive then it must be followed by a qualitative analysis with a Dragger tube.

The following actions must be taken after the initial positive test result:

1. Inform the Drilling Supervisor
2. Start adding H2S Scavenger to the active mud system. Active soluble sulphides can be analysed and monitored because of their characteristic of readily regenerating hydrogen Sulphide gas in reaction with mild acid. By adding citric acid to whole mud sample in a Garrett gas train detection of Hydrogen Sulphide on lead acetate paper indicates the presence of active Sulphides in the mud. When this qualitative test proves positive a dragger tube should be used to give qualitative results.

However it must be remembered that pre-treating the mud with MILGARD Fe means that no Sulphides will be delectable until all the scavenger has reacted, i.e.: it is impossible to detect small influxes of Hydrogen Sulphide in a system pre- treating with MILGARD Fe .

8.6.1.1 Contingency Stock to be Kept on Hand on the Rig

The following Contingency stock of treatment chemicals must be kept on hand on the rig over and above the normal mud maintenance material requirements.

Zinc Carbonate or Oxide 120 Sxs 25 kg

Lime 200 Sxs 25 kg

As above mentioned Hydrogen Sulphide is a poisonous and dangerous acidic gas encountered in many formations. It can quickly deaden senses and can be fatal, even at low concentration. Hydrogen Sulphide is characterized by its typical “rotten egg” smell. For safety reasons, it should be neutralized immediately with caustic soda or lime to increase pH to >11.5 to form sulphide (S2-), then precipitated with a source of Zinc.

Neutralization with caustic soda:

2 NaOH + H2S 2Na+ + S2- + 2H2O (at pH>11.5)

Neutralization with maintaining excess lime:

2 Ca(OH)2 + H2S Ca2+ + S2- + 2H2O (at pH>11.5)

Converting Hydrogen Sulphide into just sulphide by increasing pH is not a permanent reaction. If the pH were drop into the acid region, sulphide will be converting back into the poisonous H2S form. For this reasons H2S should always be precipitated by adding a source of Zinc such as Zinc Oxide.

Removal by precipitation with treatments of (ZnO):

H2S + ZnO ZnS + H2O

if H2S gas detected in mud following action should be taken to:

3. Immediately rise up the pH to at least 11.5 to 12 with caustic soda.
4. Buffer the pH with lime.
5. Begin treatment with Zinc oxide to remove sulphide (S2-) from the system. (1 lb/bbl treatment of Zinc oxide removes approximately 1100 mg/lit of sulphide. No more than 2 lb/bbl of Zinc oxide should be used to pre-treat the system for H2S)

If it appears that H2S bleeding in (flowing) form the formation, the mud density should be increased to shut off the flow of gas into the well-bore.

8.7 Health, Safety & Enviroment

Drilling Engineer Rules & Responsibilities

- Attend Operators/Contractors safety meetings and advise on all HSE matters pertaining to our products. The requirements for Drilling Fluids Engineers is not only to attend but to contribute to Safety meetings onsite is also important, including giving
- Presentations on the fluids and chemicals being used.
- Give toolbox / pre-tour talks on chemical safety. ƒ Take part in risk assessments relevant to fluids, in particular for the first use of new systems e.g. SBM / OBM etc.,
- Follow all procedures related to HSE and follow all well site directives issued by the operator and / or drilling contractor.
- Ensure correct and updated safety posters are in place in the mud and sack rooms or mixing areas for land rigs.
- Ensure MSDS are up to date and easy to locate.

- Use engineering controls and personal protective equipment, as
- Appropriate.
- Use rig specific HSE Observation system e.g. STOP.

Hazard & Risk Assessment

The use of chemicals in the workplace presents hazards and risks to personnel involved in their handling and application. In order to minimize these hazards risk assessments are performed and HSE control measures and management systems are established to achieve the following:-

- Identify hazards
- Safely manage those hazards.
- Identify risks
- Where possible, eliminate those risks through control / engineering measures e.g. ventilation and collection of dust.
- Where not possible, manage those risks through processes including the use of PPE.
- Provide training and awareness systems designed to achieve the above and promote continuous improvement.

Hazardous Materials

- The effect on a person of a hazardous material depends on:
- The nature of the hazardous material.
- The site of the action.
- The amount of the hazardous material involved (“dose”).
- The reaction of the individual (“susceptibility”).

Hazardous Material Effects

8.7.1.1 Local Effects

- Skin and eye irritation & burns.
- Skin defatting leading to dermatitis.

8.7.1.2 Systemic Effects

- Central nervous system (headaches, nausea, dizziness).
- Cardiovascular system (CO poisoning).
- Sensitisation (allergy) and asthma.
- Teratogenic and carcinogenic.

Chemical hazardous effects may be :

- Acute - effects lasting minutes, hours or days e.g. irritation i.e. generally short term recoverable effects.
- Chronic - effects lasting weeks, months or years e.g. occupational asthma – generally long and possibly permanent effects.
- Effects may be reversible or irreversible.

Personal Protective Equipment (PPE)



- **Hard Hat**



- **Gloves, long rubber gloves for handling hazardous materials.**



- **Eye Protection, glasses, goggles or full face mask (as appropriate).**



- **Coveralls, i.e. long sleeved, fire resistant, to cover as much body skin as possible, rubber apron when handling hazardous material.**



- **Boots – recommended rubber or treated leather.**



- **Dust mask, particulate filter mask, respirator (as appropriate).**

PPE should be checked before use each time and examined on a regular basis if not in regular use. Remember – PPE requires care and maintenance.

Material Safety Data Sheet (MSDS)

An assessment is made of the physical and health hazards of each chemical supplied by our company. This information is included in a Material Safety Data Sheet (MSDS) and, in part, on container labels.

Material Safety Data Sheets contain the following information:

1. Product Identification
2. Composition, information on ingredients
3. Hazard identification
4. First aid measures
5. Fire fighting measures
6. Accidental release measures
7. Handling and storage measures
8. Exposure controls, personal protection
9. Physical and chemical properties
10. Stability and reactivity

Ensure that the following practices are followed regarding MSDS information at the workplace:

- Provide active up-to-date MSDS files covering all drilling fluid chemicals on location either on CD or paper copies. This must also include the laboratory testing chemicals.
- Distribute the MSDS files to the Well site Manager/ OIM, Medic (or designate) and sack storage / mixing areas.
- Update the MSDS file list whenever a new item is received.

Mixing Guideline

8.7.1.3 Pre – Job

- Select the chemicals to be mixed.
- Review MSDS (Material Safety Data Sheets) .
- Review the wall charts.
- Obtain appropriate tools, e.g. barrel pump.
- Inspect the condition of the chemicals to be mixed.
- Obtain appropriate PPE and WEAR IT.
- Ensure mixing personnel have clear written instructions.
- Perform a Job Hazard Analysis for any new chemicals or personnel.

8.7.1.4 Mixing

- Check that hopper is running and that the correct lines valves and pits have been selected.

- Ensure sufficient extraction and ventilation in hopper area.
- Ensure that sacks and drums are conveniently positioned and use correct lifting procedures.
- Be aware of any forklift operations.
- Clean up spills as soon as possible.
- Close the hopper any time chemicals are not being mixed.

8.7.1.5 Clean – Up

- Inform derrick man, pump man or supervisor that job is complete.
- Clean up mixing area.
- Dispose correctly of empty sacks, drums and pallet waste, e.g. Banding, wood & plastic wrapping in the correct manner.
- Ensure forklift is parked in designated area with forks lowered.

8.8 Completion Section

Completion Fluid Formulation		
Component	**Conc. (ppb)**	**Component Description**
Calcium Bromide (14.2^{ppg})	0.984 bbl/bbl	Base Fluid
Corrosion Inhibitor	1% Vol	Anti-Corrosion
H_2S Scavenger Liquid	0.5% Vol	H_2S Removal/ Anti Corrosion
Oxygen Scavenger CA	0.5% Vol	Oxygen Removal/Anto Corrosion

8.8.1.1 DRILLING FLUID FORMULATION

Kill Mud Mixing Procedure:

1. Fill tanks with calculated amount of Drill Water $0.576^{bbl./bbl.}$
2. Treat Drill Water with Caustic soda and Soda Ash by 0.3^{pbb}
3. Mix NaCl at 66^{ppb}.
4. Add Xanthan-Gum by 0.2^{pbb}.
5. Add $CaCO_3$-F (325^{ppb}) to reach MW of $\pm 106^{pcf}$
   - In case of foaming use Alcohol-Based Defoamer.

Note: Use Hi Vis pill to clean hole if necessary.
Calculation is based on per bbl. of mud.

Completion Fluid Mixing Procedure:

1. Make sure that all pits and lines involved with completion fluid are thoroughly cleaned, flushed and free of any fluid and solid.
   - Make sure that at least one large pit is kept empty for transferring ready brine into after given time for settling. After transferring one pit to empty pit, the first one is to be flushed and cleaned for accommodating the brine of next ready pit, and so on to the last pit.
   - Depends on the rig facilities and structure, mud engineer on board may find it better to prepare the fluids pit-by-pit.
2. Transfer the content of $CaBr_2$ drums into designated pits.

- Record the number of drums poured in each tank for accurate volume in it as the base of calculation for adding other additives into; do not rely on pit gauge, sounding or mud logging sensor, but have a look on them for comparison and/or correlation if found useful.

3. Mix Corrosion Inhibitor into pits at concentration of 3.5^{pbb} (1% by volume).
4. Mix H_2S Scavenger Liquid into pits at concentration of 0.5% by Volume.
5. Do not add Oxygen Scavenger CA at this stage as it should be added directly from the top of the pit, just prior to pump the fluid into the well.
6. Shut down all agitators and mixing pumps, and leave the mixed fluid in the pits for at least 12 hours to let any solid particle inside (mostly insoluble Lime) settles down.
7. Start transferring the clear fluid of the first pit to the empty pit by means of an air-powered Wilden pump.
   - Make sure that the suction hose of the Wilden pump is not put deep close to the bottom of the pit, otherwise, settled solids are sucked in.
8. After finishing the first pit, stop Wilden pump, wash, flush and clean the pit, use it as the empty pit for accommodating the fluid of second pit using the and same above method.

Note: all $CaBr_2$ brine inside the well must be retrieved by the time of displacing the well by kill mud. In case of need, filtration must be applied to retrieved amount of $CaBr_2$.
Based on well readings and the operation, mud weight may be varied.

8.8.1.2 DOWNHOLE LOSSES

As perforation had been done in well AZN#85 & there is exposure to reservoir, risk of any probable formation losses must be taken into consideration.
Losses will be minimized/avoided by the planned use of properly sized bridging/weight material (and Polymer) in conjunction with appropriate drilling practices. If seepage losses are noted on the well no specific action will be taken but close monitoring of the loss rate is to be made. For moderate loss rates addition of $CaCO_3$ 50 micron and 150 micron is recommended to ensure that there is adequate correctly sized material to act as bridging agents.

Seepage losses Less than 10 $^{bbl./hr}$.

No specific action will be taken, but close monitoring of the probable loss rate is made.

Losses from 10-30 $^{bbl./hr}$.

- Load $CaCO_3$-M (150-300 micron) in system up to 5^{ppb}

***If loss not cured and even tends to increase, then:**

- Pump 40-60^{bbl} of pill made by 7-10^{ppb} $CaCO_3$ 150-300µ & 7-10^{ppb} $CaCO_3$ 600µ

Losses from 30-50 $^{bbl./hr}$.

- Load $CaCO_3$-M (150-300 micron) in system up to 5^{ppb}
- Since some of the $CaCO_3$-M will be discarded over the shakers, $CaCO_3$-M to compensate the separated amount on the shakers.

***If loss not cured and even tends to increase, then:**

- Pump 40-60^{bbl} of pill made by
- 10-20^{ppb} $CaCO_3$ 200-300µ
- 10-20^{ppb} $CaCO_3$ 600 µ
- 10^{pbb} Oyster Shell-F
- 10^{pbb} Oyster Shell-M.

Losses from 50-100 $^{bbl/hr}$.

- Load $CaCO_3$-M (150-300 micron) in system up to 5^{ppb}
- Load $CaCO_3$-C (600-100 micron) in system up to 5^{ppb}
- Since some of the $CaCO_3$-M&C will be discarded over the shakers, add $CaCO_3$-M&C to compensate the separated amount on the shakers.

***If loss not cured and even tends to increase, then:**

- Pump 40-60^{bbl} of the following pill:

- 10-20^{ppb} $CaCO_3$ 200-300 micron
- 10-20^{ppb} $CaCO_3$ 600 micron
- 7-10^{ppb} of each $CaCO_3$ 1000 micron & $CaCO_3$ 2000 micron
- 5-10^{pbb} Oyster Shell -F
- 5-10^{pbb} Oyster Shell -M

Losses more than 100 $^{bbl/hr}$.

High concentrated LCM pill of various sizes of Calcium Carbonate and Oyster Shell-F&M&C should be pumped and spotted and gently squeezed over the loss zone.

Spotting Procedure

1. After spotting LCM pill, pull the bit above the calculated top of the pill. Wait 30 minutes and monitor the annulus.
2. After 30 minutes, attempt to fill the annulus, if not already full. If the hole can be filled, close BOP and squeeze 10^{bbl} LCM pill into formation without exceeding 200^{psi}.
3. If the hole remains stable, continue to drill ahead.
4. Should the LCM pill fail to reduce the losses or the hole cannot be filled, then mix and spot another 50^{bbl} of LCM pill.
5. Attempt to fill the annulus. If successful, close the BOP and squeeze 20^{bbl} of LCM pill into the formation without exceeding 200^{psi}.

6. If a squeeze pressure develops, hold the pressure for 1 hour before running the drill string back to bottom.

Notes:

- LCM pills must be thoroughly homogenised and viscous enough before being pumped.
- Pumping high-concentrated LCM pill is subject to use of open-end DP or Mule-Shoe.
- Remove filter screens from the drill pipe (if there is any) and if necessary, remove strainers from the mud pump suction lines.

8.8.1.3 ENGINEERING COMMENTS

Keep watching and control on well to record loss.
All $CaBr_2$ brine inside the well must be retrieved by the time of displacing the well by kill mud. In case of need, filtration must be applied to retrieved amount of $CaBr_2$.

8.8.1.4 DRILLING ISSUES

There is no drilling objective in AZN-85 workover well; mud should be compatible with reservoir and work over operation.
Solid Control equipment

- Use shale shaker to remove debris and LCM particle in case of use.
- Personal Safety

8.8.1.5 MUD TYPE

Inhibited CaBr2 brine (Calcium Bromide Brine) will be applied as completion fluid. And simple salt saturated kill mud to kill the well and clean out inside 9 5/8", 7" Liner.

Displacement Design
The combined effect of scrapers being run in the hole, the high flow rate will help to remove/disperse general debris.
At least 150^m of pill/spacer coverage or 3 min contact time at the highest pump rates should be considered.
Saturated $CaCl_2$ brine can be applied as spacer/Hi-Vis pill while displacing.
Pit levels should be monitor throughout the displacement operation to catch and prevent an influx entering the well contributing to an incident.
All $CaBr_2$ brine inside the well must be retrieved by the time of displacing the well by kill mud. In case of need, filtration must be applied to retrieved amount of $CaBr_2$.

No.	Description	Volume [bbl.]	Comment
1	Viscous pill/spacer	50	Clean pill
2	Displace to inhibited CaBr2 Brine at 106pcf (Completion Fluid)		
3	Continue pumping Completion Fluid until cleanliness specification has been met.		

All CaBr2 retrieved from well must be stored in separated tanks. All $CaBr_2$ left on surface, even dead volume of tanks, must be transferred back to drums at end of well. Not even a drop of $CaBr_2$ is not allowed to be wasted or dump in whole operation duration.

It is recommended to use desander tank as active pit of $CaBr_2$ to minimize dead volume and return path.

8.8.1.6 VOLUME ACCOUNTING & ESTIMATED REQUIRED MATERIAL

Work over /Kill Mud Fluid				AZADEGAN #127
Fluid Density (pcf)	106		**Fluid System**	**Water base Work Over Fluid /Salt Saturated**
Hole Size (inch)	Completion Phase			
Casing Depth (m)	3939		Solids Control Equip Eff.%	
Casing OD/ ID (inch)	7	6.184	Well TD (m)	4150
Liner Depth (m)	4150		Liner Lap (m)	3839
Liner OD / ID (inch)	4-1/2	3.92	Liner Length (m)	311

Volumes Information	Qty (bbl)
Surface Volume [Considering dead volumes of pits and lines while transferring]	350
Cased Hole Volue [Capacity of 3150m 9⅝" Casing + Capacity of 603m 7" Liner+Capacity of 311m 4-1/2" Liner]	850
Total New Volume Required (bbl)	**1200**

Estimated Chemical Consumption and Cost based on 106pcf MW							
PRODUCT	UNIT SIZE	UNIT PRICE (EURO)	CONC'n	Expected Consump. for Mixing	Expected Consump. for Treating	Total Qty Expected To Be Used	TOTAL COST (EURO)
Drill Water	bbl	N/A	0.576 bbl/bbl	700		700	N/A
Lime	25 Kg	N/A	0.3	7	25	32	N/A
Soda Ash	50 Kg	N/A	0.3	4	16	20	N/A
Sodium Chloride	1 MT	N/A	63	37	18	55	N/A
Xanthan-Gum	25 Kg	N/A	0.2	5	20	25	N/A
Calcium Carbonate	1 MT	N/A	325	190	30	220	N/A
Hematite	1.5 MT	N/A	As Needed	As Needed	As Needed	As Needed	N/A

*Above calculation are based on normal hole condition
*Above Calculated is Based on **106pcf**

Completion Fluid				AZADEGAN #127
Fluid Density (pcf)	103 - 106		**Fluid System**	**CaBr2 Brine**
Hole Size (inch)	Completion Phase			
Casing Depth (m)	3939		Solids Control Equip Eff.%	
Casing OD/ ID (inch)	7	6.184	Well TD (m)	4150
Liner Depth (m)	4150		Liner Lap (m)	3839
Liner OD / ID (inch)	4-1/2	3.92	Liner Length (m)	311

Volumes Information	Qty (bbl)
Surface Volume [Considering dead volumes of pits and lines while transferring]	250
Cased Hole Volue [Capacity of 3782m 9⅝" Casing + Capacity of 426m 7" Liner]	850

Total New Volume Required (bbl)	1100

Estimated Chemical Consumption and Cost based on 74-75pcf MW							
PRODUCT	UNIT SIZE	UNIT PRICE (EURO)	CONC'n	Expected Consump. for Mixing	Expected Consump. for Treating	Total Qty Expected To Be Used	TOTAL COST (EURO)
Calcium Bromide (14.2 ppg)	55 Gal	N/A	0.984bbl /bbl	910	62	972	N/A
Corrosion Inhibitor	55 Gal	N/A	1% Vol	9	1	10	N/A
H2S Scavenger Liquid	55 Gal	N/A	0.5% Vol	5	3	8	N/A
Oxygen Scavenger CA	55 Gal	N/A	0.5% Vol	5	3	8	N/A

*Above calculation are based on normal hole condition

*Above Calculated is based on **106**^{pcf}

8.8.1.7 ESTIMATED CONTINGENCY MATERIAL

During all the phase keep a minimum safety stock of below items:

Chemical Type	Unit	Qty	Function
$CaCO_3$ M & C	1 MT BB	10 of each	Loss Circulation Material
Pipe-Freeing Agent	55 Gal DM	4	Environment-Friendly Stuck Pipe Agent
Lime	25 Kg Sx	80	CO_2 Contamination
Zinc Carbonate	25 Kg Sx	120	H_2S Scavenger
Sodium Bicarbonate	25 Kg Sx	40	Cement Contamination
Citric Acid	25 Kg Sx	40	Cement Contamination
Defoamer A	55 Gal DM	4	Anti-Foam
Bio-Cide	55 Gal DM	4	Anti-Bacteria
Oyster shell F/M/C	25 Kg Sxs	160each	Loss circulation material

Drilling Detergent	55Gal	4	Bit Balling Reducer
Oxygen Scavenger S	25Kg	30	Oxygen Scavenger, Polymer Extender
pH Buffer Agent	55Gal	4	Polymer Temperature Stabilizer
Corrosion Inhibitor	55Gal	4	Corrosion Inhibitor
Lubricant	55Gal	8	Biodegradable lubricator

8.8.1.8 MATERIAL REQUIREMENTS FOR EACH WELL

Item	QTY	Unit	Description	Function
1	972	55 Gal Dr	Calcium Bromide (14.2 ppg)	CaBr2 brine
2	55	1 MT	Salt	Weighting Agent
3	224	1 MT	Calcium Carbonate 75 Mic	Weighting Agent
4	32	25 Kg Sx	Caustic Soda	PH Control
5	20	50 Kg Sx	Soda Ash	Hardness reducer
6	25	25 Kg Sx	Xanthan-Gum	Viscosifier
9	80	25 Kg Sx	Lime	PH Control
10	160	25 Kg Sx	Shell Fish	LCM
11	160	25 KG Sx	Limestone Chips	LCM
12	10	55 Gal Dr	Corrosion Inhibitor	Completion Fluid Corrosion Inhibitor
13	8	55 Gal Dr	H2S SCAV	Completion Fluid H2S SCAV
14	8	55 Gal Dr	OX SCAV	Completion Fluid OX SCAV
15	1	55 Gal Dr	BIO-CIDE	Anti-bacterial
16	12	55 Gal Dr	PH BUFFER	PH Stabilizer

8.9 Cement Program

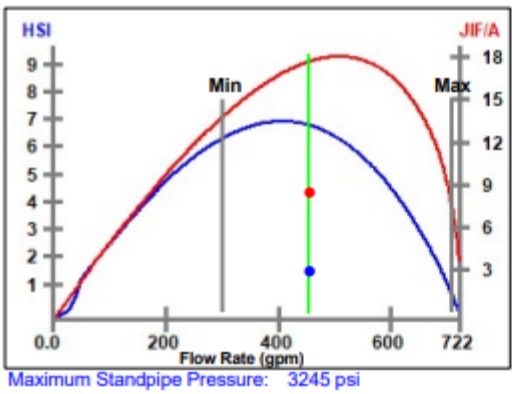
Cement program will be issued in due time.

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9. Bit & Hydraulic Program

9.1 8 – ½" Hole Section

Hydraulics Optimization Bit Run Report



Maximum Standpipe Pressure: 3245 psi

Well Information

Well Name:
AZNS-127
Drilling Contractor:
OEOC OEOC-208
Operator:
MSA
Prepared For:

Location:

Prepared By:
BBK

Hydraulic Details @ 3939 m

Flow Rate / Hole Diameter	52.9 gpm/in
System Pressure Loss	1740 psi
Downhole Motor / MWD Pressure Drop	450 psi
Bit Nozzle Pressure Drop	350 psi
Annular Pressure Drop	272 psi
Pump Input Hydraulic Power (M.E.=85%)	549 hp
Bit Hydraulic Horsepower	92 hp
Jet Impact Force	491 lbs
Impact Force/ Hole Area	8.66 lbs/in²
Hydrostatic Head	7621 psi
Equivalent Circulating Density	88.03 lbs/ft³
Cuttings Slip Velocity (Chip Size = .30 in)	13.7 m/min
Surface Equipment Length	46 m X 76.2 mm
Buoyant Collar Weight	35451 lbs

Bit Run Data

Bit Number	1
Bit Size	8 1/2 in
Bit Type	PDC

Recommended Hydraulics

Nozzles	13\13\13\13\13\13 32nds
Total Flow Area	0.77 in²

	IN	OUT
BIT HSI	1.53	1.62
JIF/Area	8.15	8.66 lbs/in²
Nozzle Velocity	57	57 m/sec
Flow Rate	450	450 gpm
SPP	1727	2090 psi
% Pressure @ Bit	19.1%	16.8%
Measured Depth	3337	3939 m
TVD	3337	3939 m
Mud Weight	80	85 lbs/ft³
Plastic Viscosity	10	15 cp
Yield Point	12	20 lbs/100 ft²

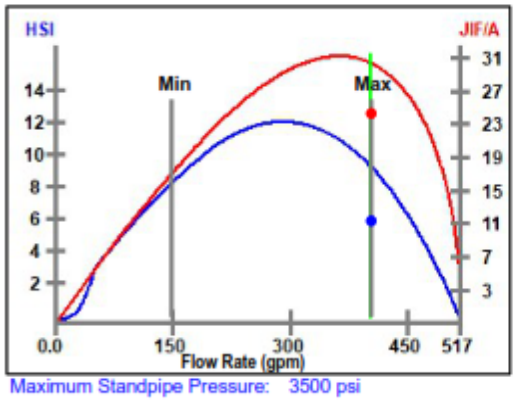
Drill String And Annular Velocity Data @ 3939 m

Depth m	OD X ID in	Length m	Hole Size in	Bore Loss psi	Ann. Loss psi	Ann. Vel. m/min	Est. Chip Vel. m/min	Crit Vel. m/min	Flow Type
0-3237	DP 5 x 4.24	3237	8.535	602.9	219.6	70	57	87	LAM
3237-3664	DP 5 x 4.24	427	8.500	79.5	29.6	71	57	87	LAM
3664-3691	HWDP 5 x 3	27	8.500	26.6	1.9	71	57	87	LAM
3691-3696	6.25 x 2.25	5	8.500	19.7	.8	101	88	92	TRANS
3696-3920	HWDP 5 x 3	224	8.500	221.1	15.5	71	57	87	LAM
3920-3929	DC 6.75 x 3	9	8.500	8.9	2.4	126	112	95	TRANS
3929-3939	DC 6.75 x 2.813	10	8.500	13.5	2.7	126	112	95	TRANS

9.2 6 – 1/8" Hole Section

Reed-Hycalog Schlumberger

Hydraulics Optimization Bit Run Report



Maximum Standpipe Pressure: 3500 psi

Hydraulic Details @ 4150 m

Flow Rate / Hole Diameter	65.1 gpm/in
System Pressure Loss	2263 psi
Bit Nozzle Pressure Drop	776 psi
Annular Pressure Drop	470 psi
Pump Input Hydraulic Power (M.E.=85%)	707 hp
Bit Hydraulic Horsepower	181 hp
Jet Impact Force	724 lbs
Impact Force/ Hole Area	24.57 lbs/in²
Hydrostatic Head	10013 psi
Equivalent Circulating Density	110.98 lbs/ft³
Cuttings Slip Velocity (Chip Size = .30 in)	10.45 m/min
Surface Equipment Length	76 m X 76.2 mm
Buoyant Collar Weight	27420 lbs

Well Information

Well Name:
AZNS-127
Drilling Contractor:
MSA OEOC 208
Operator:
OEOC
Prepared For:
PPL
Location:
Pad 54
Prepared By:
MSA/MSA

Bit Run Data

Bit Number	2
Bit Size	6 1/8 in
Bit Type	PDC

Recommended Hydraulics

Nozzles	15\15\15 32nds
Total Flow Area	0.5177 in²

	IN	OUT
BIT HSI	3.51	6.13
JIF/Area	16.63	24.57 lbs/in²
Nozzle Velocity	64	75 m/sec
Flow Rate	338	399 gpm
SPP	1841	3039 psi
% Pressure @ Bit	28.5%	25.5%
Measured Depth	3839	4150 m
TVD	3839	4150 m
Mud Weight	100	106 lbs/ft³
Plastic Viscosity	10	15 cp
Yield Point	14	24 lbs/100 ft²

Drill String And Annular Velocity Data @ 4150 m

Depth m	OD X ID in	Length m	Hole Size in	Bore Loss psi	Ann. Loss psi	Ann. Vel. m/min	Est. Chip Vel. m/min	Crit Vel. m/min	Flow Type
0-3805	DP 5 x 4.24	3805	8.535	682.8	289.4	62	52	84	LAM
3805-3806	DC 4.75 x 2.25	1	8.535	3.8	.1	59	49	83	LAM
3806-3909	DC 3.5 x 2.76	103	8.535	146.4	4.2	49	39	82	LAM
3909-3921	DC 3.5 x 2.76	12	6.125	17.1	2.0	118	108	86	TRANS
3921-3976	HWDP 3.5 x 2.25	55	6.125	209.2	9.4	118	108	86	TRANS
3976-4003	DC 4.75 x 2.25	27	6.125	102.7	25.6	199	189	94	TURB
4003-4011	DC 4.75 x 2.25	8	6.125	30.4	7.6	199	189	94	TURB
4011-4148	DC 4.75 x 2.25	137	6.125	521.2	129.8	199	189	94	TURB
4148-4150	4.75 x 2.25	2	6.125	7.6	1.9	199	189	94	TURB

10. Casing Program

10.1 Material Selection

Casing, Liner and Tubing Material selection for South Azadegan oilfield will be describe in the following.

Corrosive Environment Analysis and Strength Check

Based on the data of offset wells in this oilfield, there is a certain degree of corrosive problems in the 4 layers to be produced. Corrosive medium is principally composed of 2 parts: one is the corrosive gases like H2S and CO2 existing in the produced fluid; and the other is the high salinity formation water. The contents of main corrosive fluids in offset wells are shown in below Table:

Well No	*Reservoir formation*	*H₂S from PVT*	*CO₂ from PVT (mol%)*	*H₂S from Seperator*	*CO₂ from Separator*
AZN2	Sarvak	0.13	6.71	7.93/150psi	0.06/150psi
AZN4	Sarvak	0.06	5.44	0.02/150	6.32
	Kazhdumi	0	2.62	0	2.99/150psi
	Gadvan		2.64		
AZN6	Gadvan		2.47		2.566/400psi

1) Corrosion in this oilfield is a result of combined action of H2S, CO2 and high salinity formation water. The corrosive mechanism is complicated, and the corrosive rate on the casing is difficult to determine. Based on the NACE standard and the data from the Table, the producers result in severe corrosive environment. However, by the research on development conditions and technical indices, it can be found that although the corrosive fluid contents are high in the oilfield, the corrosive environment is not extremely serious as the following reasons:

2) In general, the temperature that cause the SSC/SCC is concentrated in the range from 20℃ to 30℃. The average depth of the horizontal section is 2,730m with the reservoir temperature of 96℃, and tensional stress on the

casing does not exist in the horizontal section. Therefore, the influence of SSC/SCC may not be considered for the casing/liner.

3) Based on the RMDP, the production GOR in the first development phase should not be higher than the initial GOR, therefore, the flowing pressure is higher than the bubble point pressure all the time, that is, degassing problems do not exist in the horizontal section and some vertical section, the H2S and CO2 have no chance of separation out of oil phase, and it is lack of prerequisite for casing corrosion.

4) Based on above all, the corrosive environment of the casing string is not severe. The optimal selection for the casing string can be stated briefly as follows.

Material Selection of Casing String

In this oilfield, the corrosion of production string obviously occurs in 2 regions. One is the region where the pressure is above the bubble point; in which degassing problems do not exist, and the corrosion environment is relatively weak. Another one is the region where the pressure is below the bubble point; because of the decline of pressure, the concentration of corrosive gas will increase and the corrosion is relatively serious. Considering the corrosive environment is not serious in the low water cut production stage and the production practice in this oilfield has proved the L80 is fit for it, So, the L80 should be taken as the material of 9-5/8″ casing, 7″, 5 1/2″ and 4-1/2″ liner. In order to prevent the hazard of gas leakage after production degassing occurs, 9-5/8″ casing, 7″ and 4-1/2″ liner should be Gas tight thread. 2) The 110S is selected as the material of 9-5/8″, 7″ and4-1/2″ casing rather than L-80 due to the high strength requirement in Fahliyan wells, what’s more the 110S is fit for sour gas environment

10.2 Casing/Liner Design Factors

Table 10-1: Casing/Liner Design Factors

Design Factors			
Pipe Body		**Connection**	
Burst	1.1	Burst	1.1
Tension	1.6	Tension	1.6
Compression	1.300	Compression	1.3
Collapse	1.1		
Triaxial	1.25		

10.3 Casing/Liner Characterize Overview

Table 10-2: Casing/Liner Characterize Overview

Casing Size (in)	Depth (m)		Weight (PPF)	Grade	Connection	Burst (psi)	Collapse (psi)	A.S. F	Triaxial	Formation Remark
	From	To								
20”	0	146	94	K-55	BTC	Already run and set				AJ
13-3/8”	0	1441	68	N-80	BTC	Already run and set				GS
9-5/8”	0	3909	53.5	T-95	TSH Blue TPCQ	Already run and set				GD
7”	3777	4226	29	110SS	TPCQ	Already run and set				FH
7”	3150	3939	29	110S	TPG2	**11220**	**8530**	-	-	GD
4 ½”	3839	4150	13.5	110S	TPG2	**12410**	**10690**	-	-	FH

7” Liner Specification (8-1/2” Hole Section)

Table 10-3: 7" Liner Specification

Design Parameters Data (7" Production Liner)	
Design Factor (Pipe) - Axial Tension:	1.600
Design Factor (Pipe) - Axial Compression:	1.300
Design Factor (Coupling) - Axial Tension:	1.600
Design Factor (Coupling) - Axial Compression:	1.300
Design Factor (Pipe) - Burst:	1.100
Design Factor (Coupling) - Burst:	1.100
Design Factor (Pipe) - Collapse:	1.100
Design Factor (Pipe) - Triaxial:	1.250
Minimum Internal Drift Diameter:	5.875 in
Single External Pressure Profile:	Yes
Temperature Deration:	No
Limit to Fracture at Shoe:	Yes
Buckling:	No
Use Burst Wall Thickness in Triaxial:	No

Initial Conditions (7" Production Liner)	
Mix-Water Density:	8.33 ppg
Lead Slurry Density:	12.76 ppg
Tail Slurry Density:	15.80 ppg
Tail Slurry Length:	0.00 m
Displacement Fluid Density:	11.36 ppg
Slackoff Force:	0 lbf
Temperatures:	Default
Vertical Depth (m)	**Temperature (°F)**
3149.76 m	234.35 °F
3419.46 m	247.62 °F
3568.22 m	254.94 °F
3908.73 m	271.70 °F

Burst Loads Data (7" Production Liner)	
Drilling Load:	**Pressure Test**
Test Pressure:	1450.00 psi
Mud Weight:	11.36 ppg
Assigned External Pressure:	Fluid Gradients (w/ Pore Pressure)
Drilling Load:	**Green Cement Pressure Test**
Test Pressure:	2500.00 psi
Mud Weight at Shoe:	11.36 ppg
TOC, MD:	3150.00 m
Lead Slurry Density:	12.76 ppg
Tail Slurry Density:	15.80 ppg
Tail Slurry Length:	0.00 m
Displacement Fluid Density:	11.36 ppg
Float Collar Depth, MD:	3910.53 m
Drilling Load:	**Drill Ahead (Burst)**
Hanger Depth, MD:	3150.00 m
TOC, MD:	3150.00 m
Shoe Depth, MD:	3939 m
MW next hole section:	11.36 ppg
ECD:	0.30 ppg
Assigned External Pressure:	Fluid Gradients (w/ Pore Pressure)
External Pressure:	**Fluid Gradients (w/ Pore Pressure)**
TOC, MD:	3150.00 m
Prior Shoe, MD:	3337.00 m
Mud Weight Above TOC:	11.36 ppg
Fluid Gradient Below TOC:	8.33 ppg
Wellhead Pressure:	0.00 psi
Pore Pressure In Open Hole:	No

Collapse Loads Data (7" Production Liner)	
Drilling Load:	**Full/Partial Evacuation**
Mud Weight:	11.36 ppg
Mud Level, MD:	3500.00 m
Assigned External Pressure:	Fluid Gradients (w/ Pore Pressure)
Drilling Load:	**Cementing**
Mud Weight at Shoe:	11.36 ppg
TOC, MD:	3150.00 m
Lead Slurry Density:	12.76 ppg
Tail Slurry Density:	15.80 ppg
Tail Slurry Length:	0.00 m
Displacement Fluid Density:	11.36 ppg
Float Collar Depth, MD:	3915 m
Assigned External Pressure:	Fluid Gradients (w/ Pore Pressure)
Drilling Load:	**Drill Ahead (Collapse)**
Hanger Depth, MD:	3150.00 m
TOC, MD:	3150.00 m
Shoe Depth, MD:	3939 m
MW next hole section:	11.36 ppg
ECD:	0.30 ppg
Assigned External Pressure:	Fluid Gradients (w/ Pore Pressure)
External Pressure:	**Fluid Gradients (w/ Pore Pressure)**
TOC, MD:	3150.00 m
Prior Shoe, MD:	3337.00 m
Fluid Gradient Above TOC:	11.36 ppg
Fluid Gradient Below TOC:	11.36 ppg
Wellhead Pressure:	0.00 psi
Pore Pressure In Open Hole Below TOC:	No

4 – 1/2” Liner Specification (5 – 7/8” Hole Section)

Table 10-4: 4-1/2" Liner Specification

Design Parameters Data (4 1/2" Production Liner)	
Design Factor (Pipe) - Axial Tension:	1.600
Design Factor (Pipe) - Axial Compression:	1.300
Design Factor (Coupling) - Axial Tension:	1.600
Design Factor (Coupling) - Axial Compression:	1.300
Design Factor (Pipe) - Burst:	1.100
Design Factor (Coupling) - Burst:	1.100
Design Factor (Pipe) - Collapse:	1.100
Design Factor (Pipe) - Triaxial:	1.250
Minimum Internal Drift Diameter:	3.116 in
Single External Pressure Profile:	Yes
Temperature Deration:	No
Limit to Fracture at Shoe:	Yes
Buckling:	No
Use Burst Wall Thickness in Triaxial:	No

Initial Conditions (4 1/2" Production Liner)	
Mix-Water Density:	8.33 ppg
Lead Slurry Density:	15.80 ppg
Displacement Fluid Density:	14.04 ppg
Slackoff Force:	0 lbf
Temperatures:	Default
Vertical Depth (m)	**Temperature (°F)**
3809.20 m	266.80 °F
4223.70 m	**287.20 °F**

Burst Loads Data (4 1/2" Production Liner)	
Drilling Load:	**Gas Kick Profile**
Influx Depth, MD:	4150 m
Kick Volume:	50.0 bbl
Kick Intensity	0.50 ppg
Maximum Mud Weight:	14.04 ppg
Kick Gas Gravity:	0.70 (0.1419 psi/ft @ 231.64 °F & 10467.98 psi)
Fracture Gradient (MD= 4225.50 m):	12735.30 psi
Fracture Margin of Error:	0.00 ppg
Drill Pipe OD:	3.500 in
Collar OD:	
Collar Length:	
Assigned External Pressure:	Fluid Gradients (w/ Pore Pressure)
Drilling Load:	**Pressure Test**
Test Pressure:	1450.00 psi
Mud Weight:	14.04 ppg
Assigned External Pressure:	Fluid Gradients (w/ Pore Pressure)
Drilling Load:	**Green Cement Pressure Test**
Test Pressure:	2500.00 psi
Mud Weight at Shoe:	14.04 ppg
TOC, MD:	3839.00 m
Lead Slurry Density:	15.80 ppg
Displacement Fluid Density:	14.04 ppg
Float Collar Depth, MD:	4126 m
External Pressure:	**Fluid Gradients (w/ Pore Pressure)**
TOC, MD:	3839.00 m
Prior Shoe, MD:	3939 m
Mud Weight Above TOC:	14.04 ppg
Fluid Gradient Below TOC:	8.33 ppg
Wellhead Pressure:	0.00 psi
Pore Pressure In Open Hole:	No

Collapse Loads Data (4 1/2" Production Liner)	
Drilling Load:	**Full/Partial Evacuation**
Mud Weight:	14.04 ppg
Mud Level, MD:	3900.00 m
Assigned External Pressure:	Fluid Gradients (w/ Pore Pressure)
Drilling Load:	**Cementing**
Mud Weight at Shoe:	14.04 ppg
TOC, MD:	3839.00 m
Lead Slurry Density:	15.80 ppg
Displacement Fluid Density:	14.04 ppg
Float Collar Depth, MD:	4126 m
Assigned External Pressure:	Fluid Gradients (w/ Pore Pressure)
Production Load:	**Above/Below Packer**
Packer Depth, MD:	3115.00 m
Perforation Depth, MD:	4050.00 m
Pore Pressure at Perforation Depth:	9205.87 psi
Density Above Packer:	14.17 ppg
Density Below Packer:	7.08 ppg
Fluid Drop Above Packer:	No
Assigned External Pressure:	Fluid Gradients (w/ Pore Pressure)
External Pressure:	**Fluid Gradients (w/ Pore Pressure)**
TOC, MD:	3839 m
Prior Shoe, MD:	3939 m
Fluid Gradient Above TOC:	14.04 ppg
Fluid Gradient Below TOC:	14.04 ppg
Wellhead Pressure:	0.00 psi
Pore Pressure In Open Hole Below TOC:	No

11. BOPs, Wellhead and Completion Schematic

11.1 BOPs Specification and Configuration

Table 11-1: AZNS-127 Well BOP Specification

Casing String	Wellhead	BOP	Hole Size
13-3/8" Intermediate casing	Section "B" Casing Head Spool 20-3/4", 3K BTM x 13-5/8", 5K TOP	13-5/8", 5K Annular BOP 13-5/8", 10K Double Rams (Pipe Ram & Shear Ram) 13-5/8", 10K Single Ram 13-5/8", 10K Drilling Spool 13-5/8", 10K Spacer Spool Adapter Flange 13-5/8", 5K BTM x 10K Top	12-1/4″
9-5/8" Production casing	Section "C" Tubing Head Spool 13-5/8", 5K BTM x 11", 10K TOP	13-5/8", 5K Annular BOP 13-5/8", 10K Double Rams (Pipe Ram & Shear Ram) 13-5/8", 10K Single Ram 13-5/8", 10K Drilling Spool 13-5/8", 10K Spacer Spool Adaptor Flange 11", 10K BTM x 13-5/8" 10K TOP	8-1/2"

12-1/4” Hole Section BOPs Arrangement

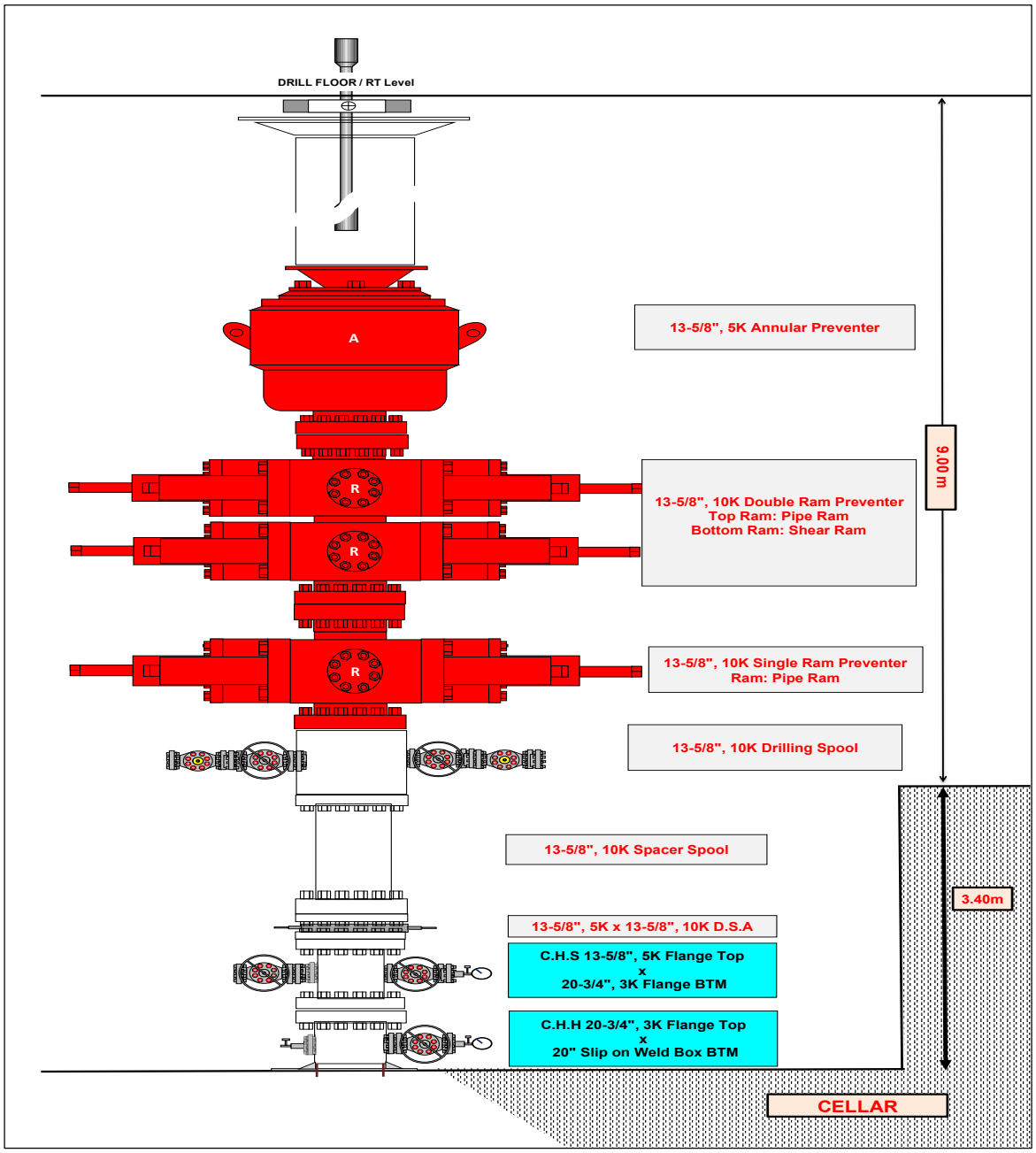


Figure 11-1: 12-1/4" Hole Section BOPs

8-1/2" Hole Section BOPs Arrangement

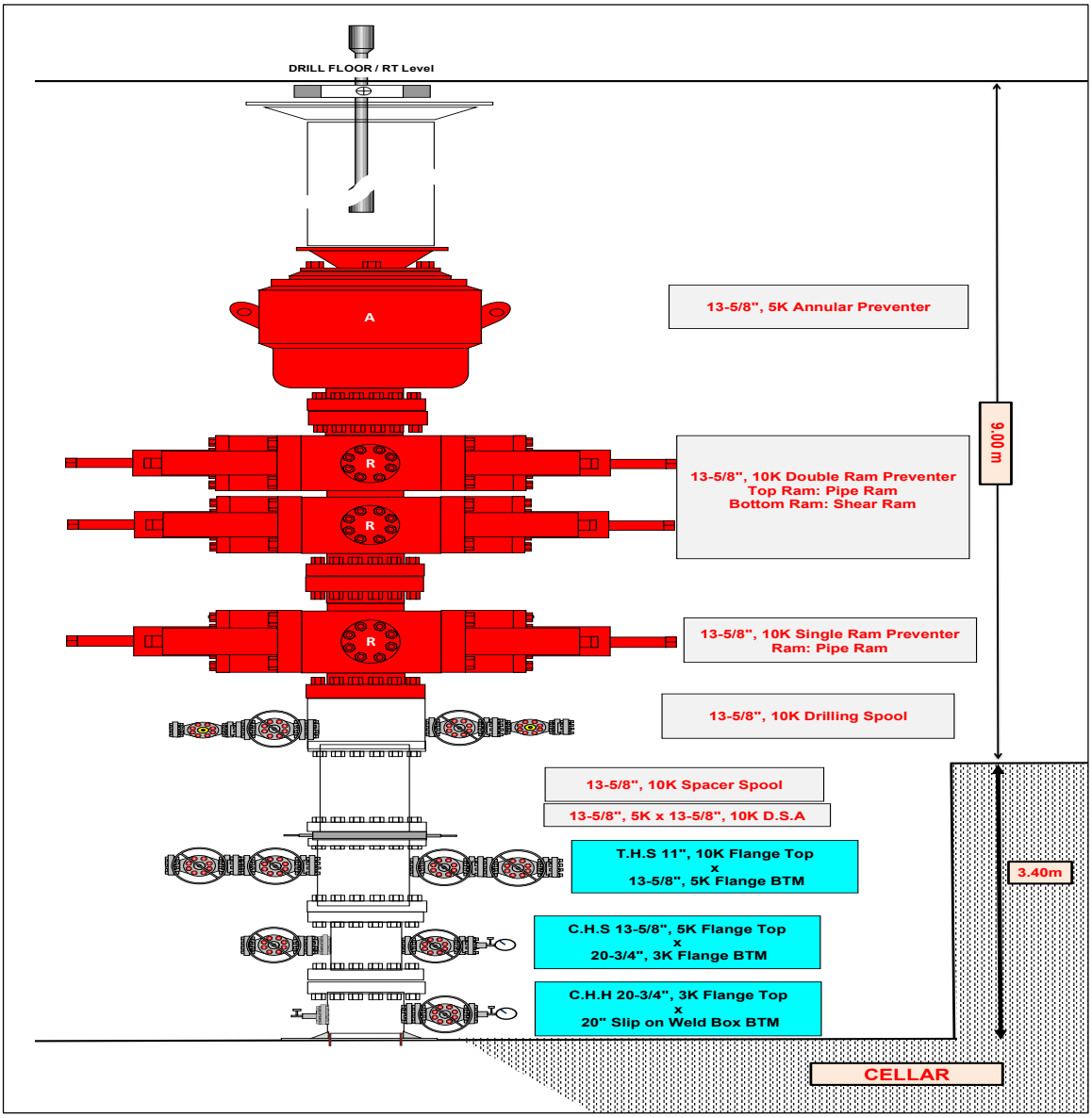


Figure 11-2: 8-1/2" Hole Section BOPs

Kill and Choke Manifold Arrangement

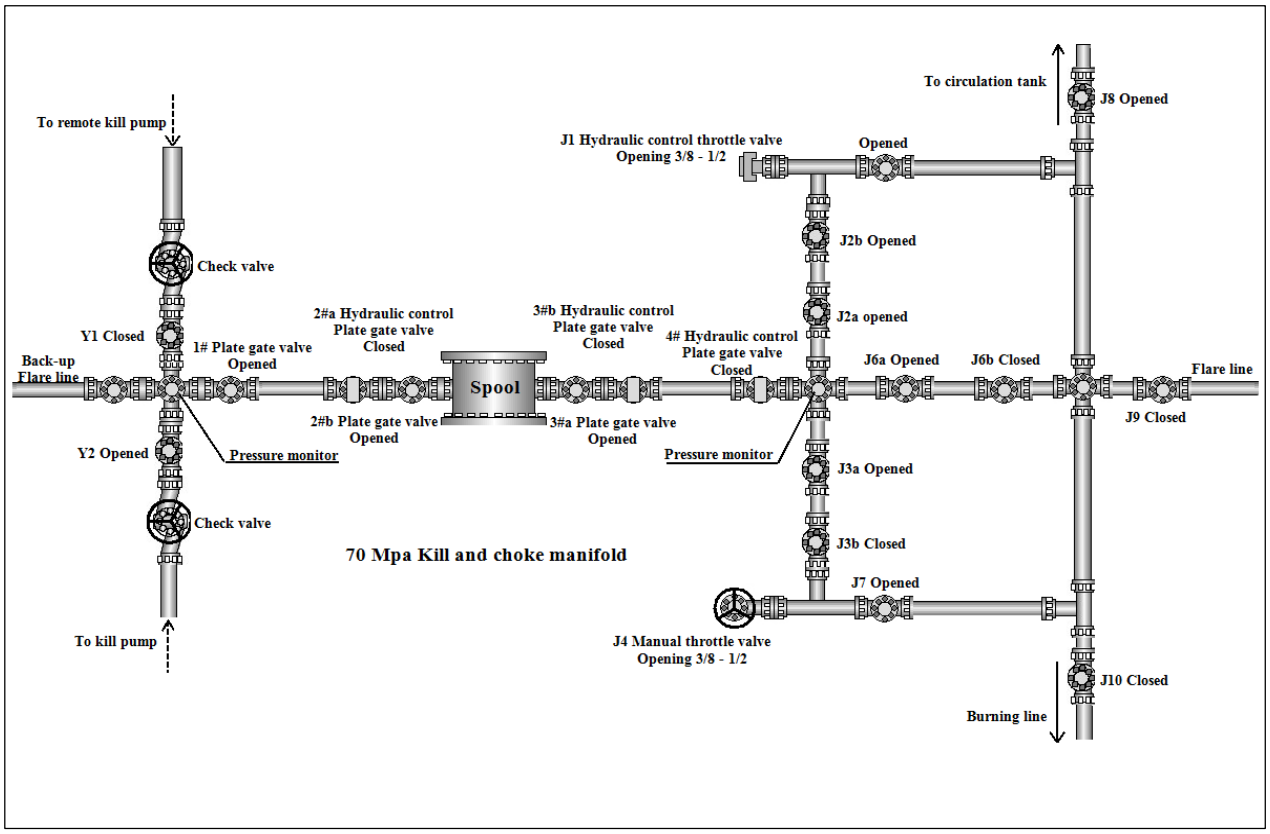


Figure 11-3: Kill and Choke Manifold Arrangement

11.2 Wellhead

Wellhead Material Selection Criteria

A. Selection of material class of Wellhead & X-mas Tree:

Table 11-2: H2S & CO2 Content in AZNS Offset Wells

Well No	Reservoir formation	H2S from PVT (mol%)	CO2 from PVT (mol%)
AZN2	Sarvak	0.13	6.71
AZN4	Sarvak	0.06	5.44
	Kuzhdumi	0	2.62
	Gadvan		2.64
AZN6	Gadvan		2.47

Considering the information above, API 6A 19th & NACE MR-0175 (2003) /ISO 1516, and Cl- content which is being most concerned, the South AZADEGA X-mas tree material and the tubing hanger should be FF0.5 class. As to the casing head, it's barely exposed to resident fluid, or only for quite a short time, when it comes to desert working condition, according to API 6A19th & NACE-MR-0175 (2003) /ISO 1516, the tubing spool should be EE0.5, the casing head material class should be DD0.5 (ANSI 4130) , stainless steel cladding in ring groove

B. Selection of PSL (product specification level):

According to API 6A 19th, The final PSL for wellhead and X-mas tree are as below:

Table 11-3: PSL for Wellhead and X-mas Tree

Item	Specification
Casing Head Housing assembly	PSL-1
Casing spool assembly	PSL-1
Tubing spool assembly	PSL-3
Tubing hanger	PSL-3G
X-Tree assembly	PSL-3G

C. Selection of PR (Performance Requirement)

According to API Specification 6A 19th edition, performance requirements are specific and unique to the product in the as-shipped condition. All products shall be designed to perform according to the requirements of API 6A clauses 4 and clauses 10 while in the pressure and temperature ranges and corrosion environment. Other requirements include load capability, cycles, and operating force or torque. There are two performance requirement levels: PR1 and PR2. Valves operating as safety valves shall be of the PR2 performance level and meet the requirements of Annex I in API 6A For SA wells, the guideline for PR selection is that all parts not contacting well fluids in normal production period are PR1, all parts contacting well fluids are PR2 as follow, Casing head housing and casing head spool assembly - PR1. Tubing head spool, tubing hanger and X-mas tree - PR2.

D. Temperature Class

Environment temp:
Well temp: 0.0244×4100+27.384=127℃
According to API 6A 19th, temp should be U (-18℃-121℃)

Table 11-4: Temperature Class for Wellhead and X-mas Tree in AZNS Oilfield

API TEMP. CLASSIFICATION	OPERATING RANGE ◦C
U	-18 TO 121

E. Designed Pressure

Base on wells data of this project, and API spec 6A 19TH, considering the planning casing program, Wellhead and X-mas tree system for the project is designed as followed:

Table 11-5: Wellhead and X-mas tree Designed Pressure in AZNS Oilfield

Item Well type	Casing Head Housing	Casing Spool	Tubing Spool	X-mas Tree
Savark	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 5K × 13 5/8" 5 K	5K
Gadvan / Fahliyan	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 10K × 13 5/8" 5 K	10K
Water disposal wells	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 5K × 13 5/8" 5K	5K

Item	Specification
Casing Head Housing assembly	MC:DD-0.5 3000PSI PSL-2 PR-2, TC: U
Casing Spool assembly	MC:DD-0.5 5000PSI PSL-2 PR2, TC: U
Tubing Spool assembly	MC: FF-05 5000PSI PSL-3, PR-2, TC: U (Savark & Water disposal wells)
	MC: FF-0.5 10000PSI PSL-3G, PR-2, TC: U (K&G&F wells)
Tubing hanger	MC: FF 0.5 disposal wells 5000PSI PSL-3G, PR2, TC: U (Savark & water)
	MC: FF 0.5 10000PSI PSL-3G, PR2, TC: U (K&G&F wells)
X-Tree assembly	MC: FF 0.5 5000PSI PSL-3G, PR-2, TC: U (Savark & Water disposal wells)
	MC: FF 0.5 10000PSI PSL-3G, PR-2, TC: U (K&G&F wells)

Wellhead Schematic

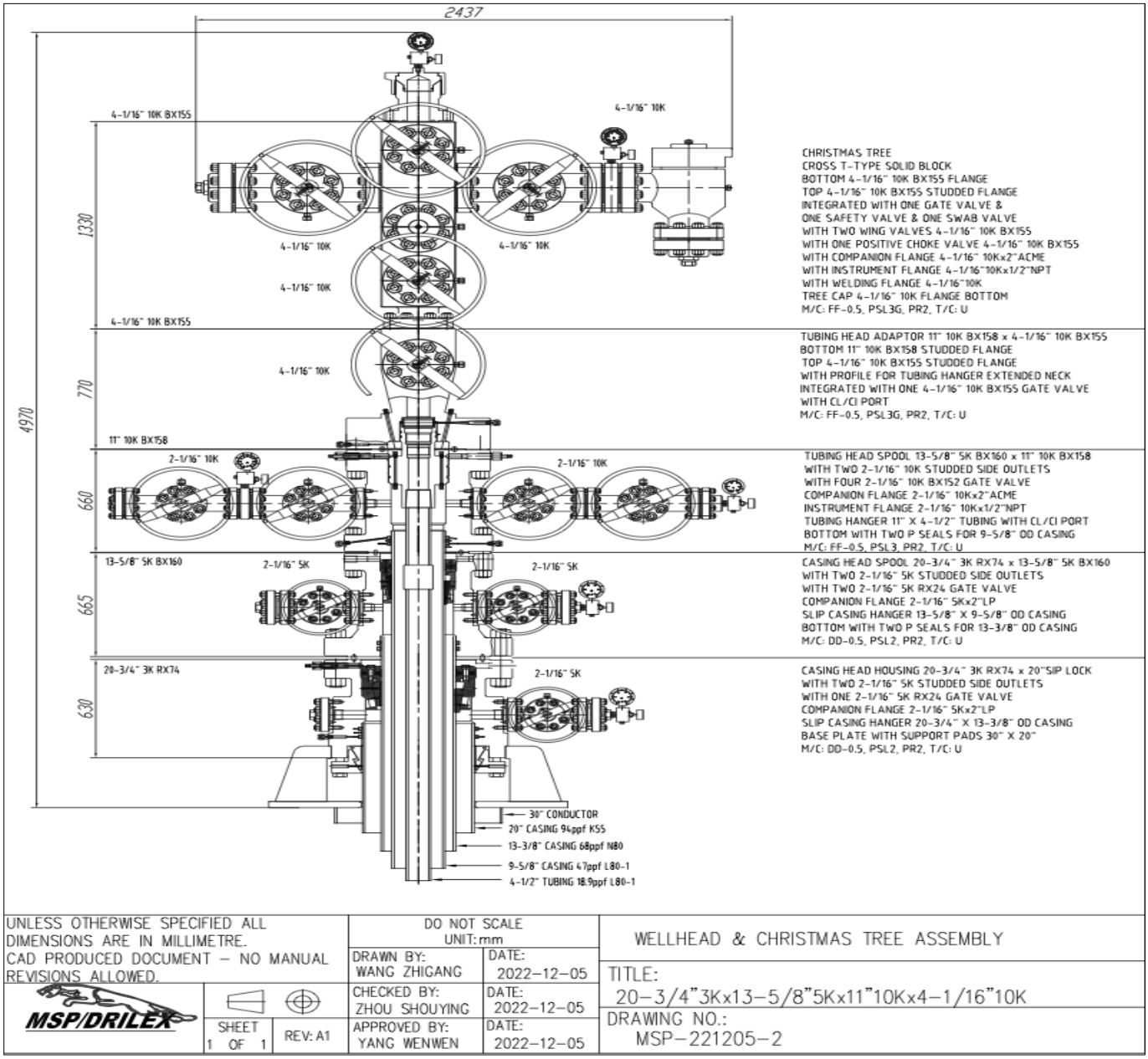


Figure 11-4: AZNS-127 Well Head Schematic

Wellhead Specification

Table 11-6: AZNS-040 Wellhead Specification

20-3/4" x 13-5/8" x 11" x 4-1/16" x 10,000 psi WP for Fahliyan wells	
Section A	**Casing head housing** for 20" (94lb/ft) OD casing x 20 3/4" 3,000 psi API flange **has hold down bolts for wear bushing,** with 2 x 2- 1/16" 3000 psi studded side outlets, test ports and seals. One side outlet has **one** 2 1/16"x 3,000 psi manual gate valve, 2" LP threaded companion flange and 2" bull plug threaded for ½" NPT and fitted with 0-3,000 psi gauge and needle valve and, the other side to have 2 1/16" x 3,000 psi x 2" LP threaded flange and 2" bull plug. Casing slip and seal assembly self-energizing to hang 13 3/8" (68lb/ft) OD casing. This section is installed on 20" casing stub by way of **slip lock** with landing plate on 30" conductor pipe. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 630mm** Material: Forged alloy steel 4130 Automatic casing slip material: Low alloy steel 4130
Section B	**Casing head spool** with 13 5/8" 5,000 psi top flange x 20 3/4" 3,000 psi bottom flange has **hold down bolts for wear bushing** and bottom preparation to seal on 13 3/8" OD casing with test ports to test annular space and casing seals. One side outlet has **one** 2 1/16"x 5,000psi manual gate valve, 2" LP threaded companion flange and 2" bull plug on one side and, the other side has **one** 2 1/16"x 5,000psi manual gate valve, 2" LP threaded companion flange, 2" bull plug threaded for ½" NPT and fitted with 0-5,000psi gauge and needle valve. Casing slip and seal assembly self-energizing to hang 9 5/8" (47lb/ft) OD casing. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 665mm** Material: Forged low alloy steel 4130 13 5/8"x 9 5/8" automatic casing slip material: Low alloy steel 4130
Section C	**Tubing head spool** with 11" 10,000 psi top flange x 13 5/8" 5,000 psi bottom flange has **hold down bolts for wear bushing** and bottom preparation to seal on 9 5/8" OD casing with test ports to test annular space and casing seals C/W T.D.Screws with 410SS material Side outlets have 1.900" sharp vee VR threads on both sides, to have **two** 2 1/16"x 10,000psi manual gate valve, 2"ACME 10K female thread companion flange, 2" bull plug threaded for ½"NPT and fitted with 0-10,000psi gauge and needle valve on one side and, the other side to have the first 2 1/16"x 10,000psi manual gate valve, 2 1/16" 10,000 psi instrument flange installed with 0ne ½"NPT and fitted with 0-10,000psi gauge and needle valve, and the second 2 1/16" x 10,000 psi manual gate valve, 2" ACME 10K female thread companion flange and 2" bull plug.(manual G.valve to be non-rising stem w/back seat capability). Clear bore for passing 8-1/2" BHA. The 11" x 4 1/2" **tubing hanger** has 4 1/2" 18.9ppf premium conn. Top & bottom thread one continuous 1/4" NPT control line exit for SSSV, two channels for **3/8"** chemical injection line exit, one 1/4" line exit for DHPG. 4" type "H" BPV thread preparation, one metal seal & two S-SEALS to pack off the upper neck and two S-SEAL to pack off the body of hanger, and tonging preparation on the lower neck. Tubing hanger shall have factory-torqued double pin tubing pup joint 4 1/2" 18.9 ppf, L80, New Vam approximate length 1600mm to facilitate makeup. **Material FF-0.5, PSL-3, PR-2 with exception to tubing hanger: FF 0.5, PSL-3G, PR-2, Temp U. Height: 850mm**

	Material of body: Forged low alloy steel 410SS G. valves, material of seats/gates & stem: Forged alloy steel AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.
Section D	**Tubing head adaptor (w/one integral manual G. valve)** 11" 10,000psi studded flange x 4 1/16" 10,000psi full bore manual gate valve with bottom preparation to seal on tubing hanger (one metal seal & 2 S. seal) and with test ports to test annular space and seals, top prepared with one continuous 1/4" NPT control line exit, one 1/4" line exit for DHPG and two channels for **3/8"** chemical injection line exit. (Manual G. valve to be non-rising stem w/back seat capability) The T-TYPE **Christmas tree** consists of the following: (1) **Solid block valve (T-Type):** 4 1/16" 10000psi bottom flange for tubing adaptor. 4 1/16" 10000psi lower manual gate valve, 4 1/16" 10000psi upper gate valve with SSV w/ hydraulic actuator, including wire line cutting function & lock open cap, (2) 4 1/16" 10000psi manual swab valve. With two 4 1/16" 10000psi studded outside for wing valves. With one 4 1/16" 10000psi studded top flange for tree cap. (3) one wing valve 4 1/16" 10,000 psi full-bore manual gate valve, one **instrument flange** with one 1/2" NPT ports c/w one 0-10,000 psi pressure gauge and needle valve installed and one 4 1/16"10,000psi Cage type positive choke with 90 deg outlet.(With 2 no of each Wrench): positive choke with 2 sets of bean sizes 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 128/64 inch and with welding flange prepared for 4" pipe(specification to be confirmed). (4) The other wing with one 4 1/16" 10,000psi full-bore manual gate valve and one companion flange w/2" ACME 10K female thread. (5) Quick union type Tree Cap above, with 4 1/16" 10,000psi flange at bottom, ACME thread union on top, including 1/2"NPT port for needle valve and pressure gauge. **Material FF-0.5, PSL-3G, PR-2, Temp U. Height of integral tubing head adaptor and gate valve=800mm**. Material of body: 410ss G. valves, material of seats/gates & stem: Forged alloy steel, AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.

11.3 Wellhead Specification and Configuration Section "A"

Casing Head Housing
20″ Slip On Lock x 20-3/4″ 3k FLG with two 2-1/16″ 3K Studded Side Outlet

List of Section A

Item	Description (Equipment and ancillary tools)	P/N & S/N	M/C	T/C	PSL	Qty.
1	20-3/4" 3K Casing head housing	P/N: 2891253 S/N: N130020	DD-0.5	U	PSL1	1 set
2	Casing hanger. Nom. 20-3/4" x 13-3/8"	P/N: 111005 S/N: N130014	DD-0.5	U	PSL1	1 set
3	Gate valve assembly 2-1/16" 3000psi	P/N: 002108A S/N: N120630	DD-0.5	U	PSL1	1 set

Drawing of Section A

A	B	C	D	E
13275mm	885 mm	3260 mm	511 mm	482 mm

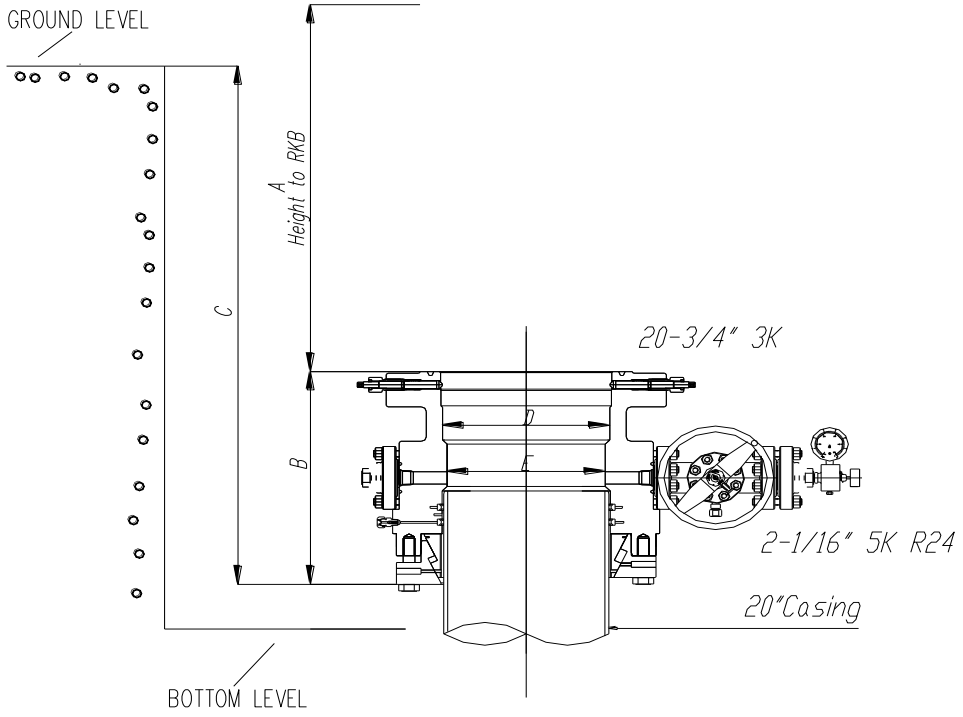


Figure 11-5: AZNS-040 Wellhead Section "A"

11.4 Wellhead Specification and Configuration Section "B"

Casing Head Spool
20-3/4″ 3k FLG x 13-5/8″ 5k FLG with two 2-1/16″5K Studded Side Outlet

List of Section B

Item	Description (Equipment and ancillary tools)	P/N & S/N	M/C	T/C	PSL	Qty
1	Casing head spool 20-3/4" 3000psi x 13-5/8" 5000psi	P/N: 2891254 S/N: N130011	DD-0.5	U	PSL1	1 set
2	Casing hanger Nom. 13-5/8" x 9-5/8"	P/N: 111001 S/N:	DD-0.5	U	PSL1	1 set
3	Gate valve assembly 2-1/16" 5000psi	P/N: 002108A S/N: N130285	DD-0.5	U	PSL1	1 set
4	Gate valve assembly 2-1/16" 5000psi	P/N: 002108A S/N: N120038	DD-0.5	U	PSL1	1 set

Drawing of Section A

A	B	C	D	E	F
12610mm	665 mm	234 mm	134.1 mm	334 mm	320 mm

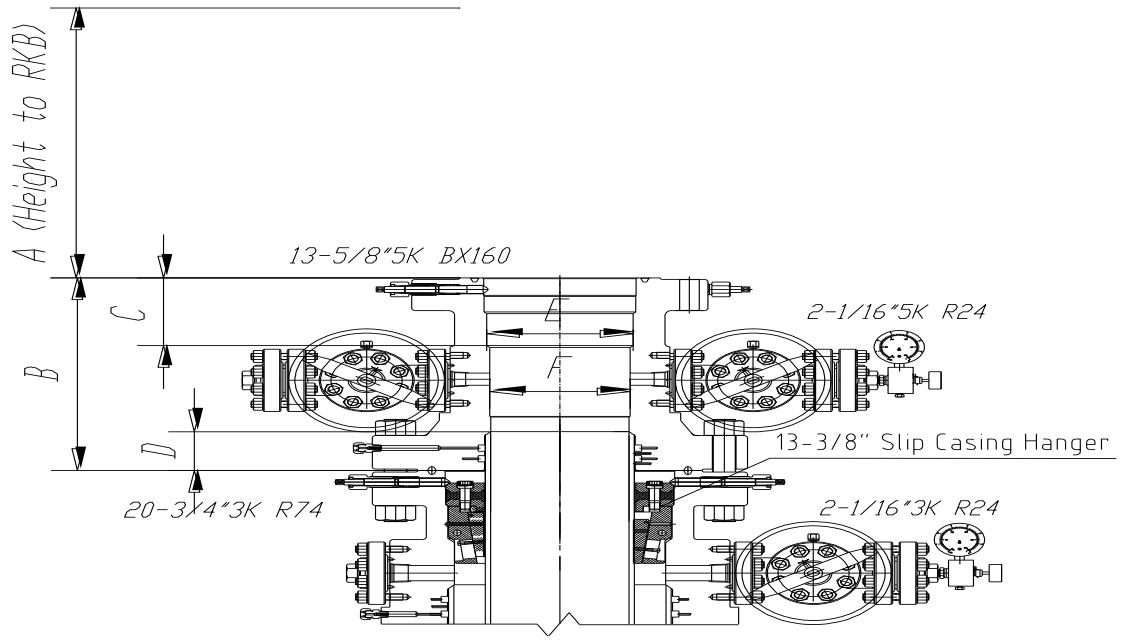


Figure 11-6: Wellhead Section "B"

11.5 Completion String Specification and Schematic

Typical Fahliyan Vertical Well Completion String Specification

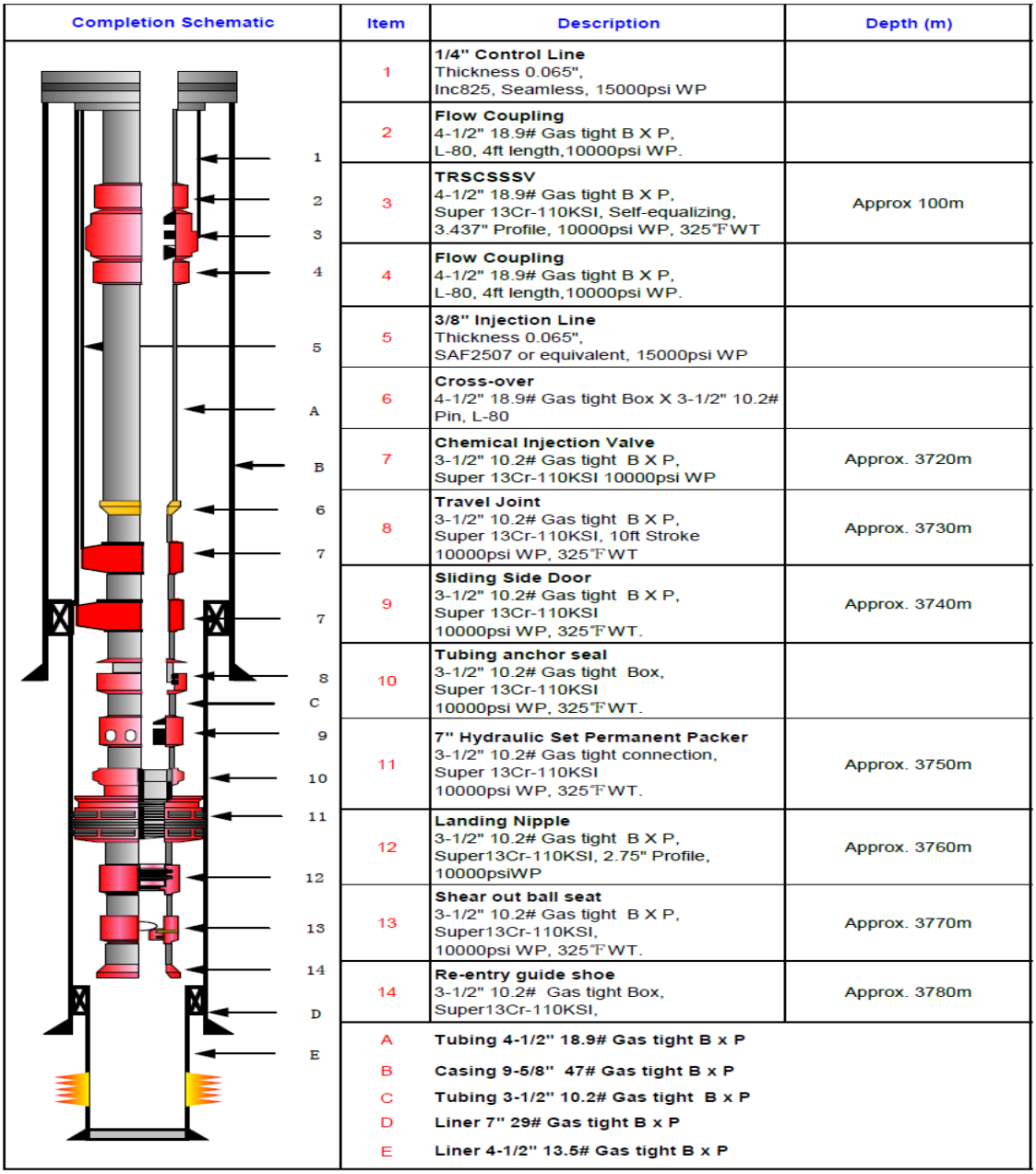
NOTE: Exact Completion string specification and schematic will be issued in due time with Completion program.

Table 11-7: Typical Fahliyan well Completion Specification

No.	Fahliyan wells (10,000 psi)	
1	1/4" Control Line	Thickness: 0.049", INC 825, 120m/Roll, 15,000Psi WP, type: Seamless
2	3/8" Injection Line. encapsulated	Thickness: 0.049", SAF2507, 4000m/Reel, 15,000Psi WP
3	32# Hydraulic Oil	200 Liter Per Tank
4	Fluted Adaptor	4-1/2" 18.9ppf, PREMIUM Pin X Pin, S13CR-110, 10,000 Psi WP
5	4-1/2" Flow Coupling	4-1/2" 18.9ppf, PREMIUM Box X Pin, S13CR-110, 6 FT, 10,000 Psi WP
6	4-/2" Tubing Retrievable Safety Valve. self-equalizing, flapper type	4-1/2" 18.9ppf PREMIUM Box x Pin, S13CR-110, Min ID: 3.437", 10,000 Psi WP, 325 F WT
7	4-1/2" Flow Coupling	4-1/2" 18.9 ppf, PREMIUM Box X Pin, S13CR-110, Min ID: 3.4", Length: 4 FT, WP: 10,000 Psi
8	Cross-over	4-1/2" 18.9#, premium Box X 3-1/2" 10.2# PREMIUM Pin, S13CR-110, Length:24"
9	3-1/2" Travel Joint	3-1/2" 10.2ppf PREMIUM B x P, Stroke Length: 10ft， S13CR-110, WP: 10,000 Psi, WT: 300 F
10	3-1/2" Chemical Injection Nipple, with permanent dual check valve	3-1/2" 10.2ppf PREMIUM B x P, S13CR-110， WP: 15,000Psi
11	Chemical Injection dual check Valve	S13CR-110, WP: 15,000Psi, WT: 325 F
12	3-1/2" Sliding side door, with integral profile	3-1/2" 10.2ppf PREMIUM B x P, S13CR-110, Min. ID: 2.75" Selective Profile, Open Down, WP:10,000 Psi, WT:300 F
13	7" x 3 ½" Hydraulic set Permanent Packer. With anchor latch and MOE	7" 26-29 PPF casing, 3-1/2" 10.2ppf PREMIUM B x P, S13CR-110, WP:10,000 Psi, WT:300 F
14	2.75"No-Go Landing nipple	3-1/2" 10.2ppf PREMIUM B x P, S13CR-110, 2.75" No-Go Profile, WP:10,000 Psi
15	3-1/2" Re-entry guide	3-1/2" 10.2ppf PREMIUM Box Up, S13CR-110, WP 10,000 psi
16	Fitting for safety valve	Size 1/4" male × 1/4" Tube OD, WP 10000Psi, Material: INC 825
17	Fitting for injection nipple	Size 3/8" male × 3/8" Tube, WP 15,000Psi, Material: SAF2507

No.	Fahliyan wells (10,000 psi)	
18	4-1/2" Cross-coupling tubing protector	Matched 4-1/2” 18.9ppf tubing, fit for 9-5/8" 47 to 53.5lb/ft Casing, including 4 slots, two for 1/4" control line, two for 3/8" injection line, Carbon steel.
19	3-1/2” Cross-coupling tubing protector	Matched 3-1/2” 10.2ppf PREMIUM, fit for 7” 29 to 35lb/ft Casing, including 4 slots, two for 1/4” control line, two for 3/8” injection line, Carbon steel.

Typical Fahliyan Vertical Well Completion Schematic



Item	Description	Depth (m)
1	**1/4" Control Line** Thickness 0.065", Inc825, Seamless, 15000psi WP	
2	**Flow Coupling** 4-1/2" 18.9# Gas tight B X P, L-80, 4ft length,10000psi WP.	
3	**TRSCSSSV** 4-1/2" 18.9# Gas tight B X P, Super 13Cr-110KSI, Self-equalizing, 3.437" Profile, 10000psi WP, 325°F WT	Approx 100m
4	**Flow Coupling** 4-1/2" 18.9# Gas tight B X P, L-80, 4ft length,10000psi WP.	
5	**3/8" Injection Line** Thickness 0.065", SAF2507 or equivalent, 15000psi WP	
6	**Cross-over** 4-1/2" 18.9# Gas tight Box X 3-1/2" 10.2# Pin, L-80	
7	**Chemical Injection Valve** 3-1/2" 10.2# Gas tight B X P, Super 13Cr-110KSI 10000psi WP	Approx. 3720m
8	**Travel Joint** 3-1/2" 10.2# Gas tight B X P, Super 13Cr-110KSI, 10ft Stroke 10000psi WP, 325°F WT	Approx. 3730m
9	**Sliding Side Door** 3-1/2" 10.2# Gas tight B X P, Super 13Cr-110KSI 10000psi WP, 325°F WT.	Approx. 3740m
10	**Tubing anchor seal** 3-1/2" 10.2# Gas tight Box, Super 13Cr-110KSI 10000psi WP, 325°F WT.	
11	**7" Hydraulic Set Permanent Packer** 3-1/2" 10.2# Gas tight connection, Super 13Cr-110KSI 10000psi WP, 325°F WT.	Approx. 3750m
12	**Landing Nipple** 3-1/2" 10.2# Gas tight B X P, Super13Cr-110KSI, 2.75" Profile, 10000psiWP	Approx. 3760m
13	**Shear out ball seat** 3-1/2" 10.2# Gas tight B X P, Super13Cr-110KSI, 10000psi WP, 325°F WT.	Approx. 3770m
14	**Re-entry guide shoe** 3-1/2" 10.2# Gas tight Box, Super13Cr-110KSI,	Approx. 3780m
A	**Tubing 4-1/2" 18.9# Gas tight B x P**	
B	**Casing 9-5/8" 47# Gas tight B x P**	
C	**Tubing 3-1/2" 10.2# Gas tight B x P**	
D	**Liner 7" 29# Gas tight B x P**	
E	**Liner 4-1/2" 13.5# Gas tight B x P**	

Figure 11-7: Typical Fahliyan Vertical Well Completion String Schematic

12. Waste Management Program

12.1 Drilling Waste Management Design

The contractor is responsible for designing the optimum practices for treating the produced waste during the operation. The initial technical waste management proposal should be included the practices for minimizing, treating and recovering of the wastes.
According to the below categories:

- Water based mud containing minerals and natural synthetic polymers, and
- Cuttings contaminated with WBM.

The contractor is encouraged to offer possible practices to minimize the drilling fluids waste and its regeneration.

Water based mud containing minerals, naturals and synthetic polymers

- **Primary operation**

The proposed primary solids control package should be able to control drilling fluid properties, reducing of mud materials usage against conventional solid control equipment's, and significantly reduce the number of excess fluids generated throughout drilling each phase of the well.

- **Secondary Operation**

All excess mud and spillage (water based) should be collected in rig site pits and free surface fluid should be treated using a built for propose flocculation station (has to join with centrifuge) consisting of integral dosing and pumps along with a centrifuge for efficient, quick and controlled separation of the solids before the cleaned fluids is returned to active system or buffer tanks.

This system should be able to remove colloidal sized particles that cannot be removed with primary solids control system.

This system may use to clean up the remaining water-based mud or decrease weight of oil base mud at the end of the drilling fluids programmed.

11.1.2 Cuttings contaminated with WBM

Spud mud and water-based mud cuttings should be fixed and be ready for transport to disposal zone. For fixing the cutting, 100-150 kg cement shall be used for each cubic meters of cutting as well as 3-4-liter Sodium Silicate for each cubic meter of cutting. (Consumption of cement and sodium silicate varies in different formations). Gypsum is not allowed for cutting fixation.

This method is physically Bonds Contaminants to Prevent Leaching to Environment. Final product will pass can test and in case of approval by company man it'll send to landfill.

12.2 Waste discharge and handling

The Company is responsible for designing of required pits and The Contractor is responsible for procedure for burring of waste cuttings in disposal zone; supervision and managing of requirement operation for deposit oil waste cutting.
The contractor is responsible for treating all the generated waste in site without any discharges in environment or damages. It is also the responsibility of the contractor for handling/discharge of generated waste after treatment.
The contractor manages and supervises the solids and fluid waste streams loading and off-loading on/from the designated areas and responsible for the loader along with the responsibility of the storage pits, tanks, etc.

12.3 Laboratory equipment's and analysis

The contractor should prepare suitable laboratory analysis equipment's to perform complete analysis services of waste matter. The provided equipment's shall make this opportunity to study the mentioned environmental policy of waste analysis.

12.4 Reports and comments

The contractor provides the following reports and documentations on contract duration:

- Daly and monthly waste management report.
- Final waste management report, including:
- Summarized amount of generated waste on each section,
- Total treated waste,
- Total chemical consumption,
- Cost monitoring report, and
- Special activities.

12.5 The equipment's installation and commission

. **First location**

The Maximum days for installation and commissioning of all equipment and necessary piping will be 7 days after rig spud. Contractor is responsible for any delay to installation and commissioning of equipment.

. **Other locations**

The Installation of all equipment should be completed during rig spud. Contractor is responsible for any delay to installation and commissioning of equipment.

12.6 Clean up process

After drilling operations and drilling rig removal, the well area, waste pit, and corral should be completely cleaned. The contractor shall remove the fixed cuttings to pre-determine locations (disposal) which can protect the environment is very effective.

12.7 Supplying services

Waste treatment package . Dewatering unit . Filter press
Transfer pump package . Air diaphragm pump . Submersible pump . Centrifugal pump
Storage tank . Waste water storage tank
Cutting treatment . Loader
Waste transportation . Cutting transport truck . Tunker truck for waste water transfer
Other . Wash gun . Laboratory and cabin

12.8 Waste management policy

Waste management services should be utilized to minimize generated waste and contaminated discharged cuttings during drilling operations. This helps keep the environment clean and ensures that our operations are in compliance with DOE (Department of Environment) legislations.

We provide cost-effective and optimum solutions from environmentally acceptable procedures to achieve zero-discharge plans and improve fluids recovery. The outcome reduces operational costs and enables recycling.

The waste is analyzed to study various parameters such as chloride, nitrate, and phosphate. Other analysis is carried out to monitor treatment processes on a regular basis during waste generation. Our waste management personnel are fully trained experienced and have the

competencies in place to provide the best reliable services with concentrated support and the least liabilities.

12.9 Drilling waste management steps

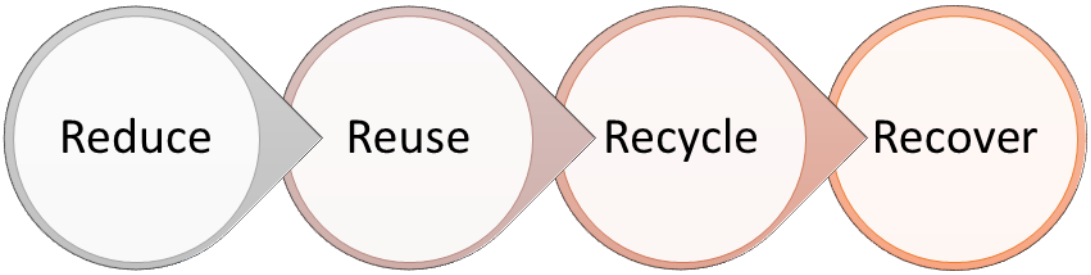


Figure 12-1 _ Drilling waste management procedure.

12.10 Calculation

11.10.1 Waste water treatment and disposal

Dewatering System

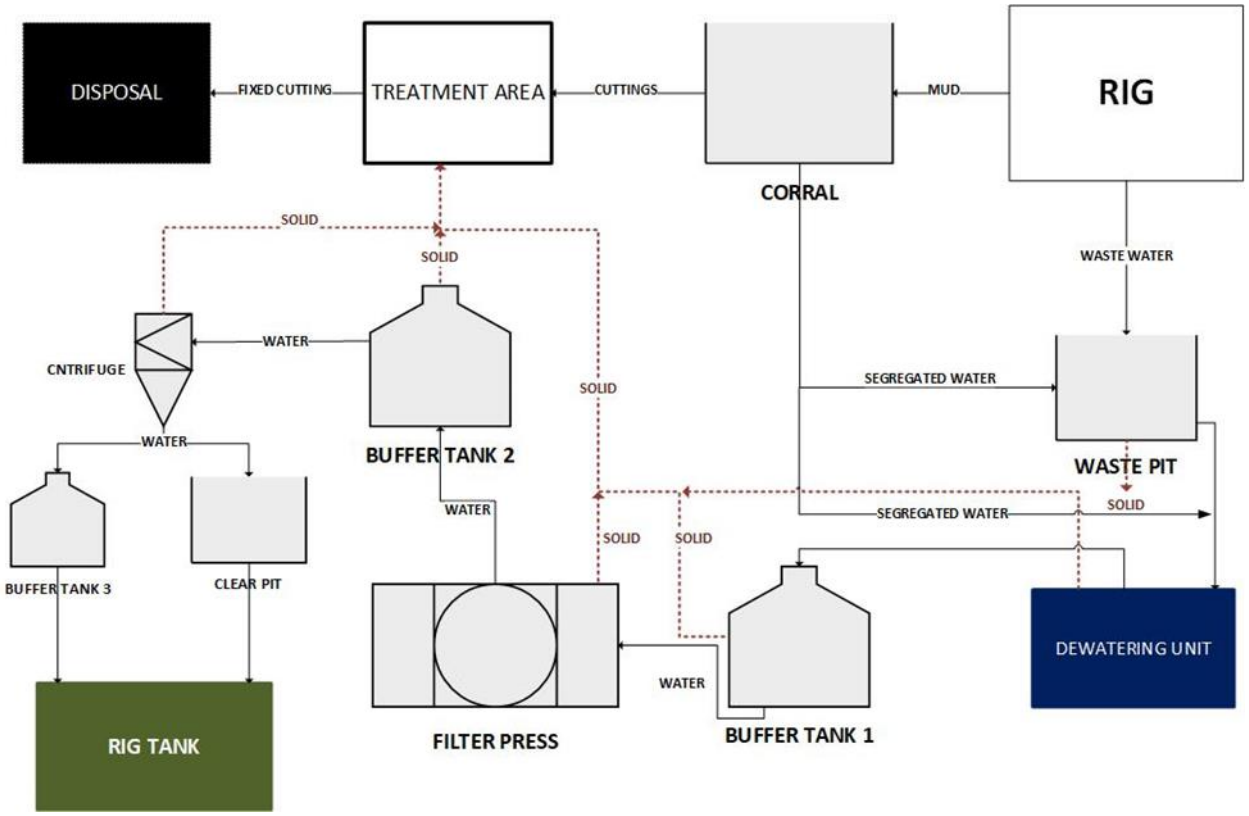


Figure 12-2_ Dewatering system flowchart.

Estimated Values

- Assumed 400 bbls in mud tank.
- The produced daily waste water of 50 m^3 and 60% mud are roughly estimated.

	unit			Total
Hole Size	Inch	**8.5**	**5.875**	-
Casing Size	Inch	7	4 ½	-
Depth	Meter	3911	4226	-
Mud system	-	Salt saturated KCl, etc. mud	Salt saturated KCl, etc. mud	-
Mud weight	PCF	75-85	100-106	-
Interval	Meter	761	406	-
Operating duration	Day(s)	40	38	**78**
Estimaded mud volume	M^3			
	bbl			
Estimated Treating waste water	M^3			
	bbl			

Table 12-1 _ Estimated treating waste water.

Dewatering Chemicals

	Application	Required concentration
Poly Aluminum chloride	Coagulant	0.5~1.5 Kg/m^3
Flocculent polymer	Flocculant	0.1~0.3 Kg/m^3
Hydrated lime	Coagulant	0.25~0.5 Kg/m^3
Soda-ash	Harness control	0.5 ~ 0.6 Kg/m^3 decreasig 1000 mg/Lit of Ca, at average
Caustic soda	pH control	0.5 ~ 0.6 Kg/m^3 Contion depended
Biocide	Microvial activity control	20 ml/ m^3

Table 12-2 _ Dewatering chemical.

Treated Water Properties

Item	Amount
pH	7~9
Total hardness	< 800 mg/litr
Turbidity	< 50 NTU
Nitrate	< 50 mg/litr
Aluminum	< 3 mg/litr
TSS	< 85 mg/litr
Sulfate	< 50 mg/litr
Ca^{2+}	< 600 mg/litr

Table 12-3 _ Treated water properties.

Material Consumption Estimation

TBA

12.11 Cutting treatment

Cutting Removal Process



Figure 12-3 _ Cutting management process.

Estimated Cutting Volume

TBA

Estimated Fixation Volume

TBA

12.12 Waste Management Equipment's

Flocculation unit

This unit, has the bellow specifications:

- Fully automated system,
- Full 316 stainless steel material,
- Eqipped with high pressure level and pH sensors,
- Equipped with digital flow meter and totalizer to monitor flow rate, and
- Touch panel technology to control prose.



Figure 12-4 _ Flocculation unit.

400^{bbl} Dewatering unit

This tank provides more than 400 bbls capacity to store contaminated water for using in dewatering process and ability to adding chemicals such lime from hopper. Two Mixer 25 hp and Jet System give ability to holding any type of mud for hours without any stealing.



Figure 12-5 _ Conditional tank.

400^{bbl} Storage tank

Each tank provides more than 400 bbls capacity to store clear water for recycling.

High Performance Filter Unit

Filtration plays an important role in the natural treatment of waste water as it percolates through the soil. It is also a major part of most water treatment. Groundwater that has been softened, or treated through iron and manganese oxidation, requires filtration to remove floc created by coagulation or oxidation processes. Since surface water is subject to run-off and does not undergo natural filtration.

it must be filtered to remove particles and impurities. Filtration can be compared to a sieve or micro- strainer that traps suspended material between the grains of filter media. However, since most suspended particles can easily pass through the spaces between grains of the filter media, straining is the least important process in filtration. Filtration primarily depends on a combination of complex physical and chemical mechanisms, the most important being adsorption. Adsorption is the process of particles sticking onto the surface of the individual filter grains or onto the previously deposited materials. Forces that attract and hold particles to the grains are the same as those that work in coagulation and flocculation. In fact, coagulation and flocculation may occur in the filter bed, especially if coagulation and flocculation before filtration was not properly controlled. Incomplete coagulation can cause serious problems in filter operation.



Figure 12-6 _ High Performance Filter Unit.

Centrifugal pump

This pump can be utilized in different situation. 3 x 4 pump can be used for transferring processed water back to rig water tanks for reuse.



Figure 12-7 _ Centrifugal pump.

Submersible pump

These pumps designed to clear and waste water from water pits and tanks to designated areas or tanks. They are very easy to handle and can be very useful in case of transferring from bottom of tanks or pits.



Figure 12-8 _ Submersible pump.

200 bar wash down guns

These wash guns minimized water consumption and can be used on the shakers to unplug screens or washing al the area on the rig from rig floor to mud pits.



Figure 12-9_ 200 bar wash down guns.

Fluid transfer diaphragm pump

This pump works with air pressure and has the responsibility of transferring mud and contaminated water.
Three inches diaphragm pump transfers mud and waste from cellar to coral to keep cellar empty.



Figure 12-10_ 3" Diaphragm pump.

Two inches diaphragm pumps transfer waste water and mud between different spots on location. They also can be utilizing to transfer mud from mud tanks dead volume to treatment tanks.



Figure 12-11_ Diaphragm pump

These pumps designed to transfer clear and waste water from water pits and tanks to designated areas or tanks. They are very easy to handle and can be very useful in case of transferring from bottom of tanks or pits.

DWM laboratory cabin

Lab equipment for DWM is consisting of:

- Retort kit,
- API Filtrate test kit,
- Testing reagents and equipment for Chloride, Calcium, Hardness and PH, and
- Hatch Water Test kit





Figure 12-12_ Hatch Water Test kit

This kits analys all conserned parameters, including aluminum, iron, nitrat, nitrite, zinc, phosphate, sulfate, sulfide, copper and cyanide concentration.

Storage container





Figure 12-13 _ Storage container.

Loader



Figure 12-14 _ Loader.

13. Health, Safety and Environment

13.1 General Quality, Health, Safety & Environmental Requirements

- PPL/PEDEC's committed to conducting operations in a manner which minimizes the risks to the health and safety of those involved in, and affected by, such operations. PPL/PEDEC's also committed to conducting its operations in a manner which minimizes the impact on natural and 'built' environments in the area of operations.
- This section sets out minimum Quality, Health, Safety and Environmental (QHSE) requirements for the WORK that MSA and subcontractors will be required to comply with whilst conducting operations for PPL/PEDEC.
- PPL/PEDEC may perform audits and assessments of MSA's and its subcontractors' performance under the CONTRACT. PPL/PEDEC reserves the right without prior notice to conduct quality, health, safety, security and environmental audits of the operations.
- MSA shall at all times be responsible for performing the WORK in a manner so as to protect against loss of life, injury to people, damage to property and the environment. The detailed manner means and methods of performing the WORK shall be under the control and direction of MSA.
- Prior to the start of Operations, certain QHSE processes must be completed. These QHSE processes include developing the final QHSE plan format and frequency of any QHSE audits, permitting and licensing and PEDEC's approvals. MSA shall ensure that sufficient resources are dedicated to this task to ensure completion prior to mobilization.
- PPL/PEDEC may at any time stop the WORK or any portion of the WORK without penalty to PPL/PEDE Cif the WORK is not conducted in accordance with the Quality, Health, Safety or Environmental guidelines specified in this CONTRACT or by the laws of the appropriate jurisdiction. PPL/PEDEC reserves the right to require MSA to remove any person from the operations in the case of non-compliance with QHSE procedures. PPL/PEDEC may also prohibit the use of equipment and plant in cases where that equipment or plant does not meet the required standards or is deemed unsafe.
- MSA shall ensure that activities associated with this CONTRACT are conducted in compliance with applicable national and local HSE laws and regulations, MSA's standards and PEDEC's requirements and IADC documents.

- Limits of Authority
- PEDEC's Representative(s) do not have the authority to relax health, safety or environment standards. If they judge the operation to be inadequate with regards to any aspect of quality, health, safety, security or the environment, then the operation may be immediately suspended.
- MSA shall be required to develop a project-specific QHSE plan to manage the WORK and associated risks. This plan shall, as a minimum, address topics included in the list detailed in Item 7. Where existing MSA QHSE publications detail the topics listed, it is sufficient to either include them verbatim, or provide reference to where they may be found in MSA's QHSE Manuals. The final project-specific QHSE plan shall satisfy MSA standards, PPL/PEDEC requirements and applicable HSE laws and regulations.
- MSA should establish an effective HSE system and implement it strictly.
- MSA must adhere to the standards and regulations of Health, Safe and Environment constituted by the local government.
- MSA must strictly adhere to the procedures / standards and regulations of Health, Safety and Environment promulgated by NIOC.
- MSA should comply with all applicable local government regulations including guidelines and recommended practices and any other Iranian authorities having jurisdiction.
- All personnel should respect the local culture, traditions and social values of the indigenous population of the Iran and South Azadegan region.
- MSA shall hire qualified and experienced worker and familiar his personnel with hazards especially with H2S and Mine and UXO (unexploded ordinance) according to related procedure.
- MSA shall conduct proper training about UXO (unexploded ordinance) and mine awareness to all personnel

13.2 Health & Safety Management

As per national/local and client (PPL/PEDEC) standards and MSA/OEOC HSE plan. Following are minimum requirement for the AZNS project:

- MSA/OEOC shall take all necessary safety measures in relation to the WORK to be provided and shall conduct itself and its work force in such a way as to comply at all times with the provisions of the national and local health and safety regulations pertinent to the contracted Scope of WORK.

- MSA/OEOC shall also comply with all and any of PPL/PEDEC'S safety standards and requirements as particularly applicable to the Operating Area and/or as may be imposed by PPL/PEDEC from time to time.
- MSA/OEOC and its subcontractors shall take all necessary steps to provide a safe and healthy working environment for employees, PPL/PEDEC's personnel and third parties in the performance of the WORK.
- MSA/OEOC shall operate a management system that ensures:
- MSA/OEOC's and sub contractors' personnel operate safe systems of work at all times for the performance of the WORK.
- Hazards associated with the work are identified and appropriate actions are taken to prevent accidents.
- Risks to safety and health of personnel involved in the WORK are identified and assessed, and any actions and measurements are implemented to either eliminate or reduce these risks to a level that is considered as low as reasonably practicable, given the nature of the WORK.
- Personnel are aware of the identified hazards and risks to their safety and health and to the safety and health of others. The detailed safe working procedures established by MSA/OEOC to ensure that the WORK will be conducted with due regard to the QHSE aspects, shall be detailed or referenced in the QHSE plan. Copies of the source documents shall be provided to PPL/PEDEC if the procedures are referenced only.

QHSE Representative:

- MSA/OEOC shall provide a dedicated, on-crew QHSE Representative. The primary responsibility of the QHSE Representative shall include, but not be limited to, the maintenance and monitoring of the implementation of HSE guidelines and procedures and where required provide HSE awareness training for the crew. Such activities shall be approved and actively supported by MSA/OEOC at corporate management level.

Training and Competence:

- MSA/OEOC shall ensure that its employees and those of its subcontractors are trained and competent to perform the WORK in a safe and environmentally responsible manner. Such training and demonstration of competence will include but not be limited to vocational, safety, emergency, environmental and other training required by regulatory agencies and IADC/OGP guidelines. MSA/OEOC's senior staff shall have received training in safety management.

Accident/Incident Reporting:

- MSA/OEOC shall report all incidents. Any hazardous incident/near miss involving PPL/PEDEC, MSA/OEOC's, or any third-party personnel, plant or equipment, shall be reported to PPL/PEDEC, whether or not injury to personnel or damage/loss to plant or equipment resulted. The following incidents shall be verbally reported immediately to the on-site PPL/PEDEC Representative(s): damage to property or equipment belonging to PPL/PEDEC, third parties, MSA/OEOC or its subcontractors; injury to personnel of PPL/PEDEC, third parties, subcontractors or MSA/OEOC; near misses which may have had serious potential consequences; environmental damage; media attention. An initial written report, containing factual information only, shall be prepared by MSA/OEOC and delivered to the PPL/PEDEC Representative within twenty-four (24) hours of the incident.
- PPL/PEDEC reserves the right to participate in the investigation of any incident arising out of performance of the WORK.
- MSA/OEOC and all subcontractors shall have an accident and incident reporting system, details of which shall be included in the QHSE plan.
- MSA/OEOC shall maintain and forward to PPL/PEDEC, a monthly summary of its safety performance.
- MSA/OEOC shall prepare and forward to PPL/PEDEC, a weekly report stating all incidents and near misses that occurred during that week.
- All MSA/OEOC and subcontractor personnel and PPL/PEDEC Representative shall attend safety meetings on a regular basis. PPL/PEDEC Representative shall be given a copy of the minutes of the meetings.

Safety Equipment & Personal Protective Equipment (PPE) requirements:

- MSA/OEOC shall ensure that appropriate PPE compliant to national standards is supplied to MSA/OEOC's personnel involved in the WORK. PPL/PEDEC intends to use water-based mud and hence MSA/OEOC's Personnel that may be exposed to this must be provided with appropriate PPE.
- MSA/OEOC shall be solely liable for all costs in equipping its own and all other personnel on contract to, or under the control of, MSA/OEOC with the required safety equipment. MSA/OEOC will provide to all MSA/OEOC's Personnel appropriate PPE throughout the WORK. Any defective PPE will be replaced without delay. Personnel without the correct PPE will not be allowed to continue work, until replacement PPE has been issued. If any person is disallowed to work owing to inadequate PPE, then PPL/PEDEC will deduct his day- rate from MSA/OEOC's invoice.

- MSA/OEOC shall detail in the QHSE plan the PPE proposed for issue to the crew. This proposal should include the arrangements for replacement (due to wear and tear and damage) of this equipment.
- This well is committed to develop for oil in a responsible manner that promotes employees' and local inhabitants' health.
- A trained Paramedic and proper MEDIVAC equipment and ambulance shall be stationed on the drilling rig during the drilling campaign.
- An emergency medical evacuation plan shall be coordinated, and shall keep in touch with emergency response centers manned by qualified doctor 24-hours per day in Tehran or Ahwaz city.
- Medical emergency report and investigation shall be performed as soon as possible in accordance with Medical Emergency Procedures.
- MSA/OEOC shall, as a minimum, conduct the WORK and provide personnel, equipment, facilities and services consistent with good oil field practices and OGP Guidelines. An assessment of the Contract Area with regard to "Environmental and Occupational Health Risks" and "Local Medical Support" shall be provided by MSA/OEOC in the QHSE plan. This assessment and the resulting "Health Risk Assessment Score" will be discussed with PPL/PEDEC, and a final assessment will be agreed.
- To ensure the fitness of those personnel engaged to undertake the WORK, MSA/OEOC shall conduct periodic medical examinations of its staff and ensure that its subcontractors' personnel are also subject to periodic medical examination. As a minimum this shall include pre-hire medical examinations of temporary local personnel and bi-annual medical examinations of its permanent and expatriate staff. Evidence should be made available to PPL/PEDEC to verify an appropriate program of immunizations consistent with the recommendations of the World Health Organization for the area of operations.
- MSA/OEOC shall be responsible for the cost of medical examinations as required to provide PPL/PEDEC with medical certificates which confirm the fitness of MSA/OEOC's assigned personnel for the WORK. Each drilling rig shall be equipped with a hospital, doctor and an ambulance type-2 level.

Drug Abuse and Control:

- MSA/OEOC warrants that it shall obtain valid written consent of all Personnel to a search of property of such Personnel when PPL/PEDEC suspects that such Personnel have in their possession drugs and/or other prohibited substances or any other item likely to prejudice safety or security. PPL/PEDE will provide a representative to attend the search.

- MSA/OEOC warrants that it shall obtain valid written consent of all Personnel to medical testing for the presence of drugs or other prohibited substances.
- MSA/OEOC further agrees that any Personnel who refuse to provide such consent or consents or withdraws such consent or consents shall at PPL/PEDEC's option be removed from the WORK and shall not, at PPL/PEDEC's option, be assigned to any future services to be provided by MSA/OEOC for PPL/PEDEC.
- MSA/OEOC further agrees that it shall immediately notify PPL/PEDEC in the event that MSA/OEOC becomes aware that any of its Personnel has been found in possession of drugs and/or any other prohibited substance. MSA/OEOC is required to notify PPL/PEDEC in the event that any Personnel provided for the WORK have previously been found in possession, or have abused drugs and/or other substances prohibited at Location. The acceptance of such Personnel for the WORK shall be at the sole discretion of PPL/PEDEC.

Camp Standards:

- MSA/OEOC shall abide by all statutory Iranian governmental health and safety regulations, and maintain acceptable standards for cleanliness and hygiene for quarters, camps and the Operating Area(s) in general. MSA/OEOC shall detail the standards and facilities proposed for its camp in the QHSE plan.

Emergency Preparedness:

- MSA/OEOC and its subcontractors shall have emergency preparedness and response plans plus procedures available at all times throughout the duration of the CONTRACT.
- MSA/OEOC shall take all necessary measures to ensure that:

I. Potential emergency situations have been identified and assessed;
II. An emergency response plan, including clear responsibilities, is established;
III. MSA/OEOC's management, employees and third parties are aware of response procedures and responsibilities;
IV. Regular drills and exercises are carried out to test the emergency response organization and procedures and that any resultant issues are properly addressed.

MSA/OEOC shall notify PPL/PEDEC immediately with respect to any emergency situation and shall provide regular updates on emergency planning procedures, tests etc.

MSA/OEOC shall detail in the QHSE plan a list of potential contingencies for which emergency response procedures are planned. The QHSE plan should also detail a schedule of proposed exercises and drills to verify the

effectiveness of the proposed arrangements in place. Development and implementation of the above plan shall incorporate any PPL/PEDEC assigned personnel to the project.

13.3 Environmental Management

- COMPANY shall comply with any and all laws, rules and regulations of governmental agencies having jurisdiction, which now exist or may be promulgated during the term of the CONTRACT, relating to the control and prevention of damage to the environment.
- Without prejudice to above, COMPANY shall ensure that all personnel use the highest prevailing standards of care and diligence to prevent unlawful disposal of all waste oil, water and other waste materials or fluids which may accumulate or be caused by COMPANY's performance of the CONTRACT. COMPANY shall not dispose of any objects, scrap, waste, rubbish, equipment, fluids or fuel into the surrounding area or allow anything to be stored loose where it can be driven into surrounding areas by wind. COMPANY shall take all reasonable precautions to prevent pollution of any nature or kind from its performance of the WORK. COMPANY shall have contingency plans and procedures in place for such events.
- In the event of any disposal prohibited by Item 3.2 above, whether by dumping, spilling, emission or otherwise arising from COMPANY's performance or non-performance of the CONTRACT, COMPANY shall be fully liable for and bear the costs of removal of any such item(s) dumped, spilled, emitted or otherwise disposed regardless of the sole, concurrent or contributory negligence or breach of duty (statutory or otherwise) of PPL/PEDEC. In the event of any such disposal, COMPANY shall immediately undertake all reasonable remedial and protective measures required in such circumstances.
- COMPANY shall notify PPL/PEDEC immediately with respect to any pollution, loss, damage, claim or demand (or occurrence which may give rise to the same) resulting from the CONTRACT. COMPANY shall report to PPL/PEDEC any incidents of non-compliance with legislative and regulatory environmental requirements that occur during the performance of the WORK. Without relieving COMPANY of any of its obligations under Item 3.2, if COMPANY does not promptly proceed to perform its obligations under this Item 3, PPL/PEDEC may take part to any degree it deems necessary in the control and removal of any pollution, dumping, spillage or contamination which is the responsibility of COMPANY under this Item 3.2, or which is due to any act or omission of COMPANY. COMPANY shall reimburse PPL/PEDEC for costs arising from such actions of PPL/PEDEC upon receipt of invoice from PPL/PEDEC. PPL/PEDEC shall have the right to deduct such costs from any sum due or becoming due to COMPANY under the CONTRACT.

- COMPANY shall operate a management system that is able to demonstrate compliance with environmental laws, rules and regulations applicable to its activities, products and services.
- MSA/OEOC and its subcontractors shall take all necessary steps to ensure that the environmental impacts of their work are responsibly managed and shall take all necessary measures to ensure that:
- COMPANY's personnel are aware of the environmental impact of the COMPANY's operations and are sufficiently trained and competent to perform the WORK.
- COMPANY's personnel operate systems, which will ensure that significant environmental impacts of WORK are identified and managed in accordance with a commitment to legal compliance and continual improvement.
- The environmental risks of operations associated with the WORK are evaluated and appropriate actions are taken to prevent accidents and reduce pollution, and that contingency plans are in place in the event of an incident.
- COMPANY will have adequate processes for monitoring the environmental performance of their WORK.
- COMPANY will, as a minimum, conduct the WORK and provide personnel, equipment, facilities and services consistent with IADC Environmental Guidelines. COMPANY's proposals for environmental protection should be detailed in the QHSE Plan. This should include, but not be limited to:

- procedures and precautions to minimize damage to crops and vegetation, the local ecology and disturbance of wildlife and farm animals;
- recycling and waste disposal;
- Clean-up and restoration of seismic lines, Rig sites access roads and trails, fixed sites occupied during the survey.

- An ecological team shall be set up by COMPANY, with the purpose of collecting all waste, rubbish, food scraps, office waste and other discarded material from camps, created by the WORK. Such activities shall be performed on a daily basis. Such material shall be appropriately disposed of, according to PPL/PEDEC approved methods or existing methods established by COMPANY, which are proven to be acceptable to PPL/PEDEC. Well location and camp site shall be delivered to PPL/PEDEC without any waste, rubbish, food scraps, office waste and other discarded material. PPL/PEDEC has the right to hold Completion certificate till cleaning the well location and camp site by COMPANY.
- The access road and well location construction shall be accorded with construction program of well location. It minimizes any impacts on the local residents, prevent all soil contamination, prevent all ground water contamination, and prevent all surface water contamination. Cutting and waste water pit shall be lining and drainage system shall be good and to avoid farm and season canal contamination.
- Management regulation for transportation truck, drilling operation machine and tools shall be carried out strictly to avoid damage farm and trees, etc.

- Well site shall be flat and clean. All the equipment, materials, drilling tools shall be put in order.
- Control the rig noise to decrease the influence to surroundings while drilling at the inhabitation area.
- COMPANY shall perform on site segregation of wastes prior to transfer to the transfer area and/or delivery to the transportation COMPANY in accordance with the following waste hazard classification.
- Non-Hazardous Wastes (Re-usable, Recyclable and Recoverable Wastes)
- Ferrous and non-ferrous metals, plastics, and wood wastes, PETs and potentially paper and glass
- Materials shall be separately segregated and provision may be made for size reduction and containerization equipment.
- The COMPANY/Generators are encouraged to seek for recycle/re-use alternatives based on the Local available facilities, for final disposal of these materials.
- It should be noted that landfill disposal of metal waste materials and wood wastes shall be prohibited. For wood wastes re-use of and/or salvage of commercial timber, or deliver/release to
- local community are recommended.

Hazardous Wastes:

- Hazardous Waste, such as clinic waste, oil waste, waste produced by painting, insulation works shall be duly segregated.

Facilities of environment protection supplied to well site:

- Mud purifying circulation system
- Cooling water spraying and circulating system for drilling pump
- Special recovery tank for deposed oil
- Reserve house for valuable mud chemicals
- Automatic mud weighting supply system
- BOP stack is equipped according to drilling design
- Define a separation area for gathering all waste according to waste management procedure.
- Requirement of environment protection on location Sewage pit Area of the sewage pit: 1500 m2
- Prior to spud mud pit and sewage pit must comply with the standard. Install cooling water spraying and circulating system for drilling pump and sewage circulating system for cleaning rig floor and equipment.
- Install BOP stack and pressure test shall be operated according to the drilling design to avoid blowout.
- Install oil supply lines and check whether there is any leakage.

Management of environment protection on location:

- In drilling operation keep mud and sewage from overflowing. If it happens, clean up timely.

- Mud chemicals shall be stored according to the standard and shall not be scattered around well site. If mud chemicals scatter while loading or unloading, clean it up at once. The sack of mud chemicals shall be recovered. Do not dispose it on site.
- Enhance water management at well location and do not wash equipment with fresh water. Stop water flowing for long time. Overhead tanks shall be utilized.
- Oil management shall be carefully at the well site. No oil dirt on site, no dripping and leaking at oil tank and manifold system area.
- After the well is completed, reinforce the sewage pit， solidify the disposed mud and have environment procedure fulfilled.
- Install fuel supply lines and keep them in good condition.
- All the drilling fluid materials shall be recovered and shall not be left at the well site after completion.
- The environmental protection personnel from COMPANY and operator side shall check and accept the well site environment according to the regulation after completion. Only well site environment is qualified, the well can be delivered.
- Transition Area for Hazardous Waste shall also be provided by each COMPANY in his allocated area,
- PPL/PEDEC shall approve each area before use as well. The transition area of hazardous waste shall be:

- Paved, fenced, marked and illuminated
- Equipped with appropriate Fire Extinguisher and other emergency response equipment (e.g., Eye-Wash and Emergency Shower, appropriate First Aid kits, neutralizer substances etc.).
- In compliance with the requirements of handling, transportation and storage of hazardous waste (based on usage chemical MSDS).
- Configured to accommodate a safe physical separation of incompatible waste materials (e.g., acids, and alkalis, organic materials and strong oxidizing agents).
- Capable enough to provide storage for the entire range of hazardous wastes generated during the work.
- The area shall be provided with Spill Containment and means of preventing run off to and from the area.
- All waste containers to be stored shall be appropriately labeled.
- Hazardous Waste shall be transferred in a timely manner, within a maximum period of 48 hours, for any period in excess of this permission shall be granted from PPL/PEDEC.

- Standard of utilization and discharging of wastes on location
- Conduct drilling operation according to the design, drilling mud is needed before spud.
- Sewage that cannot be recovered shall be put into the pit, do not discharge
- Make full use of waste water. Water in mud pump and water brake must be utilized by circulating. Water that is used to wash drill floor, shall be recovered after deposition. The recovery rate of waste water shall reach 85%.

- Disposed oil shall be recovered entirely. Recovery rate shall be 100%. Dumping of waste oil shall be prohibited.
- In drilling operation, non-chrome thinner high temperature mud system shall be used often if engineering is qualified.
- While drilling in special area adopt mud creation inhabitation techniques to decrease the waste mud creation.
- After completion, all the mud chemicals, bentonite, clays shall be recovered and shall not be left at the well site. Mud in purifying tank shall also be recovered after completion.
- In case of any flammable or noxious gas encountered, steps shall be taken at once to prevent kicks and blowout. Gas shall be guided into ignition equipment to burn it away.
- Burning material that may result in serious smog or horrible smell at the well site is not allowed. For cement or barite that may result in granular dust pollution, operation shall be performed in a closed system.
- For abandoned well, filled with cement plugs to seal the oil, gas or water formation.
- COMPANY/Generators are required to transport all generated wastes to the location/facilities designated, and licensed by authorities and/or approved/verified by the PPL/PEDEC (as appropriate).
- All wastes shall be transferred by approved Sub- COMPANY (which could be a 3rd party Sub-COMPANY of the COMPANY/Generator itself). The Waste Transporter and any other waste management service COMPANY shall be licensed in accordance with local regulations and are in compliance with such regulations. PPL/PEDEC shall verify and approve the Waste Transportation COMPANY by checking licenses and qualifications, audit and inspections, verification of human resources and received trainings etc.
- The transportation COMPANY shall ensure that all wastes are transported in a safe and responsible manner.

13.4 Security

As per PPL/PEDEC as main client and MSA/OEOC agreed plan.

- Security for the WORK shall be arranged and administered by the MSA/OEOC. MSA/OEOC shall be fully advised in writing covering all aspects of security upon. Company shall liaise with PPL/PEDEC QHSE to develop operational and contingency plans prior to Mobilization.
- The Security guards shall be appropriately housed and provided for. Transport for the Security guards to perform field activities shall be made available according to standard Iranian Oilfield procedures.

- All MSA/OEOC's personnel involved in the WORK shall possess the necessary security clearance documentation required by the Iranian authorities to work within the Operating Area.

13.5 Quality

1. Ensure that relevant experiences from the appraisal and production wells (if any) are captured and included in the operation programs /procedures.
2. Check lists to be prepared for all equipment, available for PPL/PEDEC prior to start of operations.
3. Procedures planned for all operations, available for PPL/PEDEC prior to start operations.
4. High focus and follow up of KPIs (Key Performance Indicators) by NIOC procedures.
5. Drilling risk assessment.
6. All goods and equipment shall be checked as per engineering design and inspection certificates, PPL/PEDEC approval shall be obtained before running any goods in hole.
7. Routine reports shall be submitted to PPL/PEDEC and any commented shall be corrected and informed engineering team.
   - Rig Specification shall be submitted to PPL/PEDEC any commented shall be corrected and informed engineering team.
   - Rig Layout shall be submitted to PPL/PEDEC any commented shall be corrected and informed engineering team.
   - Rig certification (COF) and record shall be submitted to PPL/PEDEC any commented shall be corrected and informed engineering team.
   - Apply PPL/PEDEC inspection work instructions.
   - Drilling String inspection must be done before and during operation based on client request and its certificates and record shall be submitted to PPL/PEDEC.
   - Pre-Spud check list must be verified by PPL/PEDEC before operation.
   - NDT certificates and record shall be submitted to PPL/PEDEC.

13.6 Meetings

HSE meetings shall be convened on a monthly basis with an agenda to discuss the general issues about HSE, any change in the contractor, in the nature and risk of the activities, HSE training program, safety analysis of the activities, as well as investigation and analysis of the accidents. The Contractor's site manager and HSE Representative

shall attend such meetings and report to the Client's site manager and HSE representative.

Weekly HSE meeting shall be conducted with the presence of MSA/OEOC HSE & Rig Contractor HSE Supervisor.

13.7 List of Minimum Required Contents

As per below tables:

Table 13-1: Minimum Onshore Rig HSE Requirements

Minimum Onshore Rig HSE Requirements	
ITEM/DESCRIPTION	**Specification**
General Safety Equipment	
Muster station sign board	2
Warning Sign (H2S Danger, Danger zone...)	As needed.
MSDS For all Chemicals	As needed.
T- Card Board	1 Panel in entrance of Rig & 1 Panel in Camp
Wind Socks	3
Fall Arrestor	Minimum 6 sets with different length Install in (Monkey board, 2 for Monkey Board ladders, Crown, In mast for hanging block, Under Rig floor)
Bug Blower (Explosion proof)	Minimum 3 no. Set in (Rig Floor, Cellar area, Spare)
Explosion Proof Torches	Minimum 6 (Company man, Tool pusher, HSE, Medic, Maintenance crew, Mud Crew)
Rescue Equipment	As needed.
Security Well Marker	By MSA/OEOC HSE Concern
Derrick man's escape system I.e. REM chute	By MSA/OEOC HSE Concern

Emergency shower with eye wash to be situated at assigned positions	Minimum 3 sets (Rig floor, chemical matters storage and handling areas)
Portable eye wash set	Minimum 5 sets
Life Ring	2
Spill Kit	3
EX Radio	At least 7
First Aid Kits	Rig Floor, Medic room, Camp Boss, Client Office, Store, Camp Galley
Rig Site Training Materials	Required
Personnel Protective Equipment	
Coverall+ Safety Helmet + Safety boots for all personnel on location and for visitors.	As needed.
Safety Eye protection & Ear Protections (Ear muff & Ear Plug)	As needed.
Anti-Acid (Apron, Gloves, Boot, Face Shield, chemical suits) for all operational personnel	As needed.
Fire Fighting Equipment	
Fire extinguishers: "Powder-50 kg"	7(In Rig site) 4(In Camp site)
Fire extinguishers: "Powder-12 kg" for all fire potential areas.	By MSA/OEOC HSE Concern
Fire extinguishers "CO2- 9 KG" For all electrical fire potential areas.	By MSA/OEOC HSE Concern
Fire extinguishers "Foam-45 Liter" For fire potential areas where required.	4 No. wheeled (Set in Diesel tanks, Engine generators in Rig site and Camp site)
Fire extinguishers "ABC 20 lbs" for all fire potential areas where required.	By MSA/OEOC HSE Concern
Mobile Foam Package (Fiberglass Body-Z4 Inductor) + 1000 Liter FFFP 3% Foam + S4 Low expansion Nuzzle (B.S)	By MSA/OEOC HSE Concern
Fire hoses with (B.S) couplings placed around the location as per firefighting plan.	By MSA/OEOC HSE Concern
Fire blankets for galley, Welding & Cutting operations	As needed.
Fire Fighting Suits	Minimum 4 sets in each Rig site and Camp site

Water pump diesel, capable of charging the fire main in case of deep well failure.	1(Rig site)1(Camp site)
Medical Equipment	
Minimum requirements for clinic and ambulance and relative medical equipment in accordance with fallowing requirements: 1. IOGP publication, Standards for local medical support, Report No: 6.44/222 2. Managing health for field operations in oil and gas activities, IOGP Report Number 343 (Revision – Version 1/ October 2011)	Required
Stretcher	Scoop Stretcher (1No.) Basket Stretcher with lifting belts and hooks) (1No.) Long Back Board with spider straps and immobilizer) (1No.) Army Stretcher(2No.)
Ambulance	1No. (B Type)

Table 13-2: Minimum Rig-less HSE Requirements for South Azadegan Project

Minimum Rig-less HSE Requirements for South Azadegan Project	
ITEM/DESCRIPTION	**Specification**
General Safety Equipment	
Eye wash & safety Shower to be situated at assigned position	(1) chemical storage and handling area
Muster station Sign board	2No.
Warning Sign (H2S Danger, Danger zone, ...)	Required
MSDS For all Chemicals	Required
T- Card Board	Required
First Aid Kits	Hospital, All offices
Wind Socks	2No.
Air Fans (Explosion proof)	1No.
Explosion Proof Torches	Required
Rescue Equipment	Required
Spill kit	3
EX Radio	At least 6

Security Well Marker	Required
Rig Site Training Materials	Required
Personnel Protective Equipment	
Coverall + Safety Helmet + boots for all personnel on location and for visitors.	Required
Safety Eye protection & Ear Protections (Ear Plug)	Required
Anti-Acid (Apron, Gloves, Boot, Face Shield, chemical suits) for all operational personnel	Required
Fire Fighting Equipment	
fire extinguishers: "Powder-50 kg"	At least 3No.
Fire extinguishers: "Powder-12 kg" for all fire potential areas.	Required
Fire extinguishers "CO2- 9 KG" For all electrical fire potential areas.	Required
Fire extinguishers "ABC 20 lbs" for all fire potential areas where required.	Required
Fire hoses with (B.S) couplings placed around the location as per firefighting plan.	Required
Fire blankets for galley, Welding & Cutting operations	Required
Fire truck (10000 Liter Water-1000 Liter Foam) Or (Firefighting systems)	1No.
Fire Fighting Suits	As per fire fighters' number
***The following equipment recommended meanwhile the risk can be retained in case that the company risk assessment considering the risk is so far as reasonably foreseeable. The measures and precautions also need to be as low as reasonably practicable.**	
Medical Equipment	
Minimum requirements for clinic and ambulance and relative medical equipment in accordance with fallowing requirements: 1. IOGP publication, Standards for local medical support, Report No: 6.44/222 2. Managing health for field operations in oil and gas activities, IOGP Report Number 343 (Revision – Version 1/ October 2011)	Required
Stretcher	Scoop Stretcher (1No.) Back Board (1No.) Army Stretcher (2No.)
Ambulance	1(B Type)

13.8 Well Control

1. Installation Requirements of BOP Stack and Control System:

- The distance between BOP control lines and relief lines shall be not less than 1m. Steel Gap Bridge should be placed in such a way to protect lines from damage by vehicle crossing. Sundries shall not be packed on the control line shield, on which welding operations are forbidden.
- The air supply shall be separated from that of driller's console and directly connected to the main air supply system, and shall be equipped with liquid/gas separators. Excessive bending and pinching on air hoses are forbidden.
- Electric power to BOP control system shall be directly from the main switch of the power board, and controlled by independent switches.
- Accumulators shall be in good condition, charged at its rated value, and ready for work at any time.

2. Installation Requirements of Choke Lines and Kill Lines, and Relief Lines:

a) Beside the drilling spool, one manual and one HCR gate valves on the side of chock manifold and two manual gate valves on the side of kill manifold shall be equipped, and the manual gate valve close to the spool shall be always open.

b) The high-pressure co-flex hose shall be installed as straight as possible and fixed by anchor after installation.

c) Relief lines (burning line and bleed line):

- No field welding is permitted on relief lines and the connecting style of relief lines must be flanged.
- The arrangement of relief lines shall take wind directions, dwellings, roads, oil tank areas, power lines and other facilities into consideration.
- While the two lines go in the same trend, the distance between the lines shall be at least 0.3m, and the two lines shall be fixed independently.
- The outlets of bleed line connecting with choke manifold shall be in a safe area which is at least 300m. The bleed line on the kill manifold side connecting with drilling spool shall be prepared and connected when necessary (about 70m from the wellhead).
- At every 10m~15m, turning points, and outlets, the lines shall be supported and fixed by cement foundations with anchor bolts or ground anchors. Hanging

portions shall be supported firmly. If crossing the stumbling blocks such as brooks or ponds is needed, metal bridging supports are necessary.

- Pre-buried anchor bolts for cement foundations shall not be less than 20mm or 3/4" in diameter, and greater than 0.5m in length.

3. Pressure test value and requirements:

a) A pressure test shall be conducted successfully where there is no pressure decline in 5 minutes for a low-pressure test and a minimum of 10 minutes for a high-pressure test.

b) Prior to each high-pressure test, a low-pressure test with 200-300 psi shall be carried out to verify sealing capability at the circumstance of low pressure and avoid washing out sealing elements.

c) High pressure test should be to the rated working pressure of ram preventers or the rated working pressure of the wellhead, whichever is the lowest, or to the rating required in DRILLING ROGRAM. Variable bore rams should be initially pressure tested on the largest and smallest OD pipe sizes that may be used during the OPERATION. Annular preventers, with a joint of drill pipe installed, should be tested to 70% of their working pressure. The high test for Kelly cocks, kill lines and choke lines should be to their rated working pressures, each part in the choke manifold is tested respectively to its rated working pressures, the relief lines be pressure tested to no less than 1,450psi (10Mpa).

d) When casing cup tester is used, the above test pressure should not exceed 80% of the internal yield pressure (burst pressure) of the exposed casing.

e) The control system of BOP shall be pressure tested to 3,000psi (21Mpa) for its reliability before putting into use.

f) Except that the BOP control system is tested with hydraulic oil, other well control equipment shall be pressure tested by clear water.

g) Testing tools

- All required sizes of the plug type testers, and cup type testers should be available for pressure testing purpose.
- When the cup type tester(s) is used for pressure test, the connection of the cup testers has to be checked, to ensure it will stand the load during the high pressures test. For surface casing, the cup tester is to be set in middle of the first casing joint in the top, so the first casing connection (between wellhead and

casing head housing) below the well head is to be tested too while testing BOP and wellhead.

4. Pressure test frequency:

All ram type preventers, annular BOP, valves, choke and kill lines, choke manifold, Kelly valves and safety valves should be pressure tested at the following frequencies:

- Prior to installation where possible.
- After installation of wellhead and BOP assembly and prior to drilling.
- After the disconnection or repair of any pressure containment seal in the BOP stack, choke line, or choke manifold, but limited to the affected component.
- Prior to drilling into a confirmed or suspected high-pressure zone.
- At any time requested by the COMPANY REPRESENTATIVE.
- In any case, every one month or during the first trip after the one-month interval with a maximum interval of 45 days. This frequency is not required for shear rams. As a minimum, the shear rams are to be tested prior to resume drilling operation after 9-5/8" casing string has been set and adequate WOC.

5. Work Requirement of Well Control Equipment:

- With shut-in casing pressure lower than 1000psi, while snubbing with annular preventer, drill tools shall have 18-deg shoulder tool joints, and the snubbing velocity shall not be greater than 0.2m/s.
- After shut off the annular preventer or ram preventer, with the shut-in casing pressure not greater than 2000psi, drill tools are allowed to pull up and down with the velocity less than 0.2m/s, however, rotating or tool joint passing through are forbidden.
- While drilling through pay zone and thereafter, BOP function (close-open) test for the BOPs shall be performed regularly.
- Kill manifolds shall not be used for drilling fluid fill-in.
- The maximum allowed shut-in casing pressure shall be remarkably indicated on choke manifolds.
- All gate valves of well control manifolds shall be tagged with serial numbers, and be indicated with on/Off.

- In case shear rams were used, after shearing the drill pipe with shear rams, retrieve the upper sheared drill pipe, lower a drill pipe joint with TIW valve installed into the hole and close pipe rams on it to prevent the possibility of shear ram's failure.

6. Management of Well Control Equipment:

- Well control equipment used by the drilling crew shall be managed and operated by designated personnel with clear post responsibilities.
- Well site shall have specified component storeroom and rubber element storeroom with air conditioners respectively. The temperature of storerooms shall meet storage requirements of components and rubber elements.
- At least 1 set is available as back-up for every type ram assembly in the well site. Ram preventer side door seals and ram rubber, every type gasket and bolt etc. should be enough for backup.

7. Killing Method and Shut-in Method:

- Driller's method is recommended for well killing.
- The hard Shut-in method is recommended for shutting in the well in Fahliyan reservoir.

8. Well Control during drilling:

a) Shut in Procedure in While on Bottom

- Position Kelly (if in use) and tool joint so that tool joints are clear of sealing elements.
- Shut off pumps, check for flow.
- If flow is noted, close the annular preventer and open remote-controlled choke line valve (HCR).
- If the well has been closed using the annular preventer, check space out (pipe rams vs. toll joint relative position) and close pipe rams and ram locks
- Bleed off pressure between pipe rams and annular preventer (if possible).
- Record shut-in drill pipe and casing pressures.
- Record volume gain and mark pits.
- Weigh mud in suction pits.
- Check blowout preventer and manifold for leaks.
- Check flowline and choke exhaust for flow.
- Check accumulator pressure.

b) Shut in Procedure in While Tripping

- If a flow is detected, install full opening safety valve and close same.
- Position tool joints clear of the sealing elements and close annular preventer and open remote-controlled choke line valve (HCR).
- Make up grey valve (dart type inside BOP on full open safety valve). Then Make up Kelly or top drive (insert a pup joint or single between safety valve and top drive) and open safety valve.
- Record casing pressure.
- Check blowout preventer, stack, and manifold for leaks.
- Check valves for correct position.
- Check accumulator pressure.
- Begin stripping to bottom.

c) Shut in Procedure for Empty Wells

- Try to RIH with drill pipe as more as possible if hole condition permits, and the drill pipe must be installed a Kelly cock or safety valve.
- Close the annular preventer and open remote-controlled choke line valve (HCR).
- If the well has been closed using the annular preventer, check space out (pipe rams vs. tool joint relative position) and close pipe rams and ram locks
- Bleed off pressure between pipe rams and annular preventer (jf possible).
- Record casing pressure

9. Well Control During Running Casing:

The well control while running casing is important and critical, best practices shall be followed, including but not limited to:

- Change out pipe rams to casing rams prior to running casing into hydrocarbon bearing formation, if possible.
- The crossover between casing and drill pipe has to be available, and good practice is to make it up with a joint of DP and keep on V-door.

10. Shut in Procedure During Logging:

Cable shall be pull out of the hole immediately when overflow is found during logging. If pit gain exceeds the specified amount, the cable shall be cut immediately and treat the overflow in accordance with the shut-in procedure for empty wells. Continuing to pull the cable through the closed annular preventer is not recommended.

11. Well Control During Testing:

- No difference to while DRILLING OPERATION, since either there is testing string in the well (same as Shut-in procedure while on bottom), or tripping the string. (Same as shut-in procedure while tripping).
- Likely there is some crude in the testing string, and precaution has to be taken while circulating the crude out of hole during killing operations to ensure it is properly handled without causing environmental impact.
- To minimize the formation damage, the killing fluids used shall be weighted brine if possible, however in well control case when it occurs, killing well safely will take priority rather than protecting formation from damage on selection of killing fluids.

12. Well Control During Stimulation:

- Similar to during testing.
- For any residual or alive acid to be circulated out during killing operations, measures have to be taken to treat the returned acid properly to avoid personnel from getting hurt, and without causing any environmental damage.

13. Well Control Drills:

Drills shall be conducted only when hole conditions and other rig activities permit. Drills should be scheduled so that crews can practice drills while on bottom, while tripping in and while out of the hole. Each operating shift shall perform at least one BOP drill under different situations per month.

14. Basic Requirement of Well Control:

- For 13-5/8”, 10K BOP, BOP should be schemed as follows: upper rams 5”, middle rams are shear rams, lower rams is variable rams (2-7/8” to 5”). For easily connecting, armored fire-resistant 10000psi and ID 4-1/16" flexible hoses are recommended as kill/choke lines which connected with BOP directly.
- The BOP stack and choke manifold, kill line and choke line, HCR and manual gate valves, inside BOP (including kelly cock) have to be H2S trimmed and stand the working temperature from -10 degrees C (in case of doing N2 for DST) to the maximum up to 100 deg C (estimated highest temperature condition).
- The BOP equipment shall be visually inspected, function tested, and hydrostatically pressure tested during commissioning and Acceptance of Drilling Unit prior to Commencement Date, then upon installation on a wellhead, prior to drilling any hole section with a potential hydrocarbon bearing reservoir, after breaking any BOP connection (such as after installation of an

additional wellhead section after running casing), and at least every 14-21 days while the BOP is installed on the Well.

- CONTRACTOR shall submit the current well control procedures manual of CONTRACTOR to COMPANY for review and approval by COMPANY. Valid IWCF Well Control Certificates (or IADC equivalent) must be held by all tool pushers, Drillers (optional) at Drilling Supervisor level and Drillers and Assistant Drillers at Driller level, as further specified in CONTRACTOR Personnel HSE Requirements.
- The accumulator must be of sufficient capacity so that it is possible to close one (1) ram and the bag, open a ram and the bag, re-close a ram and the bag and retain 1200psi accumulator pressure.
- The "OPEN" and "CLOSED" operating positions of BOP activators of the rams, annular, and remote actuated valves must be clearly identified on the BOP master control panel and remote panels, in English in addition to the native language of the majority of the senior drilling crew.
- Shear-ram BOP activator must have a flap-cover to ensure only pre-meditated and deliberated activation and to prevent accidental WORK.
- All lines and connections in the choke and kill manifold, standpipe manifold, choke and kill lines and high-pressure mud lines should be welded hard-piping with weld-on FMC Weco "hammer-lug" 1502 unions (or Western equivalent, such as Anson) or weld-on API flanges. The well control system should not include any threaded screw-on hammer-lug unions, flanges, or other threaded screw-on connections.
- All adapter or spacer spools between the wellhead and BOP must be manufactured in accordance with API 6A, and must be complete with all required studs, nuts, and ring gasket seals for both lower and upper connections; minimum spare parts should include two (2) OEM spare studs complete with nuts and two spare OEM ring gasket seals for both bottom and top connections.
- There should be a wide-bore gas vent line from the "poor-boy" degasser; the line and line outlet must be securely anchored, the line outlet must be a safe distance from the well cellar, rig floor, choke manifold, and mud logging unit.
- Well Control Equipment Welding: All welds in the well control system must be NDT inspected and records maintained of all welding and weld inspections, including choke and kill manifold, standpipe manifold, mud pump discharge manifold, choke and kill lines.
- Well Control Rig Floor Information: The following must be posted on the rig floor (in the Driller's shack / "dog-house"): Well Shut-In Procedure, Well Information Sheet (showing casing and hole sizes, depths, and such like), Kill Sheet (including most recent slow circulation rate pressures, formation integrity

leak-off test information and MAASP, present mud weight, present drill string and well volumes, and such like).

- The derrick substructure clear height (rotary table to ground level) needs to be enough for fitting in above BOP configuration (including a tubing head and a bell nipple). This need to be double checked in rig tender process. According to the calculation of height of BOP stack, the substructure clear height is at least 9 m.
- In the case if drilling 6-1/8" hole becomes necessary, or using 3-1/2" drill pipe for the cleanup run (including drill 7" shoe track), then taper drilling string (5" DP + 3-1/2" DP) is to be run, so a pair variable ram be installed in the single ram preventer. The defect installing variable ram in single ram preventer is that the ram may be damaged and the string fall in hole bottom due to the ram hanging up too much weight when the drill pipe be cut. If the pipe ram be installed in the single ram preventer and the variable ram be installed in the double ram preventer, it would be very dangerous due to the variable ram failure in frequently using.

13.9 H2S Monitoring and Prevention Measure

As per PPL/PEDEC SOW South Azadegan oilfield contains H_2S bearing formations, precautions shall be taken in place especially in 12-1/4" & 8-1/2" hole section (for typical Fahliyan vertical well) while drilling, completion and well testing operation.
H_2S can be released any time while drilling, completion and well testing operation, therefore:

- The Rig shall be prepared for H_2S (Training, procedures and protective gear).
- Make sure everyone on the Rig should be made aware of the hazard presented by H_2S during their induction when they first arrive on the rig.
- The mud shall be treated to protect the drill pipe and the casing.
- All training program shall be recorded and reported.
- H_2S emergency drill must be conducted frequently.
- Adequate sensors and alarms must be installed for the scope of monitoring the H_2S releases.

14. H2S Monitoring & Control

NOTE: Final plan for H2S Monitoring and Control will be issued in due time by Service company instruction and Rig Contractor standards based on Oilfield safety and NIOC standards.
Minimum H2S Requirements for Rig site will be as following:

Table 14-1: Minimum Rig H2S Requirements for South Azadegan Project

Minimum Rig H_2S Requirements for South Azadegan Project	
Gas Detection Equipment	
H_2S Service cabin	1 No.
H_2S Unit location point	Comply with MSA/OEOC HSE
Main Monitoring Panel	1 No. - 8 channels
Max. number of channels	8 No.
Number of H_2S channels	4 No.
Measuring range	0 - 100
Adjustable alarm siren	Minimum 4 set (Min 120db voice load)
Measuring range	0-25 PPM
Adjustable alarm setting levels	2 No. (low 10 PPM and high 15 PPM)
Gas concentration reading displays	As sensors head number
Electrical Classification	Required
Certification	Required
Monitor panel installation site	H2S Unit
Automatic switch-on back up battery (min. 12 hr. power supply)	1
SO_2 Equipment	
Portable Single Detector	2 No.
H_2S Sensors	
H_2S Sensor installation points	Comply with MSA/OEOC HSE
Electrical Classification	For hazardous area
Certification	Required

Spare sensor head	2 No.
Response gas concentration	10 ppm low & 15 ppm High
LEL Sensors	
LEL Sensor installation points	Comply with MSA/OEOC HSE
Electrical Classification	For hazardous area
Certification	Required
Response gas concentration	20 % low & 40% High
Spare sensor head	Required
LEL Sensor installation points	Comply with MSA/OEOC HSE
Electrical Cables (connecting sensors with monitor panels)	
Suitable for hazardous area	as much as required
Electrical Classification	as much as required
Visual-Acoustic H_2S Alarms	
Low level visual alarms	Yellow flashing lights
High level visual alarms	Steady red lights
Low level acoustic alarms	Intermittent horn
High level acoustic alarms	Continuous horn
Above items suitable for hazardous area	Required
Number of alarm stations	5 No.
Alarm points location	comply with MSA/OEOC HSE
Calibration kit	At least 2 No.
PORTABLE H_2S MONITORING SYSTEM	
Hand pump colorimetric tubes H_2S detectors	1 No.
H_2S colorimetric tubes ranges	Req., comply with MSA/OEOC HSE
H_2S colorimetric tubes quantity	Req., comply with MSA/OEOC HSE
Personal electronic H2S detectors	Min. 6 No
Multi Gas Portable Detectors	
Quantity	4 (two main two spare)

Measuring range	0 – 100 ppm
Alarm setting levels	two
Certification	Required
Calibration kit	Required
Electrical Classification	For hazardous area
Certification	Required
Respiratory Protection Equipment	
Air Compressors	
Quantity	2 No. (Electrical- Diesel)
Approved for air breathing	Required
Gross weight	Required
Flow rate capacity	480 lit./min each
Charging pressure	3200
Filtration and purifier device	Specify approval
Certification	As per EN-132
Alarm for contaminating gases	Required
Auto shut down for contaminating gases	Required
Air compressor driving system	Electrical
Suitable for refilling bottles of air breathing apparatuses	Required
Air test kit	At least 1 No.
FIXED CASCADE AIR BREATHING SYSTEM	
The fixed cascade air breathing system complies with code type	Specify
Certification details	Required
No. of racks	At least 3*8
Dimensions (LxWxH) each rack	required
Bottles certified as pressure vessel	Required
Total air storage capacity	50 Liter

At max. working pressure of	3300 Psi
Air Pressure regulator	Req., victory Type
Safety/relief valve	Required
Low pressure alarm device	Required
Stainless steel piping/lines	Required
Air Breathing Delivery Stations	
Number of stations	At least 7 No.
Stations locations	Comply with MSA/OEOC HSE
Each Fixed Station Consisting of:	
Outlet manifold	4-man manifold 3 No & 6-man manifold 2 No
Connection outlets	Required
Hose lines	20(25 feet)20(15 feet)
PORTABLE AIR BREATHING APPARATUSES	
Self-Contained Breathing apparatuses (SCBA 45 min)	25
EEBA (20 min.)	20
Mask type	Full face (positive pressure)
Compliance with specifications	Comply w/ EN 137
Approval Authority	required
Air bottle working pressure	required
Air bottle certificate	required
Air regulator	required
Low Pressure audible alarm	Required
Suitable to be quick connected to cascade outlet manifold	Required
Stowage suitcase	Required
SCBA Radio Communication	Required
Spare Bottles for SCBA	At least 3
BREATHING APPARATUSES location points	comply with MSA/OEOC HSE

15. Requirement

15.1 8-1/2”” Hole Section

Table 15-1: 8-1/2" Hole Section Requirement

Item No.	Supplier	8-1/2” Hole Section Requirement Description	Quantity	Unit
1	MSA/OEOC	8-1/2" Milled Tooth Bit, 4-1/2" Reg Pin, IADC Code: 217	1	Ea.
2	MSA/OEOC	8-1/2 Bottom Bit, 4-1/2" Reg Pin, IADC Code: 537	1	Ea.
3	MSA/OEOC	8-1/2” PDC Bit, 4-1/2” Reg Pin, Matrix Body 13mm, 6 Blades, IADC Code: M423	1	Ea.
4	MSA/OEOC	8-1/2” PDC Bit, 4-1/2” Reg Pin, Matrix Body 16mm, 6 Blades, IADC Code:M433	1	Ea
4	MSA/OEOC	6-1/8” Bottom Bit, 3-1/2” Reg Pin, IADC Code: 537	1	Ea
5	MSA/OEOC	Nozzle (14/32”) W/ O-Ring & Snap Ring for 8-1/2” Tri-Cone Bit	2	Set
6	MSA/OEOC	Nozzle (15/32”) W/ O-Ring & Snap Ring for 8-1/2” Tri-Cone Bit	2	Set
7	MSA/OEOC	Nozzle (13/32”) for 8-1/2” PDC Bit W/ O-Ring	2	Set
8	MSA/OEOC	Nozzle (14/32”) for 8-1/2” PDC Bit W/ O-Ring	2	Set
9	MSA/OEOC	Nozzle (15/32”) W/ O-Ring & Snap Ring for 6-1/8” Tri-Cone Bit	2	Set
10	MSA/OEOC	Nozzle (16/32”) W/ O-Ring & Snap Ring for 6-1/8” Tri-Cone Bit	2	Set
11	Rig Cont.	Nozzle Installation & Removal Tool for Tri-Cone Bit	1	Ea.
12	MSA/OEOC	Nozzle Wrench for 8-1/2” PDC Bit	1	Ea.

Item No.	Supplier	8-1/2” Hole Section Requirement Description	Quantity	Unit
13	Rig Cont.	8-1/2” Tri-Cone Bit Breaker	1	Ea.
14	MSA/OEOC	8-1/2” PDC Bit Breaker	1	Ea.
15	Rig Cont.	6-1/8” Tri-Cone Bit Breaker	1	Ea.
16	Rig Cont.	8-1/2” Ring Gauge	1	Ea.
17	Rig Cont.	6-1/8” Ring Gauge	1	Ea.
18	Rig Cont.	8-1/2” Junk Sub, 4-1/2” Reg Pin-Box	1	Ea.
19	Rig Cont.	6-1/8” Junk Sub, 3-1/2” Reg Pin-Box	1	Ea.
20	Rig Cont.	6-1/8" Bit Sub, 4-1/2” Reg Box-Box	1	Ea.
21	Rig Cont.	4-3/4” Bit Sub, 3-1/2” Reg Box-Box	1	Ea.
22	Rig Cont.	6-1/2" Spiral Drill Collar, 4-1/2” Reg Pin-Box	12	Jt.
23	Rig Cont.	8-1/2" String Stabilizer (IBS, Spiral) 4-1/2” Reg Pin-Box	1	Ea.
24	Rig Cont.	6-1/2” Drilling Jar, 4-1/2” Reg Pin-Box	1	Ea.
25	Rig Cont.	XOS, 4-1/2” Reg Pin x 4-1/2” IF Box	1	Ea.
26	Rig Cont.	4-3/4” Spiral Drill Collar, 3-1/2” IF Pin-Box	18	Jt.
27	Rig Cont.	4-3/4” Drilling Jar, 3-1/2” IF Pin-Box	1	Ea.
28	Rig Cont.	XOS, 3-1/2” IF Pin x 4-1/2” IF Box	1	Ea.
29	Rig Cont.	5” HWDP, 49.7 PPF, 4-1/2” IF Pin-Box	9	Jt.

Item No.	Supplier	8-1/2" Hole Section Requirement Description	Quantity	Unit
30	Rig Cont.	5" DP, G-105/S-135, 19.5 PPF, 4-1/2" IF Pin-Box	As Required/1000	Mt
31	Rig Cont.	3-1/2" HWDP, 23.2 PPF, 3-1/2" IF Pin-Box	6	Jt.
32	Rig Cont.	3-1/2" DP, G-105, 13.3 PPF, 3-1/2" IF Pin-Box	± 400	Mt
33	Rig Cont.	9-5/8" CSG Scraper (43.5-53.5#), 4-1/2" Reg Pin-Box	1	Ea.
34	Rig Cont.	7" CSG Scraper (32-35#), 3-1/2" Reg Pin-Box	1	Ea.
35	Rig Cont.	Float Valve for 6-1/2" Bit Sub	2	Ea.
36	Rig Cont.	Float Valve for 4-3/4" Bit Sub	2	Ea.
37	Rig Cont.	Pipe Screen for inside 5" DP	3	Ea.
38	MSA/OEOC	X.O.S 7" , 29 ppf, 110SS, Premium Gas Tight Connection pin * 4-1/2" IF Box	1	Ea.
39	MSA/OEOC	Cement "G", 110 lb./sx.	As Req.	
40	MSA/OEOC	Cement Additives	As Req.	
41	MSA/OEOC	7" Float Shoe, 110ss, 29#, Premium Gas Tight Con.	1	Ea.
42	MSA/OEOC	7" Float Collar, 110ss, 29#, Premium Gas Tight Con.	1	Ea.
43	MSA/OEOC	7" LDC, 110ss, 29#, Premium Gas Tight Con.	1	Ea.

Item No.	Supplier	8-1/2” Hole Section Requirement Description	Quantity	Unit
44	PEDEC	7” Liner, 110ss, 29#, Premium Gas Tight Con.	900	Mt
45	MSA/OEOC	Hydraulic Set, Rotating Liner Hanger, 7", 29-32 ppf,110ss, Premium Gas Tight Conn., Tandem Cone, with Integral/Split Weight Set Top Packer, 10 ft PBR, Load Capacity 2,000 KN, Packer Seal Capacity 10K, Temp. 150 ºC, with Liner Setting Sleeve c/w DPOB and Setting Tool thread, to Set Inside 9-5/8", 47-53.5 ppf Casing.	1	Set
46	Service Co.	7” Liner Hanger Running Tool	1	Ea.
47	MSA/OEOC	Liner Wiper Plug for 7” Liner, 110ss, 29#, Premium Gas Tight Con.	1	Ea.
48	Service Co.	Polish Mill for 7” Liner, 110ss, 29#, Premium Gas Tight Con. PBR	1	Ea.
49	MSA/OEOC	5” DP, G-105/S-135, 19.5 PPF Pump Down Plug (Dart)	1	Ea.
50	MSA/OEOC	7" Coupling, 110ss, 29#, Premium Gas Tight Con.	3	Ea.

Item No.	Supplier	8-1/2” Hole Section Requirement Description	Quantity	Unit
51	MSA/OEOC	API Drift for 7” Liner, 110ss, 29#, Premium Gas Tight Con.	1	Ea.
52	MSA/OEOC	8-1/2” x 7” Bow Type Centralizer	0	Ea.
53	MSA/OEOC	8-1/2” x 7” Rotating Type Centralizer C/W Nail and Pins	15	Ea.
54	MSA/OEOC	S.N. Packer 7" 29 ppf, 110SS	1	Set as Req.
55	MSA/OEOC	9-5/8” x 7” Rigid Type Centralizer c/w Nails, Non-Welded	3	Ea.
56	MSA/OEOC	7” Stop Collar c/w Spiral Nails	30	Ea.
57	MSA/OEOC	API CSG Thread Lock Compound (Baker Lock)	3	Can.
58	MSA/OEOC	API CSG Thread Sealing Compound (Dope)	3	DR.
59	Service Co.	Drilling Fluid Service (Material, Lab. & Engineering)	As Required	
60	Service Co.	Mud Logging Service	As Required	
61	Service Co.	Waste Management Service	As Required	
62	Service Co.	Directional Drilling Service (Only MWD unit, MWD Tools, MWD Engineer)	As Required	

Item No.	Supplier	8-1/2" Hole Section Requirement Description	Quantity	Unit
63	Service Co.	Fishing service	As Required	
64	Service Co.	H2S Service	As Required	
65	Service Co.	Casing Running Service	As Required	
66	Service Co.	Cementing Service (Material, Pump Truck, Silo & Engineering)	As Required	
67	Service Co.	Well Head Service	As Required	
68	Service Co.	Liner Hanger Service	As Required	
69	Service Co.	Wireline Logging Service (Logging and Perforation)	As Required	
70	Service Co	Dry Testing & DST/TCP Service	As Required	
71	Service Co	Coiled Tubing Service	As Required	
72	Service Co	Stimulation & N2 Lifting Services	As Required	
73	Service Co	Completion Service	As Required	
74	Service Co	Slick line Service	As Required	

- **Qty & Spec to be confirmed with Contact's SOW.**
- **In case of any shortage or discrepancy, the Contract's SOW will be reference.**

15.2 6-1/8” Hole Section

Table 15-2: 6-1/8" Hole Section Requirement

Item No.	Supplier	6-1/2” Hole Section Requirement Description	Quantity	Unit
1	MSA/OEOC	6-1/8” PDC Bit, 3-1/2” Reg Pin, Matrix Body 16mm, 6 Blades, IADC Code: M433	1	Ea.
2	MSA/OEOC	6-1/8” Milled Tooth Bit, 3-1/2” Reg Pin, IADC Code: 217	1	Ea.
3	MSA/OEOC	Nozzle (15/32”) W/ O-Ring & Snap Ring for 6-1/8” Tri-Cone Bit	2	Set.
4	MSA/OEOC	Nozzle (16/32”) W/ O-Ring & Snap Ring for 6-1/8” Tri-Cone Bit	2	Set.
5	MSA/OEOC	Nozzle (13/32”) for 6-1/8” PDC Bit W/ O-Ring	2	Set.
6	MSA/OEOC	Nozzle (14/32”) for 6-1/8” PDC Bit W/ O-Ring	2	Set.
7	MSA/OEOC	Nozzle Wrench for 6-1/8” PDC Bit	12	Ea.
8	MSA/OEOC	6-1/8” PDC Bit Breaker	1	Ea.
9	Rig Cont.	6-1/8” Ring Gauge	1	Ea.
10	Rig Cont.	4-3/4” Junk Sub, 3-1/2” Reg Pin-Box	1	Ea.
11	Rig Cont.	6-1/8" Bit Sub, 4-1/2” Reg Box-Box	1	Ea.
12	Rig Cont.	4-3/4" Spiral Drill Collar, 4-1/2” Reg Pin-Box	12	Jt.
13	Rig Cont.	4-3/4” Drilling Jar, 4-1/2” Reg Pin-Box	1	Ea.
14	Rig Cont.	XOS, 4-1/2” Reg Pin x 4-1/2” IF Box	1	Ea.
15	Rig Cont.	5” HWDP, 49.7 PPF, 4-1/2” IF Pin-Box	9	Jt.
16	Rig Cont.	5” DP, G-105/S-135, 19.5 PPF, 4-1/2” IF Pin-Box	As Required/1000	Mt

Item No.	Supplier	6-1/2" Hole Section Requirement Description	Quantity	Unit
17	Rig Cont.	3-1/2" HWDP, 23.2 PPF, 3-1/2" IF Pin-Box	6	Jt.
18	Rig Cont.	3-1/2" DP, G-105, 13.3 PPF, 3-1/2" IF Pin-Box	± 400	Mt
19	Rig Cont.	7" CSG Scraper (29-32#), 3-1/2" Reg Pin-Box	1	Ea.
20	Rig Cont.	Float Valve for 6-1/8" Bit Sub	2	Ea.
21	Rig Cont.	Float Valve for 4-3/4" Bit Sub	2	Ea.
22	Rig Cont.	Pipe Screen for inside 5" DP	3	Ea.
23	MSA/OEOC	X.O.S 7" , 29 ppf, 110SS, Premium Gas Tight Connection pin * 4-1/2" IF Box	1	Ea.
24	MSA/OEOC	Cement "G", 110 lb./sx.	As Req.	
25	MSA/OEOC	Cement Additives	As Req.	
26	MSA/OEOC	4-1/2" Float Shoe, 110ss, 13.5#, Premium Gas Tight Con.	1	Ea.
27	MSA/OEOC	4-1/2" Float Collar, 110ss, 13.5#, Premium Gas Tight Con.	1	Ea.
28	MSA/OEOC	4-1/2" LDC, 110ss, 13.5#, Premium Gas Tight Con.	1	Ea.
29	PEDEC	4-1/2" Liner, 110ss, 13.5#, Premium Gas Tight Con.	480	Mt
30	MSA/OEOC	Hydraulic Set, Rotating Liner Hanger, 4-1/2", 13.5 ppf,110ss, Premium Gas Tight Conn., Tandem Cone, with Integral/Split Weight Set Top Packer, 10 ft PBR, Load Capacity 2,000 KN, Packer Seal Capacity 10K, Temp. 150 ºC, with Liner Setting Sleeve c/w DPOB and Setting Tool thread, to Set Inside 7", 29 ppf Casing.	1	Set
31	Service Co.	4-1/2" Liner Hanger Running Tool	1	Ea.
32	MSA/OEOC	Liner Wiper Plug for 4-1/2" Liner, 110ss, 13.5#, Premium Gas Tight Con.	1	Ea.

Item No.	Supplier	6-1/2” Hole Section Requirement Description	Quantity	Unit
33	Service Co.	Polish Mill for 4-1/2” Liner, 110ss, 13.5#, Premium Gas Tight Con. PBR	1	Ea.
34	MSA/OEOC	5” DP, G-105/S-135, 19.5 PPF Pump Down Plug (Dart)	1	Ea.
35	MSA/OEOC	4-1/2" Coupling, 110ss, 13.5#, Premium Gas Tight Con.	3	Ea.
36	MSA/OEOC	API Drift for 4-1/2” Liner, 110ss, 13.5#, Premium Gas Tight Con.	1	Ea.
37	MSA/OEOC	6-1/8” x 4-1/2” Bow Type Centralizer	0	Ea.
38	MSA/OEOC	6-1/8” x 4-1/2” Rotating Type Centralizer C/W Nail and Pins	8	Ea.
39	MSA/OEOC	S.N. Packer 4-1/2" 13.5 ppf, 110SS	1	Set as Req.
40	MSA/OEOC	7” x 4-1/2” Rigid Type Centralizer c/w Nails, Non-Welded	3	Ea.
41	MSA/OEOC	7” Stop Collar c/w Spiral Nails	16	Ea.
42	MSA/OEOC	API CSG Thread Lock Compound (Baker Lock)	3	Can.
43	MSA/OEOC	API CSG Thread Sealing Compound (Dope)	3	DR.
44	Service Co.	Drilling Fluid Service (Material, Lab. & Engineering)	As Required	
45	Service Co.	Mud Logging Service	As Required	
46	Service Co.	Waste Management Service	As Required	
47	Service Co.	Directional Drilling Service (Only MWD unit, MWD Tools, MWD Engineer)	As Required	
48	Service Co.	Fishing service	As Required	
49	Service Co.	H2S Service	As Required	

Item No.	Supplier	6-1/2" Hole Section Requirement Description	Quantity	Unit
50	Service Co.	Casing Running Service	As Required	
51	Service Co.	Cementing Service (Material, Pump Truck, Silo & Engineering)	As Required	
52	Service Co.	Well Head Service	As Required	
53	Service Co.	Liner Hanger Service	As Required	
54	Service Co.	Wireline Logging Service (Logging and Perforation)	As Required	
55	Service Co	Dry Testing & DST/TCP Service	As Required	
56	Service Co	Coiled Tubing Service	As Required	
57	Service Co	Stimulation & N2 Lifting Services	As Required	
58	Service Co	Completion Service	As Required	
59	Service Co	Slick line Service	As Required	

- **Qty & Spec to be confirmed with Contact's SOW.**
- **In case of any shortage or discrepancy, the Contract's SOW will be reference.**

16. Past Revision Corrections

Drilling Engineer							
1	**Well Information:** • Contingency Plan مربوط به نصب SN Packer در برنامه زمانبندی می بایست حذف گردد.	11	CM	• Contingency Planمربوط به نصب SN Packer در برنامه زمانبندی اولیه مورد تایید کارفرمای محترم، لحاظ گردیده بود لذا با توجه به ست کردن پکر رشته تکمیلی در آستری 7 اینچ، امکان حذف این مورد نمی‌باشد. • از برنامه زمان بندی حذف گردید.	بودن چاه و ایجاد Re-entryبا توجه به حفره جدید، راندن آستری 7 اینچ و سیمان کاری آن بر عهده پیمان کار می باشد، لذا در صورت سیمان کاری نامناسب یا هرگونه مشکل در لبه آستری عواقب بر عهده پیمان کار می باشد، بنابرین مدت زمان اضافی برای هر گونه مشکل در این عملیات لحاظ نمی گردد	CM	
2	**Time Breakdown:** • Contingency plan #6 از برنامه می بایست حذف گردد	66	CM	• از برنامه حذف گردید.	بودن چاه و ایجاد Re-entryبا توجه به حفره جدید، راندن آستری 7 اینچ و سیمان کاری آن بر عهده پیمان کار می باشد، لذا در صورت سیمان کاری نامناسب یا هرگونه مشکل در لبه آستری عواقب بر عهده پیمان کار می باشد، بنابرین مدت زمان اضافی برای	CM	

					هر گونه مشکل در این عملیات لحاظ نمی گردد		
3	**Time Breakdown:** • در شروع حفاری هر دو حفره FIT می بایست لحاظ گردد.		CM	• در جلسات برگزار شده در شرکت متن به تاریخ های دوم و دهم تیر ماه 1403علی رغم وجود FIT در برنامه پیشنهادی این شرکت،اخذ FIT توسط کارفرمای محترم حذف گردید لذا با توجه به نهایی شدن برنامه زمانبندی در جلسه مذکور امکان اخذ FIT بدون منظور کردن زمان وجود ندارد. • علی رغم دستور کارفرمای طرح در جلسه مشترک مبنی بر عدم نیاز به انجام FIT، این تست در برنامه گنجانده شد.	منوط به تایید کارفرما	AN	
4	**Time Breakdown:** • جدول Contingency plan از برنامه حذف گردد.	67	CM	• از برنامه زمان بندی حذف گردید.	بودن چاه و ایجاد Re-entryبا توجه به حفره جدید، راندن آستری 7 اینچ و سیمان کاری آن بر عهده پیمان کار می باشد، لذا در صورت سیمان کاری نامناسب یا هرگونه مشکل در لبه آستری عواقب بر عهده پیمان کار می باشد، بنابرین مدت زمان اضافی برای		

					هر گونه مشکل در این عملیات لحاظ نمی گردد		
5	**Time Breakdown:** • Dry test برای لبه آستری "4 1/2 لحاظ گردد.		CM	• با توجه به قرار گیری لبه این آستری زیر پکررشته تکمیلی، موضوع در جلسه با شرکت متن مطرح گردید، لذا جهت کاهش هزینه و زمان پروژه با تأیید کارفرما حذف گردید.	منوط به تایید کارفرما	AN	
6	**Time Breakdown:** • جدول Contingency plan از برنامه حذف گردد.	67	CM	• از برنامه زمان بندی حذف گردید.	بودن چاه و ایجاد Re-entryبا توجه به حفره جدید، راندن آستری 7 اینچ و سیمان کاری آن بر عهده پیمان کار می باشد، لذا در صورت سیمان کاری نامناسب یا هرگونه مشکل در لبه آستری عواقب بر عهده پیمان کار می باشد، بنابرین مدت زمان اضافی برای هر گونه مشکل در این عملیات لحاظ نمی گردد	CM	
7	**Sequence of Operation (8 ½”):** • Contingency plan از برنامه حذف گردد.	71	CM	• از برنامه زمان بندی حذف گردید..	بودن چاه و ایجاد Re-entryبا توجه به حفره جدید، راندن آستری 7 اینچ و سیمان کاری آن بر عهده پیمان کار می باشد، لذا در صورت سیمان کاری نامناسب یا هرگونه	CM	

					مشکل در لبه آستری عواقب بر عهده پیمان کار می باشد، بنابرین مدت زمان اضافی برای هر گونه مشکل در این عملیات لحاظ نمی گردد		
8	**Casing/Liner Characterize Overview:** • با توجه به تاکید کارفرما (نمایندگان فنی کارفرما محترم شرکت مهندسی توسعه نفت) مطابق جلسه فنی مورخ 1403 /04/10 استفاده و تایید از آستری 7 اینچ و متراژ لازمه منوط به تایید کارفرما می باشد • پوندژ و گرید آستری "1/2 4 و "7 با شرح کار مغایرت دارد، اصلاح شود. • قابل ذکر است در فرایند QC کالا مطابقت کالا و نوع کانکشن پیشنهادی پیمانکار با شرایط درخواستی بررسی خواهد شد	150	CM	• انجام شد. • پونداژ ها مطابق موجودی انبار کارفرمای محترم می باشد. • کالا ها مطابق موجودی انبار کارفرما می باشد.	منوط به تایید کارفرما	AN	
9	**Table of comment:** • جدول کامنت های ارسالی به انتهای برنامه اضافه شود.						
Fluid							

10	PCF-103چگالی سیال حفاری در حفره" 5 7/8 از 100-106 اصلاح گردد.PCF 100 به در صفحه ی 101 نیز تصحیح گردد.		CM	• اصلاح شد.			